

Process Costing

24 problems · workbook pages 8 to 16

What this module actually is

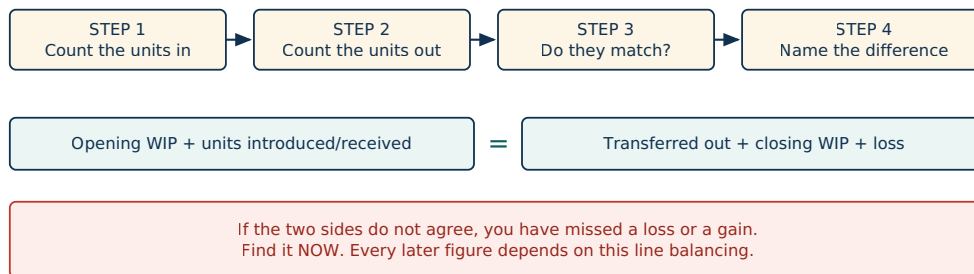
Process costing is used where a product is made in a **continuous chain of stages** — sugar, chemicals, oil, paper, cement. You cannot point at one unit and say “this one cost ₹40”, because every unit goes through the same pipe. So we do the only thing possible: **total the cost of a process and divide by the units it produced**. That single sentence is the whole module.

Everything else in these 24 problems is one of four complications layered on top of that idea:

Complication	What changes	Problems
Nothing — plain chain	Total each process, divide by units, carry the total forward to the next process.	1-4
Some units are lost	Part of the input disappears (evaporation, spoilage). Expected loss is <i>normal</i> and its cost is absorbed by the good units. Anything worse is <i>abnormal loss</i> ; anything better is <i>abnormal gain</i> . Both are taken out of the process at the good-unit rate.	5-9
Two or more products emerge	One process throws out several saleable products. The joint cost has to be split between them — usually by working <i>backwards</i> from selling price.	10-14
Each process adds profit	A process sells to the next at cost + profit. The profit inside unsold stock is not real yet, so a <i>reserve for unrealised profit</i> must be removed.	15-16
Some units are half-finished	Closing work-in-progress is 60% done. You convert part-finished units into <i>equivalent</i> finished units before dividing.	17-24

The one habit that solves every problem in this module

Before you write a single figure, do this on your rough sheet. It takes forty seconds and it is the difference between a full-marks answer and a mess.



The units reconciliation. Write it for every process, every time.

THE THREE RATE FORMULAS — THIS IS ALL THE ARITHMETIC THERE IS

$$\text{Cost per unit (no loss)} = \frac{\text{Total cost of the process}}{\text{Units produced}}$$

$$\text{Cost per good unit (with loss)} = \frac{\text{Total cost} - \text{Scrap value of normal loss}}{\text{Input units} - \text{Normal loss units}}$$

Learn this one properly. Normal loss is removed from BOTH the top and the bottom. Abnormal loss and abnormal gain are then valued at this same rate — never at a different one.

$$\text{Cost per unit (part-finished stock)} = \frac{\text{Cost of that element}}{\text{Equivalent units of that element}}$$

Material, labour and overhead each get their own rate, because they are each at a different stage of completion.

Q1 Three processes, overhead on wages

Plain chain · p.8

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

There are three distinct processes A, B and C. During the four-week period 1,000 units are produced and the following information is available.

Particulars	Process A	Process B	Process C
Direct materials	2,000	1,000	—
Direct wages	1,500	700	800
Direct expenses	300	100	—

The indirect production cost were ₹4,500 and these are to be apportioned on the basis of direct wage cost. Prepare the process accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

Three things to notice, and they decide the whole answer:

- **“1,000 units are produced”** — one figure for all three processes. So there is **no loss anywhere**. 1,000 units go in, 1,000 come out of every process. That means no normal loss working, no abnormal loss — the easiest kind.
- **“apportioned on the basis of direct wage cost”** — the question hands you the basis. You do not have to guess it. Whenever a question gives you a lump overhead and a basis, your first working note is always the same: find the ratio, then split.
- **Process C has no materials and no direct expenses**. Do not panic and do not leave the row out — write a dash. An examiner wants to see that you looked and found nothing.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Add up the given basis across all processes (here: total direct wages).
2. Overhead rate = Total indirect cost ÷ Total basis. Keep it as a plain number or a percentage — whichever divides cleanly.
3. Multiply each process's basis by that rate. **Check the three pieces add back to the original lump sum** before you go on.
4. Open Process A. Debit its own four costs. Total it. That total transfers out to B.
5. Open Process B. **First debit is the transfer from A**, then B's own costs. Total, transfer to C.
6. Open Process C the same way. Its total goes to Finished Stock.
7. Divide each process total by 1,000 to show cost per unit. The last one is the answer the examiner is looking for.

WORKING NOTES

W1 Apportionment of indirect production cost

Total direct wages = 1,500 + 700 + 800 = **₹3,000**

Overhead rate = ₹4,500 ÷ ₹3,000 = **1.5 times the wages** (i.e. 150% of direct wages)

Process	Direct wages	× 1.5	Overhead absorbed
A	1,500	1,500 × 1.5	2,250
B	700	700 × 1.5	1,050
C	800	800 × 1.5	1,200
Total	3,000		4,500

Cross-check: 2,250 + 1,050 + 1,200 = 4,500 equals the indirect cost given. If this does not tally, stop and re-divide.

Process A Account

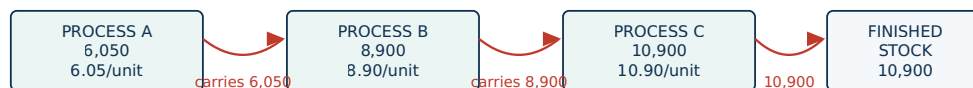
Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Direct materials	1,000	—	2,000	By Transfer to Process B 6,050 ÷ 1,000 units = ₹6.05 per unit	1,000	6.05	6,050
To Direct wages	—	—	1,500				
To Direct expenses	—	—	300				
To Indirect production cost W1: 1,500 × 1.5	—	—	2,250				
Total	1,000	—	6,050	Total	1,000	—	6,050

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A ← total of Process A a/c	1,000	6.05	6,050	By Transfer to Process C $8,900 \div 1,000 \text{ units} = ₹8.90 \text{ per unit}$	1,000	8.90	8,900
To Direct materials	—	—	1,000				
To Direct wages	—	—	700				
To Direct expenses	—	—	100				
To Indirect production cost W1: 700×1.5	—	—	1,050				
Total	1,000	—	8,900	Total	1,000	—	8,900

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B ← total of Process B a/c	1,000	8.90	8,900	By Transfer to Finished Stock $10,900 \div 1,000 = ₹10.90 \text{ per unit}$	1,000	10.90	10,900
To Direct materials	—	—	—				
To Direct wages	—	—	800				
To Direct expenses	—	—	—				
To Indirect production cost W1: 800×1.5	—	—	1,200				
Total	1,000	—	10,900	Total	1,000	—	10,900



Each box adds only its OWN cost; the arrow carries the whole running total forward.

The chain. Notice the cost per unit can only go up as you move right.

THE TRAP / WHERE MARKS ARE LOST

- Students write Process B's cost per unit as $2,850 \div 1,000 = ₹2.85$, using only B's own costs. Wrong. The unit leaving B has A's cost inside it too. **The transfer-in is a debit like any other.**
- Forgetting to show **units** in the account. Marks are given for the units column, and it is free — every figure is 1,000.
- Apportioning the overhead on the wrong basis (materials, or equally between three processes). The question said direct wage cost. Read it, underline it, use it.

FINAL ANSWER

Cost per unit — Process A	₹ 6.05
Cost per unit — Process B	₹ 8.90
Cost per unit — Process C (final)	₹ 10.90
Total cost of 1,000 units	₹ 10,900

State the final cost per unit in words as your closing line: "The cost of production is ₹10,900 for 1,000 units, i.e. ₹10.90 per unit."

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details show the cost of the three processes of manufacturing. The production of each process is passed on to the next till completion.

Particulars	Process A	Process B	Process C
Wages and materials	60,800	24,000	58,500
Works overheads	11,200	10,500	12,000
Production in units	72,000	75,000	96,000
Opening stock (units from preceding process on 1st July)	—	8,000	33,000
Closing stock (units from preceding process on 31st July)	—	2,000	11,000

READ THE QUESTION FIRST — WHAT TO SPOT

This problem looks like Q1 but it is not, and one phrase tells you so: “**units from preceding process**”.

- The opening and closing stock in Process B is **not half-finished work**. It is a pile of finished A-units waiting to be worked on. So it is valued at **A's rate**, not B's. Same for C — its stock is B-units, valued at B's rate.
- Because there is stock lying on both sides, the units B actually *consumed* are not the units it *received*. You must work the consumption out.
- “Production in units” is the **output** of that process. Divide by this figure to get the rate — never by the units consumed.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. **Process A first** — it has no stock, so rate = total cost ÷ production.
2. For B and C, find the units consumed: **Opening stock + Received from previous process – Closing stock**.
3. Value those consumed units at the **previous process's rate**. That is the transfer-in cost.
4. Add the process's own wages, materials and works overheads.
5. Rate for this process = that grand total ÷ **its own production in units**.
6. Now write the account. Debit side: opening stock, transfer-in, own costs. Credit side: closing stock (at the *previous* rate) and transfer-out (at *this* rate). The two sides must balance to the rupee.

WORKING NOTES

W1 Process A rate (no stock, so straight division)

Total cost = 60,800 + 11,200 = **₹72,000**

Rate = 72,000 ÷ 72,000 units = **₹1.00 per unit**

W2 Units consumed by Process B and Process C

Particulars	Process B	Process C
Opening stock of previous-process units	8,000	33,000
Add: Received from previous process	72,000	75,000
Units available	80,000	1,08,000

Particulars	Process B	Process C
Less: Closing stock of previous-process units	(2,000)	(11,000)
Units consumed in the process	78,000	97,000

Received by B = A's production 72,000. Received by C = B's production 75,000.

W3 Rate for Process B

Cost of units consumed = $78,000 \times ₹1.00 = 78,000$
Add own cost = $24,000 + 10,500 = 34,500$
Total = **₹1,12,500**
Rate = $1,12,500 \div 75,000$ units produced = **₹1.50 per unit**

W4 Rate for Process C

Cost of units consumed = $97,000 \times ₹1.50 = 1,45,500$
Add own cost = $58,500 + 12,000 = 70,500$
Total = **₹2,16,000**
Rate = $2,16,000 \div 96,000$ units produced = **₹2.25 per unit**

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Wages and materials	72,000	—	60,800	By Transfer to Process B W1: rate ₹1.00	72,000	1	72,000
To Works overheads	—	—	11,200				
Total	72,000	—	72,000	Total	72,000	—	72,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock 8,000 A-units @ ₹1.00 ← A's rate	8,000	1	8,000	By Closing stock 2,000 A-units @ ₹1.00 — still A's rate	2,000	1	2,000
To Transfer from Process A	72,000	1	72,000	By Transfer to Process C W3: 75,000 @ ₹1.50	75,000	1.50	1,12,500
To Wages and materials	—	—	24,000	(3,000 units consumed but not yielded — process shrinkage, no scrap value given)	3,000	—	—
To Works overheads	—	—	10,500				
Total	80,000	—	1,14,500	Total	80,000	—	1,14,500

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock <i>33,000 B-units @ ₹1.50 ← B's rate</i>	33,000	1.50	49,500	By Closing stock <i>11,000 B-units @ ₹1.50</i>	11,000	1.50	16,500
To Transfer from Process B	75,000	1.50	1,12,500	By Transfer to Finished Stock <i>W4: 96,000 @ ₹2.25</i>	96,000	2.25	2,16,000
To Wages and materials	—	—	58,500	<i>(1,000 units shrinkage)</i>	1,000	—	—
To Works overheads	—	—	12,000				
Total	1,08,000	—	2,32,500	Total	1,08,000	—	2,32,500



Why Process B's stock is valued at A's rate, not B's.

THE TRAP / WHERE MARKS ARE LOST

- Valuing B's opening/closing stock at B's own rate of ₹1.50. It has not been through B yet — it is worth ₹1.00. This single error breaks both sides of the account.
- Dividing by units consumed (78,000) instead of units produced (75,000). The rate must describe a unit that came out.
- Panicking when the units do not balance. Here 78,000 are consumed but only 75,000 emerge. That gap is real shrinkage. No scrap value is given, so it carries no cost — simply show it and move on. Do not invent a normal-loss working the question never asked for.

FINAL ANSWER

Cost per unit — Process A	₹ 1.00
Cost per unit — Process B	₹ 1.50
Cost per unit — Process C (final)	₹ 2.25
Cost transferred to Finished Stock	₹ 2,16,000

Both B and C accounts balance exactly — 1,14,500 and 2,32,500. Always show the totals; a balancing account is self-proving.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information relating to an article which undergoes three processes for manufacture, show the cost of each process and the cost per article.

Particulars	Process A	Process B	Process C
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250

The indirect expenses amounted to ₹21,250 and the number of articles produced during the month was 240 units.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same shape as Q1 — one output figure (240 articles), so **no losses**.
- **The basis for splitting the indirect expenses is NOT stated.** This is deliberate. The accepted convention when no basis is given is to apportion on **direct labour**. Say so in one line in your answer — “indirect expenses apportioned on the basis of direct labour in the absence of any other basis” — and you protect the marks.
- Test whether your assumed basis is right by checking the split comes out in round figures. Here labour gives exactly 25%. That is the examiner confirming your choice.
- Two answers are wanted: **cost of each process** and **cost per article**. Give both, clearly labelled.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the chosen basis — labour: 20,000 + 50,000 + 15,000.
2. Rate = 21,250 ÷ that total. Express as a percentage if it is clean.
3. Split, then cross-check the pieces add back to 21,250.
4. Build the three process accounts in a chain, transferring the running total.
5. Divide each process total by 240 to get cost per article at that stage.
6. Final line: cost per article = Process C total ÷ 240.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: direct labour)

Total labour = 20,000 + 50,000 + 15,000 = **₹85,000**

Rate = 21,250 ÷ 85,000 = 0.25 = **25% of direct labour**

Process	Labour	× 25%	Indirect expenses
A	20,000	20,000 × 0.25	5,000
B	50,000	50,000 × 0.25	12,500
C	15,000	15,000 × 0.25	3,750
Total	85,000		21,250

Cross-check: 5,000 + 12,500 + 3,750 = 21,250 — and every figure is a round number, which confirms labour was the intended basis.

Statement of Cost of Each Process and Cost per Article

Particulars	Process A	Process B	Process C
Cost of previous process (transferred in)	—	69,000	1,62,000
Materials consumed	37,500	12,500	50,000
Labour	20,000	50,000	15,000
Direct expenses	6,500	18,000	6,250
Indirect expenses (W1)	5,000	12,500	3,750
Total cost of the process	69,000	1,62,000	2,37,000
Cost per article (÷ 240)	287.50	675.00	987.50

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials consumed	240	—	37,500	By Transfer to Process B $69,000 \div 240 = ₹287.50$	240	287.50	69,000
To Labour	—	—	20,000				
To Direct expenses	—	—	6,500				
To Indirect expenses W1	—	—	5,000				
Total	240	—	69,000	Total	240	—	69,000

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	240	287.50	69,000	By Transfer to Process C $1,62,000 \div 240 = ₹675.00$	240	675	1,62,000
To Materials consumed	—	—	12,500				
To Labour	—	—	50,000				
To Direct expenses	—	—	18,000				
To Indirect expenses W1	—	—	12,500				
Total	240	—	1,62,000	Total	240	—	1,62,000

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	240	675	1,62,000	By Transfer to Finished Stock $2,37,000 \div 240 = ₹987.50$	240	987.50	2,37,000
To Materials consumed	—	—	50,000				
To Labour	—	—	15,000				
To Direct expenses	—	—	6,250				
To Indirect expenses W1	—	—	3,750				
Total	240	—	2,37,000	Total	240	—	2,37,000

WHY THIS WORKS

The statement and the three accounts say exactly the same thing. If the question says “show the cost of each process” the statement alone is enough; if it says “prepare process accounts” you must draw the accounts. When in doubt give the statement first — it is faster to write and it makes the accounts almost automatic.

FINAL ANSWER

Cost of Process A ₹ 69,000

Cost of Process B ₹ 1,62,000

Cost of Process C ₹ 2,37,000

Cost per article ₹ 987.50

Q4 By-product and residue

Chain + by-product · p.8-9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A particular brand of chemical passed through three processes during the week ended 15th January. 600 tins were produced. The following information is disclosed by each process.

Particulars	Process A	Process B	Process C
Materials	4,000	2,000	1,500
Labour expenses	3,000	2,500	2,300
Direct expenses	6,000	200	500
Cost of tins	—	—	2,030
Packing charges	—	—	325

The indirect expenses were ₹1,600. The by-product is sold for ₹240 (Process B). The residue was sold for ₹125.50 (Process C).

Prepare the process A/c with cost of production of the product per tin.

READ THE QUESTION FIRST — WHAT TO SPOT

- Again 600 tins throughout — **no process loss**.
- Two new items: a **by-product sold for 240** and a **residue sold for 125.50**. Both are **deducted** from the cost of the process in which they arise. They are not income and they do not go to Profit & Loss — they reduce the cost that the good output has to carry.
- “Cost of tins” and “packing charges” belong to Process C only. They are ordinary debits — do not treat them as anything special.
- The indirect expense basis is again not stated, so apportion on labour and say so.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Apportion the indirect expenses on labour (state the assumption).
2. Build Process A normally.
3. In Process B, debit everything, then **credit the by-product sale of 240**. The amount that transfers to C is the total *after* that deduction.
4. In Process C, debit everything including tins and packing, then **credit the residue sale of 125.50**.
5. Cost of production = final balance of Process C. Divide by 600 tins.

WORKING NOTES

W1 Apportionment of indirect expenses (basis: labour)

Total labour = 3,000 + 2,500 + 2,300 = ₹7,800

Rate = $1,600 \div 7,800 = 0.205128$ per rupee of labour

Process	Labour	Working	Indirect expenses
A	3,000	$3,000 \times 0.205128$	615.38
B	2,500	$2,500 \times 0.205128$	512.82
C	2,300	$2,300 \times 0.205128$	471.80
Total	7,800		1,600.00

The rate does not divide neatly here, so carry two decimals and force the last figure to make the total exactly 1,600. Write “difference of ₹0.01 adjusted in Process C” if you like — examiners accept it and it shows you checked.

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	600	—	4,000	By Transfer to Process B	600	22.69	13,615.38
To Labour expenses	—	—	3,000				
To Direct expenses	—	—	6,000				
To Indirect expenses W1	—	—	615.38				
Total	600	—	13,615.38	Total	600	—	13,615.38

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	600	22.69	13,615.38	By Sale of by-product deducted from cost, NOT income	—	—	240
To Materials	—	—	2,000	By Transfer to Process C $18,828.20 - 240 = 18,588.20$	600	30.98	18,588.20
To Labour expenses	—	—	2,500				
To Direct expenses	—	—	200				
To Indirect expenses W1	—	—	512.82				
Total	600	—	18,828.20	Total	600	—	18,828.20

Process C Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process B	600	30.98	18,588.20	By Sale of residue deducted from cost	—	—	125.50
To Materials	—	—	1,500	By Cost of production transferred $25,715.00 - 125.50 = 25,589.50$	600	42.65	25,589.50
To Labour expenses	—	—	2,300				
To Direct expenses	—	—	500				
To Cost of tins	—	—	2,030				
To Packing charges	—	—	325				
To Indirect expenses W1	—	—	471.80				
Total	600	—	25,715	Total	600	—	25,715

THE TRAP / WHERE MARKS ARE LOST

- Adding the by-product sale to the credit of Profit & Loss instead of deducting it inside the process. In a **by-product of small value** the standard treatment is to credit the process. That is what “(Process B)” in the question is telling you.
- Deducting the residue from Process B or from the total instead of from Process C. The question names the process for each — use it.
- Rounding the per-tin figure too early. Carry the paise through and round only at the last line.

FINAL ANSWER

Total cost of production ₹ **25,589.50**

Number of tins **600**

Cost of production per tin ₹ **42.65**

Closing line: “Cost of production is ₹25,589.50 for 600 tins, i.e. ₹42.65 per tin.”

Q5 Normal loss and abnormal loss

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare a Process Account and Abnormal Loss Account from the following information.

Particulars	
Input of raw material	1,000 units @ ₹20 per unit
Direct material	₹4,200
Direct wages	₹6,000
Production overheads	₹6,000
Actual output transferred to Process II	900 units
Normal loss	5%
Value of scrap per unit	₹8

READ THE QUESTION FIRST — WHAT TO SPOT

This is the first problem with a loss, and it is the template for Q5–Q9. Four figures decide everything:

- **Input 1,000 units.** Normal loss is a percentage *of input* — so 5% of 1,000 = 50 units. Not 5% of output.
- **Expected output = 1,000 – 50 = 950 units.** Write this down. It is the figure you compare the actual against.
- **Actual output is 900**, which is *less* than expected 950. So 50 units have gone missing beyond what was allowed — that is **abnormal loss**. If actual had been more than 950 it would have been abnormal gain.
- **Scrap is worth 8 per unit.** Normal loss units are sold as scrap, so they do carry a value — and that value is removed from the cost before you divide.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits — every cost that entered the process.
2. Normal loss in **units** = given % × input units. Normal loss in **rupees** = those units × scrap rate.
3. Expected output = Input – Normal loss units.
4. Compare actual output with expected output. Shortfall = abnormal loss; excess = abnormal gain.
5. **Cost per good unit** = (Total cost – Scrap value of normal loss) ÷ (Input units – Normal loss units). Show this fraction in full.
6. Value the transfer-out AND the abnormal loss at that **same** rate.
7. Write the process account — it must balance. Then open the Abnormal Loss account: debit the process value, credit the scrap realised, and the balance goes to Profit & Loss.

WORKING NOTES

W1 Total cost debited to the process

Raw material 1,000 × ₹20 = 20,000
Direct material = 4,200
Direct wages = 6,000
Production overheads = 6,000
Total = ₹36,200

W2 The units reconciliation

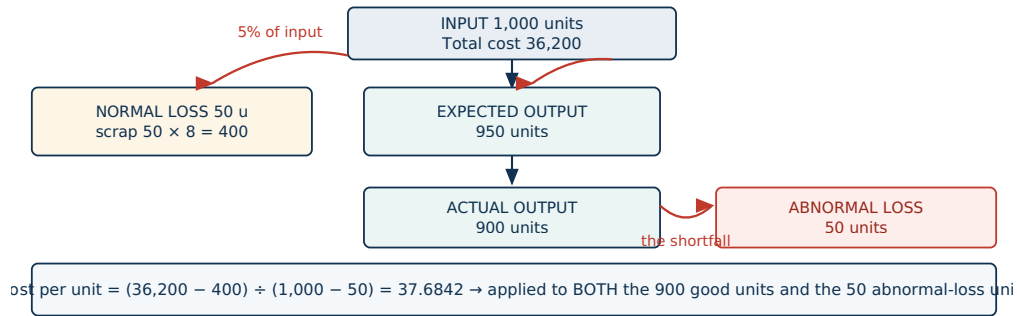
Normal loss = 5% of 1,000 = **50 units**, scrap value 50 × ₹8 = **₹400**
Expected output = 1,000 – 50 = **950 units**
Actual output = 900 units
Abnormal loss = 950 – 900 = **50 units**

W3 Cost per good unit — the formula that matters most

$$\text{₹36,200} - \text{₹400} \div 1,000 - 50 = \text{₹35,800} \div 950 = \text{₹37.6842 per unit}$$

Normal loss is taken out of the top (its scrap value) and out of the bottom (its units). That is how the cost of the normal loss gets loaded onto the good units.

Transferred out 900 × 37.6842 = **₹33,915.79**
Abnormal loss 50 × 37.6842 = **₹1,884.21**



Normal loss is expected and free. Abnormal loss is the shortfall against expectation.

Process I Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Raw material	1,000	20	20,000	By Normal loss 50 units × ₹8 scrap	50	8	400
To Direct material	—	—	4,200	By Abnormal loss W3: 50 × 37.6842	50	37.68	1,884.21
To Direct wages	—	—	6,000	By Transfer to Process II W3: 900 × 37.6842	900	37.68	33,915.79
To Production overheads	—	—	6,000				
Total	1,000	—	36,200	Total	1,000	—	36,200

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c ← from the process account	50	37.68	1,884.21	By Sale of scrap 50 units × ₹8 — abnormal loss units are also sold as scrap	50	8	400
Total	50	—	1,884.21	By Profit & Loss A/c balance — the real loss to the business	—	—	1,484.21
				Total	50	—	1,884.21

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process I A/c	50	8	400	By Cost Ledger Control A/c (cash from scrap)	50	8	400
Total	50	—	400	Total	50	—	400

THE TRAP / WHERE MARKS ARE LOST

- Dividing by 1,000 instead of 950. You would get ₹35.80 and every later figure would be wrong. **The denominator is always input minus normal loss units.**
- Forgetting to deduct the 400 scrap value from the numerator.
- Valuing the abnormal loss at the scrap rate of 8. No — abnormal loss is valued at the **good-unit rate**. It only realises 8 when sold, and the difference is the loss you take to Profit & Loss.
- Taking the whole 1,884.21 to Profit & Loss. Only the **net** figure of 1,484.21 goes there, after crediting the 400 the scrap actually fetched.

FINAL ANSWER

Cost per good unit **₹ 37.6842**

Transferred to Process II (900 units)	₹ 33,915.79
Abnormal loss (50 units, at cost)	₹ 1,884.21
Scrap realised on abnormal loss	₹ 400.00
Net abnormal loss charged to Profit & Loss	₹ 1,484.21

Note how the Process account, the Abnormal Loss account and the Normal Loss account all close cleanly. Show all three unless the question asks for fewer.

Q6 Loss, gain and neither — all in one

Loss & gain · p.9

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The product of a company passes through 3 distinct processes. The following information is obtained from the accounts for the month ending January 31, 2024.

Particulars	Process A	Process B	Process C
Direct material	7,800	5,940	8,886
Direct wages	6,000	9,000	12,000
Production overheads	6,000	9,000	12,000

3,000 units @ ₹3 each were introduced to Process I. There was no stock of materials or work in progress. The output of each process passes directly to the next process and finally to Finished Stock A/c. The following additional data is obtained:

Process	Output	Normal loss %	Realisable value of scrap
Process 1	2,850	5%	2
Process 2	2,520	10%	4
Process 3	2,250	15%	5

Prepare Process Cost Account, Normal Loss Account and Abnormal Gain or Loss Account.

READ THE QUESTION FIRST — WHAT TO SPOT

This is **the** model problem of the module. It contains all three possibilities in one question, which is exactly why examiners like it.

- **Process 1 ends with no abnormal item at all** — actual output equals expected. Do not force an abnormal figure just because the other processes have one.
- **Process 2 has an abnormal loss** (output less than expected).
- **Process 3 has an abnormal gain** (output MORE than expected). An abnormal gain is entered on the **debit** side of the process account — the opposite side from a loss. This single fact is the most commonly dropped mark in the whole module.
- Normal loss % is always applied to **the units entering that process**, which is the previous process's output — not the original 3,000.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Work the processes strictly in order. Each one needs the previous one's output *and* its rate.
2. For each process: total the debits, compute normal loss units and scrap value, find expected output, compare with actual.
3. Rate = (Total cost – normal loss scrap) ÷ (Input units – normal loss units).

4. **Loss** → credit the process with the shortfall at that rate. **Gain** → debit the process with the excess at that rate.
5. Open one Normal Loss A/c for all three processes together. Debit each process's scrap value into it.
6. If there is an abnormal gain, the normal loss did not fully happen — so **credit Normal Loss A/c and debit Abnormal Gain A/c** with the gain units × that process's scrap rate.
7. Close Abnormal Loss and Abnormal Gain to Profit & Loss.

WORKING NOTES

W1 Process 1

Total cost = $(3,000 \times ₹3 = 9,000) + 7,800 + 6,000 + 6,000 = ₹28,800$

Normal loss = 5% of 3,000 = **150 units**; scrap $150 \times ₹2 = ₹300$

Expected output = $3,000 - 150 = 2,850$ units. Actual = 2,850 units.

Difference = NIL — no abnormal loss or gain in Process 1.

$$\text{Rate} = \frac{28,800 - 300}{3,000 - 150} = \frac{28,500}{2,850} = ₹10.00 \text{ per unit}$$

W2 Process 2

Total cost = $(2,850 \times ₹10 = 28,500) + 5,940 + 9,000 + 9,000 = ₹52,440$

Normal loss = 10% of **2,850** = **285 units**; scrap $285 \times ₹4 = ₹1,140$

Expected output = $2,850 - 285 = 2,565$ units. Actual = 2,520 units.

Abnormal LOSS = 2,565 - 2,520 = 45 units

$$\text{Rate} = \frac{52,440 - 1,140}{2,850 - 285} = \frac{51,300}{2,565} = ₹20.00 \text{ per unit}$$

Abnormal loss $45 \times 20 = ₹900$

Transfer to Process 3 $2,520 \times 20 = ₹50,400$

W3 Process 3

Total cost = $(2,520 \times ₹20 = 50,400) + 8,886 + 12,000 + 12,000 = ₹83,286$

Normal loss = 15% of **2,520** = **378 units**; scrap $378 \times ₹5 = ₹1,890$

Expected output = $2,520 - 378 = 2,142$ units. Actual = 2,250 units.

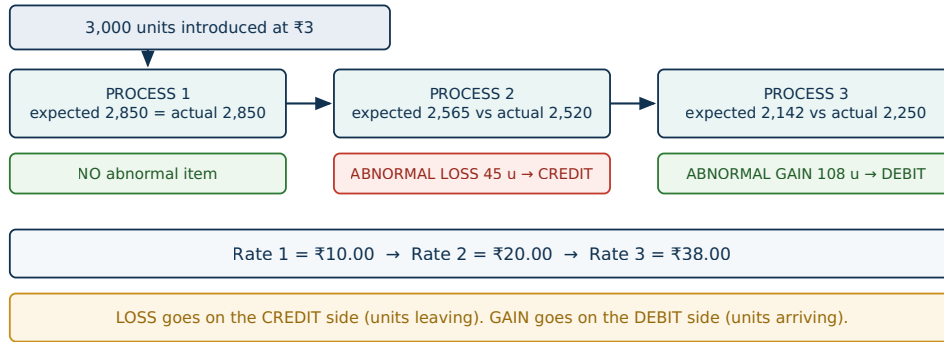
Abnormal GAIN = 2,250 - 2,142 = 108 units (actual is MORE than expected)

$$\text{Rate} = \frac{83,286 - 1,890}{2,520 - 378} = \frac{81,396}{2,142} = ₹38.00 \text{ per unit}$$

Note the rate is still built on EXPECTED output (2,142), not on the actual 2,250. The gain is then valued at this rate and added back.

Abnormal gain $108 \times 38 = ₹4,104$ (debit side)

Transfer to Finished Stock $2,250 \times 38 = ₹85,500$



One question, all three outcomes. Learn this picture and Q5 to Q9 all become the same problem.

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	3,000	3	9,000	By Normal loss $150 \text{ u} \times ₹2$	150	2	300
To Direct material	—	—	7,800	By Transfer to Process 2 $W1: 2,850 \times ₹10$	2,850	10	28,500
To Direct wages	—	—	6,000				
To Production overheads	—	—	6,000				
Total	3,000	—	28,800	Total	3,000	—	28,800

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 1	2,850	10	28,500	By Normal loss $285 \text{ u} \times ₹4$	285	4	1,140
To Direct material	—	—	5,940	By Abnormal loss $W2: 45 \times ₹20$	45	20	900
To Direct wages	—	—	9,000	By Transfer to Process 3 $W2: 2,520 \times ₹20$	2,520	20	50,400
To Production overheads	—	—	9,000				
Total	2,850	—	52,440	Total	2,850	—	52,440

Process 3 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process 2	2,520	20	50,400	By Normal loss $378 \text{ u} \times ₹5$	378	5	1,890
To Direct material	—	—	8,886	By Transfer to Finished Stock $W3: 2,250 \times ₹38$	2,250	38	85,500
To Direct wages	—	—	12,000				
To Production overheads	—	—	12,000				
To Abnormal gain $W3: 108 \times ₹38$ — DEBIT side	108	38	4,104				
Total	2,628	—	87,390	Total	2,628	—	87,390

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	150	2	300	By Abnormal Gain A/c <i>108 u × ₹5 — the Process 3 normal loss that never happened</i>	108	5	540
To Process 2 A/c	285	4	1,140	By Cost Ledger Control A/c (scrap actually sold)	705	—	2,790
To Process 3 A/c	378	5	1,890				
Total	813	—	3,330	Total	813	—	3,330

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 2 A/c	45	20	900	By Cost Ledger Control A/c <i>45 u × ₹4 scrap</i>	45	4	180
Total	45	—	900	By Profit & Loss A/c <i>balance = the real loss</i>	—	—	720
				Total	45	—	900

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c <i>108 u × ₹5 — scrap income forgone</i>	108	5	540	By Process 3 A/c	108	38	4,104
To Profit & Loss A/c <i>balance = the real gain</i>	—	—	3,564				
Total	108	—	4,104	Total	108	—	4,104

THE TRAP / WHERE MARKS ARE LOST

- **Putting the abnormal gain on the credit side.** Think about it physically: extra units have *arrived* in the process, so they must be debited like any other input. If your Process 3 account will not balance, this is why.
- Applying the 15% normal loss to 3,000 instead of 2,520. Each percentage applies to that process's own input.
- Forgetting the entry **Abnormal Gain A/c → Normal Loss A/c 540**. Because 108 units did not perish, the company also did not earn scrap on them — so the gain must be reduced by $108 \times ₹5$. Without this the Normal Loss account will not close.
- Using the actual output 2,250 as the denominator for the rate. The denominator is always **input minus normal loss** = 2,142.

FINAL ANSWER

Cost per unit — Process 1	₹ 10.00
Cost per unit — Process 2	₹ 20.00
Cost per unit — Process 3	₹ 38.00
Abnormal loss (Process 2, 45 units)	₹ 900
Abnormal gain (Process 3, 108 units)	₹ 4,104
Transferred to Finished Stock (2,250 units)	₹ 85,500

Net loss to P&L on abnormal loss ₹ 720

Net gain to P&L on abnormal gain ₹ 3,564

Every one of the six accounts closes exactly. If yours do not, check the abnormal gain is on the debit side and that the 540 entry is present.

Q7 Abnormal gain only

Loss & gain · p.10

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Product A is obtained after it passes through three distinct processes. 2,000 kgs of material at ₹5 per kg were issued to Process 1, direct wages amounted to ₹900 and other production overheads incurred was ₹500. Normal loss is estimated at 10% of input. The wastage is sold at ₹3 per kg. The actual output is 1,850 kgs. Prepare Process 1 A/c, Normal Loss A/c and Abnormal Gain A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Only **Process 1** is asked for. The question mentions three processes to set the scene — ignore the other two. Answer exactly what is asked.
- Expected output = 2,000 – 200 = 1,800 kg. Actual is **1,850 kg — higher**. The question even names the account it wants: **Abnormal Gain A/c**. That is your confirmation before you calculate anything.
- Because there is a gain, remember the two-part consequence: debit the process with the gain at cost, and transfer the forgone scrap value to Normal Loss A/c.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total the debits.
2. Normal loss = 10% of 2,000 = 200 kg; scrap value = 200 × 3.
3. Expected output 1,800 kg vs actual 1,850 kg → abnormal gain 50 kg.
4. Rate = (11,400 – 600) ÷ (2,000 – 200).
5. Debit the process with 50 kg at that rate; credit the transfer of 1,850 kg at that rate.
6. Normal Loss A/c: debit 200 kg at scrap rate; credit 50 kg to Abnormal Gain A/c; the rest is scrap sold.
7. Abnormal Gain A/c: debit the forgone scrap and the balance to P&L; credit the process figure.

WORKING NOTES

W1 Cost and units

Material 2,000 × ₹5 = 10,000

Direct wages = 900 | Production overheads = 500

Total cost = ₹11,400

Normal loss = 10% of 2,000 = **200 kg**; scrap 200 × ₹3 = **₹600**

Expected output = 1,800 kg | Actual output = 1,850 kg

Abnormal gain = 1,850 – 1,800 = 50 kg

$$\text{Rate} = \frac{11,400 - 600}{2,000 - 200} = \frac{10,800}{1,800} = \text{₹6.00 per kg}$$

Abnormal gain 50 × 6 = **₹300**

Transfer out 1,850 × 6 = **₹11,100**

Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	5	10,000	By Normal loss 200 kg × ₹3	200	3	600
To Direct wages	—	—	900	By Transfer to Process 2 W1: 1,850 × ₹6	1,850	6	11,100
To Production overheads	—	—	500				
To Abnormal gain W1: 50 kg × ₹6	50	6	300				
Total	2,050	—	11,700	Total	2,050	—	11,700

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	200	3	600	By Abnormal Gain A/c 50 kg × ₹3 — loss that did not occur	50	3	150
Total	200	—	600	By Cost Ledger Control A/c (scrap sold)	150	3	450
				Total	200	—	600

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c scrap income forgone on 50 kg	50	3	150	By Process 1 A/c	50	6	300
To Profit & Loss A/c balance = net gain	—	—	150				
Total	50	—	300	Total	50	—	300

WHY THIS WORKS

The net gain reaching Profit & Loss is only ₹150, not ₹300. The process saved ₹300 of cost by producing 50 extra kg, but it also lost the ₹150 of scrap money those 50 kg would have fetched. Gain is always **cost saved minus scrap forgone**.

FINAL ANSWER

Cost per kg	₹ 6.00
Transferred out (1,850 kg)	₹ 11,100
Abnormal gain at cost (50 kg)	₹ 300
Scrap value forgone	₹ 150
Net abnormal gain to Profit & Loss	₹ 150

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through 2 distinct processes A and B. The normal wastage of each process is as follows: Process A – 3% of products entering the process; Process B – 5% of products entering the process.

The wastage of Process A was sold at 50 paise per unit and that of Process B at ₹1 per unit. 10,000 units were issued to Process A at ₹2 per unit. The other expenses were as follows:

Particulars	Process A	Process B
Materials	2,000	3,000
Wages	10,000	16,000
Overheads	2,100	2,375
Actual output (units)	9,500	9,100

Prepare process accounts and other ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- “3% of products entering the process” — so Process B’s 5% is on the 9,500 units it receives, not on the original 10,000.
- Process A: expected 9,700, actual 9,500 → **abnormal loss 200 units**.
- Process B: expected 9,025, actual 9,100 → **abnormal gain 75 units**.
- Two different scrap rates — 50 paise in A and ₹1 in B. Keep them apart; using one rate for both is a common slip.
- “and other ledger accounts” means: Normal Loss A/c, Abnormal Loss A/c and Abnormal Gain A/c. Draw all three.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Process A: total debits, normal loss 3% of 10,000, expected output, compare, rate, then value the loss and the transfer.
2. Carry A's output *and* A's rate into Process B as the transfer-in.
3. Process B: normal loss 5% of the units received (9,500), then the same sequence.
4. Build the three ledger accounts, remembering the Abnormal Gain → Normal Loss entry.

WORKING NOTES

W1 Process A

Total cost = $(10,000 \times ₹2 = 20,000) + 2,000 + 10,000 + 2,100 = ₹34,100$

Normal loss = 3% of 10,000 = **300 units**; scrap $300 \times ₹0.50 = ₹150$

Expected output 9,700 | Actual 9,500 → **abnormal loss 200 units**

$$\text{Rate} = \frac{34,100 - 150}{10,000 - 300} = \frac{33,950}{9,700} = ₹3.50 \text{ per unit}$$

Abnormal loss $200 \times 3.50 = ₹700$

Transfer to B $9,500 \times 3.50 = ₹33,250$

W2 Process B

Total cost = 33,250 + 3,000 + 16,000 + 2,375 = ₹54,625

Normal loss = 5% of 9,500 = 475 units; scrap 475 × ₹1 = ₹475

Expected output 9,025 | Actual 9,100 → **abnormal gain 75 units**

$$\text{Rate} = \frac{54,625 - 475}{9,500 - 475} = \frac{54,150}{9,025} = \text{₹6.00 per unit}$$

Abnormal gain 75 × 6 = ₹450 (debit side)

Transfer to Finished Stock 9,100 × 6 = ₹54,600

Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	10,000	2	20,000	By Normal loss 300 u × ₹0.50	300	0.50	150
To Materials	—	—	2,000	By Abnormal loss W1: 200 × ₹3.50	200	3.50	700
To Wages	—	—	10,000	By Transfer to Process B W1: 9,500 × ₹3.50	9,500	3.50	33,250
To Overheads	—	—	2,100				
Total	10,000	—	34,100	Total	10,000	—	34,100

Process B Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process A	9,500	3.50	33,250	By Normal loss 475 u × ₹1	475	1	475
To Materials	—	—	3,000	By Transfer to Finished Stock W2: 9,100 × ₹6	9,100	6	54,600
To Wages	—	—	16,000				
To Overheads	—	—	2,375				
To Abnormal gain W2: 75 × ₹6 — DEBIT	75	6	450				
Total	9,575	—	55,075	Total	9,575	—	55,075

Normal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	300	0.50	150	By Abnormal Gain A/c 75 u × ₹1 (B's scrap rate)	75	1	75
To Process B A/c	475	1	475	By Cost Ledger Control A/c (scrap sold)	700	—	550
Total	775	—	625	Total	775	—	625

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process A A/c	200	3.50	700	By Cost Ledger Control A/c $200 \times ₹0.50$	200	0.50	100
Total	200	—	700	By Profit & Loss A/c	—	—	600
				Total	200	—	700

Abnormal Gain Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Normal Loss A/c	75	1	75	By Process B A/c	75	6	450
To Profit & Loss A/c	—	—	375				
Total	75	—	450	Total	75	—	450

THE TRAP / WHERE MARKS ARE LOST

- Using ₹0.50 as the scrap rate in Process B. B's wastage sells for ₹1. Two processes, two rates.
- Taking 5% of 10,000 in Process B. It receives only 9,500.
- Netting the abnormal loss of Process A against the abnormal gain of Process B. They are separate accounts and separate P&L entries — never set one off against the other.

FINAL ANSWER

Cost per unit — Process A	₹ 3.50
Cost per unit — Process B	₹ 6.00
Abnormal loss (A, 200 units)	₹ 700 — net to P&L ₹ 600
Abnormal gain (B, 75 units)	₹ 450 — net to P&L ₹ 375
Transferred to Finished Stock (9,100 units)	₹ 54,600

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following details are extracted from the costing records of an oil mill for the year ended 31st March, 2015. Purchase of 5,400 tonnes of coconut ₹2,20,000.

Particulars	Crushing	Refining	Finishing
Cost of labour	2,750	1,100	1,650
Electric power	660	396	264
Sundry material	110	2,200	—
Repairs to machinery	308	363	154
Steam	600	495	495
Factory expenses	1,452	726	242

Cost of sacks ₹8,250. 3,200 tonnes of crude oil were produced. 2,600 tonnes of oil produced by the refining process. 2,550 tonnes of refined oil were finished for delivery. Coconut sacks sold ₹440. 1,925 tonnes of coconut residue sold ₹12,100. Loss in weight in crushing 275 tonnes. 500 tonnes of by-products obtained from refining process ₹7,425.

Show the accounts for (a) coconut crushing process, (b) refining process, (c) finishing process, to arrive at the cost per tonne of each process and the total cost per tonne of the finished oil.

READ THE QUESTION FIRST — WHAT TO SPOT

A long question, but it is only Q1 with three deductions. The one thing that makes it easy is the **units reconciliation** — do it first and the whole answer falls out.

- **Crushing:** 5,400 t in. Out: crude oil 3,200 + residue 1,925 + loss in weight 275 = 5,400 . Perfect — no abnormal item.
- **Refining:** 3,200 t in. Out: refined oil 2,600 + by-product 500 = 3,100, so loss in weight = **100 t** (you have to work this out; the question does not give it).
- **Finishing:** 2,600 t in, 2,550 t out, so loss in weight = **50 t**.
- Three items are **credited** to their process because they are saleable: sacks 440 and residue 12,100 in crushing, by-product 7,425 in refining.
- Loss in weight is pure evaporation. It has **no scrap value**, so it carries no rupee figure — only tonnes. Show the tonnes and put a dash in the amount column.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Reconcile tonnes for each stage first. Work out any loss in weight the question has not given you.
2. Debit the process with the material coming in plus all its own expenses.
3. Credit any saleable output that is *not* the main product (sacks, residue, by-product) at its sale value.
4. Credit the loss in weight in tonnes only, amount nil.
5. The balance is the cost of the main output. Divide by its tonnage for cost per tonne.
6. Carry that balance forward as the debit of the next process.

WORKING NOTES

W1 Tonnage reconciliation — do this before anything else

Stage	In (t)	Main output (t)	Other output (t)	Loss in weight (t)
Crushing	5,400	3,200 crude oil	1,925 residue	275 (given)
Refining	3,200	2,600 refined oil	500 by-product	100 (derived)
Finishing	2,600	2,550 finished oil	—	50 (derived)

Refining loss = $3,200 - 2,600 - 500 = \mathbf{100 \text{ tonnes}}$

Finishing loss = $2,600 - 2,550 = \mathbf{50 \text{ tonnes}}$

W2 Cost per tonne at each stage

$$\begin{aligned} \text{Crushing} &= \frac{2,21,590}{3,200 \text{ t}} = \mathbf{₹69.25} \\ \text{Refining} &= \frac{2,19,445}{2,600 \text{ t}} = \mathbf{₹84.40} \\ \text{Finishing} &= \frac{2,22,250}{2,550 \text{ t}} = \mathbf{₹87.16} \end{aligned}$$

(a) Coconut Crushing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Coconut purchased	5,400	—	2,20,000	By Sale of sacks	—	—	440
To Cost of sacks	—	—	8,250	By Sale of coconut residue	1,925	—	12,100
To Cost of labour	—	—	2,750	By Loss in weight evaporation — tonnes only, no value	275	—	—
To Electric power	—	—	660	By Crude oil to Refining W2: 2,34,130 – 440 – 12,100	3,200	69.25	2,21,590
To Sundry material	—	—	110				
To Repairs to machinery	—	—	308				
To Steam	—	—	600				
To Factory expenses	—	—	1,452				
Total	5,400	—	2,34,130	Total	5,400	—	2,34,130

(b) Refining Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Crude oil from Crushing	3,200	69.25	2,21,590	By Sale of by-products	500	—	7,425
To Cost of labour	—	—	1,100	By Loss in weight W1: derived as 100 t	100	—	—
To Electric power	—	—	396	By Refined oil to Finishing W2: 2,26,870 – 7,425	2,600	84.40	2,19,445
To Sundry material	—	—	2,200				
To Repairs to machinery	—	—	363				
To Steam	—	—	495				
To Factory expenses	—	—	726				
Total	3,200	—	2,26,870	Total	3,200	—	2,26,870

(c) Finishing Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Refined oil from Refining	2,600	84.40	2,19,445	By Loss in weight W1: derived as 50 t	50	—	—
To Cost of labour	—	—	1,650	By Finished oil (cost of delivery) W2: 2,22,250 ÷ 2,550 t	2,550	87.16	2,22,250
To Electric power	—	—	264				
To Repairs to machinery	—	—	154				
To Steam	—	—	495				
To Factory expenses	—	—	242				
Total	2,600	—	2,22,250	Total	2,600	—	2,22,250

THE TRAP / WHERE MARKS ARE LOST

- Treating the sacks, residue and by-product as income in Profit & Loss. They reduce the cost of the process that produced them.
- Giving the loss in weight a rupee value. It evaporated — there is nothing to sell. Tonnes only.
- Not deriving the refining and finishing losses. If you skip them the tonnage columns will not add up and the examiner will see it immediately.
- Dividing the crushing cost by 5,400 tonnes. The cost belongs to the 3,200 tonnes of crude oil that actually came out.

FINAL ANSWER

Cost per tonne — Crushing (crude oil)	₹ 69.25
Cost per tonne — Refining (refined oil)	₹ 84.40
Cost per tonne — Finishing (finished oil)	₹ 87.16
Total cost of 2,550 tonnes delivered	₹ 2,22,250

Closing line: “The total cost per tonne of the finished oil is ₹87.16.”

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In manufacturing the main product a company processes the incidental waste into two by-products A and B. From the following data relating to the products prepare a comparative profit & loss statement showing individual cost and other details. The total cost up to separation point was ₹3,10,400.

Particulars	Main product	Product A	Product B
Sales	8,00,000	64,000	96,000
Cost after separation	80,000	12,800	14,400
Estimated net profit (% of sales)	?	20%	30%
Estimated selling expenses (% of sales)	20%	10%	15%

READ THE QUESTION FIRST — WHAT TO SPOT

- The main product's profit is a **question mark**. That tells you the whole plan: work out the two by-products completely, take their joint-cost shares away from ₹3,10,400, and whatever is left belongs to the main product.
- You are given **profit % and selling-expense % for A and B**. That is the signal for the **reverse-cost method** — start at sales and climb down.
- All percentages here are **of sales**, not of cost. Read the bracket every time.
- "Comparative profit & loss statement" means one table with a column per product and a total column. Give the total column — it is where the cross-check lives.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. For each by-product, start with sales. Subtract profit (% of sales). Subtract selling expenses (% of sales). What remains is that product's **total cost**.
2. Subtract its cost after separation. What remains is its **share of the joint cost**.
3. Add the by-products' shares. Subtract from the total joint cost. The balance is the **main product's share**.
4. Now build the main product's column forwards: joint share + cost after separation + selling expenses = total cost; sales – total cost = profit.
5. Express the main product's profit as a % of its sales — that is the missing '?'.
6. Cross-check: every column's total cost + profit must equal its sales.



The reverse-cost ladder. Every joint / by-product problem in Q10 to Q14 is this one picture.

WORKING NOTES

W1 Product A — climbing down from sales

Sales = 64,000

Less: Net profit 20% of 64,000 = (12,800)
 Less: Selling expenses 10% of 64,000 = (6,400)
Total cost = 44,800
 Less: Cost after separation = (12,800)
Share of joint cost = ₹32,000

W2 Product B — same ladder

Sales = 96,000
 Less: Net profit 30% of 96,000 = (28,800)
 Less: Selling expenses 15% of 96,000 = (14,400)
Total cost = 52,800
 Less: Cost after separation = (14,400)
Share of joint cost = ₹38,400

W3 Main product's share is whatever is left

Total joint cost = 3,10,400
 Less: Product A's share (W1) = (32,000)
 Less: Product B's share (W2) = (38,400)
Main product's share of joint cost = ₹2,40,000

Comparative Profit and Loss Statement

Particulars	Main product	Product A	Product B	Total
Sales	8,00,000	64,000	96,000	9,60,000
Share of joint cost (W1-W3)	2,40,000	32,000	38,400	3,10,400
Cost after separation	80,000	12,800	14,400	1,07,200
Selling expenses	1,60,000	6,400	14,400	1,80,800
Total cost	4,80,000	51,200	67,200	5,98,400
Net profit (Sales – Total cost)	3,20,000	12,800	28,800	3,61,600
Net profit as % of sales	40%	20%	30%	37.67%

Selling expenses: main 20% of 8,00,000 = 1,60,000 | A 10% of 64,000 = 6,400 | B 15% of 96,000 = 14,400.

THE TRAP / WHERE MARKS ARE LOST

- Taking the profit % on **cost** instead of on sales. Here everything is on sales.
- Forgetting to subtract selling expenses on the way down. If you skip them, A's joint share comes to 38,400 instead of 32,000 and the whole answer shifts.
- Apportioning the joint cost by sales value or by weight. The question gives you profit and selling-expense percentages precisely so you use the reverse-cost method.

FINAL ANSWER

Joint cost — Main product	₹ 2,40,000
Joint cost — Product A	₹ 32,000
Joint cost — Product B	₹ 38,400
Main product's net profit	₹ 3,20,000 (40% of sales)
Total profit of all three	₹ 3,61,600

Q11 Selling expenses have to be derived

Joint & by-product · p.11

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A factory producing an article 'X' also produces product 'Y' which is further processed into a finished product. The joint cost of manufacturing is: Materials 50,000; Labour 30,000; Overheads 20,000 — total ₹1,00,000. The subsequent costs are as follows:

Particulars	X	Y
Materials	30,000	15,000
Labour	14,000	10,000
Overheads	6,000	5,000
Total	50,000	30,000
Selling price	1,60,000	80,000
Estimated profit on selling price	25%	20%

Assume that the selling and distribution expenses are in proportion to the selling price. Show how you would apportion the joint cost of manufacturing and prepare a statement showing the cost of production of both products and the ledger accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- This one has a hidden step. The **selling and distribution expenses are not given** — you must derive the total, then split it.
- How to derive it: total sales 2,40,000 minus total profit gives total cost. Take away the costs you already know (joint 1,00,000 + subsequent 80,000) and the gap must be the S&D expenses.
- "In proportion to the selling price" — so split in the ratio $1,60,000 : 80,000 = 2 : 1$.
- Then it is the ordinary reverse-cost ladder for each product.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Compute each product's profit from the given % of selling price. Add them.
2. Total cost = Total sales — Total profit.
3. **S&D expenses = Total cost — joint cost — subsequent costs.** This is the derived figure.
4. Split the S&D expenses in the ratio of selling prices.
5. For each product climb down: Sales — profit = total cost; then — S&D — subsequent cost = share of joint cost.
6. Check the two joint shares add back to 1,00,000.

WORKING NOTES**W1 Deriving the selling & distribution expenses**

Profit on X = 25% of 1,60,000 = 40,000

Profit on Y = 20% of 80,000 = 16,000

Total profit = 56,000

Total sales = 1,60,000 + 80,000 = 2,40,000

Total cost = 2,40,000 – 56,000 = **1,84,000**
 Known costs = joint 1,00,000 + subsequent 80,000 = 1,80,000
 $\therefore$ **S&D expenses = 1,84,000 – 1,80,000 = ₹4,000**

Split 2 : 1 $\rightarrow$ X = 4,000 $\times$ 2/3 = **2,666.67** | Y = 4,000 $\times$ 1/3 = **1,333.33**

W2 Apportionment of the joint cost

Particulars	Product X	Product Y	Total
Selling price	1,60,000	80,000	2,40,000
Less: Estimated profit	(40,000)	(16,000)	(56,000)
Total cost	1,20,000	64,000	1,84,000
Less: Selling & distribution (W1)	(2,666.67)	(1,333.33)	(4,000)
Less: Subsequent cost	(50,000)	(30,000)	(80,000)
Share of joint cost	67,333.33	32,666.67	1,00,000

Cross-check: 67,333.33 + 32,666.67 = 1,00,000 exactly the joint cost given. That agreement is your proof the method is right.

Statement of Cost of Production and Profit

Particulars	Product X	Product Y	Total
Share of joint cost (W2)	67,333.33	32,666.67	1,00,000
Subsequent cost — materials	30,000	15,000	45,000
Subsequent cost — labour	14,000	10,000	24,000
Subsequent cost — overheads	6,000	5,000	11,000
Cost of production	1,17,333.33	62,666.67	1,80,000
Selling & distribution expenses	2,666.67	1,333.33	4,000
Total cost	1,20,000	64,000	1,84,000
Sales	1,60,000	80,000	2,40,000
Profit	40,000	16,000	56,000
Profit as % of selling price	25%	20%	23.33%

Product X Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned	—	By Sales	—
To Materials	—		
To Labour	—		
To Overheads	—		
To Selling & distribution W1: 4,000 $\times$ 2/3	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product Y Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned	—	By Sales	—
To Materials	—		
To Labour	—		
To Overheads	—		
To Selling & distribution W1: $4,000 \times 1/3$	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

WHY THIS WORKS

The reason the S&D expenses can be derived at all is that the question fixes the profit on *both* products. Once profit is fixed, total cost is fixed, and any cost you have not been told about must be the difference. Whenever a joint-cost question gives you every profit percentage but leaves one cost unnamed, this is the trick being tested.

FINAL ANSWER

Joint cost apportioned to X	₹ 67,333.33
Joint cost apportioned to Y	₹ 32,666.67
Selling & distribution expenses (derived)	₹ 4,000
Cost of production — X	₹ 1,17,333.33
Cost of production — Y	₹ 62,666.67
Total profit	₹ 56,000

Q12 When the printed figures do not reconcile

Joint & by-product · p.11-12

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In a manufacturing concern a certain product is manufactured which also yields 2 by-products. The joint expenses of manufacture are: Materials 8,500; Labour 9,000; Overheads 7,500 — total ₹25,000. Subsequent expenses are:

Particulars	A	B	C
Materials	2,500	1,200	1,400
Labour	1,900	1,600	2,000
Overheads	1,500	900	1,050
Total	5,900	3,700	4,450

The selling prices are A — 30,000; B — 20,000 and C — 15,000. Estimated profits on selling prices are A — 40%; B — 30% and C — 25%. Show the apportionment of joint cost and prepare the product accounts.

READ THE QUESTION FIRST — WHAT TO SPOT

- Same ladder as Q10 and Q11, but with **no selling expenses** mentioned — so the ladder is shorter: Sales – profit – subsequent cost = share of joint cost.
- **Read the warning below before you write your answer.** The figures in this question do not reconcile, and knowing how to handle that is worth more than getting a tidy number.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. For each product: Sales – profit (% of sales) = total cost.
2. Total cost – subsequent cost = **notional share of joint cost**.
3. Add the three notional shares and **compare with the actual joint cost**.
4. If they agree, you are finished. **If they do not agree** (as here), apportion the *actual* joint cost in the *ratio* of the notional shares.
5. Recompute each product's real profit using its apportioned share, and state that the actual profit differs from the estimate.

WORKING NOTES

W1 Notional share of joint cost by reverse cost

Particulars	A	B	C	Total
Selling price	30,000	20,000	15,000	65,000
Less: Estimated profit (40% / 30% / 25%)	(12,000)	(6,000)	(3,750)	(21,750)
Total cost	18,000	14,000	11,250	43,250
Less: Subsequent expenses	(5,900)	(3,700)	(4,450)	(14,050)
Notional share of joint cost	12,100	10,300	6,800	29,200

W2 The figures do not reconcile — and here is how to deal with it

Notional shares add to = **29,200**

Actual joint expenses are only = **25,000**

Difference = **₹4,200** — the estimated profits are more optimistic than the actual cost structure allows.

The joint cost that actually exists is 25,000, so that is all you can apportion. Apportion it in the **ratio of the notional shares** 12,100 : 10,300 : 6,800.

Product	Notional share	Working	Apportioned joint cost
A	12,100	$25,000 \times 12,100 / 29,200$	10,359.59
B	10,300	$25,000 \times 10,300 / 29,200$	8,818.49
C	6,800	$25,000 \times 6,800 / 29,200$	5,821.92
Total	29,200		25,000.00

Statement of Cost and Actual Profit

Particulars	A	B	C	Total
Apportioned joint cost (W2)	10,359.59	8,818.49	5,821.92	25,000
Subsequent expenses	5,900	3,700	4,450	14,050

Particulars	A	B	C	Total
Total cost	16,259.59	12,518.49	10,271.92	39,050
Sales	30,000	20,000	15,000	65,000
Actual profit	13,740.41	7,481.51	4,728.08	25,950
Actual profit as % of sales	45.80%	37.41%	31.52%	39.92%
Estimated profit % (from the question)	40%	30%	25%	—

Product A Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product B Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

Product C Account

Particulars	Amount ₹	Particulars	Amount ₹
To Joint cost apportioned W2	—	By Sales	—
To Subsequent expenses	—		
To Profit & Loss A/c (profit)	—		
Total	—	Total	—

THE TRAP / WHERE MARKS ARE LOST

Please read this one carefully — it is about your exam, not just this sum.

- The notional shares total 29,200 but the joint expenses are 25,000. As printed, the question cannot give a tidy answer. This happens in workbooks.
- The **wrong** response is to write 12,100 / 10,300 / 6,800 as your apportionment. Those figures add to 29,200, which is ₹4,200 more joint cost than the company actually incurred.
- The **right** response is what W2 does: apportion the real 25,000 in the ratio of the notional shares, then show the actual profit percentages and note that they exceed the estimates. Write one sentence saying so. An examiner rewards a student who spots the inconsistency and handles it openly.
- **Do check this one with your professor** — if he teaches the plain notional-share answer, give that instead. Either way, understand why the two differ.

FINAL ANSWER

Apportioned joint cost — A	₹ 10,359.59
Apportioned joint cost — B	₹ 8,818.49
Apportioned joint cost — C	₹ 5,821.92
Total actual profit	₹ 25,950
Notional shares (for the ratio only)	12,100 : 10,300 : 6,800

Q13 Three joint products, full ladder

Joint & by-product · p.12

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Bright Chemical Ltd. electrolyses common salt to obtain 3 joint products — caustic soda, chlorine and hydrogen. During a costing period, the expenditure relating to the inputs of the common process amounted to ₹3,50,000. After separation, expenses amounting to ₹1,60,000, ₹75,000 and ₹10,000 were incurred for caustic soda, chlorine and hydrogen respectively. The entire production was sold and ₹3,75,000, ₹2,50,000 and ₹60,000 were realised for caustic soda, chlorine and hydrogen respectively. The selling expenses were estimated at 5% of realisation from sale. The profit is 15%, 10% and 5% of realisation from sale of caustic soda, chlorine and hydrogen respectively.

Draw a columnar statement showing the apportionment of joint cost and the profitability of each product.

READ THE QUESTION FIRST — WHAT TO SPOT

- The full four-rung ladder this time: Sales — profit — selling expenses — separation expenses = share of joint cost.
- Selling expenses are **5% of sales for all three** — one rate, three amounts.
- These are **joint products**, not by-products. That changes nothing about the arithmetic; it only means all three are commercially important, so none is credited to another.
- Like Q12, the derived shares here **do not** add up to the joint cost given. The same ratio treatment applies. Do not be alarmed — be systematic.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Build the ladder in a single columnar table — one column per product, one total column.
2. Row 1 Sales. Row 2 less profit. Row 3 less selling expenses (5% of sales). Row 4 gives total cost.
3. Row 5 less separation expenses. Row 6 gives the notional joint-cost share.
4. Total row 6 and compare with the actual joint cost of 3,50,000.
5. Apportion the actual 3,50,000 in the ratio of the notional shares.
6. Rebuild each column forwards to show the actual profitability, and comment on how it compares with the estimate.

WORKING NOTES

W1 The reverse-cost ladder, all three products together

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Sales realisation	3,75,000	2,50,000	60,000	6,85,000
Less: Profit (15% / 10% / 5% of sales)	(56,250)	(25,000)	(3,000)	(84,250)
Less: Selling expenses (5% of sales)	(18,750)	(12,500)	(3,000)	(34,250)
Total cost	3,00,000	2,12,500	54,000	5,66,500

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Less: Expenses after separation	(1,60,000)	(75,000)	(10,000)	(2,45,000)
Notional share of joint cost	1,40,000	1,37,500	44,000	3,21,500

W2 Reconciling with the actual joint cost

Notional shares total = **3,21,500**

Actual joint expenditure = **3,50,000**

Shortfall = **₹28,500** — the joint process cost more than the estimated profits allow for, so the actual profits will be *lower* than estimated.

Product	Notional share	Working	Apportioned joint cost
Caustic soda	1,40,000	$3,50,000 \times 1,40,000 / 3,21,500$	1,52,410.58
Chlorine	1,37,500	$3,50,000 \times 1,37,500 / 3,21,500$	1,49,688.96
Hydrogen	44,000	$3,50,000 \times 44,000 / 3,21,500$	47,900.46
Total	3,21,500		3,50,000.00

Columnar Statement of Apportionment and Profitability

Particulars	Caustic soda	Chlorine	Hydrogen	Total
Apportioned joint cost (W2)	1,52,410.58	1,49,688.96	47,900.46	3,50,000
Expenses after separation	1,60,000	75,000	10,000	2,45,000
Selling expenses (5% of sales)	18,750	12,500	3,000	34,250
Total cost	3,31,160.58	2,37,188.96	60,900.46	6,29,250
Sales realisation	3,75,000	2,50,000	60,000	6,85,000
Actual profit / (loss)	43,839.42	12,811.04	(900.46)	55,750
Actual profit as % of sales	11.69%	5.12%	(1.50%)	8.14%
Estimated profit % (question)	15%	10%	5%	—

WHY THIS WORKS

Look at the hydrogen column: an actual **loss** of ₹900.46 against an estimated 5% profit. That is a real finding, not an error — the joint process cost ₹28,500 more than the estimates assumed, and hydrogen, having the smallest sales, cannot carry its share. Write that sentence in your answer. It shows you can read a statement instead of only filling it in.

FINAL ANSWER

Apportioned joint cost — Caustic soda **₹ 1,52,410.58**

Apportioned joint cost — Chlorine **₹ 1,49,688.96**

Apportioned joint cost — Hydrogen **₹ 47,900.46**

Total actual profit **₹ 55,750**

Notional shares (ratio basis) **1,40,000 : 1,37,500 : 44,000**

State clearly that the notional shares totalled ₹3,21,500 against actual joint cost of ₹3,50,000, and that you apportioned the actual figure in that ratio.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A by-product 'Beta' is derived in the course of manufacturing product 'Alpha'. The by-product is further processed for sale. From the following data available from the records, prepare an account showing the cost per kg of the product 'Alpha' and the by-product 'Beta'.

Particulars	Joint expenses	Separate — Alpha	Separate — Beta
Materials	10,000	6,000	500
Labour	7,000	5,000	2,000
Overheads	2,500	1,500	600
Total	19,500	12,500	3,100

The quantities produced during the period under consideration were: Alpha — 1,000 kgs and Beta — 500 kgs. The selling price was ₹120/kg on which the profit earned was 30%.

READ THE QUESTION FIRST — WHAT TO SPOT

- A **by-product** problem: Alpha is the main product, Beta is incidental. The normal treatment is to work out Beta's share of the joint cost by reverse cost and leave the balance for Alpha.
- Total cost available to share out = joint 19,500 + Alpha's own 12,500 + Beta's own 3,100 = **₹35,100**. Keep that number in your head — it is the ceiling on every cost figure in this problem.
- **Now look at the selling price.** 500 kg of Beta at 120 would be 60,000 of sales, and 1,000 kg of Alpha at 120 would be 1,20,000. Either one is far bigger than the total cost of 35,100. So the reverse-cost ladder is going to break. Read W2 before you attempt this in the exam.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Identify the main product and the by-product.
2. Apply the reverse-cost ladder to the **by-product**: Sales – profit – its own separate expenses = its share of the joint cost.
3. Joint cost less the by-product's share = the main product's share.
4. Main product cost = its joint share + its own separate expenses. Divide by its kg for cost per kg. Same for the by-product.
5. **Always sanity-check** that no share is negative and that the shares add back to the joint cost. If they do not, the data is faulty — see W2.

WORKING NOTES

W1 The reverse-cost attempt (taking ₹120/kg as Beta's price)

Beta sales = 500 kg × 120 = 60,000

Less: Profit 30% of 60,000 = (18,000)

Total cost of Beta = 42,000

Less: Beta's separate expenses = (3,100)

Beta's share of joint cost = **38,900**

But the whole joint cost is only **19,500** → Alpha's share would be **19,500 – 38,900 = (₹19,400)**, a negative cost.

W2 THE DATA AS PRINTED CANNOT BE SOLVED — AND WHAT TO DO ABOUT IT

A cost can never be negative. Whichever product you assign the ₹120/kg to, the sales value dwarfs the total cost of ₹35,100, so the ladder produces an impossible share. Check the figures against the original in your workbook — most likely the selling price, the profit percentage or the joint expenses has a typographical error.

In the exam, do this: write one line stating the inconsistency, then solve the problem on the next most defensible basis and say which basis you used. You will be given credit for the method. The standard fallback for a by-product whose selling price is unusable is to apportion the joint cost on **physical quantity**, which is shown in W3.

W3 Workable solution — joint cost apportioned on physical quantity

Quantities Alpha 1,000 kg : Beta 500 kg = **2 : 1**

Alpha's share = $19,500 \times 2/3 = \text{₹}13,000$

Beta's share = $19,500 \times 1/3 = \text{₹}6,500$

Particulars	Alpha	Beta	Total
Share of joint cost (2 : 1)	13,000	6,500	19,500
Add: Separate expenses	12,500	3,100	15,600
Total cost	25,500	9,600	35,100
Quantity produced (kg)	1,000	500	1,500
Cost per kg	25.50	19.20	—

Alpha Account (main product)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Share of joint cost W3: $19,500 \times 2/3$	1,000	13	13,000	By Cost of production transferred $25,500 \div 1,000 \text{ kg}$	1,000	25.50	25,500
To Separate materials	—	—	6,000				
To Separate labour	—	—	5,000				
To Separate overheads	—	—	1,500				
Total	1,000	—	25,500	Total	1,000	—	25,500

Beta Account (by-product)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Share of joint cost W3: $19,500 \times 1/3$	500	13	6,500	By Cost of production transferred $9,600 \div 500 \text{ kg}$	500	19.20	9,600
To Separate materials	—	—	500				
To Separate labour	—	—	2,000				
To Separate overheads	—	—	600				
Total	500	—	9,600	Total	500	—	9,600

FINAL ANSWER

Alpha — total cost	₹ 25,500
Alpha — cost per kg	₹ 25.50
Beta — total cost	₹ 9,600

Beta — cost per kg

₹ 19.20

Basis used

Physical quantity 2 : 1 (see W2 for why)

Ask your professor about this one. If he wants the reverse-cost method, the corrected selling price is needed. Show W1, W2 and W3 exactly as laid out above and you will not lose method marks either way.

Q15 Profit as % of cost, plus unrealised profit

Inter-process profit · p.12-13

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through two processes X and Y. The output of Process X is charged to Process Y at a price which includes profit of **20% on actual cost** and the output of Process Y is charged to Finished Stock A/c at a price which includes **10% profit on actual cost**. The following data is provided for the month of July.

Particulars	X	Y
Material (2,500 units)	1,250	—
Labour	625	500
Overheads	1,875	750
Indirect material	—	1,250

There was no partly finished WIP. Out of the finished stock, 1,500 units had been sold for ₹7,500. Prepare the Process A/c and Finished Stock A/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the **inter-process profit** topic. Each process sells to the next at a mark-up, so the transfer price is bigger than the true cost.
- **“20% on actual cost”** — profit is 20% of cost, so if cost is 3,750 the profit is 750. (Contrast Q16, where profit is a % of the transfer price — a different calculation.)
- There is **no WIP inside the processes**, which makes this the easy version. The only stock is in Finished Stock: 2,500 units produced, 1,500 sold, so **1,000 units left**.
- That closing stock of 1,000 units carries profit added by X and by Y which the company has not actually earned yet, because the goods are still on the shelf. That is the **unrealised profit**, and a reserve must be created for it.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Total Process X's own costs. That is its **actual cost**.
2. Add the stated profit percentage **of that cost**. The sum is the transfer price to Y.
3. Debit Process Y with the transfer price (not the cost), add Y's own costs, total to get Y's actual cost.
4. Add Y's profit percentage of that total. The sum transfers to Finished Stock.
5. In Finished Stock, work out the per-unit transfer value, split it between units sold and units in hand, and compute the profit on sale.
6. **Unrealised profit** = total profit loaded by the processes $\times \frac{\text{closing stock units}}{\text{total units produced}}$. Create the reserve for it.

WORKING NOTES

W1 Process X

Material 1,250 + Labour 625 + Overheads 1,875

Actual cost = ₹3,750

Profit 20% of 3,750 = **₹750**

Transfer price to Process Y = ₹4,500

W2 Process Y

Transfer from X 4,500 + Indirect material 1,250 + Labour 500 + Overheads 750

Actual cost = ₹7,000

Profit 10% of 7,000 = **₹700**

Transfer to Finished Stock = ₹7,700 for 2,500 units

W3 Finished Stock

Value per unit = $7,700 \div 2,500 = \text{₹}3.08$

Cost of 1,500 units sold = $1,500 \times 3.08 = \text{₹}4,620$

Closing stock 1,000 units = $1,000 \times 3.08 = \text{₹}3,080$

Profit on sale = $7,500 - 4,620 = \text{₹}2,880$

W4 Reserve for unrealised profit — the marks everyone drops

Profit loaded by the processes = $750 (X) + 700 (Y) = \text{₹}1,450$

True cost of the 2,500 units = $7,700 - 1,450 = 6,250$ (check: $3,750 + 1,250 + 500 + 750 = 6,250$ ✓)

Closing stock is 1,000 of 2,500 units = **40%**

$$\text{Reserve} = \frac{1,000 \text{ units}}{2,500 \text{ units}} \times \text{₹}1,450 = \text{₹}580$$

Equivalently $3,080 \times 1,450 / 7,700 = 580$. Both routes give the same figure.

Process X Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,500	—	1,250	By Transfer to Process Y W1: cost 3,750 + profit 750	2,500	1.80	4,500
To Labour	—	—	625				
To Overheads	—	—	1,875				
Actual cost	—	—	3,750				
To Profit — 20% on cost W1: 20% of 3,750	—	—	750				
Total	2,500	—	4,500	Total	2,500	—	4,500

Process Y Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process X	2,500	1.80	4,500	By Transfer to Finished Stock W2: cost 7,000 + profit 700	2,500	3.08	7,700
To Indirect material	—	—	1,250				
To Labour	—	—	500				
To Overheads	—	—	750				
Actual cost	—	—	7,000				
To Profit — 10% on cost W2: 10% of 7,000	—	—	700				
Total	2,500	—	7,700	Total	2,500	—	7,700

Finished Stock Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Transfer from Process Y	2,500	3.08	7,700	By Sales	1,500	5	7,500
To Profit & Loss A/c (profit on sale) W3: 7,500 – 4,620	—	—	2,880	By Closing stock W3: 1,000 × ₹3.08	1,000	3.08	3,080
Total	2,500	—	10,580	Total	2,500	—	10,580

THE TRAP / WHERE MARKS ARE LOST

- **Confusing “20% on cost” with “20% on transfer price”.** On cost: profit = $0.20 \times \text{cost}$. On transfer price: profit = $0.20 \times \text{TP}$, which means profit = $0.25 \times \text{cost}$. Q16 uses the second form — compare the two side by side once and you will never mix them again.
- Forgetting the reserve for unrealised profit entirely. It is a separate, named figure and questions ask for it explicitly.
- Charging Process Y with X's cost of 3,750. Y is charged the *transfer price* of 4,500 — that is the whole point of the topic.

FINAL ANSWER

Process X — actual cost / transfer price	₹ 3,750 / ₹ 4,500
Process Y — actual cost / transfer price	₹ 7,000 / ₹ 7,700
Profit on sale of 1,500 units	₹ 2,880
Closing stock (1,000 units)	₹ 3,080
Reserve for unrealised profit	₹ 580
Total profit shown (750 + 700 + 2,880)	₹ 4,330
Real profit after the reserve	₹ 3,750

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A product passes through three processes to completion, known as A, B and C. The output of each process is charged to the next process at a price calculated to give a profit of **20% on the transfer price**. The output of Process C is charged to finished stock on a similar basis. There was no partly finished WIP in any process on December 31st, on which day the following information was obtained.

Particulars	Process A	Process B	Process C
Materials	4,000	6,000	2,000
Labour	6,000	4,000	8,000
Stock: 31st Dec	2,000	4,000	6,000

There was no stock in hand on Jan 1st and overheads were ignored. Of the goods passed into finished stock, ₹4,000 remained in hand on Dec 31st and the balance has been sold for ₹36,000. Show the Process A/c and calculate reserve for unrealised profits.

READ THE QUESTION FIRST — WHAT TO SPOT

- **“20% on the transfer price”** — not on cost. So if the transfer price is 10,000 the profit is 2,000 and the cost is 8,000. Working backwards: Transfer price = Cost ÷ 0.80.
- **Every process has closing stock this time.** That stock is deducted *before* you load the profit, because you only charge profit on what actually leaves.
- Because stock sits in B and C, and those stocks contain profit passed on from earlier processes, you must track **cost and profit in separate columns**. This is the **three-column format** — Cost | Profit | Total. Use it and this problem is straightforward; try to do it in one column and you will not be able to find the reserve.
- Process A's own stock contains **no** profit — nothing has been marked up into it yet. Only B, C and Finished Stock hold unrealised profit.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. Draw three columns for every process: **Cost, Profit, Total**.
2. Enter the transfer-in split into its cost and profit parts, then the process's own costs (all in the Cost column — own costs carry no profit).
3. Total the three columns.
4. Deduct closing stock. Split it in the **same cost : profit ratio as the total debit**: stock cost part = stock $\times \frac{\text{total cost}}{\text{total debit}}$.
5. What remains is the cost of goods transferred. Transfer price = that cost ÷ 0.80; the difference is the profit for this process.
6. Repeat for the next process. In Finished Stock do the same split for the goods in hand.
7. **Reserve = the Profit column of every closing stock added together.**
8. Cross-check: total apparent profit – reserve = the realised profit from Finished Stock.

WORKING NOTES

W1 Turning “20% on transfer price” into arithmetic

$$\text{Transfer price} = \frac{\text{Cost of goods transferred}}{0.80} \quad \text{and} \quad \text{Profit} = 20\% \text{ of that price}$$

Because Cost = 80% of the transfer price. Note this is the same as 25% on cost — but always work from the price, as the question states it.

W2 Splitting each closing stock into cost and profit

Process	Total debit	of which Cost	Closing stock	Stock — cost part	Stock — profit part
A	10,000	10,000	2,000	2,000	Nil
B	20,000	18,000	4,000	$4,000 \times 18/20 = 3,600$	400
C	30,000	24,400	6,000	$6,000 \times 24,400/30,000 = 4,880$	1,120
Finished Stock	30,000	19,520	4,000	$4,000 \times 19,520/30,000 = 2,602.67$	1,397.33
Total reserve					2,917.33

Process A Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Materials	4,000	—	4,000
To Labour	6,000	—	6,000
Total debit	10,000	—	10,000
Less: Closing stock	(2,000)	—	(2,000)
Cost of goods transferred	8,000	—	8,000
Add: Profit — 20% on transfer price $8,000 \div 0.80 = 10,000$; profit 2,000	—	2,000	2,000
Transferred to Process B	8,000	2,000	10,000

Process B Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process A	8,000	2,000	10,000
To Materials	6,000	—	6,000
To Labour	4,000	—	4,000
Total debit	18,000	2,000	20,000
Less: Closing stock W2: 4,000 split 18/20	(3,600)	(400)	(4,000)
Cost of goods transferred	14,400	1,600	16,000
Add: Profit $16,000 \div 0.80 = 20,000$; profit 4,000	—	4,000	4,000
Transferred to Process C	14,400	5,600	20,000

Process C Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process B	14,400	5,600	20,000
To Materials	2,000	—	2,000
To Labour	8,000	—	8,000

Particulars	Cost ₹	Profit ₹	Total ₹
Total debit	24,400	5,600	30,000
Less: Closing stock W2: 6,000 split 24,400/30,000	(4,880)	(1,120)	(6,000)
Cost of goods transferred	19,520	4,480	24,000
Add: Profit $24,000 \div 0.80 = 30,000$; profit 6,000	—	6,000	6,000
Transferred to Finished Stock	19,520	10,480	30,000

Finished Stock Account

Particulars	Cost ₹	Profit ₹	Total ₹
To Transfer from Process C	19,520	10,480	30,000
Less: Closing stock in hand W2: 4,000 split 19,520/30,000	(2,602.67)	(1,397.33)	(4,000)
Cost of sales	16,917.33	9,082.67	26,000
Sales	—	—	36,000
Profit on sale $36,000 - 26,000$	—	10,000	10,000
Total profit realised	—	19,082.67	—

Reconciliation — proof that the reserve is right

Particulars	Amount ₹
Profit shown by Process A	2,000
Profit shown by Process B	4,000
Profit shown by Process C	6,000
Profit on sale in Finished Stock	10,000
Total apparent profit	22,000.00
Less: Reserve for unrealised profit (W2)	(2,917.33)
Actual (realised) profit	19,082.67
Agrees with the Finished Stock A/c profit column ✓	19,082.67

WHY THIS WORKS

The reserve exists because the company has already recorded ₹22,000 of profit, but ₹2,917.33 of that is sitting inside goods it still owns. You cannot count profit on goods you have not sold. Removing the reserve brings the books back to the profit actually earned — and the fact that it reconciles to the paisa is your proof the whole answer is right.

FINAL ANSWER

Process A — profit / transfer price	₹ 2,000 / ₹ 10,000
Process B — profit / transfer price	₹ 4,000 / ₹ 20,000
Process C — profit / transfer price	₹ 6,000 / ₹ 30,000
Profit on sale	₹ 10,000
Total apparent profit	₹ 22,000
Reserve for unrealised profit	₹ 2,917.33
Actual realised profit	₹ 19,082.67

Reserve built up as: A nil + B 400 + C 1,120 + Finished Stock 1,397.33 = ₹2,917.33.

Equivalent Production

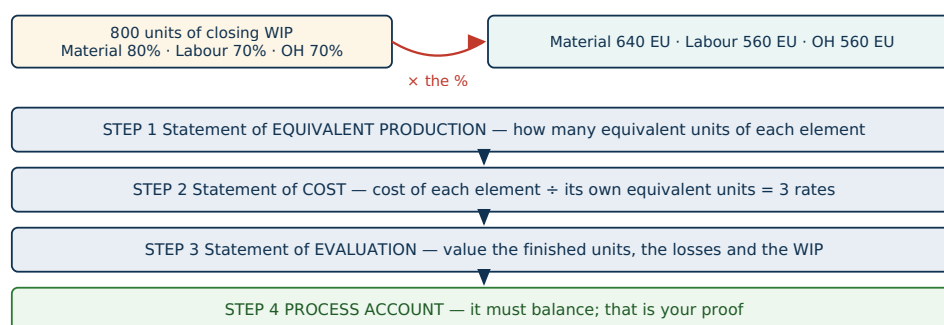
Problems 17 to 24 · workbook pages 13 to 16

THE SINGLE IDEA BEHIND ALL EIGHT PROBLEMS

Until now every unit was either finished or lost. Now some units are **half-finished** at the end of the month. You cannot divide cost by “3,000 finished units + 800 half-finished units”, because those 800 have not absorbed a full unit’s worth of cost.

So you convert them. **800 units that are 70% complete are treated as 560 complete units.** That converted figure is called an **equivalent unit**, and once you have it the problem collapses back into the ordinary “total cost ÷ units” you already know.

The one extra wrinkle: material, labour and overhead are usually at *different* stages of completion. So you do the conversion **three times** and get **three separate rates**.



Four statements, always in this order, in every one of Q17 to Q24. Never skip one.

The rules for the equivalent-production statement

What appears in the output column	Material	Labour	Overhead	Why
Units completed & transferred	100%	100%	100%	They are finished. Full cost in every element.
Normal loss	Nil	Nil	Nil	This is the rule students forget. Normal loss gets zero equivalent units — its cost is absorbed by the good units, so giving it units would double-count.
Abnormal loss	its % of completion	its %	its %	It is a real loss of real cost, so it must carry equivalent units. If the question says the scrapped units were 100% complete, use 100%.
Abnormal gain	100%	100%	100%	Always 100%, and it is deducted in the statement, not added.

What appears in the output column	Material	Labour	Overhead	Why
Closing WIP	its stated %	its stated %	its stated %	The whole reason the topic exists.
Opening WIP — FIFO only	the % still <i>to do</i>	the % to do	the % to do	Under FIFO the opening WIP was partly done last month, so only the balance belongs to this month. 40% complete means 60% to do.

FIFO VERSUS AVERAGE — THE ONLY DIFFERENCE THAT MATTERS

	FIFO method	Average method
Opening WIP units in the statement	Only the unfinished portion (the % still to do)	Not shown separately — the opening units are inside the completed figure at 100%
Opening WIP cost	Kept apart. Added back only at the evaluation stage, on top of the cost of finishing it.	Merged with this month's cost before dividing.
Which costs go into the rate	This month's costs only	Opening WIP cost + this month's costs
Use it when	The question says FIFO, or gives the opening WIP's degree of completion element by element	The question says average, or gives the opening WIP cost as one lump you cannot split

Q21 and Q22 are FIFO. Q23 is average. **Q24 asks for both** on the same data, which is the best possible way to see the difference — work it and compare the two totals.

Q17 The four statements, cleanest case

Equivalent production · p.13

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In Process A on March 1st, there was no opening work-in-progress. During the month of March, 2,000 units of material was issued at a cost of ₹18,000. Labour and overheads totalled to ₹9,000 and ₹6,600 respectively. On 31st March 1,500 units were completed and transferred to the next process. Of the remaining 500 units which are incomplete, degree of completion was: Material 100%, Labour 60% and Overheads 30%.

Prepare (a) Statement of equivalent production (b) Statement of cost (c) Statement of evaluation (d) Process account.

READ THE QUESTION FIRST — WHAT TO SPOT

- **No opening WIP and no losses.** The simplest possible version — the perfect one to learn the four statements on.
- Units: 2,000 in = 1,500 transferred + 500 closing WIP. Balanced, so nothing is lost.
- **Material is 100% complete.** That is normal — material is usually issued in full at the start, so closing WIP has all its material but only part of its labour and overhead.
- Three different percentages means **three different rates**. Do not average them into one.

(a) Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	1,500	100	1,500	100	1,500	100	1,500
Closing work-in-progress	500	100	500	60	300	30	150
Total equivalent units			2,000		1,800		1,650

(b) Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material	18,000	2,000	9.00
Labour	9,000	1,800	5.00
Overhead	6,600	1,650	4.00
Total cost per unit			18.00

(c) Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	1,500 units × ₹18.00	27,000
Closing work-in-progress		
Material	500 EU × ₹9.00	4,500
Labour	300 EU × ₹5.00	1,500
Overhead	150 EU × ₹4.00	600
Value of closing WIP		6,600
Total		33,600

(d) Process A Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Material	2,000	—	18,000	By Transfer to next process <i>1,500 × ₹18</i>	1,500	18	27,000
To Labour	—	—	9,000	By Closing WIP <i>statement (c)</i>	500	—	6,600
To Overheads	—	—	6,600				
Total	2,000	—	33,600	Total	2,000	—	33,600

THE TRAP / WHERE MARKS ARE LOST

- Dividing all three costs by 2,000. Only material has 2,000 equivalent units.
- Valuing closing WIP at $500 \times \text{total rate } 18 = 9,000$. Wrong — you must value each element on **its own** equivalent units, giving 6,600.
- Skipping the statement of evaluation and jumping to the account. The three statements each carry marks of their own.

FINAL ANSWER

Cost per unit — Material / Labour / Overhead **₹ 9.00 / ₹ 5.00 / ₹ 4.00**

Total cost per unit **₹ 18.00**

Value of 1,500 units transferred **₹ 27,000**

Value of closing WIP (500 units) **₹ 6,600**

Process account total **₹ 33,600**

Proof: $18,000 + 9,000 + 6,600 = 33,600 = 27,000 + 6,600$

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

During January 2,000 units were introduced into Process 1. The normal loss was estimated at 5% on input. At the end of the month 1,400 units had been produced and transferred to the next process. 460 units were incomplete and 140 units had been scrapped. It was estimated that incomplete units had reached a stage in production as follows: Material 75% complete, Labour 50% complete and overheads 50% complete.

The cost of 2,000 units introduced was ₹5,800. Direct material introduced during the process amounted to ₹1,440. Direct labour cost was ₹3,340. Production overheads incurred were ₹1,670. Units scrapped realised ₹1 each. The units scrapped have passed through the processes so were 100% complete as regards material, labour and overheads.

You are required to (a) prepare a statement of equivalent production (b) evaluate the cost of abnormal loss, finished goods and closing stock and (c) prepare the Process 1 a/c and abnormal loss a/c.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units: 2,000 = 1,400 transferred + 460 WIP + 140 scrapped. Balanced
- **Normal loss = 5% of 2,000 = 100 units.** Actually scrapped = 140. So **abnormal loss = 40 units.**
- **Normal loss gets NIL equivalent units** in the statement. Abnormal loss gets 100% in every element, because the question tells you the scrapped units were fully processed.
- The scrap money of ₹1 per unit applies to the **normal loss units only** when you reduce the material cost. The abnormal loss units also fetch ₹1 each, but that goes in the Abnormal Loss account, not in the cost statement.

(a) Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	1,400	100	1,400	100	1,400	100	1,400
Normal loss (5% of 2,000)	100	—	Nil	—	Nil	—	Nil
Abnormal loss	40	100	40	100	40	100	40
Closing work-in-progress	460	75	345	50	230	50	230
Total equivalent units			1,785		1,670		1,670

(b1) Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 5,800 + 1,440 – 100 scrap	7,140	1,785	4.00
Labour	3,340	1,670	2.00
Overhead	1,670	1,670	1.00
Total cost per unit			7.00

Material cost = cost of units introduced 5,800 + material added 1,440 = 7,240

Less: scrap value of **normal loss** 100 units × ₹1 = (100)

Net material cost = ₹7,140

(b2) Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	1,400 × ₹7.00	9,800
Abnormal loss	40 × ₹7.00	280
Closing work-in-progress		
Material	345 EU × ₹4.00	1,380
Labour	230 EU × ₹2.00	460
Overhead	230 EU × ₹1.00	230
Value of closing WIP		2,070
Total		12,150

(c) Process 1 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Units introduced	2,000	—	5,800	By Normal loss 100 units × ₹1	100	1	100
To Direct material	—	—	1,440	By Abnormal loss 40 × ₹7	40	7	280
To Direct labour	—	—	3,340	By Transfer to next process 1,400 × ₹7	1,400	7	9,800
To Production overheads	—	—	1,670	By Closing WIP statement (b2)	460	—	2,070
Total	2,000	—	12,250	Total	2,000	—	12,250

Abnormal Loss Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Process 1 A/c	40	7	280	By Cash / Cost Ledger (scrap) 40 × ₹1	40	1	40
Total	40	—	280	By Profit & Loss A/c	—	—	240
				Total	40	—	280

THE TRAP / WHERE MARKS ARE LOST

- **Giving the normal loss 100 equivalent units.** It gets nil. This is the single most common error in the topic.
- Deducting the scrap of all 140 scrapped units (140) from material cost instead of only the normal loss 100.
- Forgetting that the abnormal loss must be valued at the full rate of 7.00 even though it only sells for 1.00.

FINAL ANSWER

Equivalent units — Material / Labour / Overhead	1,785 / 1,670 / 1,670
Cost per unit — Material / Labour / Overhead	₹ 4.00 / ₹ 2.00 / ₹ 1.00
Total cost per unit	₹ 7.00
Finished goods (1,400 units)	₹ 9,800
Abnormal loss (40 units)	₹ 280 — net to P&L ₹ 240
Closing stock (460 units)	₹ 2,070

Proof: 12,250 debits = 100 + 280 + 9,800 + 2,070 = 12,250

Q19 Closing WIP has to be derived

Equivalent production · p.14

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare statement of equivalent production, statement of cost and process a/c from the following information.

Particulars	
Units introduced	7,600
Output (units)	6,000
Process cost — Materials	₹ 14,560
Process cost — Labour	₹ 21,360
Process cost — Overheads	₹ 14,240
Degree of completion for closing WIP — Materials	80%
Degree of completion for closing WIP — Labour	70%
Degree of completion for closing WIP — Overheads	70%

READ THE QUESTION FIRST — WHAT TO SPOT

- The closing WIP figure is **not given** — you must derive it: 7,600 introduced – 6,000 output = **1,600 units** of closing WIP.
- No losses are mentioned and the units balance exactly, so there is no normal or abnormal loss working at all.
- No opening WIP either, so the FIFO / average question does not arise.

Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	6,000	100	6,000	100	6,000	100	6,000
Closing WIP $7,600 - 6,000$	1,600	80	1,280	70	1,120	70	1,120
Total equivalent units			7,280		7,120		7,120

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material	14,560	7,280	2.00
Labour	21,360	7,120	3.00
Overhead	14,240	7,120	2.00
Total cost per unit			7.00

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	$6,000 \times ₹7.00$	42,000
Closing work-in-progress		

Particulars	Working	Amount ₹
Material	1,280 EU × ₹2.00	2,560
Labour	1,120 EU × ₹3.00	3,360
Overhead	1,120 EU × ₹2.00	2,240
Value of closing WIP		8,160
Total		50,160

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	7,600	—	14,560	By Transfer to next process	6,000	7	42,000
To Labour	—	—	21,360	By Closing WIP	1,600	—	8,160
To Overheads	—	—	14,240				
Total	7,600	—	50,160	Total	7,600	—	50,160

FINAL ANSWER

Closing WIP (derived) **1,600 units**

Equivalent units — Material / Labour / Overhead **7,280 / 7,120 / 7,120**

Cost per unit **₹ 2.00 + ₹ 3.00 + ₹ 2.00 = ₹ 7.00**

Transferred out (6,000 units) **₹ 42,000**

Closing WIP **₹ 8,160**

Proof: 14,560 + 21,360 + 14,240 = 50,160 = 42,000 + 8,160

Q20 Normal loss, no abnormal item

Equivalent production · p.14

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare the process account from the following. Number of units introduced into the process — 4,000 units. Units completed and transferred — 3,000 units. Work-in-progress at the end of the period — 800 units (stage of completion material 80%, labour 70% and overheads 70%). Normal loss at the end of the period estimated 200 units (scrap value at ₹1 per unit). Value of materials ₹7,480, wages ₹10,680 and overheads ₹7,120.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units: 4,000 = 3,000 transferred + 800 WIP + 200 normal loss. Balanced exactly, so **there is no abnormal loss or gain** — do not invent one.
- Normal loss again takes **nil equivalent units**, and its scrap value of 200 × ₹1 = ₹200 comes off the material cost.

Statement of Equivalent Production

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred	3,000	100	3,000	100	3,000	100	3,000
Normal loss	200	—	Nil	—	Nil	—	Nil

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Closing WIP	800	80	640	70	560	70	560
Total equivalent units			3,640		3,560		3,560

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 7,480 – 200 scrap	7,280	3,640	2.00
Wages	10,680	3,560	3.00
Overhead	7,120	3,560	2.00
Total cost per unit			7.00

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	3,000 × ₹7.00	21,000
Closing work-in-progress		
Material	640 EU × ₹2.00	1,280
Wages	560 EU × ₹3.00	1,680
Overhead	560 EU × ₹2.00	1,120
Value of closing WIP		4,080
Total		25,080

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Materials	4,000	—	7,480	By Normal loss 200 × ₹1	200	1	200
To Wages	—	—	10,680	By Transfer to next process	3,000	7	21,000
To Overheads	—	—	7,120	By Closing WIP	800	—	4,080
Total	4,000	—	25,280	Total	4,000	—	25,280

FINAL ANSWER

Equivalent units — Material / Wages / Overhead	3,640 / 3,560 / 3,560
Cost per unit	₹ 2.00 + ₹ 3.00 + ₹ 2.00 = ₹ 7.00
Transferred out (3,000 units)	₹ 21,000
Closing WIP (800 units)	₹ 4,080
Normal loss scrap	₹ 200
Proof: 25,280 = 200 + 21,000 + 4,080 = 25,280	

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company follows process costing and manufactures a product in one process. The work-in-process at the end of each month is valued according to **FIFO** method. At the beginning of the month of January, the inventory of WIP showed 400 units, 40% completed, valued as follows: Materials ₹3,600; Labour ₹3,400; Overheads ₹1,000 — total ₹8,000.

In the month of January, materials were purchased for ₹75,000. Wages and overheads in the month amounted to ₹79,800 and ₹21,280 respectively. **Actual issue of materials to production was ₹68,500.** Finished production taken into the stock in the month was 2,500 units. There was no loss in the process. At the end of the month, the WIP inventory was 500 units and 80% complete as regards materials and 60% complete as regards labour and overhead.

You are required to compute equivalent production and prepare process account.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The ₹75,000 of purchases is a deliberate trap.** Purchases are not a process cost. Only what is **issued to production** — ₹68,500 — enters the process. Use 68,500 and ignore 75,000.
- **FIFO is stated explicitly.** So the opening WIP of 400 units, already 40% done, needs only the remaining **60%** of work this month → 240 equivalent units in each element.
- Units introduced must be derived: 2,500 completed + 500 closing WIP – 400 opening WIP = **2,600 units introduced.**
- Under FIFO the **opening WIP cost of ₹8,000 is kept out of the rate.** It is added back only when you value the finished goods.

WORKING NOTES

W1 Unit flow

Opening WIP = 400 units | Completed = 2,500 units | Closing WIP = 500 units

Units introduced = 2,500 + 500 – 400 = 2,600 units

Of the 2,500 completed, 400 came from opening WIP → **2,100 were introduced and completed this month**

Statement of Equivalent Production — FIFO

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening WIP — work still to do 40% done, so 60% remains	400	60	240	60	240	60	240
Introduced and completed this month	2,100	100	2,100	100	2,100	100	2,100
Closing WIP	500	80	400	60	300	60	300
Total equivalent units			2,740		2,640		2,640

Statement of Cost — this month's costs only

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material issued, not purchased	68,500	2,740	25.0000
Labour	79,800	2,640	30.2273

Element	Cost ₹	Equivalent units	Cost per unit ₹
Overhead	21,280	2,640	8.0606
Total cost per unit			63.2879

The labour and overhead rates do not come out round in this problem. Carry four decimal places through the evaluation and the process account will still balance to the rupee — which is shown below.

Statement of Evaluation

Particulars	Working	Amount ₹
1. Opening WIP completed (400 units)		
Cost brought forward	given	8,000.00
Material to finish	240 EU × ₹25.0000	6,000.00
Labour to finish	240 EU × ₹30.2273	7,254.55
Overhead to finish	240 EU × ₹8.0606	1,934.55
Total for the 400 units		23,189.10
2. Introduced and completed (2,100 units)		
At full cost	2,100 × ₹63.2879	1,32,904.55
3. Closing WIP (500 units)		
Material	400 EU × ₹25.0000	10,000.00
Labour	300 EU × ₹30.2273	9,068.18
Overhead	300 EU × ₹8.0606	2,418.18
Value of closing WIP		21,486.36
Total		1,77,580.01

Process Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening WIP	400	—	8,000	By Finished goods transferred 23,189.10 + 1,32,904.55	2,500	—	1,56,093.65
To Materials issued	2,600	—	68,500	By Closing WIP	500	—	21,486.36
To Wages	—	—	79,800				
To Overheads	—	—	21,280				
Total	3,000	—	1,77,580	Total	3,000	—	1,77,580.01

THE TRAP / WHERE MARKS ARE LOST

- Using the ₹75,000 **purchases** figure instead of the ₹68,500 **issued** figure. Purchases go to the stores account, not the process.
- Giving the opening WIP 400 equivalent units. Under FIFO it gets only the 240 that represent work done *this* month.
- Adding the opening WIP cost of ₹8,000 into the numerator before dividing. Under FIFO it stays out of the rate and is added back at evaluation.

FINAL ANSWER

Units introduced (derived)	2,600
Equivalent units — Material / Labour / Overhead	2,740 / 2,640 / 2,640
Cost per unit — Material	₹ 25.0000
Cost per unit — Labour	₹ 30.2273
Cost per unit — Overhead	₹ 8.0606
Finished goods (2,500 units)	₹ 1,56,093.65
Closing WIP (500 units)	₹ 21,486.36

Proof: $8,000 + 68,500 + 79,800 + 21,280 = 1,77,580$ and $1,56,093.65 + 21,486.36 = 1,77,580.01$ (1 paisa rounding).

Q22 FIFO with scrap and two material inputs

Equivalent production · p.15

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare Process 2 account from the following. Opening stock 600 units at ₹1,050 (degree of completion material 80%, labour 60%, overheads 60%). Transfer from Process 1 — 11,000 units at ₹5,500. Transfer to Process 3 — 8,800 units. Direct material added in Process 2 — ₹2,410. Direct labour ₹7,155, Production overhead ₹9,540. Units scrapped — 1,200 (degree of completion material 100%, labour 70% and overheads 70%). Closing stock 1,600 units (degree of completion material 70%, labour 60%, overheads 60%). There was a normal loss in the process at 10% of production. Units scrapped would realise 50 paise per unit.

READ THE QUESTION FIRST — WHAT TO SPOT

- Units in: opening 600 + received 11,000 = 11,600. Units out: 8,800 + 1,200 scrapped + 1,600 closing = 11,600
- “Normal loss at 10% of production” is ambiguous wording. The standard reading is 10% of the units introduced into the process from Process 1 = 10% of 11,000 = **1,100 units**. State your assumption in one line. Abnormal loss then = $1,200 - 1,100 = 100$ units.
- The opening stock’s degrees of completion are given element by element, so **use FIFO**.
- Two material figures must be added: the ₹5,500 transferred in and the ₹2,410 added here. Then take off the normal loss scrap of $1,100 \times ₹0.50 = ₹550$.

WORKING NOTES

W1 Loss analysis (assumption stated)

Normal loss = 10% of 11,000 units introduced = **1,100 units**

Units actually scrapped = 1,200 units

Abnormal loss = $1,200 - 1,100 = 100$ units

Scrap credit taken to cost statement = $1,100 \times ₹0.50 = ₹550$

W2 FIFO unit flow

Opening stock 600 units, so of the 8,800 transferred out, 600 came from opening stock

Introduced and completed = $8,800 - 600 = 8,200$ units

Statement of Equivalent Production — FIFO

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening stock — work still to do Mat 20%, Lab 40%, OH 40%	600	20	120	40	240	40	240
Introduced and completed	8,200	100	8,200	100	8,200	100	8,200
Normal loss (W1)	1,100	—	Nil	—	Nil	—	Nil
Abnormal loss (W1)	100	100	100	70	70	70	70
Closing stock	1,600	70	1,120	60	960	60	960
Total equivalent units			9,540		9,470		9,470

Statement of Cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material 5,500 + 2,410 – 550	7,360	9,540	0.7715
Labour	7,155	9,470	0.7555
Overhead	9,540	9,470	1.0074
Total cost per unit			2.5344

Statement of Evaluation

Particulars	Working	Amount ₹
1. Opening stock completed (600 units)		
Cost brought forward	given	1,050.00
Material to finish	120 EU × ₹0.7715	92.58
Labour to finish	240 EU × ₹0.7555	181.33
Overhead to finish	240 EU × ₹1.0074	241.77
Sub-total		1,565.68
2. Introduced and completed (8,200 units)		
At full cost	8,200 × ₹2.5344	20,782.28
Transferred to Process 3 (8,800 units)		22,347.96
3. Abnormal loss (100 units)		
Material / Labour / Overhead	100×0.7715 + 70×0.7555 + 70×1.0074	200.56
4. Closing stock (1,600 units)		
Material	1,120 EU × ₹0.7715	864.07
Labour	960 EU × ₹0.7555	725.32
Overhead	960 EU × ₹1.0074	967.10
Value of closing stock		2,556.49
Total		25,105.01

Process 2 Account

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock	600	—	1,050	By Normal loss $1,100 \times ₹0.50$	1,100	0.50	550
To Transfer from Process 1	11,000	0.50	5,500	By Abnormal loss	100	—	200.56
To Direct material added	—	—	2,410	By Transfer to Process 3	8,800	—	22,347.96
To Direct labour	—	—	7,155	By Closing stock	1,600	—	2,556.49
To Production overhead	—	—	9,540				
Total	11,600	—	25,655	Total	11,600	—	25,655.01

THE TRAP / WHERE MARKS ARE LOST

About the ambiguous wording. “Normal loss at 10% of production” could be read as 10% of units introduced (1,100), 10% of total input including opening stock (1,160), or 10% of output transferred (880). I have used **10% of units introduced = 1,100**, which is the most common convention. In the exam, **write down which basis you have used** in one line. Method marks are given for a stated, consistent assumption; they are lost only for silence.

FINAL ANSWER

Normal loss / Abnormal loss	1,100 units / 100 units
Equivalent units — Material / Labour / Overhead	9,540 / 9,470 / 9,470
Cost per unit	₹ 0.7715 + ₹ 0.7555 + ₹ 1.0074 = ₹ 2.5344
Transferred to Process 3 (8,800 units)	₹ 22,347.96
Abnormal loss (100 units)	₹ 200.56
Closing stock (1,600 units)	₹ 2,556.49

Proof: debits 1,050 + 5,500 + 2,410 + 7,155 + 9,540 = 25,655 and credits 550 + 200.56 + 22,347.96 + 2,556.49 = 25,655.01

Q23 The average method

Equivalent production · p.15-16

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following figures relate to a single industrial process. Opening stock 10,000 units — Material ₹2,250, Wages ₹650, Overheads ₹400, total ₹3,300. Units introduced 40,000 units — Material ₹9,250, Labour ₹4,600, Overheads ₹3,100.

During the period 30,000 units were completed and 20,000 units remained in process. The degree of completion of closing stock of WIP was: Materials 100%, Labour 25%, Overheads 25%. Make necessary computations and prepare process account by using the **average method**.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The question says average method**, so everything changes from Q21 and Q22:
- The opening stock does **not** appear as a separate line in the equivalent production statement. Its 10,000 units are simply inside the 30,000 completed, at 100%.
- The opening stock **cost is merged** with this month's cost before you divide. Material becomes 2,250 + 9,250 = 11,500, and so on.
- Units: 10,000 + 40,000 = 50,000 = 30,000 completed + 20,000 WIP no losses.

Statement of Equivalent Production — Average method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred <i>includes the opening 10,000 at 100%</i>	30,000	100	30,000	100	30,000	100	30,000
Closing WIP	20,000	100	20,000	25	5,000	25	5,000
Total equivalent units			50,000		35,000		35,000

Statement of Cost — opening cost merged with current cost

Element	Cost ₹	Equivalent units	Cost per unit ₹
Material <i>2,250 + 9,250</i>	11,500	50,000	0.23
Wages <i>650 + 4,600</i>	5,250	35,000	0.15
Overheads <i>400 + 3,100</i>	3,500	35,000	0.10
Total cost per unit			0.48

Statement of Evaluation

Particulars	Working	Amount ₹
Units completed & transferred	30,000 × ₹0.48	14,400
Closing work-in-progress		
Material	20,000 EU × ₹0.23	4,600
Wages	5,000 EU × ₹0.15	750
Overheads	5,000 EU × ₹0.10	500
Value of closing WIP		5,850
Total		20,250

Process Account (average method)

Particulars	Units	Rate	Amount ₹	Particulars	Units	Rate	Amount ₹
To Opening stock	10,000	—	3,300	By Transfer to finished stock	30,000	0.48	14,400
To Material	40,000	—	9,250	By Closing WIP	20,000	—	5,850
To Labour	—	—	4,600				
To Overheads	—	—	3,100				
Total	50,000	—	20,250	Total	50,000	—	20,250

WHY THIS WORKS

Notice how much shorter this is than Q21. Under the average method you never separate “work done last month” from “work done this month” — you pool everything and take one average rate. That is why it is the easier method, and why examiners often specify FIFO instead.

FINAL ANSWER

Equivalent units — Material / Wages / Overheads **50,000 / 35,000 / 35,000**

Cost per unit **₹ 0.23 + ₹ 0.15 + ₹ 0.10 = ₹ 0.48**

Transferred out (30,000 units) **₹ 14,400**

Closing WIP (20,000 units)

₹ 5,850

Proof: $3,300 + 9,250 + 4,600 + 3,100 = 20,250 = 14,400 + 5,850$

Q24 FIFO and average on the same data

Equivalent production · p.16

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prepare the statement of equivalent production from the following under **FIFO and Average Method**.

Opening work-in-progress — 2,000 units (completed as material 80%, labour 60% and overheads 60%). Units introduced and completed 5,000 units. Units remaining as closing work-in-progress — 3,000 units (completed as material 80%, labour and overheads 60% each). There are no process losses.

READ THE QUESTION FIRST — WHAT TO SPOT

- No costs are given, so **only the two statements are wanted** — no cost statement, no evaluation, no process account. Answer exactly what is asked.
- Total units completed = opening WIP 2,000 (finished first) + 5,000 introduced and completed = **7,000 units**. Units introduced = 5,000 + 3,000 closing = **8,000 units**.
- This is the best problem in the module for **seeing** the FIFO/average difference, because the same data gives two different answers. Work both and compare the totals.

(a) Statement of Equivalent Production — FIFO method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Opening WIP — work still to do Mat 20%, Lab 40%, OH 40%	2,000	20	400	40	800	40	800
Introduced and completed	5,000	100	5,000	100	5,000	100	5,000
Closing WIP	3,000	80	2,400	60	1,800	60	1,800
Total equivalent units			7,800		7,600		7,600

(b) Statement of Equivalent Production — Average method

Output / Particulars	Units	Material %	Material EU	Labour %	Labour EU	Overhead %	Overhead EU
Units completed & transferred 2,000 opening + 5,000 = 7,000, all at 100%	7,000	100	7,000	100	7,000	100	7,000
Closing WIP	3,000	80	2,400	60	1,800	60	1,800
Total equivalent units			9,400		8,800		8,800

The difference, side by side

Element	FIFO equivalent units	Average equivalent units	Difference	Why
Material	7,800	9,400	1,600	Average counts the opening WIP's material at 100% (2,000) instead of only the 20% still to do (400). Difference = $2,000 - 400 = 1,600$.
Labour	7,600	8,800	1,200	2,000 at 100% instead of 800 → difference 1,200.

Element	FIFO equivalent units	Average equivalent units	Difference	Why
Overhead	7,600	8,800	1,200	Same as labour.

WHY THIS WORKS

Average always gives the **larger** number of equivalent units, because it counts the whole of the opening WIP instead of just the unfinished part. Larger equivalent units with a larger cost pool (opening cost is merged in too) means the two methods usually give similar rates — but never identical ones. Whenever a question mentions opening WIP, your first job is to find out which method it wants.

FINAL ANSWER

FIFO — Material / Labour / Overhead **7,800 / 7,600 / 7,600**

Average — Material / Labour / Overhead **9,400 / 8,800 / 8,800**

Units completed **7,000**

Units introduced **8,000**

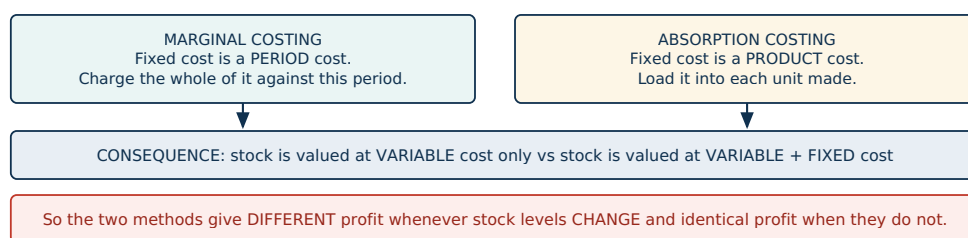
Under FIFO only the 400 / 800 / 800 of opening-WIP work belongs to this period; under average the full 2,000 units are counted at 100%.

Marginal Costing and Absorption Costing

20 problems · workbook pages 21 to 27

The one difference that creates this whole module

There is exactly **one** disagreement between marginal costing and absorption costing, and every problem in this module is a consequence of it:



Fix this picture in your head. It answers every 'why is the profit different?' question.

If ...	Then ...
Production = Sales (no change in stock)	Both methods give the same profit.
Production > Sales (stock going up)	Absorption profit is HIGHER , because some of this period's fixed cost has been parked inside the closing stock instead of being charged.
Production < Sales (stock coming down)	Absorption profit is LOWER , because last period's fixed cost comes out of opening stock and gets charged now, on top of this period's.

Absorption profit – Marginal profit = Fixed overhead in closing stock – Fixed overhead in opening stock

This is the reconciliation. Every marginal-versus-absorption problem in this module can be checked with this one line. If your two profits differ by anything else, you have made an error.

The two families of problem in Module 2

Family	What you are asked to do	Problems
A. Profit statements	Prepare a profit statement under marginal costing, under absorption costing, or both, and reconcile them. Watch stock movements and the under / over absorption of fixed overhead.	1, 15-20

Family	What you are asked to do	Problems
B. Cost-Volume-Profit	Contribution, P/V ratio, break-even point, margin of safety, required sales for a target profit. Pure formula work — the same six formulas every time.	2-14

Profit statements

Use these steps for Q1, 15, 16, 17, 18, 19 and 20

The two statement formats — learn the skeletons, fill in the numbers

MARGINAL COSTING statement	ABSORPTION COSTING statement
Sales	Sales
Less: Opening stock at VARIABLE cost per unit	Less: Opening stock at FULL cost per unit
Add: Variable cost of production	Add: Full cost of production variable + fixed absorbed
Less: Closing stock at VARIABLE cost	Less: Closing stock at FULL cost
= Variable cost of sales	= Cost of sales
CONTRIBUTION = Sales – variable cost of sales	GROSS PROFIT = Sales – cost of sales
Less: ALL fixed costs the actual amount for the period	Less / Add: Under / over absorption of fixed production overhead
	Less: Fixed selling, admin and distribution costs
= PROFIT	= PROFIT

THE SEVEN STEPS, EVERY TIME

1. **Work out the stock movement in units first.** Opening + Production – Sales = Closing. Write this line down before anything else — it tells you immediately which method will show the higher profit.
2. **Variable cost per unit** = direct material + direct labour + variable overheads. (Variable *selling* overhead is a cost of *selling*, so it never goes into stock value.)
3. **Fixed overhead absorption rate** = $\frac{\text{Budgeted fixed production overhead}}{\text{Budgeted / normal output in units}}$. Use **normal or budgeted** output, never actual.
4. **Full cost per unit** = variable production cost + that absorption rate.
5. Build the marginal statement. Charge the **whole actual fixed cost** for the period.
6. Build the absorption statement. Then compute **absorbed = actual production × rate** and compare with the **actual fixed overhead**. Absorbed more than actual = **over-absorption** (add it back); absorbed less = **under-absorption** (deduct it).
7. **Reconcile.** The difference between the two profits must equal the fixed overhead inside the change in stock. If it does not, find your error before writing the answer.

THE THREE ERRORS THAT COST THE MOST MARKS

- **Valuing stock at full cost in the marginal statement.** Under marginal costing stock carries variable cost only. This is the definition of the method.

- **Dividing fixed overhead by actual output to get the absorption rate.** The rate is fixed in advance from *budgeted* or *normal* output. That is precisely why under and over absorption exist — if you used actual output there would never be any.
- **Including variable selling overhead in the stock value.** Unsold goods have not been sold, so no selling cost has been incurred on them. It is deducted after contribution, not before.

Cost-Volume-Profit

Use these formulas for Q2 to Q14

EVERYTHING STARTS FROM ONE EQUATION

$$\text{Sales} - \text{Variable cost} = \text{CONTRIBUTION} = \text{Fixed cost} + \text{Profit}$$

Read it both ways. Left to right it tells you what contribution IS. Right to left it tells you what contribution DOES: it first pays off the fixed cost, and whatever is left over is profit. Break-even is simply the point where contribution has exactly paid the fixed cost and nothing remains.

Every one of the thirteen problems in this family is that equation rearranged. There is nothing else to learn.

The six formulas — and what each one is really asking

You want ...	Formula	In plain words
Contribution	$\text{Sales} - \text{Variable cost}$ or $\text{Fixed cost} + \text{Profit}$	What is left from the selling price after paying the costs that grow with volume.
P/V ratio	$\frac{\text{Contribution}}{\text{Sales}} \times 100$	How many paise out of every rupee of sales survives to help pay the fixed cost. It does not change when volume changes — only when price or variable cost changes.
BEP in units	$\frac{\text{Fixed cost}}{\text{Contribution per unit}}$	How many units must be sold before the fixed cost is fully paid.
BEP in rupees	$\frac{\text{Fixed cost}}{\text{P/V ratio}}$	The same point expressed as a sales value. Use this when the question gives you totals and no unit figures.
Margin of safety	$\frac{\text{Actual sales} - \text{BEP sales}}{\text{P/V ratio}}$ or $\frac{\text{Profit}}{\text{P/V ratio}}$	How far sales can fall before you start making a loss. The cushion.
Sales for a target profit	$\frac{\text{Fixed cost} + \text{Target profit}}{\text{P/V ratio}}$	Treat the target profit exactly like an extra slice of fixed cost that also has to be paid for.

WHEN THE QUESTION GIVES YOU TWO PERIODS AND NO COSTS

Q7 and Q8 give only sales and profit for two periods. You cannot see the fixed cost or the variable cost at all. There is a standard route out, and it works because **fixed cost does not change between the two periods**, so every extra rupee of contribution shows up as an extra rupee of profit.

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100$$

1. Find the P/V ratio from the change in profit over the change in sales.
2. Take any one period. Contribution = Sales × P/V ratio.

3. **Fixed cost = Contribution – Profit** for that same period.

4. Now you have everything — compute BEP and margin of safety normally.

5. **Check** your fixed cost against the *other* period. If it does not give that period's profit too, you have gone wrong.

ROUNDING THE P/V RATIO — A REAL WARNING

When the P/V ratio is a recurring decimal such as 44.44% or 54.55%, **do not round it to 44% or 54% before dividing**. In Q11 parts (v) and (vi) that rounding shifts the break-even figure by several hundred rupees. Keep the ratio as a fraction — 8,000/18,000 rather than 0.44 — and divide once at the end.

Q1 Both statements at normal volume

Statements · p.21

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The basic production data of ABC Ltd are as follows: Normal volume of production = 19,500 units per period. Sale price \$4 per unit. Variable cost \$2 per unit. Fixed cost \$1 per unit. Total fixed cost = \$19,500 (\$1 × 19,500 units, normal). Selling and distribution costs have been omitted. The opening and closing stocks consist of both finished goods and equivalent units of work-in-progress.

Present the profit and loss statement under Absorption and Marginal costing method.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Read the question again and notice what is missing.** You are given the normal volume, the three unit costs and the price — but **no actual production figure and no actual sales figure** for any period. Without those, no stock movement can be computed.
- So the only statement that can be prepared from the data as printed is the one at **normal volume**, where production = sales = 19,500 units and there is no change in stock.
- That turns out to be a useful answer rather than a wasted one, because it demonstrates the headline rule of the module: **when stock does not change, the two methods give exactly the same profit.**
- The sentence about opening and closing stocks is telling you what *would* happen if figures were supplied. The illustration at the end shows it.

(a) Marginal Costing Profit Statement — at normal volume of 19,500 units

Particulars	Per unit \$	Total \$
Sales	4.00	78,000
Less: Variable cost	(2.00)	(39,000)
Contribution	2.00	39,000
Less: Fixed cost charged in full as a period cost		(19,500)
Profit		19,500

(b) Absorption Costing Profit Statement — at normal volume of 19,500 units

Particulars	Per unit \$	Total \$
Sales	4.00	78,000
Less: Cost of sales		
Variable production cost	2.00	39,000
Fixed production overhead absorbed 19,500 × \$1	1.00	19,500
Total cost of sales	3.00	(58,500)
Profit	1.00	19,500

Absorbed overhead = 19,500 units × \$1 = \$19,500 = the actual fixed cost, so there is **no under or over absorption** at normal volume.

WHY THIS WORKS

Both methods give \$19,500. That is not a coincidence and it is the point of the problem: at normal volume the fixed overhead absorbed into production exactly equals the fixed overhead incurred, so nothing is left sitting in stock and the two profits must agree.

The illustration below shows the two cases where they diverge. Learn the direction: **stock up → absorption profit higher; stock down → absorption profit lower.**

What would happen if production and sales differed — an illustration

	Production = Sales 19,500 / 19,500	Production > Sales 19,500 / 17,500	Production < Sales 17,500 / 19,500
Closing stock (units)	Nil	2,000	Nil (2,000 opening used up)
Marginal costing profit	19,500	\$15,500	\$19,500
Absorption costing profit	19,500	\$17,500	\$17,500
Difference	Nil	+2,000	−2,000
Explanation	no stock change	2,000 units × \$1 fixed cost parked in stock	2,000 units × \$1 released from stock

THE TRAP / WHERE MARKS ARE LOST

Note on the data. The question gives no actual production or sales figures, so a full multi-period comparison cannot be prepared from it. If your professor expects a version with stock movements, he will supply the production and sales for each period — the method is exactly the one shown in Q16 and Q20. Say in one line that you have prepared the statement at normal volume, and you protect the marks.

FINAL ANSWER

Contribution per unit	\$ 2.00
Marginal costing profit	\$ 19,500
Absorption costing profit	\$ 19,500
Difference	Nil — production equals sales, so no fixed cost is held in stock

Q2 Straight marginal cost statement

CVP · p.21-22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information, find out the amount of profit earned during the year using marginal cost technique: Fixed Cost ₹5,00,000; Variable Cost ₹10 per unit; Selling Price ₹15 per unit; Output level 1,50,000 units.

READ THE QUESTION FIRST — WHAT TO SPOT

- Everything is per unit except the fixed cost, so multiply the two unit figures by the output of 1,50,000 and put the fixed cost in as a single lump.
- “Using marginal cost technique” means **present it as a statement ending in contribution then profit** — not just the final number. The layout carries marks.
- Always show the **per-unit column beside the total column**. It is free marks and it makes checking trivial.

Marginal Cost Statement

Particulars	Per unit ₹	Total ₹
Sales $1,50,000 \times ₹15$	15	22,50,000
Less: Variable cost $1,50,000 \times ₹10$	(10)	(15,00,000)
Contribution	5	7,50,000
Less: Fixed cost		(5,00,000)
Profit		2,50,000

FINAL ANSWER

Contribution per unit ₹ 5

Total contribution ₹ 7,50,000

Profit ₹ 2,50,000

Cross-check with the marginal cost equation: Contribution = Fixed cost + Profit $\rightarrow 7,50,000 = 5,00,000 + 2,50,000$

Q3 Fixed cost hidden inside the BEP

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Find the profit from the following data: Sales ₹80,000; Marginal Cost ₹60,000; Break-even point ₹60,000.

READ THE QUESTION FIRST — WHAT TO SPOT

- “Marginal cost” is just another name for **variable cost**. So contribution = 80,000 – 60,000 = ₹20,000.
- The fixed cost is **not given** — you have to extract it from the break-even point. That is the whole trick of this problem.
- Work the BEP formula **backwards**: if $\text{BEP} = \text{Fixed cost} \div \text{P/V ratio}$, then $\text{Fixed cost} = \text{BEP} \times \text{P/V ratio}$.

WORKING NOTES

W1 Contribution and P/V ratio

Contribution = Sales – Marginal cost = 80,000 – 60,000 = **₹20,000**
P/V ratio = 20,000 ÷ 80,000 = **25%**

W2 Recovering the fixed cost from the break-even point

$$\text{BEP (in ₹)} = \frac{\text{Fixed cost}}{\text{P/V ratio}} \rightarrow \text{Fixed cost} = \text{BEP} \times \text{P/V ratio}$$

Fixed cost = 60,000 × 25% = **₹15,000**

Marginal Cost Statement

Particulars	Amount ₹
Sales	80,000
Less: Marginal (variable) cost	(60,000)
Contribution	20,000
Less: Fixed cost W2	(15,000)
Profit	5,000

FINAL ANSWER

Contribution **₹ 20,000**
P/V ratio **25%**
Fixed cost (derived) **₹ 15,000**
Profit **₹ 5,000**

Check: margin of safety = 80,000 – 60,000 = 20,000; profit = MOS × P/V ratio = 20,000 × 25% = 5,000 — a second route to the same answer, worth showing.

Q4 Break-even point in units and in rupees

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following particulars, calculate the break-even point: Variable cost per unit ₹12; Fixed Expenses ₹60,000; Selling Price per unit ₹18.

READ THE QUESTION FIRST — WHAT TO SPOT

- Unit figures are given, so use the **units** form of the BEP formula: Fixed cost ÷ contribution per unit.
- Give **both** answers — units and sales value. The question says “break-even point”, and a complete answer states it both ways.

WORKING NOTES

Contribution per unit = $18 - 12 = ₹6$

$$\text{BEP (units)} = \frac{\text{Fixed cost}}{\text{Contribution per unit}} = \frac{60,000}{6} = \mathbf{10,000 \text{ units}}$$

BEP (sales value) = $10,000 \text{ units} \times ₹18 = \mathbf{₹1,80,000}$

Alternative route to the same value: P/V ratio = $6 \div 18 = 33.33\%$, so BEP = $60,000 \div 0.3333 = ₹1,80,000$

FINAL ANSWER

Contribution per unit	₹ 6
BEP in units	10,000 units
BEP in sales value	₹ 1,80,000
P/V ratio	33.33%

Q5 Working the BEP formula backwards

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following particulars, find out the selling price per unit if B.E.P. is to be brought down to 9,000 units. Variable cost per unit ₹75; Fixed Expenses ₹2,70,000; Selling Price per unit ₹100.

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the BEP formula run **backwards**. You are told the answer (9,000 units) and asked for the selling price that produces it.
- First find where you currently stand: at ₹100 the contribution is ₹25 and the BEP is $2,70,000 \div 25 =$ **10,800 units**. The company wants it down to 9,000, which means each unit must contribute **more**.
- Let the new selling price be x . Then contribution per unit $= x - 75$, and $9,000 = 2,70,000 \div (x - 75)$. Solve for x .

WORKING NOTES

W1 The present position

Present contribution $= 100 - 75 = ₹25$ per unit

Present BEP $= 2,70,000 \div 25 =$ **10,800 units**

Target BEP $= 9,000$ units $\rightarrow$ the price must rise

W2 Solving for the new selling price

Required contribution per unit $= \frac{\text{Fixed cost}}{\text{Target BEP units}} = \frac{2,70,000}{9,000} =$ **₹30**

New selling price $= \text{Variable cost} + \text{Required contribution}$
 $= 75 + 30 =$ **₹105 per unit**

Proof: at ₹105 the contribution is ₹30, so BEP $= 2,70,000 \div 30 = 9,000$ units exactly as required.

FINAL ANSWER

Present contribution / BEP **₹ 25 per unit / 10,800 units**

Required contribution per unit **₹ 30**

New selling price **₹ 105 per unit**

Increase needed **₹ 5 per unit, i.e. 5%**

Q6 How a price cut moves the break-even point

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following data, calculate break-even point expressed in terms of units and also the new B.E.P., if the selling price is reduced by 10%.

Particulars	Amount
Fixed Expenses	
Depreciation	₹ 1,00,000
Salaries	₹ 1,00,000
Variable Expenses	
Materials	₹ 3 per unit
Labour	₹ 2 per unit
Selling Price	₹ 10 per unit

READ THE QUESTION FIRST — WHAT TO SPOT

- Add the two fixed items into one figure and the two variable items into one per-unit figure. Do this before you touch the formula.
- **A 10% cut in selling price does not change the fixed cost or the variable cost.** Only the contribution shrinks — and because contribution is the denominator, the break-even point rises. Expect a bigger answer for the second part.
- Notice how sharp the effect is: a 10% price cut here raises the break-even by 25%. That is the lesson of the problem, and worth one closing sentence in your answer.

WORKING NOTES

W1 Present break-even point

Fixed expenses = 1,00,000 + 1,00,000 = **₹2,00,000**

Variable cost per unit = 3 + 2 = **₹5**

Contribution per unit = 10 – 5 = **₹5**

$$\text{BEP} = \frac{2,00,000}{5} = \mathbf{40,000 \text{ units}}$$

W2 New break-even point after a 10% price cut

New selling price = 10 – 10% = **₹9**

Variable cost is unchanged at ₹5

New contribution per unit = 9 – 5 = **₹4**

$$\text{New BEP} = \frac{2,00,000}{4} = \mathbf{50,000 \text{ units}}$$

The effect of the price cut, side by side

	Present	After 10% price cut	Change
Selling price per unit	10	9	–10%
Variable cost per unit	5	5	no change
Contribution per unit	5	4	–20%
P/V ratio	50%	44.44%	–5.56 points
Break-even point (units)	40,000	50,000	+25%

WHY THIS WORKS

A 10% cut in price caused a 20% fall in contribution and a 25% rise in the break-even point. The reason is that the whole of the price cut comes out of contribution — the variable cost does not fall to share the pain. This is why price cuts are so much more dangerous than they look, and saying so in one line turns a numerical answer into an interpreted one.

FINAL ANSWER

Contribution per unit — present / new ₹ 5 / ₹ 4

Present BEP 40,000 units

New BEP after 10% price cut 50,000 units

BEP in value — present / new ₹ 4,00,000 / ₹ 4,50,000

Q7 Two periods, no costs given

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The sales turnover and profits during two periods are as under:

Period	Sales	Profit
Period I	₹ 20 lakhs	₹ 2 lakhs
Period II	₹ 30 lakhs	₹ 4 lakhs

Calculate the P/V ratio, fixed cost, break-even point and margin of safety.

READ THE QUESTION FIRST — WHAT TO SPOT

- You are given **no costs at all** — only sales and profit. This is the two-period type from Playbook B.
- The key insight: **fixed cost is the same in both periods**. So the extra ₹2 lakhs of profit in Period II came entirely from the extra ₹10 lakhs of sales. Every rupee of that increase is pure contribution.
- Once you have the P/V ratio you can work out the contribution of either period, and fixed cost falls out as contribution minus profit.
- **Always verify with the other period**. If your fixed cost does not also reproduce Period I's profit, you have made an error.

WORKING NOTES

W1 P/V ratio from the change

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100 = \frac{₹2,00,000}{₹10,00,000} \times 100 = \mathbf{20\%}$$

W2 Fixed cost

Take Period I: Contribution = Sales × P/V ratio = 20,00,000 × 20% = ₹4,00,000

Fixed cost = Contribution – Profit = 4,00,000 – 2,00,000 = **₹2,00,000**

Verification on Period II: contribution = 30,00,000 × 20% = ₹6,00,000; profit = 6,00,000 – 2,00,000 = ₹4,00,000 matches the given figure.

W3 Break-even point and margin of safety

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{2,00,000}{0.20} = \mathbf{₹10,00,000}$$

MOS Period I = 20,00,000 – 10,00,000 = **₹10,00,000** (50% of sales)

MOS Period II = 30,00,000 – 10,00,000 = **₹20,00,000** (66.67% of sales)

Marginal Cost Statement for both periods

Particulars	Period I ₹	Period II ₹
Sales	20,00,000	30,00,000
Less: Variable cost 80% of sales	(16,00,000)	(24,00,000)
Contribution 20% of sales	4,00,000	6,00,000
Less: Fixed cost W2 — same in both periods	(2,00,000)	(2,00,000)
Profit	2,00,000	4,00,000

FINAL ANSWER

P/V ratio 20%

Fixed cost ₹ 2,00,000

Break-even sales ₹ 10,00,000

Margin of safety — Period I ₹ 10,00,000 (50%)

Margin of safety — Period II ₹ 20,00,000 (66.67%)

Q8 Break-even from two years' results

CVP · p.22

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following data are obtained from the records of a company:

Particulars	First Year ₹	Second Year ₹
Sales	80,000	90,000
Profit	10,000	14,000

Calculate the break-even point.

READ THE QUESTION FIRST — WHAT TO SPOT

- Exactly the same shape as Q7. Change in profit ₹4,000 over change in sales ₹10,000 gives the P/V ratio straight away.
- The question asks only for the break-even point, but **show the fixed cost and the P/V ratio on the way** — those are the working marks.

WORKING NOTES

W1 P/V ratio

$$\text{P/V ratio} = \frac{\text{Change in profit}}{\text{Change in sales}} \times 100 = \frac{14,000 - 10,000}{90,000 - 80,000} \times 100 = \frac{4,000}{10,000} \times 100 = \mathbf{40\%}$$

W2 Fixed cost

$$\text{First year contribution} = 80,000 \times 40\% = \mathbf{₹32,000}$$

$$\text{Fixed cost} = 32,000 - 10,000 = \mathbf{₹22,000}$$

Verification: second year contribution = 90,000 × 40% = ₹36,000; 36,000 – 22,000 = ₹14,000 matches.

W3 Break-even point

$$\text{BEP} = \frac{22,000}{0.40} = \mathbf{₹55,000}$$

Marginal Cost Statement for both years

Particulars	First Year ₹	Second Year ₹
Sales	80,000	90,000
Less: Variable cost (60% of sales)	(48,000)	(54,000)
Contribution (40% of sales)	32,000	36,000
Less: Fixed cost	(22,000)	(22,000)
Profit	10,000	14,000

FINAL ANSWER

P/V ratio **40%**

Fixed cost	₹ 22,000
Break-even point	₹ 55,000
Margin of safety — first year	₹ 25,000
Margin of safety — second year	₹ 35,000

Q9 Margin of safety from profit alone

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company earned a profit of ₹30,000 during a particular year. If the marginal cost and selling price of a product are ₹8 and ₹10 per unit respectively, find out the amount of Margin of Safety.

READ THE QUESTION FIRST — WHAT TO SPOT

- You are not given sales or fixed cost, so you cannot find the BEP and subtract. You need the **other** margin-of-safety formula.
- Because profit is the part of contribution left over *after* the fixed cost is paid, the sales that generated that profit are pure margin of safety. So MOS = profit ÷ P/V ratio.
- In units it is even simpler: MOS units = profit ÷ contribution per unit.

WORKING NOTES

Contribution per unit = $10 - 8 = ₹2$

P/V ratio = $2 \div 10 = 20\%$

$$\text{MOS (in units)} = \frac{\text{Profit}}{\text{Contribution per unit}} = \frac{30,000}{2} = \mathbf{15,000 \text{ units}}$$

$$\text{MOS (in ₹)} = 15,000 \times ₹10 = \mathbf{₹1,50,000}$$

Or directly: MOS = Profit ÷ P/V ratio = $30,000 \div 0.20 = ₹1,50,000$. Both routes, same answer.

WHY THIS WORKS

Think about what margin of safety means physically. Break-even sales pay the fixed cost exactly and leave nothing. Every rupee of sales *beyond* that point contributes only profit. So if you know the profit and you know how much profit each rupee of sales produces — the P/V ratio — you can work backwards to how many rupees of sales sat above break-even. That is the margin of safety, and it is why you never needed the fixed cost.

FINAL ANSWER

Contribution per unit	₹ 2
P/V ratio	20%
Margin of safety in units	15,000 units
Margin of safety in value	₹ 1,50,000

Q10 Extracting variable cost from total cost

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following details find out (a) Profit Volume Ratio (b) BEP (c) Margin of Safety.

Particulars	₹
Sales	1,00,000
Total cost	80,000
Fixed Cost	20,000
Net profit	20,000

READ THE QUESTION FIRST — WHAT TO SPOT

- The **variable cost is not given directly** — you must strip it out: Variable cost = Total cost – Fixed cost = 80,000 – 20,000 = ₹60,000.
- Once you have that, build the marginal cost statement and the three answers drop out of it.
- The net profit of ₹20,000 is given, so use it to **check** your statement rather than to compute anything.

Marginal Cost Statement

Particulars	Amount ₹
Sales	1,00,000
Less: Variable cost 80,000 total cost – 20,000 fixed	(60,000)
Contribution	40,000
Less: Fixed cost	(20,000)
Profit agrees with the given figure ✓	20,000

WORKING NOTES

$$(a) \text{ P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{40,000}{1,00,000} \times 100 = \mathbf{40\%}$$

$$(b) \text{ BEP} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{20,000}{0.40} = \mathbf{₹50,000}$$

$$(c) \text{ MOS} = \text{Actual sales} - \text{BEP sales} = 1,00,000 - 50,000 = \mathbf{₹50,000}$$

Check by the other route: MOS = Profit ÷ P/V ratio = 20,000 ÷ 0.40 = ₹50,000 ✓

FINAL ANSWER

Variable cost (derived)	₹ 60,000
Contribution	₹ 40,000
(a) P/V ratio	40%
(b) Break-even point	₹ 50,000

Q11 Eight situations in one statement

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information is obtained from a company for 2016: Sales ₹20,000; Variable Cost ₹10,000; Fixed Cost ₹6,000.

(a) Find P/V Ratio, Break-even point and Margin of safety at this level, and **(b)** the effect on P/V Ratio, Break-even point and Margin of safety if: (i) 20% decrease in fixed cost; (ii) 10% increase in fixed cost; (iii) 10% decrease in variable cost; (iv) 10% increase in variable cost; (v) 10% increase in selling price together with an increase of fixed overheads by ₹1,200; (vi) 10% decrease in selling price; (vii) 10% decrease in sales price accompanied by 10% decrease in variable costs.

READ THE QUESTION FIRST — WHAT TO SPOT

- Eight columns of the same four-line statement. **Do it as one wide table**, exactly as your classmate's notes do — never as eight separate little sums, or you will lose track.
- Work out which line each change touches **before** you compute anything: a fixed-cost change touches only the FC line and leaves the P/V ratio alone; a variable-cost or selling-price change moves contribution and therefore **does** change the P/V ratio.
- In (v) the selling price rises 10% so sales become 22,000, but the **variable cost stays at 10,000** — volume has not changed, only price. Same logic in (vi).
- **In (vii) both fall 10%**, so contribution falls 10% too and the **P/V ratio is unchanged at 50%**. That is the interesting result of the whole question.

Which line does each change touch?

Case	Sales	Variable cost	Fixed cost	Does the P/V ratio change?
(i) FC down 20%	—	—	↓ 4,800	No — contribution untouched
(ii) FC up 10%	—	—	↑ 6,600	No
(iii) VC down 10%	—	↓ 9,000	—	Yes — rises to 55%
(iv) VC up 10%	—	↑ 11,000	—	Yes — falls to 45%
(v) SP up 10%, FC up 1,200	↑ 22,000	—	↑ 7,200	Yes — rises to 54.55%
(vi) SP down 10%	↓ 18,000	—	—	Yes — falls to 44.44%
(vii) SP down 10% and VC down 10%	↓ 18,000	↓ 9,000	—	No — stays exactly 50%

Complete solution — all eight situations in one statement

Particulars	(a) Present	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
Sales	20,000	20,000	20,000	20,000	20,000	22,000	18,000	18,000
Less: Variable cost	(10,000)	(10,000)	(10,000)	(9,000)	(11,000)	(10,000)	(10,000)	(9,000)
Contribution	10,000	10,000	10,000	11,000	9,000	12,000	8,000	9,000
Less: Fixed cost	(6,000)	(4,800)	(6,600)	(6,000)	(6,000)	(7,200)	(6,000)	(6,000)
Profit	4,000	5,200	3,400	5,000	3,000	4,800	2,000	3,000

Particulars	(a) Present	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)
P/V ratio	50.00%	50.00%	50.00%	55.00%	45.00%	54.55%	44.44%	50.00%
BEP sales	12,000	9,600	13,200	10,909	13,333	13,200	13,500	12,000
Margin of safety	8,000	10,400	6,800	9,091	6,667	8,800	4,500	6,000

WORKING NOTES

How the BEP row was computed in each column

Case	Fixed cost	P/V ratio	BEP = FC ÷ P/V	MOS = Sales – BEP
(a)	6,000	50.00%	12,000	8,000
(i)	4,800	50.00%	9,600	10,400
(ii)	6,600	50.00%	13,200	6,800
(iii)	6,000	55.00%	10,909	9,091
(iv)	6,000	45.00%	13,333	6,667
(v)	7,200	$12,000/22,000 = 54.5455\%$	13,200	8,800
(vi)	6,000	$8,000/18,000 = 44.4444\%$	13,500	4,500
(vii)	6,000	$9,000/18,000 = 50.00\%$	12,000	6,000

THE TRAP / WHERE MARKS ARE LOST

Important — two figures where care is needed. In cases (v) and (vi) the P/V ratio is a recurring decimal. If you round it to 0.54 and 0.44 before dividing you get BEP figures of ₹13,333 and ₹13,636 instead of the correct ₹13,200 and ₹13,500. Your classmate's handwritten notes contain exactly this rounding, so **if your answer differs from those notes in columns (v) and (vi), you are right and the notes are rounded.**

Keep the ratio as a fraction — $12,000/22,000$ and $8,000/18,000$ — and divide only once, at the end.

WHY THIS WORKS

Look across the P/V ratio row. It does not budge for (i) and (ii) because a change in fixed cost cannot affect contribution. It moves for (iii) to (vi) because each of those changes either the price or the variable cost. And in (vii) it returns to exactly 50% because price and variable cost fell by the same percentage, so their *ratio* is unchanged — the profit falls, but the efficiency of each rupee of sales does not.

FINAL ANSWER

(a) Present — P/V / BEP / MOS	50% / ₹ 12,000 / ₹ 8,000
(i) FC down 20%	50% / ₹ 9,600 / ₹ 10,400 — profit ₹ 5,200
(ii) FC up 10%	50% / ₹ 13,200 / ₹ 6,800 — profit ₹ 3,400
(iii) VC down 10%	55% / ₹ 10,909 / ₹ 9,091 — profit ₹ 5,000
(iv) VC up 10%	45% / ₹ 13,333 / ₹ 6,667 — profit ₹ 3,000
(v) SP up 10%, FC up 1,200	54.55% / ₹ 13,200 / ₹ 8,800 — profit ₹ 4,800
(vi) SP down 10%	44.44% / ₹ 13,500 / ₹ 4,500 — profit ₹ 2,000
(vii) SP and VC both down 10%	50% / ₹ 12,000 / ₹ 6,000 — profit ₹ 3,000

The best single-sentence conclusion: a fixed-cost change moves the break-even point but never the P/V ratio; a price or variable-cost change moves both.

Q12 The four standard CVP questions

CVP · p.23

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

You are given the following data: Variable Cost ₹6,00,000; Fixed Cost ₹3,00,000; Net Profit ₹1,00,000; Sales ₹10,00,000.

Find (a) P/V Ratio (b) B.E.P. (c) Profit when sales amounted to ₹12,00,000 (d) Sales required to earn a profit of ₹2,00,000.

READ THE QUESTION FIRST — WHAT TO SPOT

- All four parts come from the same statement, so **build the statement first** and answer from it.
- For (c) and (d) remember the fixed cost does **not** change with volume. That is what makes both parts one-line calculations.
- For (d), treat the target profit exactly like extra fixed cost that must also be covered.

Marginal Cost Statement (present position)

Particulars	Amount ₹
Sales	10,00,000
Less: Variable cost	(6,00,000)
Contribution	4,00,000
Less: Fixed cost	(3,00,000)
Profit agrees with the given ₹1,00,000 ✓	1,00,000

WORKING NOTES

$$(a) \text{ P/V ratio} = \frac{4,00,000}{10,00,000} \times 100 = 40\%$$

$$(b) \text{ BEP} = \frac{3,00,000}{0.40} = ₹7,50,000$$

(c) Profit when sales are ₹12,00,000

Contribution = 12,00,000 × 40% = 4,80,000

Less: Fixed cost = (3,00,000)

Profit = ₹1,80,000

Faster route: extra sales of ₹2,00,000 × 40% = ₹80,000 extra profit, so 1,00,000 + 80,000 = ₹1,80,000. Fixed cost does not change, so every extra rupee of contribution is extra profit.

(d) Sales required for a profit of ₹2,00,000

$$\text{Required sales} = \frac{\text{Fixed cost} + \text{Target profit}}{\text{P/V ratio}} = \frac{3,00,000 + 2,00,000}{0.40} = \frac{5,00,000}{0.40} = ₹12,50,000$$

FINAL ANSWER

(a) P/V ratio	40%
(b) Break-even point	₹ 7,50,000
(c) Profit at sales of ₹ 12,00,000	₹ 1,80,000
(d) Sales to earn ₹ 2,00,000	₹ 12,50,000
Margin of safety at present	₹ 2,50,000

Q13 Seven parts including tax

CVP · p.24

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following, find out (i) Profit Volume Ratio (ii) Break-even point (iii) Sales for 40% P/V Ratio (iv) Margin of Safety if actual sales were ₹3,00,000 (v) Net Profit if actual sales were ₹3,00,000 (vi) Required Sales to earn a profit of ₹70,000 (vii) Required sales to earn a net profit of ₹70,000 after tax, given tax rate = 60%.

Position of the Co. for the year 2025	₹
Sales	2,00,000
Variable Cost	1,50,000
Contribution	50,000
Fixed Cost	15,000
Profit	35,000

READ THE QUESTION FIRST — WHAT TO SPOT

- Seven parts, and the last two are the ones that separate students.
- **Part (iii) needs interpreting.** The P/V ratio does not depend on volume, so “sales for 40% P/V ratio” must mean: what sales value would give a 40% P/V ratio if the variable cost stays at ₹1,50,000? If contribution is to be 40% of sales, variable cost must be 60% of sales — so sales = $1,50,000 \div 0.60$.
- **Part (vii) is about tax.** ₹70,000 is wanted *after* 60% tax, so the profit before tax must be bigger. Gross it up first: pre-tax profit = $70,000 \div (1 - 0.60)$. Only then use the sales formula. Forgetting to gross up is the classic error.

WORKING NOTES

$$(i) \text{ P/V ratio} = \frac{50,000}{2,00,000} \times 100 = \mathbf{25\%}$$

$$(ii) \text{ BEP} = \frac{15,000}{0.25} = \mathbf{₹60,000}$$

(iii) Sales that would give a 40% P/V ratio

If contribution is 40% of sales, then variable cost is 60% of sales
Variable cost is unchanged at 1,50,000

$$\text{Sales} = \frac{1,50,000}{0.60} = \mathbf{₹2,50,000}$$

Check: at sales of 2,50,000 the contribution is $2,50,000 - 1,50,000 = ₹1,00,000$, which is exactly 40% of 2,50,000

(iv) and (v) — if actual sales were ₹3,00,000

$$\text{MOS} = 3,00,000 - 60,000 \text{ (BEP)} = \text{₹}2,40,000$$

$$\text{Contribution} = 3,00,000 \times 25\% = 75,000$$

$$\text{Net profit} = 75,000 - 15,000 = \text{₹}60,000$$

$$\text{Or: profit} = \text{MOS} \times \text{P/V ratio} = 2,40,000 \times 25\% = \text{₹}60,000$$

(vi) Sales to earn a profit of ₹70,000

$$\text{Required sales} = \frac{15,000 + 70,000}{0.25} = \frac{85,000}{0.25} = \text{₹}3,40,000$$

(vii) Sales to earn ₹70,000 after 60% tax

Step 1 — gross the profit up to a pre-tax figure:

$$\text{Profit before tax} = \frac{70,000}{1 - 0.60} = \frac{70,000}{0.40} = \text{₹}1,75,000$$

Step 2 — now use the ordinary formula on the pre-tax profit:

$$\text{Required sales} = \frac{15,000 + 1,75,000}{0.25} = \frac{1,90,000}{0.25} = \text{₹}7,60,000$$

Check: sales 7,60,000 $\times$ 25% = contribution 1,90,000; less fixed cost 15,000 = pre-tax profit 1,75,000; less tax at 60% = 1,05,000; net profit = ₹70,000

THE TRAP / WHERE MARKS ARE LOST

- **Part (vii):** dividing 70,000 by 0.25 straight away, without grossing up for tax. That gives ₹3,40,000 — the answer to part (vi), not part (vii). Tax always comes out *after* profit, so the profit you need to *earn* is always bigger than the profit you want to *keep*.
- **Part (iii):** trying to change the P/V ratio by changing volume. Volume cannot change a ratio of contribution to sales — only price or variable cost can. State your assumption that variable cost is unchanged.

FINAL ANSWER

(i) P/V ratio	25%
(ii) Break-even point	₹ 60,000
(iii) Sales for 40% P/V ratio	₹ 2,50,000
(iv) MOS at sales of ₹ 3,00,000	₹ 2,40,000
(v) Net profit at sales of ₹ 3,00,000	₹ 60,000
(vi) Sales for profit of ₹ 70,000	₹ 3,40,000
(vii) Sales for ₹ 70,000 after 60% tax	₹ 7,60,000

Q14 Merger of two plants at different capacities

CVP · p.24

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Two manufacturing companies which have the following operating details decided to merge:

Particulars	Company I	Company II
Capacity utilisation (%)	90	60
Sales (₹ in lakhs)	540	300
Variable costs (₹ in lakhs)	396	225
Fixed costs (₹ in lakhs)	80	50

Assuming that the proposal is implemented, calculate: **(a)** Break-even sales of the merged plant and the capacity utilisation at that stage; **(b)** Profitability of the merged plant at 80% capacity utilisation; **(c)** Sales turnover of the merged plant to earn a profit of ₹75 lakh; **(d)** When the merged plant is working at a capacity to earn a profit of ₹75 lakh, what percentage increase in selling price is required to sustain an increase of 5% in fixed overheads?

READ THE QUESTION FIRST — WHAT TO SPOT

This is the hardest problem in the module and it is worth learning properly, because the same structure appears in every merger question.

- **The two companies are running at different capacities** — 90% and 60%. You cannot add 540 and 300 and call it the merged sales, because those are sales at *different* levels of activity.
- **So step one is always: restate both companies at 100% capacity.** Divide sales and variable cost by the capacity percentage. Only then can you add them.
- **Fixed cost is NOT scaled up.** Fixed cost does not depend on capacity — that is what makes it fixed. Add $80 + 50 = 130$ lakh and leave it alone. This is the single most common error.
- Once you have the merged plant at 100%, everything else is ordinary CVP work.
- Part (d) is different in kind: volume stays the same and you must recover extra fixed cost purely by raising price. So the price rise, in rupees, equals the extra fixed cost; express it as a percentage of the sales at that level.

WORKING NOTES

W1 Restate both companies at 100% capacity — the essential first step

Particulars	Company I at 90%	Company I at 100%	Company II at 60%	Company II at 100%
Sales	540	$540 \div 0.90 = \mathbf{600}$	300	$300 \div 0.60 = \mathbf{500}$
Variable cost	396	$396 \div 0.90 = \mathbf{440}$	225	$225 \div 0.60 = \mathbf{375}$
Contribution	144	160	75	125
Fixed cost	80	80 NOT scaled	50	50 NOT scaled

W2 The merged plant at 100% capacity

Particulars	₹ in lakhs
Sales $600 + 500$	1,100
Less: Variable cost $440 + 375$	(815)

Particulars	₹ in lakhs
Contribution	285
Less: Fixed cost 80 + 50, unchanged	(130)
Profit at 100% capacity	155

$$\text{P/V ratio of the merged plant} = \frac{285}{1,100} \times 100 = \mathbf{25.9091\%}$$

Keep this as the fraction 285/1,100 in every later division. Rounding it to 26% will move your break-even figure by several lakh.

(a) Break-even sales and the capacity at that point

$$\text{BEP sales} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{130}{285/1,100} = \frac{130 \times 1,100}{285} = \mathbf{₹501.75 \text{ lakh}}$$

$$\text{Capacity utilisation at BEP} = \frac{501.75}{1,100} \times 100 = \mathbf{45.61\%}$$

(b) Profitability at 80% capacity utilisation

Particulars	₹ in lakhs
Sales 80% of 1,100	880.00
Less: Variable cost 80% of 815	(652.00)
Contribution 80% of 285	228.00
Less: Fixed cost	(130.00)
Profit	98.00

(c) Sales needed to earn a profit of ₹75 lakh

$$\text{Required sales} = \frac{130 + 75}{285/1,100} = \frac{205 \times 1,100}{285} = \mathbf{₹791.23 \text{ lakh}}$$

$$\text{Capacity at that level} = 791.23 \div 1,100 = \mathbf{71.93\%}$$

(d) Price increase needed to absorb a 5% rise in fixed overheads

$$\text{Increase in fixed cost} = 5\% \text{ of } 130 = \mathbf{₹6.50 \text{ lakh}}$$

Volume is unchanged, so the whole 6.50 lakh must come from a higher selling price.

$$\text{Sales at that level (from part c)} = \mathbf{₹791.23 \text{ lakh}}$$

$$\text{Required increase in selling price} = \frac{6.50}{791.23} \times 100 = \mathbf{0.82\%}$$

Check: new sales = 791.23 × 1.0082 = ₹797.73 lakh. Contribution rises by the full 6.50 (variable cost is unchanged because volume is unchanged) to 211.50; less new fixed cost of 136.50 = ₹75 lakh profit

THE TRAP / WHERE MARKS ARE LOST

- **Scaling up the fixed cost along with sales.** Fixed cost stays at 130 lakh at every capacity. If you scale it you will get every one of the four answers wrong.
- **Adding $540 + 300 = 840$ and treating that as merged sales.** Those are sales at 90% and 60% respectively. Restate to 100% first — $600 + 500 = 1,100$.
- **Rounding the P/V ratio to 26%.** It is 25.9091%. Use the fraction $285/1,100$ throughout.
- **In part (d), also raising the variable cost.** Volume has not changed, so variable cost has not changed. Only the price moves.

FINAL ANSWER

Merged plant at 100% — sales / contribution / fixed cost	₹ 1,100 L / ₹ 285 L / ₹ 130 L
P/V ratio of the merged plant	25.9091%
(a) Break-even sales	₹ 501.75 lakh
(a) Capacity utilisation at BEP	45.61%
(b) Profit at 80% capacity	₹ 98 lakh
(c) Sales for a profit of ₹ 75 lakh	₹ 791.23 lakh (71.93% capacity)
(d) Price increase for 5% higher fixed cost	0.82%

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Rain Until September Co makes a product, the Splash, which has a variable production cost of \$6 per unit and a sales price of \$10 per unit. At the beginning of September 20X0 there were no opening inventories and production during the month was 20,000 units. Fixed costs for the month were \$45,000 (production, administration, sales and distribution). There were no variable marketing costs.

Calculate the contribution and profit for September 20X0, using marginal costing principles, if sales were (a) 10,000 Splashes (b) 15,000 Splashes (c) 20,000 Splashes.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Marginal costing only** — the question says so. That makes it simple: fixed cost of \$45,000 is charged in full in every one of the three cases, no matter how much was sold.
- Contribution per unit = $10 - 6 = \$4$. Multiply by **units SOLD**, not units produced. Contribution comes from selling, not from making.
- Production is 20,000 in all three cases, so the closing inventory differs: 10,000, 5,000 and nil units. Under marginal costing it is valued at the variable cost of \$6.
- Notice case (a) makes a **loss**. Do not assume every answer is a profit — write the bracket.

Marginal Costing Profit Statement — September 20X0

Particulars	(a) Sales 10,000	(b) Sales 15,000	(c) Sales 20,000
Sales $\text{units sold} \times \10	1,00,000	1,50,000	2,00,000
Less: Variable cost of sales $\text{units sold} \times \6	(60,000)	(90,000)	(1,20,000)
CONTRIBUTION	40,000	60,000	80,000
Less: Fixed costs $\text{charged in full every time}$	(45,000)	(45,000)	(45,000)
PROFIT / (LOSS)	(5,000)	15,000	35,000
Memo: closing inventory in units	10,000	5,000	Nil
Memo: closing inventory value at \$6	60,000	30,000	Nil

WORKING NOTES

$$\text{Contribution per unit} = 10 - 6 = \$4$$

$$\text{Break-even point} = \frac{45,000}{4} = 11,250 \text{ units}$$

That single figure explains all three answers at a glance: 10,000 units is below break-even so there is a loss; 15,000 and 20,000 are above it so there is a profit.

WHY THIS WORKS

Because the fixed cost is charged in full every time, profit moves in exact step with contribution: each extra 5,000 units sold adds $5,000 \times \$4 = \$20,000$ of profit. That straight-line relationship is the whole practical value of marginal costing — a manager can read the effect of a volume change immediately, which absorption costing does not allow.

FINAL ANSWER

Contribution per unit **\$ 4**

Break-even point	11,250 units
(a) 10,000 units	Contribution \$ 40,000 — Loss \$ 5,000
(b) 15,000 units	Contribution \$ 60,000 — Profit \$ 15,000
(c) 20,000 units	Contribution \$ 80,000 — Profit \$ 35,000

Q16 Both methods, two periods, full reconciliation

Statements · p.25

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company makes and sells a single product. At the beginning of period 1 there is no opening stock of the product, for which the variable production cost is ₹4 and the sale price is ₹6 per unit. Fixed costs are ₹2,000 per period, of which ₹1,500 are fixed production costs.

Particulars	Period 1	Period 2
Sales	1,200 units	1,800 units
Production	1,500 units	1,500 units

What would be the profit in each period using (a) Absorption costing (assume normal output is 1,500 units per period) and (b) Marginal costing?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Do the stock movement first.** Period 1: $0 + 1,500 - 1,200 = 300$ units closing. Period 2: $300 + 1,500 - 1,800 = \text{nil}$ closing.
- So stock **rises** in period 1 (absorption profit will be higher) and **falls back to zero** in period 2 (absorption profit will be lower). Predict the direction before you calculate — then your answer confirms it.
- Only ₹1,500 of the ₹2,000 fixed cost is a **production** cost, so only that part is absorbed into units. The other ₹500 is a non-production fixed cost and is deducted below gross profit in the absorption statement.
- Production equals normal output (1,500) in both periods, so there is **no under or over absorption** anywhere in this problem. That is deliberate — it isolates the stock effect.

WORKING NOTES

W1 Stock movement and the absorption rate

	Period 1	Period 2
Opening stock (units)	Nil	300
Add: Production	1,500	1,500
Less: Sales	(1,200)	(1,800)
Closing stock (units)	300	Nil

$$\text{Fixed production overhead rate} = \frac{\text{₹1,500}}{1,500 \text{ units normal output}} = \text{₹1 per unit}$$

Full production cost per unit = variable 4 + fixed 1 = **₹5**

Marginal (variable) cost per unit = **₹4**

(a) Absorption Costing Profit Statement

Particulars	Period 1 ₹	Period 2 ₹
Sales units × ₹6	7,200	10,800
Opening stock at ₹5 full cost	Nil	1,500
Add: Cost of production 1,500 × ₹5	7,500	7,500
Less: Closing stock at ₹5	(1,500)	Nil
Cost of sales	(6,000)	(9,000)
Gross profit	1,200	1,800
Less: Non-production fixed costs 2,000 – 1,500	(500)	(500)
PROFIT	700	1,300

No under or over absorption: production of 1,500 units equals normal output, so absorbed overhead of $1,500 \times ₹1 = ₹1,500$ exactly equals the actual fixed production cost.

(b) Marginal Costing Profit Statement

Particulars	Period 1 ₹	Period 2 ₹
Sales	7,200	10,800
Opening stock at ₹4 variable cost	Nil	1,200
Add: Variable cost of production 1,500 × ₹4	6,000	6,000
Less: Closing stock at ₹4	(1,200)	Nil
Variable cost of sales	(4,800)	(7,200)
CONTRIBUTION	2,400	3,600
Less: Fixed costs all ₹2,000, in full	(2,000)	(2,000)
PROFIT	400	1,600

Reconciliation — the proof

Particulars	Period 1 ₹	Period 2 ₹	Total ₹
Marginal costing profit	400	1,600	2,000
Add: Fixed overhead in closing stock 300 × ₹1 / nil	300	Nil	—
Less: Fixed overhead in opening stock nil / 300 × ₹1	Nil	(300)	—
Absorption costing profit	700	1,300	2,000

Over the two periods together both methods give ₹2,000. The methods differ only in **timing**, never in total profit over the life of the stock.

WHY THIS WORKS

Follow the ₹300 through. In period 1, 300 unsold units carry ₹1 of fixed cost each into the balance sheet instead of the profit and loss account — so absorption profit is ₹300 higher. In period 2 those same units are sold, that ₹300 comes back out of stock and is charged — so absorption profit is ₹300 lower. Over both periods it cancels exactly.

That is the whole marginal-versus-absorption argument in one number: absorption costing lets profit be moved between periods by making stock. Marginal costing does not.

FINAL ANSWER

Closing stock — Period 1 / Period 2	300 units / Nil
Fixed production overhead rate	₹ 1 per unit
Absorption profit — Period 1 / Period 2	₹ 700 / ₹ 1,300
Marginal profit — Period 1 / Period 2	₹ 400 / ₹ 1,600
Total over both periods (either method)	₹ 2,000

Q17 Two products, marginal costing

Statements · p.25-26

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Mill Stream makes two products, the Mill and the Stream. Information relating to each for April 20X1 is as follows.

Particulars	Mill	Stream
Opening inventory	Nil	Nil
Production (units)	15,000	6,000
Sales (units)	10,000	5,000
Sales price per unit	\$20	\$30
Unit costs		
Direct materials	8	14
Direct labour	4	2
Variable production overhead	2	1
Variable sales overhead	2	3
Fixed costs for the month		
Production costs	\$40,000	
Administration costs	\$15,000	
Sales and distribution costs	\$25,000	

(a) Using marginal costing principles, calculate the profit in April 20X1. **(b)** Calculate the profit if sales had been 15,000 units of Mill and 6,000 units of Stream.

READ THE QUESTION FIRST — WHAT TO SPOT

- Marginal costing, so **total the fixed costs into one figure**: $40,000 + 15,000 + 25,000 = \$80,000$. It is charged in full in both parts.
- **Variable sales overhead is included in the contribution calculation** because it varies with each unit sold — but it must **never** go into the value of unsold stock. Here it does not matter for stock because we only need the profit, but note the principle.
- Contribution per unit must be found for each product separately, then multiplied by **that product's** sales units.
- In part (b), production has not changed — only sales. Since fixed cost is unchanged too, the extra profit is simply the extra contribution.

WORKING NOTES

W1 Contribution per unit for each product

Particulars	Mill \$	Stream \$
Selling price	20	30
Less: Direct materials	(8)	(14)
Less: Direct labour	(4)	(2)
Less: Variable production overhead	(2)	(1)
Less: Variable sales overhead	(2)	(3)
Contribution per unit	4	10
<i>Memo: variable production cost per unit for stock valuation</i>	14	17

W2 Total fixed costs for the month

Production 40,000 + Administration 15,000 + Sales and distribution 25,000 = **\$80,000**

(a) Marginal Costing Profit Statement — actual sales

Particulars	Mill \$	Stream \$	Total \$
Sales units	10,000	5,000	—
Contribution per unit W1	4	10	—
Total contribution	40,000	50,000	90,000
Less: Fixed costs W2			(80,000)
PROFIT			10,000

(b) Marginal Costing Profit Statement — if all production had been sold

Particulars	Mill \$	Stream \$	Total \$
Sales units	15,000	6,000	—
Contribution per unit	4	10	—
Total contribution	60,000	60,000	1,20,000
Less: Fixed costs			(80,000)
PROFIT			40,000

WORKING NOTES

Extra units sold = 5,000 Mill + 1,000 Stream

Extra contribution = $(5,000 \times 4) + (1,000 \times 10) = 20,000 + 10,000 = \mathbf{\$30,000}$

Profit rises from \$10,000 to \$40,000 — an increase of exactly \$30,000 ✓

THE TRAP / WHERE MARKS ARE LOST

- Multiplying contribution per unit by **production** units (15,000 and 6,000) in part (a). Only 10,000 and 5,000 were sold. Contribution comes from sales.
- Leaving the variable **sales** overhead out of the contribution calculation. It is a variable cost and must be deducted — Mill's contribution is \$4, not \$6.

- Splitting the fixed costs between the two products. There is no basis given and no need — under marginal costing you deduct total fixed cost from total contribution.

FINAL ANSWER

Contribution per unit — Mill / Stream	\$ 4 / \$ 10
Total fixed costs	\$ 80,000
(a) Profit at actual sales	Contribution \$ 90,000 – \$ 80,000 = \$ 10,000
(b) Profit if all output sold	Contribution \$ 1,20,000 – \$ 80,000 = \$ 40,000

Q18 Annual budget, quarterly statement

Statements · p.26

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Big Woof Co manufactures a single product, the Bark: Selling price \$180.00; Direct materials \$40.00; Direct labour \$16.00; Variable overheads \$10.00 per unit.

Annual fixed production overheads are budgeted to be \$1.6 million and Big Woof expects to produce 1,280,000 units of the Bark each year. Overheads are absorbed on a per unit basis. Actual overheads are \$1.6 million for the year. Budgeted fixed selling costs are \$320,000 per quarter. Actual sales and production for the first quarter of 20X8: **Sales 240,000 units; Production 280,000 units.** There is no opening inventory at the beginning of January.

Prepare statements of profit or loss for the quarter using (a) Marginal costing (b) Absorption costing.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Careful with the time periods.** The fixed production overhead of \$1.6 million is an **annual** figure, but you are preparing a **quarterly** statement. Divide by four: \$400,000 for the quarter. The selling cost of \$320,000 is already quarterly — do not divide it.
- The **absorption rate is built from the annual budget**: $1,600,000 \div 1,280,000$ units = \$1.25 per unit. Do not use the quarterly figures to build the rate.
- Production 280,000 exceeds sales 240,000, so closing inventory is **40,000 units** and stock is rising — therefore **absorption profit will be higher**. Predict it, then prove it.
- Absorbed overhead = $280,000 \times 1.25 = \$350,000$ against an actual quarterly charge of \$400,000, so there is an **under-absorption of \$50,000** to be deducted.

WORKING NOTES

W1 Unit costs and the absorption rate

Variable cost per unit = $40 + 16 + 10 = \$66$

Contribution per unit = $180 - 66 = \$114$

$$\text{Fixed production overhead absorption rate} = \frac{\$1,600,000 \text{ per year}}{1,280,000 \text{ units per year}} = \$1.25 \text{ per unit}$$

Full production cost per unit = $66 + 1.25 = \$67.25$

W2 Quarterly fixed costs and inventory

Fixed production overhead for the quarter = $1,600,000 \div 4 = \$400,000$

Fixed selling cost for the quarter = **\$320,000** (already quarterly)

Closing inventory = $0 + 280,000 - 240,000 = 40,000$ units

W3 Under / over absorption

Overhead absorbed = $280,000 \text{ units} \times \$1.25 = 350,000$

Actual overhead for the quarter = 400,000

UNDER-absorbed = \$50,000 — deducted in the absorption statement

(a) Marginal Costing Statement of Profit or Loss — quarter to March 20X8

Particulars	Amount \$
Sales $240,000 \times \$180$	43,200,000
Less: Variable cost of sales $240,000 \times \$66$	(15,840,000)
CONTRIBUTION $240,000 \times \$114$	27,360,000
Less: Fixed production overhead W2	(400,000)
Less: Fixed selling costs	(320,000)
PROFIT	26,640,000

(b) Absorption Costing Statement of Profit or Loss — quarter to March 20X8

Particulars	Amount \$
Sales	43,200,000
Cost of production $280,000 \times \$67.25$	18,830,000
Less: Closing inventory $40,000 \times \$67.25$	(2,690,000)
Cost of sales $240,000 \times \$67.25$	(16,140,000)
Gross profit	27,060,000
Less: Under-absorbed fixed production overhead W3	(50,000)
Less: Fixed selling costs	(320,000)
PROFIT	26,690,000

Reconciliation

Particulars	Amount \$
Marginal costing profit	26,640,000
Add: Fixed production overhead carried in closing inventory $40,000 \times \$1.25$	50,000
Absorption costing profit	26,690,000

THE TRAP / WHERE MARKS ARE LOST

- **Charging the whole \$1.6 million in a three-month statement.** It is an annual cost — only a quarter of it belongs to this quarter.
- **Dividing the \$320,000 selling cost by four.** The question says “per quarter”. Read the units on every figure.
- **Building the absorption rate from 280,000 actual units instead of 1,280,000 budgeted.** That would give \$1.43 and destroy the whole answer — and you would also find no under-absorption, which should ring a warning bell.

FINAL ANSWER

Contribution per unit	\$ 114
Absorption rate / full cost per unit	\$ 1.25 / \$ 67.25
Closing inventory	40,000 units = \$ 2,690,000
Under-absorbed overhead	\$ 50,000
(a) Marginal costing profit	\$ 26,640,000
(b) Absorption costing profit	\$ 26,690,000
Difference	\$ 50,000 = 40,000 units × \$ 1.25 ✓

Q19 Opening and closing stock at different rates

Statements · p.26-27

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Eagle makes one product. Budgeted production and sales for the year just ended were 32,000 units. Actual sales for the year were 34,000 units at a selling price of \$34 per unit, and actual production was 36,000 units. Opening inventory was 5,000 units valued at the unit cost for the preceding year. Using an absorption costing system based on budgeted levels of output, unit costs for the year and the preceding year were as follows:

Particulars	Year \$	Preceding year \$
Prime cost	24	22
Variable production overhead	3.5	1
Fixed production overhead	1.5	3
Selling and administrative overhead (fixed)	1	1
Total cost	30	27
Profit	4	3
Selling price	34	30

Fixed production overhead for the year just ended was \$48,000 (budget and actual). Actual variable costs per unit were the same as budgeted. Prepare statements of profit or loss for the year just ended on the basis that (i) an absorption costing system is used and (ii) a marginal costing system is used.

READ THE QUESTION FIRST — WHAT TO SPOT

The hardest statement problem in the module, because **opening and closing stock are valued at different rates** — last year's and this year's.

- **Opening inventory of 5,000 units is at the PRECEDING year's cost.** For absorption that is $22 + 1 + 3 = \$26$ per unit. For marginal it is $22 + 1 = \$23$ per unit. Note the fixed element was \$3 last year, not \$1.50.
- Closing inventory = $5,000 + 36,000 - 34,000 = 7,000$ units, valued at **this** year's cost: absorption $24 + 3.5 + 1.5 = \$29$; marginal $24 + 3.5 = \$27.50$.
- **Check the fixed overhead rate:** $\$48,000 \div 32,000$ budgeted units = \$1.50 which matches the table. That confirms the rate is built on budgeted output.
- Actual production 36,000 exceeds budget 32,000, so overhead is **OVER-absorbed**: $36,000 \times 1.50 = \$54,000$ absorbed against \$48,000 actual, a credit of \$6,000.
- Selling and administrative overhead is **fixed**, so it is $\$1 \times 32,000$ budgeted units = \$32,000 for the year — not $\$1 \times 34,000$.

WORKING NOTES

W1 Inventory in units

Opening 5,000 + Production 36,000 – Sales 34,000 = **Closing 7,000 units**

W2 Unit values for stock — two different years, two different rates

	Absorption \$	Marginal \$
Opening stock — preceding year rates		
Prime cost	22	22
Variable production overhead	1	1
Fixed production overhead	3	—
Per unit	26	23
Closing stock — this year's rates		
Prime cost	24	24
Variable production overhead	3.5	3.5
Fixed production overhead	1.5	—
Per unit	29	27.50

Opening stock value — absorption = 5,000 × 26 = **\$130,000** | marginal = 5,000 × 23 = **\$115,000**

Closing stock value — absorption = 7,000 × 29 = **\$203,000** | marginal = 7,000 × 27.50 = **\$192,500**

W3 Over-absorption of fixed production overhead

Absorbed = 36,000 units × \$1.50 = 54,000

Actual = 48,000

OVER-absorbed = \$6,000 — added back

(i) Absorption Costing Statement of Profit or Loss

Particulars	Amount \$
Sales 34,000 × \$34	1,156,000
Opening inventory W2: 5,000 × \$26	130,000
Add: Cost of production 36,000 × \$29	1,044,000
Less: Closing inventory W2: 7,000 × \$29	(203,000)
Cost of sales	(971,000)
Gross profit	185,000
Add: Over-absorbed fixed production overhead W3	6,000
Less: Selling and administrative overhead fixed, \$1 × 32,000 budget	(32,000)
PROFIT	159,000

(ii) Marginal Costing Statement of Profit or Loss

Particulars	Amount \$
Sales	1,156,000
Opening inventory W2: $5,000 \times \$23$	115,000
Add: Variable cost of production $36,000 \times \$27.50$	990,000
Less: Closing inventory W2: $7,000 \times \$27.50$	(192,500)
Variable cost of sales	(912,500)
CONTRIBUTION	243,500
Less: Fixed production overhead actual	(48,000)
Less: Selling and administrative overhead	(32,000)
PROFIT	163,500

Reconciliation

Particulars	Amount \$
Marginal costing profit	163,500
Add: Fixed overhead in closing inventory $7,000 \times \$1.50$	10,500
Less: Fixed overhead in opening inventory $5,000 \times \$3.00$	(15,000)
Absorption costing profit	159,000

Here the absorption profit is **lower** even though inventory rose in units — because the fixed overhead rate *halved* from \$3.00 to \$1.50. Always reconcile on the **rupee value** of fixed overhead in stock, never on units.

WHY THIS WORKS

This problem exists to break a lazy rule. Students learn “stock up → absorption profit higher” and apply it blindly. Here inventory rose from 5,000 to 7,000 units, yet absorption profit is \$4,500 **lower**.

The reason is that the fixed overhead rate fell from \$3.00 to \$1.50. The 5,000 opening units released \$15,000 of last year’s fixed cost into this year’s charge, while the 7,000 closing units only carried \$10,500 forward. The rule is really about the **value** of fixed overhead held in stock, not the number of units.

FINAL ANSWER

Closing inventory	7,000 units
Stock value — absorption: opening / closing	\$ 130,000 / \$ 203,000
Stock value — marginal: opening / closing	\$ 115,000 / \$ 192,500
Over-absorbed overhead	\$ 6,000
(i) Absorption costing profit	\$ 159,000
(ii) Marginal costing profit	\$ 163,500
Difference	\$ 4,500 = 10,500 – 15,000 ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Stoney is drafting a budget on the basis of the following data: Direct material \$10 per unit; Direct labour \$5 per unit; Variable production expenses \$8 per unit; Fixed production costs \$27,000 per month; Normal output 9,000 units per month (90% capacity); Sales price \$30 per unit.

In order to build up inventory in anticipation of an increase in demand expected later in the year, production is to exceed sales in the first three months as follows:

	Month 1	Month 2	Month 3
Production	6,500	9,000	10,000
Sales	5,000	8,500	9,500

Prepare two statements of profit or loss, each in comparative columnar form, covering each of the three months: (i) on a marginal costing basis and (ii) on a full absorption costing basis.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Track the inventory across all three months first** — the closing stock of one month is the opening stock of the next. Get this table right and the rest is mechanical.
- Absorption rate = $27,000 \div 9,000$ **normal output** = \$3 per unit. Note it is built on **normal** output, not on each month's actual production.
- That is why **each month has a different under/over absorption**: month 1 produces only 6,500 (under-absorbed), month 2 produces exactly 9,000 (nil), month 3 produces 10,000 (over-absorbed). This is the point of the problem.
- Stock rises every month, so **absorption profit is higher in every month**.

WORKING NOTES

W1 Inventory movement in units — carry it across the months

	Month 1	Month 2	Month 3
Opening stock	Nil	1,500	2,000
Add: Production	6,500	9,000	10,000
Less: Sales	(5,000)	(8,500)	(9,500)
Closing stock	1,500	2,000	2,500

W2 Unit costs

Variable production cost per unit = $10 + 5 + 8 = \$23$

Contribution per unit = $30 - 23 = \$7$

$$\text{Fixed overhead absorption rate} = \frac{\$27,000}{9,000 \text{ units normal output}} = \$3 \text{ per unit}$$

Full production cost per unit = $23 + 3 = \$26$

W3 Under / over absorption, month by month

	Month 1	Month 2	Month 3
Production (units)	6,500	9,000	10,000
Overhead absorbed at \$3	19,500	27,000	30,000
Actual fixed overhead	27,000	27,000	27,000
Under (–) / Over (+) absorbed	(7,500)	Nil	+3,000

(i) Marginal Costing Statements of Profit or Loss

Particulars	Month 1 \$	Month 2 \$	Month 3 \$
Sales $\text{units sold} \times \30	150,000	255,000	285,000
Opening stock $\text{at } \$23$	Nil	34,500	46,000
Add: Variable cost of production $\times \$23$	149,500	207,000	230,000
Less: Closing stock $\text{at } \$23$	(34,500)	(46,000)	(57,500)
Variable cost of sales	(115,000)	(195,500)	(218,500)
CONTRIBUTION $\text{units sold} \times \7	35,000	59,500	66,500
Less: Fixed production costs $\text{in full every month}$	(27,000)	(27,000)	(27,000)
PROFIT	8,000	32,500	39,500

(ii) Absorption Costing Statements of Profit or Loss

Particulars	Month 1 \$	Month 2 \$	Month 3 \$
Sales	150,000	255,000	285,000
Opening stock $\text{at } \$26$	Nil	39,000	52,000
Add: Cost of production $\times \$26$	169,000	234,000	260,000
Less: Closing stock $\text{at } \$26$	(39,000)	(52,000)	(65,000)
Cost of sales $\text{units sold} \times \26	(130,000)	(221,000)	(247,000)
Gross profit	20,000	34,000	38,000
Under (–) / Over (+) absorbed overhead W3	(7,500)	Nil	3,000
PROFIT	12,500	34,000	41,000

Reconciliation, month by month

Particulars	Month 1 \$	Month 2 \$	Month 3 \$
Marginal costing profit	8,000	32,500	39,500
Add: Fixed overhead in closing stock $\text{closing units} \times \3	4,500	6,000	7,500
Less: Fixed overhead in opening stock $\text{opening units} \times \3	Nil	(4,500)	(6,000)
Absorption costing profit	12,500	34,000	41,000
<i>Memo:</i> increase in stock (units)	1,500	500	500
<i>Memo:</i> difference in profit $= \text{increase} \times \3	4,500	1,500	1,500

WHY THIS WORKS

Month 2 is the instructive one. Production of 9,000 exactly equals normal output, so there is **no under or over absorption at all** — yet the two profits still differ by \$1,500, because stock still rose by 500 units. That separates the two effects cleanly:

- **Under/over absorption** arises when actual production differs from *normal output*.
- **The marginal-versus-absorption profit gap** arises when production differs from *sales*.

They are different things and a good answer does not confuse them.

FINAL ANSWER

Contribution per unit / full cost per unit	\$ 7 / \$ 26
Absorption rate	\$ 3 per unit (27,000 ÷ 9,000 normal)
Closing stock — months 1 / 2 / 3	1,500 / 2,000 / 2,500 units
(i) Marginal profit — M1 / M2 / M3	\$ 8,000 / \$ 32,500 / \$ 39,500
(ii) Absorption profit — M1 / M2 / M3	\$ 12,500 / \$ 34,000 / \$ 41,000
Under / over absorption — M1 / M2 / M3	(\$ 7,500) / Nil / +\$ 3,000

Cost-Volume-Profit Analysis

18 problems · workbook pages 31 to 37

What Module 3 is really about

Module 2 taught you the formulas. Module 3 asks you to **make a decision** with them. Almost every problem here is a manager saying “should I do this or not?”, and your job is to answer with a number and a recommendation.

THE SINGLE RULE THAT DECIDES EVERY PROBLEM IN THIS MODULE

Ignore fixed cost. Compare CONTRIBUTION.

Fixed cost is going to be incurred whether you accept the order or not, so it cannot help you choose. Only the amounts that CHANGE between the alternatives matter. In this module that is almost always contribution.

There are only two exceptions, and both are flagged where they arise:

- **When fixed cost itself changes** because of the decision — new machinery, an extra sales office, a shut-down saving. Then the change in fixed cost is relevant and must be brought in.
- **When capacity runs out.** Then you are not choosing whether to produce, but WHAT to produce, and the answer is contribution per unit of the scarce resource — not contribution per unit of product.

The four decision types — recognise which one you are in

Type	The question sounds like ...	What you compare	Problems
1. Special order & pricing	“An export order at a reduced price ... should we accept?” “What minimum price?”	Extra contribution from the order against any extra fixed cost it causes. Spare capacity is the key condition.	1, 2, 4, 5, 6, 11
2. Make or buy	“It costs us ₹6.25 to make; the market price is ₹4.85.”	The variable cost of making against the buying price. Fixed cost continues either way, so exclude it.	3
3. Key factor / product mix	“Material is restricted to 18,400 kg” “the limiting factor is labour”	Contribution per unit of the limiting factor , then rank and allocate.	7, 8, 9, 10
4. Shut-down	“Management plans to shut down the plant ... should it?”	Loss if you continue against loss if you shut. Shut only if the shut-down loss is smaller.	12, 13, 14

Type	The question sounds like ...	What you compare	Problems
5. Charts	"Draw the break-even chart" "draw the profit-volume graph"	Nothing — but you must plot correctly and read the BEP off the graph, then verify by calculation.	15-18

HOW TO ANSWER ANY DECISION QUESTION AND NOT LOSE MARKS

1. **Separate fixed from variable** in whatever the question gives you. Nothing can start until this is done.
2. **Find the contribution per unit** — selling price minus *all* variable costs including variable selling expenses.
3. **Check the capacity.** Is there spare capacity for the new order, or must something be given up? This single check changes the whole answer.
4. **Build a comparative statement** with one column per alternative, including the present position as a column. Never answer a decision question with loose calculations.
5. **Write the recommendation in words**, with the amount. "Accept the order, as it increases profit by ₹24,000." The sentence carries marks of its own.
6. **Mention the non-financial point** if there is an obvious one — effect on the existing market, reliability of the supplier, the workforce. One line is enough and it is often the difference between a good answer and a full-mark one.

Q1 Minimum price for a special order

Special order · p.31

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A manufacturer makes an average profit of ₹2.5 on a selling price of ₹14.5. He produces and sells 60,000 units at 60% capacity. His cost of sales per unit is: Direct Materials ₹4; Direct Wages ₹1; Factory overheads (Variable) ₹3; Selling overheads (Variable) 25 paise. Total fixed cost ₹2,25,000.

During the current year he anticipates that (a) fixed cost will increase by 10% and (b) material and labour cost will increase by 5% each. Under these circumstances he obtains an offer for a further 20% utilisation of his capacity. What minimum price would you recommend to accept the offer so as to ensure an overall profit of ₹1,60,000?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Verify the given data first.** Variable cost = $4 + 1 + 3 + 0.25 = ₹8.25$; fixed per unit = $2,25,000 \div 60,000 = ₹3.75$; total ₹12.00; profit = $14.50 - 12.00 = ₹2.50$. That tie-up proves you have read the question correctly.
- **60,000 units is 60% capacity**, so 100% = 1,00,000 units and the further 20% = **20,000 units**.
- The two cost increases apply to **everything**, existing units as well as the new order.
- This is a **working-backwards** problem: the profit is given and the price is unknown. So build the required total contribution first, take away what the existing 60,000 units provide, and whatever is left must come from the 20,000 new units.

WORKING NOTES

W1 Revised variable cost per unit

Element	Present ₹	Change	Revised ₹
Direct materials	4.00	+5%	4.20
Direct wages	1.00	+5%	1.05
Factory overheads (variable)	3.00	no change	3.00
Selling overheads (variable)	0.25	no change	0.25
Variable cost per unit	8.25		8.50

W2 Revised fixed cost and capacity

Revised fixed cost = 2,25,000 + 10% = **₹2,47,500**

60,000 units = 60% capacity → 100% = 1,00,000 units

Further 20% of capacity = **20,000 units**

W3 Working backwards to the minimum price

Required profit = 1,60,000

Add: Revised fixed cost = 2,47,500

Total contribution required = ₹4,07,500

Less: Contribution from existing 60,000 units = 60,000 × (14.50 – 8.50) = 60,000 × 6 = (3,60,000)

Contribution needed from the new 20,000 units = ₹47,500

$$\text{Contribution per unit on the offer} = \frac{47,500}{20,000} = ₹2.375$$

Minimum price = Variable cost + Required contribution = 8.50 + 2.375 = ₹10.875 per unit

Round up to ₹10.88 in practice — rounding down would leave the profit target just short.

Proof — profit statement at the recommended price

Particulars	Amount ₹
Sales — existing 60,000 × ₹14.50	8,70,000
Sales — offer 20,000 × ₹10.875	2,17,500
Total sales	10,87,500
Less: Variable cost 80,000 × ₹8.50	(6,80,000)
Contribution	4,07,500
Less: Fixed cost (revised)	(2,47,500)
Profit exactly the target ✓	1,60,000

THE TRAP / WHERE MARKS ARE LOST

- Applying the 5% increase only to the new order. The cost rise affects the whole year's production.
- Increasing the factory and selling overheads by 5% too. The question says **material and labour** only.
- Adding the old fixed cost per unit of ₹3.75 to the minimum price. The fixed cost is already covered by the existing 60,000 units, which is why the offer can be accepted at ₹10.875 — far below the normal ₹14.50 — and still increase profit.

FINAL ANSWER

Revised variable cost per unit	₹ 8.50
Revised fixed cost	₹ 2,47,500
Units in the offer (20% of capacity)	20,000
Total contribution required	₹ 4,07,500
Contribution needed per offer unit	₹ 2.375
Minimum price recommended	₹ 10.875 (say ₹ 10.88) per unit

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

ABC Company produces a single product sold at ₹75 per unit. Present production and sale is 40,000 units per month, representing 50% of capacity. Variable cost per unit ₹50; fixed costs per month ₹10 lakhs. Three proposals are on hand:

- I. Accept an export supply order for 30,000 units per month at a reduced price of ₹60 per unit, incurring an additional variable cost of ₹5 per unit towards export packing, duties etc.
- II. Increase domestic market sales by selling to a domestic chain store 30,000 units at ₹55 per unit, retaining the existing sales at the existing price.
- III. Reduce the selling price for increased domestic sale as advised by the sales department:

Reduce selling price per unit (₹)	Increase in sales expected (units)
5	10,000
8	30,000
11	35,000

Prepare a table to present the results of the above proposals and give your comments and advice.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Capacity check first.** 40,000 units = 50%, so 100% = 80,000 units per month and there are **40,000 units of spare capacity**. Every proposal must be tested against this ceiling.
- **The present position is break-even.** Contribution $40,000 \times 25 = ₹10,00,000$, which exactly equals the fixed cost. Profit is nil. Say this in your answer — it explains why the management is looking at proposals at all.
- **Proposals I and II add sales without touching the existing price.** So compute only the extra contribution and add it to the present profit of nil.
- **Proposal III cuts the price on ALL units**, existing and new. That is the crucial difference — you must recompute the whole contribution, not just the increment. Most students get this wrong.

WORKING NOTES

W1 Present position and capacity

Contribution per unit = $75 - 50 = ₹25$

Present contribution = $40,000 \times 25 = ₹10,00,000$

Less fixed cost = (10,00,000) → **present profit = NIL (break-even)**

40,000 units = 50% capacity → 100% = 80,000 units; **spare capacity 40,000 units**

W2 Proposal III — the whole price is cut, so recompute everything

Price cut ₹	New selling price	Total units	Contribution per unit	Total contribution	Within capacity?
5	70	50,000	20	10,00,000	Yes
8	67	70,000	17	11,90,000	Yes
11	64	75,000	14	10,50,000	Yes

Total units = 40,000 existing + the stated increase. Contribution per unit = new price – ₹50 variable cost.

Comparative Statement of the Present Position and All Proposals

Particulars	Present	Proposal I (export)	Proposal II (chain store)	III — cut ₹5	III — cut ₹8	III — cut ₹11
Units sold	40,000	70,000	70,000	50,000	70,000	75,000
Selling price on existing units	75	75	75	70	67	64
Selling price on additional units	—	60	55	70	67	64
Contribution — existing 40,000 units	10,00,000	10,00,000	10,00,000	8,00,000	6,80,000	5,60,000
Contribution — additional units	—	1,50,000	1,50,000	2,00,000	5,10,000	4,90,000
Total contribution	10,00,000	11,50,000	11,50,000	10,00,000	11,90,000	10,50,000
Less: Fixed cost	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)	(10,00,000)
PROFIT	Nil	1,50,000	1,50,000	Nil	1,90,000	50,000

WHY THIS WORKS

Advice: accept Proposal III with a price reduction of ₹8, giving a profit of ₹1,90,000 per month. It is the best of the six columns.

Two things are worth commenting on, and both earn marks:

- **Proposals I and II give identical profit** of ₹1,50,000. The export order sells at ₹60 but carries ₹5 of extra cost; the chain store sells at ₹55 with no extra cost. Both leave a contribution of ₹5 per unit. On financial grounds there is nothing to choose — so decide on non-financial grounds. The export order does not disturb the home market, whereas selling to a domestic chain at ₹55 risks existing customers demanding the same price. **On that basis Proposal I is safer than Proposal II.**
- **Cutting the price by ₹11 is worse than cutting it by ₹8**, even though it sells 5,000 more units. The extra volume does not make up for the contribution given away on all 75,000 units. This is the danger of chasing volume through price cuts.

THE TRAP / WHERE MARKS ARE LOST

- Treating Proposal III as an *increment* and adding only the new units' contribution. The price cut applies to the existing 40,000 units too, which is why the ₹5 cut leaves profit at nil rather than improving it.
- Forgetting to check capacity. All the options here fit inside 80,000 units, but had any exceeded it you would have had to turn away domestic sales — a completely different calculation.

FINAL ANSWER

Contribution per unit (present)	₹ 25
Present profit	Nil — the company is at break-even
Proposal I — export at ₹60	Profit ₹ 1,50,000
Proposal II — chain store at ₹55	Profit ₹ 1,50,000
Proposal III — cut ₹5 / ₹8 / ₹11	Nil / ₹ 1,90,000 / ₹ 50,000
Recommendation	Proposal III with an ₹ 8 cut — profit ₹ 1,90,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A radio manufacturing company finds that while it costs ₹6.25 to make each component X 2730, the same is available in the market at ₹4.85 each, with an assurance of continued supply. The breakdown of cost is: Materials ₹2.75 each; Labour ₹1.75 each; Other variables ₹0.50 each; Depreciation and other fixed costs ₹1.25 each — total ₹6.25 each.

(a) Should you make or buy? **(b)** What would be your decision if the supplier offered the component at ₹5.85 each?

READ THE QUESTION FIRST — WHAT TO SPOT

- The whole problem turns on one question: **which costs disappear if you stop making?**
- Materials, labour and other variables disappear — they are only incurred when you make. **Depreciation and other fixed costs do not.** The machine still depreciates whether you use it or not.
- So the relevant cost of making is $2.75 + 1.75 + 0.50 = \text{₹}5.00$, not ₹6.25. Comparing 6.25 with the market price is the trap the question is built around.

Relevant cost comparison

Particulars	Per component ₹	Relevant?
Materials	2.75	Yes — avoided if bought
Labour	1.75	Yes — avoided if bought
Other variables	0.50	Yes — avoided if bought
Relevant (marginal) cost of making	5.00	
Depreciation and other fixed costs	1.25	No — incurred anyway, so it is irrelevant to the decision
Total cost as shown in the books	6.25	— misleading for this decision

The decision under each supplier price

	(a) Supplier at ₹4.85	(b) Supplier at ₹5.85
Relevant cost of making	5.00	5.00
Cost of buying	4.85	5.85
Saving by buying / (by making)	0.15 by buying	(0.85) — make instead
Decision	BUY	MAKE

WHY THIS WORKS

(a) Buy. At ₹4.85 the supplier is cheaper than the ₹5.00 it costs in avoidable terms, so buying saves ₹0.15 per component.

(b) Make. At ₹5.85 the supplier is dearer than ₹5.00, so making saves ₹0.85 per component. Note that the company should *still make* even though the book cost of ₹6.25 looks higher than ₹5.85 — because ₹1.25 of that book cost will be incurred whether the component is made or not.

Non-financial points worth one line: the question mentions “an assurance of continued supply”, which removes the usual worry about depending on an outside supplier. Against that, buying leaves the company’s own capacity idle — if that freed capacity can be used for something else, its contribution should be added to the case for buying.

FINAL ANSWER

Relevant (avoidable) cost of making	₹ 5.00 per component
Irrelevant fixed cost excluded	₹ 1.25 per component
(a) At ₹ 4.85	BUY — saves ₹ 0.15 per component
(b) At ₹ 5.85	MAKE — saves ₹ 0.85 per component

Q4 Foreign order using idle capacity

Special order · p.32

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Indo-British Company has capacity to produce 5,000 articles but actually produces only 2,000 articles for the home market at the following costs: Materials ₹40,000; Wages ₹36,000; Factory overheads — Fixed ₹12,000, Variable ₹20,000; Administration overhead — Fixed ₹18,000; Selling and Distribution overheads — Fixed ₹10,000, Variable ₹16,000. Total cost ₹1,52,000.

The home market can consume only 2,000 articles at a selling price of ₹80 per article. An additional order for the supply of 3,000 articles is received from a foreign country at ₹65 per article. Should this order be accepted or not, if execution of this order entails an additional packing cost of ₹3,000?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Split the costs.** Variable: materials 40,000 + wages 36,000 + factory variable 20,000 + selling variable 16,000 = ₹1,12,000 for 2,000 units, i.e. **₹56 per unit**. Fixed: 12,000 + 18,000 + 10,000 = **₹40,000**.
- **Capacity is exactly right.** 2,000 + 3,000 = 5,000 = full capacity. So no domestic sales have to be sacrificed — the order uses idle capacity only.
- Because fixed cost is already fully covered by the home market, the only test is whether the ₹65 price beats the ₹56 variable cost plus the ₹3,000 packing cost.
- The packing cost of ₹3,000 is a **lump sum specific to this order**, so it is relevant even though it is not a per-unit cost.

WORKING NOTES**W1 Splitting the present cost**

Element	Total ₹	Nature	Per unit ₹
Materials	40,000	Variable	20
Wages	36,000	Variable	18
Factory overheads — variable	20,000	Variable	10
Selling and distribution — variable	16,000	Variable	8
Total variable cost (2,000 units)	1,12,000		56
Factory overheads — fixed	12,000	Fixed	—
Administration overhead — fixed	18,000	Fixed	—
Selling and distribution — fixed	10,000	Fixed	—
Total fixed cost	40,000		—
Total cost	1,52,000		76

Comparative Statement of Profitability

Particulars	Present 2,000 units ₹	Foreign order 3,000 units ₹	Total 5,000 units ₹
Sales $2,000 \times 80$ $3,000 \times 65$	1,60,000	1,95,000	3,55,000
Less: Variable cost at ₹56 per unit	(1,12,000)	(1,68,000)	(2,80,000)
Less: Additional packing cost	—	(3,000)	(3,000)
Contribution	48,000	24,000	72,000
Less: Fixed cost <i>unchanged — already borne by the home market</i>	(40,000)	—	(40,000)
PROFIT	8,000	24,000	32,000

WHY THIS WORKS

Accept the order. It adds ₹24,000 of contribution and, because the fixed cost of ₹40,000 is already fully recovered from the home market, the whole of that ₹24,000 drops straight into profit. Profit rises from ₹8,000 to ₹32,000 — a fourfold increase.

Notice that the order price of ₹65 is **below the full cost of ₹76 per unit**. A student who compares 65 with 76 will reject the order and lose the entire question. The correct comparison is 65 against the ₹56 variable cost plus the ₹1 per unit of packing.

One non-financial line: the order is from a foreign country, so it does not undercut the home market where the price stays at ₹80. That is what makes accepting it safe.

FINAL ANSWER

Variable cost per unit	₹ 56
Total fixed cost	₹ 40,000
Contribution from the order	₹ 1,95,000 – ₹ 1,71,000 = ₹ 24,000
Present profit / profit after acceptance	₹ 8,000 / ₹ 32,000
Decision	ACCEPT — profit increases by ₹ 24,000

Q5 Export order needing more capacity than exists

Special order · p.32–33

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company currently operating at 80% capacity has the following particulars: Sales ₹32,00,000; Direct materials ₹10,00,000; Direct labour ₹4,00,000; Variable overheads ₹2,00,000; Fixed overheads ₹13,00,000.

An export order has been received that would utilise half the capacity of the factory. This order cannot be split — it has to be taken in full and executed at 10% below the normal domestic prices, or rejected totally. The alternatives available are:

- (1) Reject the order and continue with domestic sales only (as at present); or
- (2) Accept the order, split capacity between overseas and domestic sales and turn away excess domestic demand; or
- (3) Increase capacity to accept the export order and maintain present domestic sales by (a) buying equipment that will increase capacity, resulting in an increase of ₹1,00,000 in fixed cost, or (b) increasing the normal wage rate by $1\frac{1}{2}$ times to increase output.

Prepare a comparative statement of profitability and suggest the best alternative.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Scale everything to a per-1% basis first.** The figures are at 80% capacity, so divide by 80 to get the value of 1% and then build any level you need. Sales are ₹40,000 per 1%; variable cost is ₹20,000 per 1%.
- **The export order needs 50% of capacity** and cannot be split. Present domestic use is 80%. So $80 + 50 = 130\%$ — more than the factory has. That conflict is the whole problem.
- Alternative 2 keeps total capacity at 80%: 50% export + only 30% domestic. You **give up** 50 percentage points of domestic sales.
- Alternative 3 runs at 130% by either buying equipment (extra fixed cost) or paying $1\frac{1}{2}$ times the wage rate on the extra output.
- **Fixed overhead of ₹13,00,000 stays the same in alternatives 1, 2 and 3(b)** — only 3(a) changes it.

WORKING NOTES

W1 Value of 1% of capacity

Element	At 80% capacity ₹	Per 1% ₹	At 50% ₹	At 30% ₹
Sales	32,00,000	40,000	20,00,000	12,00,000
Direct materials	10,00,000	12,500	6,25,000	3,75,000
Direct labour	4,00,000	5,000	2,50,000	1,50,000
Variable overheads	2,00,000	2,500	1,25,000	75,000
Total variable cost	16,00,000	20,000	10,00,000	6,00,000
Contribution	16,00,000	20,000	10,00,000	6,00,000

W2 The export order at 50% capacity, priced 10% below normal

Sales at normal price = ₹20,00,000
 Less 10% = **export sales value ₹18,00,000**
 Variable cost at 50% = ₹10,00,000
Export contribution = ₹8,00,000

W3 Alternative 3(b) — wages at $1\frac{1}{2}$ times on the extra 50%

Direct materials at 50% = 6,25,000
 Direct labour at 50% $\times 1.5 = 2,50,000 \times 1.5 = 3,75,000$
 Variable overheads at 50% = 1,25,000
Variable cost of the export order = ₹11,25,000
 Export contribution = $18,00,000 - 11,25,000 = ₹6,75,000$

Comparative Statement of Profitability

Particulars	Alt 1 Reject	Alt 2 Split 50+30	Alt 3(a) Buy equipment	Alt 3(b) Overtime wages
Capacity used	80% domestic	50% export 30% domestic	80% domestic 50% export	80% domestic 50% export
Domestic sales	32,00,000	12,00,000	32,00,000	32,00,000
Export sales W2	—	18,00,000	18,00,000	18,00,000
Total sales	32,00,000	30,00,000	50,00,000	50,00,000

Particulars	Alt 1 Reject	Alt 2 Split 50+30	Alt 3(a) Buy equipment	Alt 3(b) Overtime wages
Less: Variable cost — domestic	(16,00,000)	(6,00,000)	(16,00,000)	(16,00,000)
Less: Variable cost — export	—	(10,00,000)	(10,00,000)	(11,25,000)
Total contribution	16,00,000	14,00,000	24,00,000	22,75,000
Less: Fixed overheads	(13,00,000)	(13,00,000)	(13,00,000)	(13,00,000)
Less: Additional fixed cost for equipment	—	—	(1,00,000)	—
PROFIT	3,00,000	1,00,000	10,00,000	9,75,000
Ranking	3rd	4th — worst	1st — BEST	2nd

WHY THIS WORKS

Recommendation: Alternative 3(a) — buy the equipment and accept the export order while keeping domestic sales at 80%. Profit ₹10,00,000.

- **Alternative 2 is the worst of all**, at ₹1,00,000 — worse even than rejecting the order. Giving up 50 percentage points of full-price domestic business to take on 50 points of discounted export business destroys contribution. This is the trap the question is built around: an export order at a discount is only worth having if it uses **spare** capacity.
- 3(a) beats 3(b) by ₹25,000 because the extra fixed cost of ₹1,00,000 is cheaper than the ₹1,25,000 of extra wages. But note that 3(a) commits the company permanently while overtime can be stopped — worth one sentence if the order is a one-off.

FINAL ANSWER

Value of 1% capacity — sales / variable cost	₹ 40,000 / ₹ 20,000
Export contribution at normal wages	₹ 8,00,000
Alt 1 — reject	Profit ₹ 3,00,000
Alt 2 — split capacity	Profit ₹ 1,00,000 — worst
Alt 3(a) — buy equipment	Profit ₹ 10,00,000 — BEST
Alt 3(b) — overtime at 1½ times	Profit ₹ 9,75,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Prestige Company Private Limited, manufacturing pressure cookers, has drawn up the following budget for the year 20X5-X6: Raw materials ₹20,00,000; Labour, stores, power and other variable costs ₹6,00,000; Manufacturing overheads (Fixed) ₹7,00,000; Variable distribution costs ₹4,00,000; General overheads including selling (Fixed) ₹3,00,000 — total ₹40,00,000. Income from sales ₹50,00,000; Budgeted profit ₹10,00,000.

The general manager suggests reducing selling price by 5% and expects to achieve an additional volume of 50%. There is sufficient manufacturing capacity. A more intensive manufacturing programme will involve additional costs of ₹50,000 for production planning. It will also be necessary to open an additional sales office at a cost of ₹1,00,000 per annum.

The sales manager suggests increasing selling price by 10%, which is estimated to reduce sales volume by 10%. At the same time, savings in manufacturing overheads and general overheads of ₹50,000 and ₹1,00,000 per annum respectively are expected on this reduced volume.

Which of these two proposals would you accept and why?

READ THE QUESTION FIRST — WHAT TO SPOT

- **Split the budget into variable and fixed first.** Variable = raw materials 20,00,000 + labour and other variable 6,00,000 + variable distribution 4,00,000 = **₹30,00,000**. Fixed = 7,00,000 + 3,00,000 = **₹10,00,000**.
- **Handle volume and price separately.** Variable cost changes with **volume only**; sales value changes with **both** volume and price. Doing them in one step is where errors creep in.
- GM: volume $\times 1.50$ and price $\times 0.95$, so sales = $50,00,000 \times 1.50 \times 0.95$. Variable cost = $30,00,000 \times 1.50$ (the price cut does not change costs).
- Sales manager: volume $\times 0.90$ and price $\times 1.10$, so sales = $50,00,000 \times 0.90 \times 1.10$. Variable cost = $30,00,000 \times 0.90$.
- Then adjust fixed cost: GM **adds** 1,50,000; sales manager **saves** 1,50,000.

WORKING NOTES

W1 Splitting the budget

Element	Amount ₹	Nature
Raw materials	20,00,000	Variable
Labour, stores, power and other variable costs	6,00,000	Variable
Variable distribution costs	4,00,000	Variable
Total variable cost	30,00,000	
Manufacturing overheads	7,00,000	Fixed
General overheads including selling	3,00,000	Fixed
Total fixed cost	10,00,000	

Budgeted contribution = $50,00,000 - 30,00,000 = ₹20,00,000 \rightarrow$ P/V ratio **40%**

Budgeted profit = $20,00,000 - 10,00,000 = ₹10,00,000 \checkmark$ agrees with the question

W2 Building each proposal

General manager: sales = $50,00,000 \times 1.50 \times 0.95 = ₹71,25,000$

variable cost = $30,00,000 \times 1.50 = ₹45,00,000$

fixed cost = $10,00,000 + 50,000 + 1,00,000 = ₹11,50,000$

Sales manager: sales = $50,00,000 \times 0.90 \times 1.10 = ₹49,50,000$

variable cost = $30,00,000 \times 0.90 = ₹27,00,000$

fixed cost = $10,00,000 - 50,000 - 1,00,000 = ₹8,50,000$

Comparative Statement of Profitability

Particulars	Original budget ₹	General manager ₹	Sales manager ₹
Volume index	100%	150%	90%
Price index	100%	95%	110%
Sales	50,00,000	71,25,000	49,50,000
Less: Variable cost	(30,00,000)	(45,00,000)	(27,00,000)
Contribution	20,00,000	26,25,000	22,50,000
P/V ratio	40.00%	36.84%	45.45%
Less: Fixed cost	(10,00,000)	(11,50,000)	(8,50,000)
PROFIT	10,00,000	14,75,000	14,00,000
Increase over budget	—	+4,75,000	+4,00,000

WHY THIS WORKS

Accept the general manager's proposal. It gives ₹14,75,000 against the sales manager's ₹14,00,000 — ₹75,000 better, and ₹4,75,000 above the original budget.

But the reasoning matters more than the ₹75,000, and a full-mark answer says so:

- The GM wins on **volume**: his P/V ratio actually *falls* from 40% to 36.84% because he cut the price, yet 50% more volume more than makes up for it.
- The sales manager wins on **margin**: his P/V ratio *rises* to 45.45% and he even saves fixed cost — but on 10% less volume.
- **Risk cuts the other way.** The GM's plan depends on actually selling 50% more, which is a large assumption, and it commits ₹1,50,000 of new fixed cost that cannot easily be reversed. The sales manager's plan needs no extra spending and earns nearly as much. If you doubt the 50% volume increase, the sales manager's proposal is the safer choice — say so.

FINAL ANSWER

Variable cost / fixed cost in the budget	₹ 30,00,000 / ₹ 10,00,000
Budgeted P/V ratio	40%
General manager — profit	₹ 14,75,000 (P/V 36.84%)
Sales manager — profit	₹ 14,00,000 (P/V 45.45%)
Recommendation	General manager's proposal — ₹ 75,000 better

Key Factor and Product Mix

Use these steps for Q7, Q8, Q9 and Q10

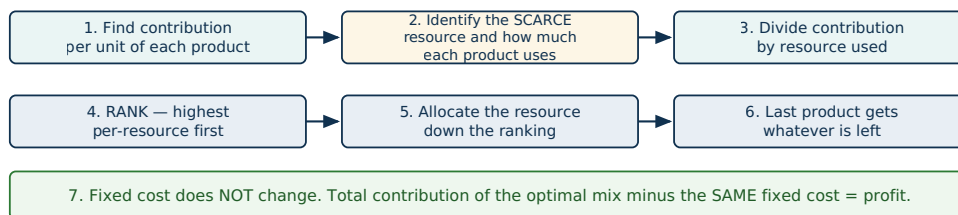
THE IDEA, AND WHY THE OBVIOUS ANSWER IS WRONG

When something runs short — material, labour hours, machine hours — you cannot make everything. You must choose. And the instinct is to make the product with the **highest contribution per unit**.

That instinct is wrong. If product A earns ₹30 of contribution but eats 15 labour hours, while product B earns only ₹16 but needs just 2 hours, then in the 15 hours that one A takes you could have made seven and a half B's and earned ₹120. B is four times better, even though its contribution per unit is half.

Rank by $\frac{\text{Contribution per unit}}{\text{Units of the scarce resource per unit}}$

This is called contribution per unit of key factor, or contribution per limiting factor. It is the entire topic in one line.



The six steps. Step 3 is the one that decides the answer; step 7 is the one students forget.

THREE THINGS TO BE CAREFUL ABOUT

- **Work out the resource per unit from the money figure.** If materials cost ₹80 per unit at ₹20 per kg, then the product uses 4 kg. Questions almost never give you the quantity directly.
- **Fixed cost stays the same in the optimal mix.** It was fixed before the shortage and it is fixed after. Do not recompute it per unit — carry the same total across.
- **Check whether demand also limits you.** You cannot make more of a product than can be sold, even if it ranks first. In Q8 and Q10 the budgeted quantity is the demand ceiling.

Q7 Three sales mixtures compared

Product mix · p.33-34

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Present the following information to show to the management (a) the marginal product cost and the contribution per unit and (b) the total contribution and profits resulting from each of the following sales mixtures.

Particulars	Product A	Product B
Direct materials per unit	₹ 10	₹ 9
Direct wages per unit	₹ 3	₹ 2
Sales price per unit	₹ 20	₹ 15

Fixed expenses ₹800. Variable expenses are allocated to products as 100% of direct wages.

Sales mixtures: **(i)** 1,000 units of A and 2,000 units of B; **(ii)** 1,500 units of A and 1,500 units of B; **(iii)** 2,000 units of A and 1,000 units of B.

READ THE QUESTION FIRST — WHAT TO SPOT

- “Variable expenses are 100% of direct wages” means variable overhead = the same amount as wages: A gets ₹3, B gets ₹2. Add it as a third variable element.
- There is **no scarce resource here** — the total is 3,000 units in every mixture. So this is not a ranking problem; you simply evaluate three given mixes and pick the best.
- Because total units are the same in all three, the best mix is automatically the one with more of the **higher-contribution** product. A earns ₹4 and B earns ₹2, so expect mixture (iii) to win.

(a) Marginal cost and contribution per unit

Particulars	Product A ₹	Product B ₹
Selling price	20	15
Less: Direct materials	(10)	(9)
Less: Direct wages	(3)	(2)
Less: Variable expenses 100% of direct wages	(3)	(2)
Marginal (variable) cost per unit	16	13
Contribution per unit	4	2
P/V ratio	20%	13.33%

(b) Total contribution and profit under each sales mixture

Particulars	(i) 1,000 A + 2,000 B	(ii) 1,500 A + 1,500 B	(iii) 2,000 A + 1,000 B
Contribution from A units × ₹4	4,000	6,000	8,000
Contribution from B units × ₹2	4,000	3,000	2,000
Total contribution	8,000	9,000	10,000
Less: Fixed expenses	(800)	(800)	(800)
PROFIT	7,200	8,200	9,200
Ranking	3rd	2nd	1st — BEST

WHY THIS WORKS

Mixture (iii) is best, giving a profit of ₹9,200.

The reason is simple once you see it: all three mixes sell exactly 3,000 units, and A contributes ₹4 against B's ₹2. So every unit of B swapped for a unit of A adds ₹2. Moving from mixture (i) to (iii) swaps 1,000 units of B for 1,000 units of A and gains exactly ₹2,000 — which is the difference between ₹7,200 and ₹9,200.

Recommendation: sell as much of A as the market will absorb. The only reason to keep B in the mix at all is if demand for A is limited to 2,000 units.

FINAL ANSWER

Marginal cost per unit — A / B	₹ 16 / ₹ 13
Contribution per unit — A / B	₹ 4 / ₹ 2
(i) Contribution / profit	₹ 8,000 / ₹ 7,200
(ii) Contribution / profit	₹ 9,000 / ₹ 8,200
(iii) Contribution / profit	₹ 10,000 / ₹ 9,200 — BEST

Q8 Material shortage — optimal mix

Key factor · p.34

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company manufactures three products. The budgeted quantity, selling prices and unit costs are as under:

Particulars	A	B	C
Raw materials (@ ₹20 per kg)	80	40	20
Direct wages (@ ₹5 per hour)	5	15	10
Variable overheads	10	30	20
Fixed overheads	9	22	18
Budgeted production (in units)	6,400	3,200	2,400
Selling price per unit (₹)	140	120	90

(i) Present a statement of budgeted profit. **(ii)** Set the optimal product mix and determine the profit if the supply of raw materials is restricted to 18,400 kgs.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Convert the material cost into kilograms.** At ₹20 per kg: A uses $80 \div 20 = 4$ kg, B uses $40 \div 20 = 2$ kg, C uses $20 \div 20 = 1$ kg. This is the step the whole of part (ii) depends on.
- **Fixed overhead per unit is given**, so total fixed cost = $(6,400 \times 9) + (3,200 \times 22) + (2,400 \times 18) = ₹1,71,200$. That total **does not change** in part (ii).
- **Check the shortage.** Budget needs $(6,400 \times 4) + (3,200 \times 2) + (2,400 \times 1) = 25,600 + 6,400 + 2,400 = 34,400$ kg, but only 18,400 kg is available — a severe shortage of 16,000 kg.
- So rank by **contribution per kg**, and remember the budgeted production is the **demand ceiling** — you cannot make more of C than 2,400 units even though it ranks first.

(i) Statement of Budgeted Profit

Particulars	A	B	C	Total ₹
Selling price per unit	140	120	90	—
Less: Raw materials	(80)	(40)	(20)	—
Less: Direct wages	(5)	(15)	(10)	—
Less: Variable overheads	(10)	(30)	(20)	—
Variable cost per unit	95	85	50	—
Contribution per unit	45	35	40	—
Budgeted units	6,400	3,200	2,400	—
Total contribution	2,88,000	1,12,000	96,000	4,96,000
Fixed overheads units × rate	(57,600)	(70,400)	(43,200)	(1,71,200)
BUDGETED PROFIT				3,24,800

(ii) Step 1 — Ranking by contribution per kg of raw material

Particulars	A	B	C
Contribution per unit ₹	45	35	40
Raw material per unit cost ÷ ₹20	4 kg	2 kg	1 kg
Contribution per kg	11.25	17.50	40.00
Rank	3rd	2nd	1st

(ii) Step 2 — Allocating the 18,400 kg down the ranking

Rank	Product	Units made	kg per unit	kg used	kg remaining
1st	C — full demand 2,400 units	2,400	1	2,400	16,000
2nd	B — full demand 3,200 units	3,200	2	6,400	9,600
3rd	A — only what is left	2,400	4	9,600	Nil
	Total			18,400	Nil

A's output falls from the budgeted 6,400 units to 2,400 units — 9,600 kg ÷ 4 kg per unit. A is cut back because it has the lowest contribution per kg.

(ii) Step 3 — Profit under the optimal mix

Product	Units	Contribution per unit ₹	Total contribution ₹
C	2,400	40	96,000
B	3,200	35	1,12,000
A	2,400	45	1,08,000
Total contribution			3,16,000
Less: Fixed overheads unchanged from part (i)			(1,71,200)
PROFIT under the optimal mix			1,44,800

WHY THIS WORKS

Look at what the ranking did. Product A has the **highest contribution per unit** at ₹45 — and it is the one cut back hardest, from 6,400 units to 2,400. Product C has the **lowest** selling price and only ₹40 of contribution, yet it is protected in full.

The reason is that A swallows 4 kg per unit while C needs only 1 kg. Per kilogram of the scarce material, C returns ₹40 and A returns just ₹11.25. When a resource is scarce, what matters is the return per unit of that resource — never the return per unit of product.

Profit falls from ₹3,24,800 to ₹1,44,800 because of the shortage. But had you kept the budgeted proportions instead of re-ranking, it would have fallen much further — the ranking saves the company money.

FINAL ANSWER

Contribution per unit — A / B / C	₹ 45 / ₹ 35 / ₹ 40
Raw material per unit — A / B / C	4 kg / 2 kg / 1 kg
Contribution per kg — A / B / C	₹ 11.25 / ₹ 17.50 / ₹ 40.00
Ranking	C first, B second, A third
(i) Budgeted profit	₹ 3,24,800
Optimal mix	C 2,400 units, B 3,200 units, A 2,400 units
(ii) Profit under optimal mix	₹ 1,44,800

Q9 Labour shortage — ranking two products

Key factor · p.34

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information in respect of Product A and Product B of a firm is given:

Particulars	Product A	Product B
Selling price per unit	₹ 75	₹ 48
Direct materials	₹ 30	₹ 30
Direct labour hours (₹0.50 per hour)	15 hours	2 hours

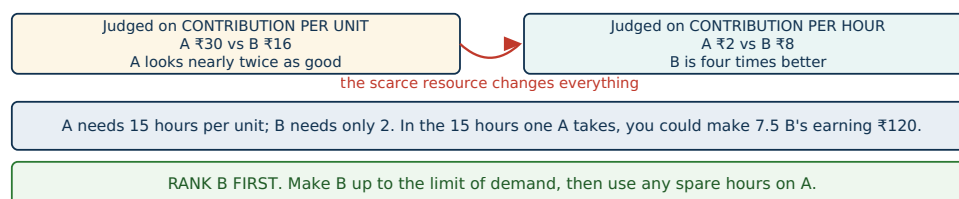
Variable overhead = 100% of direct wages. Fixed overheads = ₹3,000. Present the above information to show the profitability of products during a labour shortage.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Compute the wages from the hours.** A: 15 hours × ₹0.50 = ₹7.50. B: 2 hours × ₹0.50 = ₹1.00. Then variable overhead equals the same amounts.
- **The key factor is named for you** — “during a labour shortage”. So rank by contribution per **labour hour**.
- This problem is the clearest illustration in the module of why contribution per unit misleads. A earns nearly twice B’s contribution per unit, and is **four times worse** per hour.
- No quantities or total hours are given, so you are asked only to **show the profitability** — the ranking — not to build an optimal mix.

Statement of Profitability during a Labour Shortage

Particulars	Product A ₹	Product B ₹
Selling price per unit	75	48
Less: Direct materials	(30)	(30)
Less: Direct wages 15 × 0.50 2 × 0.50	(7.50)	(1.00)
Less: Variable overhead 100% of direct wages	(7.50)	(1.00)
Marginal cost per unit	45.00	32.00
Contribution per unit	30.00	16.00
P/V ratio	40.00%	33.33%
Direct labour hours per unit	15	2
Contribution per labour hour	2.00	8.00
Rank during a labour shortage	2nd	1st — four times better



Why contribution per unit is the wrong ranking when a resource is scarce.

WHY THIS WORKS

Product B is the more profitable during a labour shortage, earning ₹8 per labour hour against A's ₹2 — four times as much.

Recommendation: produce B up to the limit of its demand and use any remaining hours on A. Note that if labour were *not* scarce, A would be the better product on a per-unit basis and the answer would reverse — which is exactly why identifying the key factor is the first thing you do.

The fixed overhead of ₹3,000 has deliberately not been used. It is the same whichever product is made, so it cannot influence the ranking. Mention that you have excluded it and why.

FINAL ANSWER

Marginal cost per unit — A / B	₹ 45 / ₹ 32
Contribution per unit — A / B	₹ 30 / ₹ 16
Labour hours per unit — A / B	15 / 2
Contribution per labour hour — A / B	₹ 2 / ₹ 8
Recommendation	Product B — four times more profitable per scarce labour hour

Q10 Labour limit — optimal mix and profit

Key factor · p.34–35

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

In a factory producing two different kinds of articles, the limiting factor is the availability of labour. From the following information for 2016, show which product is more profitable.

Particulars	Product A (₹ per unit)	Product B (₹ per unit)
Materials	5.00	5.00
Labour — 6 hours @ ₹0.50	3.00	—
Labour — 3 hours @ ₹0.50	—	1.50
Overheads — Fixed (50% of labour)	1.50	0.75
Overheads — Variable	1.50	1.50
Total Cost	11.00	8.75
Selling Price	14.00	11.00
Total production for the month	500 units	600 units

Also set the optimal product mix and the profitability if the maximum labour hours per month is 4,200 hours.

READ THE QUESTION FIRST — WHAT TO SPOT

- **The fixed overhead is described as 50% of labour**, so it is stated per unit but it is still a fixed cost. Exclude it from the marginal cost and treat the total as a lump sum: $(500 \times 1.50) + (600 \times 0.75) = ₹1,200$.
- Marginal cost = materials + labour + variable overhead only. A = $5 + 3 + 1.50 = ₹9.50$; B = $5 + 1.50 + 1.50 = ₹8.00$.
- **Check the shortage.** Present production needs $(500 \times 6) + (600 \times 3) = 3,000 + 1,800 = 4,800$ hours, but only 4,200 are available — short by **600 hours**.
- Rank by contribution per labour hour, then allocate. Present production is the **demand ceiling** for each product.

Statement of Profitability per unit and per labour hour

Particulars	Product A ₹	Product B ₹
Selling price	14.00	11.00
Less: Materials	(5.00)	(5.00)
Less: Labour	(3.00)	(1.50)
Less: Variable overheads	(1.50)	(1.50)
Marginal cost per unit	9.50	8.00
Contribution per unit	4.50	3.00
Labour hours per unit	6	3
Contribution per labour hour	0.75	1.00
Rank	2nd	1st — more profitable

Optimal product mix within 4,200 labour hours

Rank	Product	Units	Hours per unit	Hours used	Hours left
1st	B — full demand 600 units	600	3	1,800	2,400
2nd	A — only what is left	400	6	2,400	Nil
	Total			4,200	Nil

A is cut from 500 units to 400 units — 2,400 hours ÷ 6 hours per unit. B is protected in full because it earns more per hour.

Profitability of the optimal mix

Product	Units	Contribution per unit ₹	Total contribution ₹
B	600	3.00	1,800
A	400	4.50	1,800
Total contribution			3,600
Less: Fixed overheads $(500 \times 1.50) + (600 \times 0.75)$			(1,200)
PROFIT			2,400

THE TRAP / WHERE MARKS ARE LOST

- **Including the fixed overhead in the marginal cost.** The question calls it “Overheads — Fixed” even though it is shown per unit. It must be excluded from contribution and deducted once as a total.
- **Ranking A first because its contribution of ₹4.50 is higher.** Per hour A returns only ₹0.75 against B’s ₹1.00.
- **Making more than 600 units of B** because it ranks first. 600 is all the market takes — the given production figure is the demand ceiling.

FINAL ANSWER

Marginal cost per unit — A / B	₹ 9.50 / ₹ 8.00
Contribution per unit — A / B	₹ 4.50 / ₹ 3.00
Contribution per labour hour — A / B	₹ 0.75 / ₹ 1.00
More profitable product	Product B
Optimal mix within 4,200 hours	B 600 units + A 400 units
Total contribution / fixed cost	₹ 3,600 / ₹ 1,200

Q11 Export order versus local order — same figures, different advice

Special order · p.35

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The cost of a manufacturing company for the product is: Materials ₹12.00; Labour ₹9.00; Variable expenses ₹6.00; Fixed expenses ₹18.00 — total ₹45.00. The unit of product is sold for ₹51.00.

The company's normal capacity is 1,00,000 units. The figures given above are for 80,000 units. The company has received an offer for 20,000 units @ ₹36 per unit from a foreign customer.

Advise the manufacturer on whether the order should be accepted. Also give your advice if the order is from a local merchant.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Variable cost per unit = 12 + 9 + 6 = ₹27.** The ₹18 of fixed expenses is per unit at 80,000 units, so total fixed cost = 80,000 × 18 = **₹14,40,000.**
- **Capacity check.** 80,000 + 20,000 = 1,00,000 = exactly normal capacity. So the order uses idle capacity and no existing sales are lost, and **fixed cost does not increase.**
- The offer price of ₹36 is well below the full cost of ₹45 — which is the trap. Compare it with the **₹27 variable cost**, not with ₹45.
- **The second half of the question is about business judgement, not arithmetic.** The numbers are identical for a local merchant; what changes is the risk to the existing market. Make sure you answer it in words.

Comparative Statement of Profitability

Particulars	Present 80,000 units ₹	Foreign order 20,000 units ₹	Total 1,00,000 units ₹
Sales 80,000×51 20,000×36	40,80,000	7,20,000	48,00,000
Less: Variable cost at ₹27 per unit	(21,60,000)	(5,40,000)	(27,00,000)
Contribution	19,20,000	1,80,000	21,00,000
Contribution per unit	24	9	21
Less: Fixed expenses unchanged — spare capacity used	(14,40,000)	—	(14,40,000)
PROFIT	4,80,000	1,80,000	6,60,000

WHY THIS WORKS

Advice on the foreign order: ACCEPT. Each unit earns ₹36 – ₹27 = ₹9 of contribution, and since the fixed cost of ₹14,40,000 is already fully recovered from the 80,000 domestic units, the whole ₹1,80,000 goes straight to profit. Profit rises from ₹4,80,000 to ₹6,60,000.

Advice if the order is from a local merchant: normally REJECT — and the arithmetic is not the reason. The figures are identical, so on paper the order still adds ₹1,80,000. The problem is what happens next:

- The merchant sells in the same market where the company charges ₹51. Goods bought at ₹36 can be resold below ₹51, undercutting the company's own customers.
- Existing customers who learn of the ₹36 price will demand it too. If even a fifth of the 80,000 regular units had to be repriced downward, the loss would swamp the ₹1,80,000 gained.

- A foreign market is **geographically separate**, so the low price cannot leak back. That separation is the only thing that makes the export order safe.

Accept from a local merchant **only** if the goods can be genuinely segregated — a different brand, a different region, or a binding condition on resale. Say that, and you have the full mark.

FINAL ANSWER

Variable cost per unit	₹ 27
Total fixed cost	₹ 14,40,000
Contribution per unit on the offer	₹ 36 – ₹ 27 = ₹ 9
Extra contribution from 20,000 units	₹ 1,80,000
Profit before / after	₹ 4,80,000 / ₹ 6,60,000
Foreign customer	ACCEPT — profit rises by ₹ 1,80,000
Local merchant	REJECT unless the market can be kept separate — risk of undercutting the existing ₹51 price

Shut-down Decisions

Use these steps for Q12, Q13 and Q14

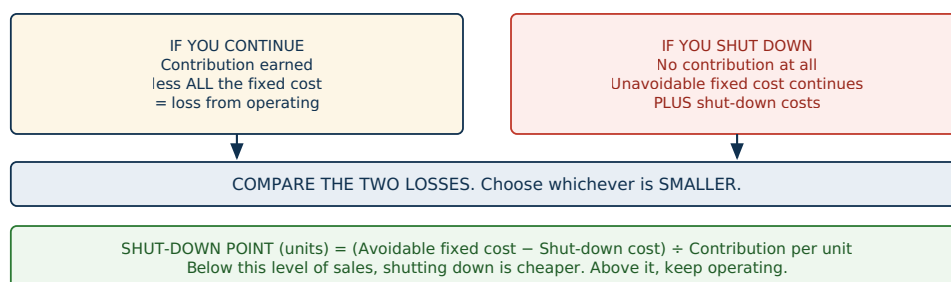
THE QUESTION IS NOT “ARE WE LOSING MONEY?”

A factory making a loss should not automatically close. Closing does not make the loss go away — some fixed costs continue whether the gates are open or shut (rent, rates, security, depreciation, key salaries), and closing itself **costs money** (redundancy, mothballing machinery, restarting later).

So the real question is: **which loss is smaller — the loss from carrying on, or the loss from stopping?**

Continue if $\text{CONTRIBUTION} > \text{Avoidable fixed cost} - \text{Shut-down cost}$

In plain words: keep going as long as the contribution you earn is bigger than the money you would actually save by stopping.



The comparison, and the formula for the exact break-point between the two.

THE FIVE STEPS

1. **Contribution per unit** = selling price – all variable costs (including variable selling expenses).
2. **Total fixed cost for the period.** Watch the period — if the fixed cost is annual and the question asks about a quarter, divide by four.
3. **Unavoidable fixed cost** = what still has to be paid after shutting. **Avoidable** = **Total** – **Unavoidable**.
4. **Loss if you continue** = Total fixed cost – Contribution. **Loss if you shut** = Unavoidable fixed cost + Shut-down cost.
5. Compare, recommend in words, then compute the **shut-down point** if asked.

THE TWO TRAPS

- **Treating the whole fixed cost as avoidable.** Read the question carefully: it always tells you how much continues, either directly ("only " + R + "20,000 can be avoided") or indirectly ("fixed cost will be reduced to " + R + "130,000").
- **Forgetting the shut-down cost.** It works *against* closing and it is easy to miss. It reduces the saving, and in the shut-down-point formula it is **subtracted** from the avoidable fixed cost.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The selling price per unit of a product is \$14. For the forthcoming period the demand will be only 5,000 units. The fixed expenses at 50% activity (5,000 units) will be \$30,000. The company is thinking of shutting down operations, in which case an additional amount of \$2,000 will have to be incurred for shutting down and only \$20,000 of the above fixed costs can be avoided.

What should be the variable cost per unit to recommend a shut-down?

READ THE QUESTION FIRST — WHAT TO SPOT

- This is the shut-down comparison run **backwards**. You are not asked which is better — you are asked what variable cost would make shutting down the better choice.
- **Work out the shut-down loss first, because it is a fixed number.** Unavoidable fixed cost = $30,000 - 20,000 = \$10,000$, plus shut-down cost \$2,000 = **\$12,000**. That is the loss if the plant closes, whatever the variable cost is.
- Then set the loss from continuing equal to \$12,000 and solve for the variable cost. That gives the **indifference point** — above it, shut down.

WORKING NOTES

W1 Loss if the plant is shut down

Total fixed expenses = \$30,000
 Less: Avoidable on shut-down = (\$20,000)
Unavoidable fixed cost = \$10,000
 Add: Shut-down costs = \$2,000
LOSS IF SHUT DOWN = \$12,000

W2 Loss if operations continue, in terms of the variable cost v

Contribution = $5,000 \times (14 - v)$
 Loss if continue = $30,000 - 5,000(14 - v)$

W3 Setting the two equal to find the indifference point

$30,000 - 5,000(14 - v) = 12,000$
 $30,000 - 12,000 = 5,000(14 - v)$
 $18,000 = 5,000(14 - v)$
 $14 - v = \frac{18,000}{5,000} = 3.60$
 $v = 14 - 3.60 = \$10.40$ per unit

Proof and interpretation

Variable cost per unit	Contribution per unit	Total contribution	Loss if continue	Loss if shut	Decision
\$10.00	\$4.00	20,000	(10,000)	(12,000)	Continue — loss is smaller
\$10.40	\$3.60	18,000	(12,000)	(12,000)	Indifferent — the break-point

Variable cost per unit	Contribution per unit	Total contribution	Loss if continue	Loss if shut	Decision
\$11.00	\$3.00	15,000	(15,000)	(12,000)	Shut down — shutting is cheaper

WHY THIS WORKS

Answer: the variable cost per unit would have to be \$10.40 or more before a shut-down could be recommended.

At exactly \$10.40 the two losses are both \$12,000 and management is indifferent. Above \$10.40 the contribution falls below the \$18,000 that operating must generate to justify itself, and closing becomes the cheaper option.

Note the shape of the answer: the plant is worth running even at a loss, as long as that loss is smaller than \$12,000. A contribution of anything above \$18,000 makes operating worthwhile even though the company never reaches profit.

FINAL ANSWER

Unavoidable fixed cost	\$ 10,000
Shut-down cost	\$ 2,000
Loss if shut down	\$ 12,000
Contribution needed to justify operating	\$ 18,000
Contribution needed per unit	\$ 18,000 ÷ 5,000 = \$ 3.60
Variable cost at which shut-down is recommended	\$ 10.40 per unit or more

Q13 Shut down, and the shut-down point in capacity terms

Shut-down · p.35

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

G Ltd. produces and sells 95,000 units of X in a year at its 80% production capacity. The selling price of the product is \$8 per unit. The variable cost is 75% of the sale price per unit. The fixed cost is \$350,000. The company is continuously incurring losses and management plans to shut down the plant. The fixed cost is expected to be reduced to \$130,000. Additional costs of plant shut-down are expected at \$15,000.

Should the plant be shut down? What is the capacity level of production at the shut-down point?

READ THE QUESTION FIRST — WHAT TO SPOT

- “Variable cost is 75% of the sale price” → \$6 per unit, so contribution is \$2 per unit. Convert the percentage into money before anything else.
- “Fixed cost expected to be reduced to \$130,000” means **\$130,000 is unavoidable** and $350,000 - 130,000 = \textbf{\$220,000 is avoidable}$.
- **95,000 units = 80% capacity**, so 100% capacity = $95,000 \div 0.80 = \textbf{118,750 units}$. You need this to answer the second part.
- The plant is loss-making either way — the question is only which loss is smaller.

Comparative Statement — continue or shut down

Particulars	Continue \$	Shut down \$
Sales 95,000 × \$8	760,000	—
Less: Variable cost 95,000 × \$6	(570,000)	—

Particulars	Continue \$	Shut down \$
Contribution 95,000 × \$2	190,000	Nil
Less: Fixed cost	(350,000)	(130,000)
Less: Shut-down costs	—	(15,000)
LOSS	(160,000)	(145,000)
Decision	Loss is larger	SHUT DOWN — saves \$15,000

WORKING NOTES

Shut-down point

Avoidable fixed cost = 350,000 – 130,000 = \$220,000

Less: Additional shut-down cost = (\$15,000)

Net saving from shutting down = \$205,000

$$\text{Shut-down point} = \frac{\text{Avoidable fixed cost} - \text{Shut-down cost}}{\text{Contribution per unit}} = \frac{205,000}{2} = \mathbf{102,500 \text{ units}}$$

100% capacity = 95,000 ÷ 0.80 = **118,750 units**

Capacity at the shut-down point = $\frac{102,500}{118,750} \times 100 = \mathbf{86.32\%}$

Consistency check: the company is operating at 80% (95,000 units), which is **below** the shut-down point of 86.32% (102,500 units). So shutting down is correct — the two answers agree, which is your proof.

WHY THIS WORKS

Yes, shut the plant down. It saves \$15,000 — the loss falls from \$160,000 to \$145,000.

The shut-down point of **102,500 units, or 86.32% of capacity**, is the more useful answer for management. It says: if sales can be lifted above 102,500 units the plant is worth running again; below that it is not. At the current 95,000 units the company is 7,500 units short of the level that would justify keeping the gates open.

One line worth adding: closing is a serious step with consequences the numbers do not show — loss of skilled workers, of customers, and the cost of restarting. If the shortfall is only 7,500 units and demand may recover, management might reasonably keep operating for a while and accept the extra \$15,000 as the price of staying in business.

FINAL ANSWER

Contribution per unit	\$ 2 (selling \$ 8 less variable \$ 6)
Avoidable / unavoidable fixed cost	\$ 220,000 / \$ 130,000
Loss if continue	\$ 160,000
Loss if shut down	\$ 145,000
Decision	SHUT DOWN — saves \$ 15,000
Shut-down point	102,500 units = 86.32% of capacity

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A Paint Manufacturing Company manufactures 200,000 medium-sized tins of 'Spray Lac Paints' annually when working at normal capacity. Its cost of manufacture per unit is \$16.40 made up as: Direct Materials \$7.80; Direct Labour \$2.10; Variable Overheads \$2.50; Fixed Overheads \$4.00.

Each tin is sold for \$21 with variable selling expenses of \$0.60 per tin. During the next quarter only 10,000 units can be produced and sold. Management plans to shut down the plant, estimating that the fixed manufacturing cost can be reduced to \$74,000 for the quarter. When the plant is operating, fixed overheads are incurred at a uniform rate throughout the year. Additional costs of plant shut-down for the quarter are estimated at \$14,000.

- (a) Express your opinion, with calculations, as to whether the plant should be shut down during the quarter.
 (b) Calculate the shut-down point for the quarter in tins.

READ THE QUESTION FIRST — WHAT TO SPOT

The trap in this problem is the **time period**. Almost everything given is annual; the question is about one quarter.

- **Annual fixed overhead** = 200,000 tins × \$4.00 = \$800,000. "Incurred at a uniform rate throughout the year" means **\$200,000 per quarter**. That is the figure to use, not 800,000.
- **Do not forget the variable selling expense of \$0.60**. It is listed separately from the manufacturing cost, so it is easy to miss — but it is variable and must come out of contribution.
- Variable cost per tin = 7.80 + 2.10 + 2.50 + 0.60 = **\$13.00**, so contribution = 21 – 13 = **\$8.00 per tin**.
- **\$74,000 is the unavoidable fixed cost** for the quarter, so avoidable = 200,000 – 74,000 = \$126,000.

WORKING NOTES

W1 Contribution per tin

Element	Per tin \$
Selling price	21.00
Less: Direct materials	(7.80)
Less: Direct labour	(2.10)
Less: Variable overheads	(2.50)
Less: Variable selling expenses	(0.60)
Total variable cost	(13.00)
Contribution per tin	8.00

W2 Fixed overhead for the QUARTER — the key step

Annual fixed overhead = 200,000 tins × \$4.00 = **\$800,000**
 Incurred uniformly, so per quarter = 800,000 ÷ 4 = **\$200,000**
 Unavoidable if shut = \$74,000 → **avoidable = \$126,000**

(a) Comparative Statement for the quarter

Particulars	Continue \$	Shut down \$
Sales $10,000 \times \$21$	210,000	—
Less: Variable cost $10,000 \times \$13$	(130,000)	—
Contribution $10,000 \times \$8$	80,000	Nil
Less: Fixed overheads for the quarter W2	(200,000)	(74,000)
Less: Additional shut-down costs	—	(14,000)
LOSS	(120,000)	(88,000)
Decision	Loss is larger by \$32,000	SHUT DOWN — saves \$32,000

WORKING NOTES

Avoidable fixed cost for the quarter = $200,000 - 74,000 = \$126,000$

Less: Additional shut-down cost = $(\$14,000)$

Net saving from shutting down = \$112,000

$$\text{Shut-down point} = \frac{112,000}{8} = \mathbf{14,000 \text{ tins}}$$

Consistency check: only 10,000 tins can be sold, which is below the shut-down point of 14,000 tins — so shutting down is correct. The two parts of the answer confirm each other.

THE TRAP / WHERE MARKS ARE LOST

- **Using \$800,000 of fixed overhead in a quarterly statement.** That is the annual figure. The question says the overhead is incurred uniformly, which is a direct instruction to divide by four.
- **Omitting the \$0.60 variable selling expense.** It sits outside the \$16.40 manufacturing cost and is easy to overlook, but it reduces contribution from \$8.60 to \$8.00 and changes the shut-down point from 13,023 to 14,000 tins.
- **Treating the whole \$200,000 as avoidable.** Only \$126,000 is; \$74,000 continues.

FINAL ANSWER

Variable cost / contribution per tin	\$ 13.00 / \$ 8.00
Fixed overhead for the quarter	\$ 200,000
Avoidable / unavoidable	\$ 126,000 / \$ 74,000
Loss if continue	\$ 120,000
Loss if shut down	\$ 88,000
(a) Opinion	SHUT DOWN — saves \$ 32,000
(b) Shut-down point	14,000 tins for the quarter

Q15 Break-even chart

Charts · p.36

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

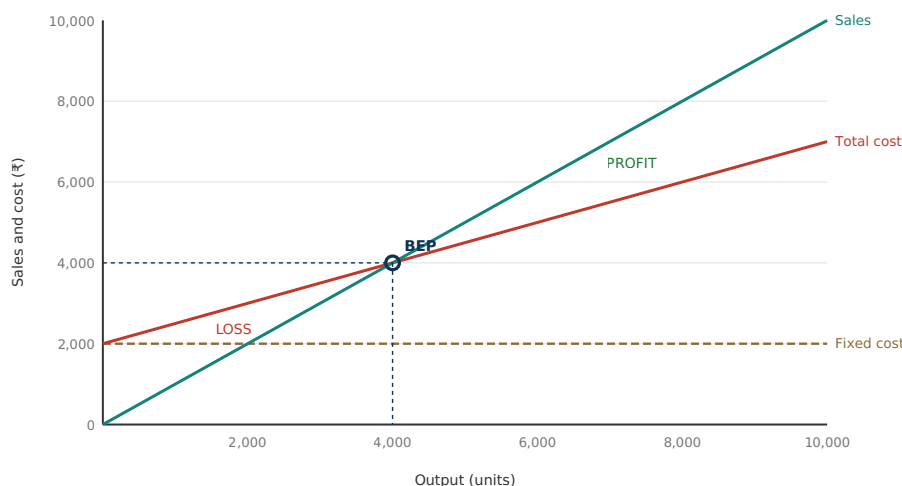
From the following information, prepare the break-even chart. Fixed Cost ₹2,000; Variable Cost ₹0.50 per unit; Sales ₹1 per unit. Units produced and sold 2,000; 4,000; 6,000; 8,000 and 10,000.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Build the table before you draw anything.** Compute sales, variable cost, total cost and profit at each of the five volumes. The chart is just those numbers plotted.
- Three lines go on the chart: a **horizontal** fixed-cost line at ₹2,000; a **total-cost** line starting at ₹2,000 on the vertical axis; and a **sales** line starting at the origin.
- **The total cost line must start at the fixed cost, not at zero.** At zero output you still pay ₹2,000. This is the single most common drawing error.
- The break-even point is where the sales line crosses the total cost line. Mark it, drop a dotted line to each axis, and label both readings.

Statement of cost and profit at each volume — the data for the chart

Units	Sales @ ₹1	Variable cost @ ₹0.50	Fixed cost	Total cost	Profit / (Loss)
2,000	2,000	1,000	2,000	3,000	(1,000)
4,000	4,000	2,000	2,000	4,000	Nil — BEP
6,000	6,000	3,000	2,000	5,000	1,000
8,000	8,000	4,000	2,000	6,000	2,000
10,000	10,000	5,000	2,000	7,000	3,000



Break-even chart. BEP = 4,000 units = ₹4,000. Note how the total-cost line begins at ₹2,000 on the vertical axis, not at the origin.

WORKING NOTES

Contribution per unit = $1.00 - 0.50 = \text{₹}0.50$

P/V ratio = $0.50 \div 1.00 = 50\%$

$$\text{BEP (units)} = \frac{2,000}{0.50} = 4,000 \text{ units} \quad \text{BEP (₹)} = \frac{2,000}{0.50} = \text{₹}4,000$$

The table, the formula and the chart all give 4,000 units. Always verify the graph by calculation — examiners award marks for the check.

THE TRAP / WHERE MARKS ARE LOST

- Starting the total-cost line at the origin. It must start at the fixed cost.
- Forgetting to label the axes and state the scale. Both carry marks.
- Not marking the profit and loss areas. The wedge left of the BEP is loss, the wedge right of it is profit — shade or label them.

- Drawing the chart and stopping. **Always verify the BEP by formula underneath.**

FINAL ANSWER

Contribution per unit / P/V ratio ₹ 0.50 / 50%

Break-even point **4,000 units = ₹ 4,000**

Loss at 2,000 units ₹ 1,000

Profit at 10,000 units ₹ 3,000

Margin of safety at 10,000 units **6,000 units = ₹ 6,000**

Q16 Break-even chart with a revised selling price

Charts · p.36

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

You are given the following data for the costing year of a factory: Budget output 1,00,000 units; Fixed Expenses ₹5,00,000; Variable Expenses ₹10 per unit; Selling Price ₹20 per unit.

Draw a break-even chart showing the break-even point. If the selling price is reduced to ₹18 per unit, what will be the new break-even point?

READ THE QUESTION FIRST — WHAT TO SPOT

- Contribution = $20 - 10 = ₹10$ per unit, so the P/V ratio is 50% and the BEP is $5,00,000 \div 10 = \mathbf{50,000 \text{ units}}$ — exactly half the budget.
- **When the price falls to ₹18 the variable cost does not change.** Only contribution falls, from ₹10 to ₹8, so the BEP rises to $5,00,000 \div 8 = \mathbf{62,500 \text{ units}}$.
- On the chart, show the second sales line as a **dashed line of lower slope** from the same origin. Two sales lines, one total-cost line, two break-even points.

WORKING NOTES

Present position — selling price ₹20

Contribution per unit = $20 - 10 = \mathbf{₹10}$ → P/V ratio **50%**

BEP = $\frac{5,00,000}{10} = \mathbf{50,000 \text{ units}}$

BEP sales value = $50,000 \times 20 = \mathbf{₹10,00,000}$

Profit at budget = $(1,00,000 \times 10) - 5,00,000 = \mathbf{₹5,00,000}$

Revised position — selling price reduced to ₹18

Contribution per unit = $18 - 10 = \mathbf{₹8}$ → P/V ratio **44.44%**

New BEP = $\frac{5,00,000}{8} = \mathbf{62,500 \text{ units}}$

New BEP sales value = $62,500 \times 18 = \mathbf{₹11,25,000}$

Profit at budget = $(1,00,000 \times 8) - 5,00,000 = \mathbf{₹3,00,000}$



	At ₹20	At ₹18	Change
Selling price per unit	20	18	−10%
Variable cost per unit	10	10	no change
Contribution per unit	10	8	−20%
P/V ratio	50.00%	44.44%	−5.56 points
Break-even point (units)	50,000	62,500	+25%
Break-even sales value	10,00,000	11,25,000	+12.5%
Profit at budgeted 1,00,000 units	5,00,000	3,00,000	−40%

On the chart you can see it directly: the dashed sales line is flatter, so it takes longer to climb above the total cost line and it crosses further to the right. That visual is exactly what a break-even chart is for — it shows a manager the consequence of a pricing decision at a glance.

Profit at budget — before / after ₹ 5,00,000 / ₹ 3,00,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following figures relate to a particular year's working at 100% capacity level in a manufacturing concern: Fixed Overheads ₹1,20,000; Variable overheads ₹2,00,000; Direct wages ₹1,50,000; Direct Materials ₹4,10,000; Sales ₹10,00,000.

Represent the above figures on a Break-even Chart and determine from the chart the break-even point. Verify your results by calculations.

READ THE QUESTION FIRST — WHAT TO SPOT

- **Only the fixed overhead is fixed.** Direct materials, direct wages and variable overheads are all variable: $4,10,000 + 1,50,000 + 2,00,000 = \text{₹}7,60,000$.
- Because no unit figures are given, plot the chart against **sales value or capacity percentage** on the horizontal axis rather than units. Either is acceptable — state which you have used.
- Contribution = $10,00,000 - 7,60,000 = \text{₹}2,40,000$, so the P/V ratio is **24%** and the BEP is $1,20,000 \div 0.24 = \text{₹}5,00,000$, which is exactly **50% of capacity**. A round answer like that is the examiner confirming your method.

WORKING NOTES

W1 Separating fixed from variable

Element	Amount ₹	Nature
Direct materials	4,10,000	Variable
Direct wages	1,50,000	Variable
Variable overheads	2,00,000	Variable
Total variable cost	7,60,000	
Fixed overheads	1,20,000	Fixed
Total cost	8,80,000	
Sales	10,00,000	
Profit	1,20,000	

W2 Verification by calculation

$$\text{Contribution} = 10,00,000 - 7,60,000 = \text{₹}2,40,000$$

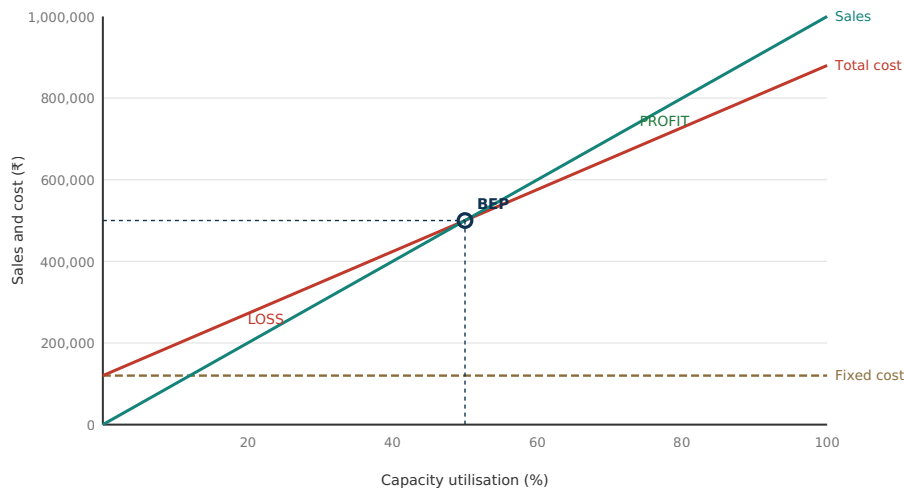
$$\text{P/V ratio} = \frac{2,40,000}{10,00,000} \times 100 = 24\%$$

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{1,20,000}{0.24} = \text{₹}5,00,000$$

$$\text{BEP as a percentage of capacity} = \frac{5,00,000}{10,00,000} \times 100 = 50\% \text{ of capacity}$$

$$\text{Margin of safety} = 10,00,000 - 5,00,000 = \text{₹}5,00,000 \text{ (50\% of sales)}$$

$$\text{Profit check} = \text{MOS} \times \text{P/V ratio} = 5,00,000 \times 24\% = \text{₹}1,20,000 \checkmark \text{ agrees with W1}$$



Break-even chart plotted against capacity. The BEP reads off the chart at 50% of capacity, i.e. ₹5,00,000 of sales — confirmed by calculation in W2.

WHY THIS WORKS

Reading the chart: the sales line and the total-cost line meet at the 50% capacity mark, which corresponds to ₹5,00,000 of sales on the vertical axis. The calculation in W2 gives exactly the same figure, which is what the question means by “verify your results”.

The chart also shows the margin of safety directly — it is the horizontal distance from the break-even point at 50% to the 100% capacity line, and the vertical gap between the two lines at 100% capacity is the profit of ₹1,20,000.

FINAL ANSWER

Total variable cost / fixed cost	₹ 7,60,000 / ₹ 1,20,000
Contribution	₹ 2,40,000
P/V ratio	24%
Break-even point	₹ 5,00,000 = 50% of capacity
Margin of safety	₹ 5,00,000 (50% of sales)
Profit at 100% capacity	₹ 1,20,000

Q18 Profit-volume graph

Charts · p.36-37

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Draw the profit-volume graph and find out the P/V Ratio with the following information: Output 3,000 units; Volume of Sales ₹7,500; Variable Cost ₹4,500; Fixed Cost ₹1,500.

READ THE QUESTION FIRST — WHAT TO SPOT

A **profit-volume graph** is not the same as a break-even chart, and the difference is worth knowing:

	Break-even chart	Profit-volume graph
What is plotted	Sales and total cost, both as lines	Profit or loss only , as one line
The vertical axis	Rupees of sales and cost, all positive	Profit above zero, loss below zero
Where the line starts	Sales at origin; cost at the fixed cost	At the loss equal to the fixed cost , below the axis

	Break-even chart	Profit-volume graph
Where the BEP appears	Where the two lines cross	Where the single line crosses the horizontal axis
Its advantage	Shows sales, cost and volume together	Much clearer for reading profit at any volume

- To draw it you need only **two points**: the loss at zero sales, which equals the fixed cost of ₹1,500 (plotted *below* the axis), and the profit at the given sales of ₹7,500, which is ₹1,500 above it. Join them with a straight line.
- Where the line cuts the horizontal axis is the break-even point.

WORKING NOTES

W1 The figures behind the graph

Particulars	Amount ₹	Per unit ₹
Sales 3,000 units	7,500	2.50
Less: Variable cost	(4,500)	(1.50)
Contribution	3,000	1.00
Less: Fixed cost	(1,500)	—
Profit	1,500	—

$$P/V \text{ ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{3,000}{7,500} \times 100 = \mathbf{40\%}$$

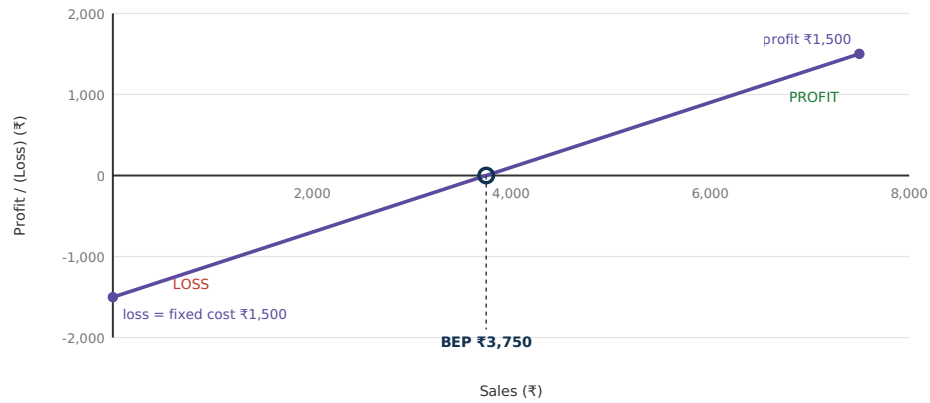
$$\text{BEP (units)} = \frac{1,500}{1.00} = \mathbf{1,500 \text{ units}}$$

$$\text{BEP (value)} = \frac{1,500}{0.40} = \mathbf{₹3,750} \quad (= 1,500 \text{ units} \times ₹2.50 \checkmark)$$

$$\text{Margin of safety} = 7,500 - 3,750 = \mathbf{₹3,750}$$

W2 The two points to plot

Point	Sales ₹	Profit / (Loss) ₹	Meaning
Start	Nil	(1,500)	At zero sales the loss is the whole fixed cost
Crossing	3,750	Nil	Break-even point
End	7,500	1,500	The actual position given in the question



Profit-volume graph. One line only. It starts at a loss equal to the fixed cost and crosses the axis at the break-even point of ₹3,750.

THE TRAP / WHERE MARKS ARE LOST

- Drawing a break-even chart instead. A P/V graph has **one line** and the axis runs below zero.
- Starting the line at the origin. At zero sales you are losing the fixed cost, so the line starts **below** the axis at ₹1,500.
- Forgetting to state the P/V ratio. The question asks for it explicitly — it is the **slope** of the line, 40%.

FINAL ANSWER

Contribution / contribution per unit	₹ 3,000 / ₹ 1.00
P/V ratio	40%
Break-even point	1,500 units = ₹ 3,750
Margin of safety	₹ 3,750 (50% of sales)
Profit at the given sales	₹ 1,500
Two points plotted	(0, -₹ 1,500) and (₹ 7,500, +₹ 1,500)

Relevant Costing

15 problems · workbook pages 39–44

What Module 4 is really about

Modules 2 and 3 gave you a cost sheet and asked for a decision. Module 4 does something harder and more useful: it hands you a cost sheet that is **deliberately misleading** and asks you to throw most of it away.

Every figure printed in a Module 4 question is there for one of two reasons — either it changes because of the decision, or it is a decoy. Your entire job is to sort them.

THE ONE TEST THAT DECIDES EVERY FIGURE IN THIS MODULE

Will this amount be DIFFERENT if I say yes rather than no?

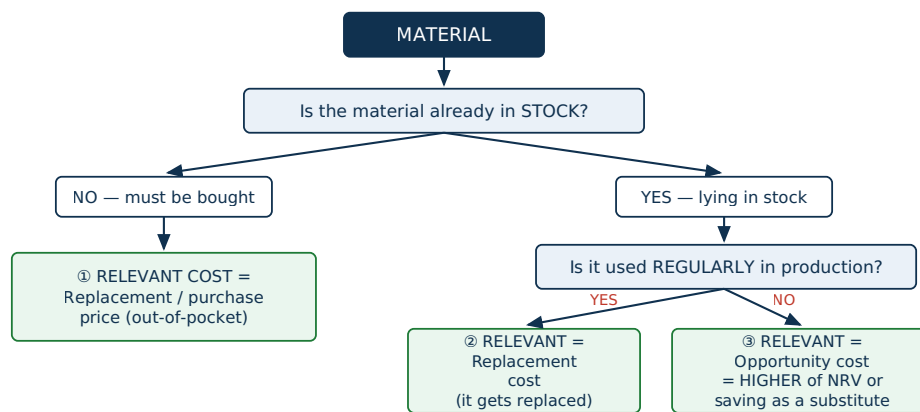
If yes, it is relevant — bring it in. If no, it is irrelevant — strike it out. Nothing else matters: not whether it is large, not whether it appears in the cost sheet, not whether the accountant has charged it to the job.

Two consequences follow, and between them they explain almost every mark in this module:

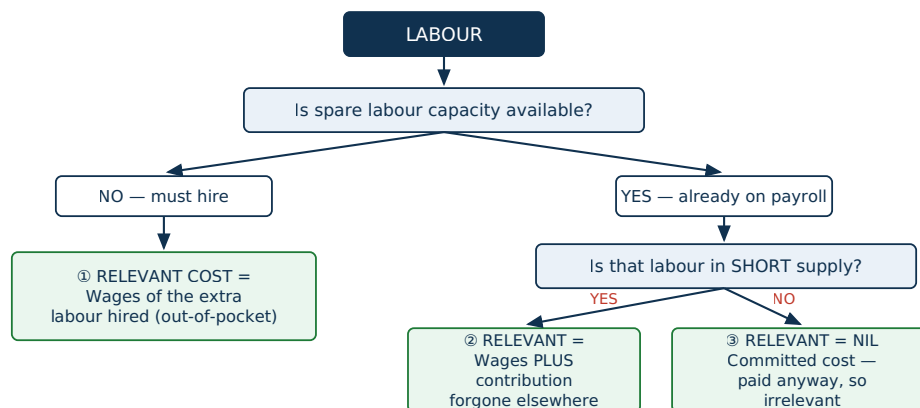
- **Money already spent can never be relevant.** It is gone. Book value, original cost, the cost of goods already manufactured — all sunk, all struck out. This feels wrong the first time, because the amounts are usually the biggest on the page. That is exactly why they are printed.
- **Money not spent can still be relevant.** If accepting the job means giving up a sale, a hire charge, or a saving, that sacrifice is a real cost of the job. This is **opportunity cost**, and it is the single most heavily examined idea in Module 4.

The two decision trees — learn these and the module is done

Nearly every line item you will be asked to classify is either a material or a labour cost. There is one tree for each. Walk the tree, land in a box, write that box's rule in the "Nature and computation" column. That is the whole method.



Decision tree for MATERIAL. Every material row in every problem of this module lands in one of the three boxes ① ② ③.



Decision tree for LABOUR. Box ③ is the one students miss: permanent staff who are simply re-deployed cost the decision nothing.

Everything else, in one table

Item in the question	Relevant?	Why
Original cost / purchase price already paid	NO	Sunk. The cash has left. Nothing you decide now can bring it back.
Book value, written-down value, net book value	NO	An accounting record of a sunk cost. It is not a cash flow and not a sacrifice.
Depreciation	NO	Non-cash, and merely the sunk cost spread over years. <i>Exception:</i> tax saved on depreciation is a real cash flow and is relevant.
Apportioned / allocated / absorbed overhead	NO	An internal book-keeping share of a cost the firm incurs anyway. Look for the words “allocated”, “apportioned”, “charged at ...% of labour”.
General fixed cost, absorbed at a normal rate	NO	Unchanged by the decision.
Specific or incremental fixed cost	YES	A fixed cost caused <i>by this decision alone</i> — a machine bought for this job, an extra supervisor, a rented warehouse. It changes, so it counts.
Variable / out-of-pocket cost of doing the job	YES	Fresh cash outflow caused by saying yes.
Realisable value given up (NRV)	YES	You could have sold it. Using it on the job sacrifices that money.
Contribution / hire charge / saving forgone	YES	Opportunity cost. The best alternative you gave up.

Item in the question	Relevant?	Why
Residual or scrap value at the end	YES	A cash inflow caused by the decision — net it off the cost.

The five problem types in this module

Type	The question sounds like ...	What you produce	Problems
1. Relevant cost of materials	A table of book value, realisable value and replacement cost for materials A, B, C, D.	One <i>Nature and computation</i> row per material, walking the material tree.	2, 10
2. Relevant cost of a machine or asset	"Bought 4 years ago for ... now has a book value of ... could be sold for ..."	Out-of-pocket running cost plus the fall in resale value as opportunity cost.	3, 4, 8, 12
3. Minimum price / accept-or-reject	"Estimate the minimum price so the company is not worse off."	Total relevant cost. That total <i>is</i> the minimum price.	7, 14, 15
4. Make or buy, with an alternative use	"... or the machine can be hired out at ..."	A column per alternative; pick the lowest <i>net</i> relevant cost.	6, 13
5. Further processing / rectification	"Rejected because of defects ... if rectified by spending ..."	Incremental revenue against incremental cost. The manufacturing cost is sunk.	9, 11
6. Opportunity cost of choosing one product	"Compute the opportunity cost for each of the products."	Contribution of the best forgone alternative, product by product.	1, 5

THE TRAP / WHERE MARKS ARE LOST

- **Answering with the total cost instead of the relevant cost.** If you find yourself using every number in the question, you have almost certainly made a mistake — Module 4 questions always contain decoys.
- **Leaving the "Nature and computation" column blank.** In this module the *reasoning* carries most of the marks. A bare figure of ₹1,750 earns little; "no other use, so opportunity cost at realisable value $700 \times ₹2.50$ " earns full marks.
- **Forgetting to state the decision.** Every problem ends with a recommendation in words. Write it, and give the amount by which one option beats the other.
- **Taking the lower of two opportunity costs.** When a material has two possible alternative uses, the sacrifice is the **better** one you gave up — so take the HIGHER. This is Q2 material D, and it is worth a mark on its own.

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

I travel from HSR to Hebbal, twenty-five kilometres from my house, for my regular work, spending ₹250 per day. On the way I pick up my employee Mr. M from Marathahalli, which is 5 kilometres from HSR. What is the cost of the pick-up of my employee Mr. M?

READ THE QUESTION FIRST — WHAT TO SPOT

- This looks like a trick question and it is — but the trick is the whole point of the module. Read it twice and one phrase decides everything.
- **“On the way”** — these three words are the answer. Marathahalli lies on a journey that is already being made. No extra kilometre is driven.
 - **“for my regular work”** — the trip happens whether or not Mr. M is picked up. The ₹250 is therefore **committed**, and unchanged by the decision.
 - The 25 km and the 5 km are given only so that you can confirm the pick-up point is *en route*. They are not there to be apportioned.
 - Apply the test: *will the ₹250 be different if I stop picking him up?* No. Therefore it is irrelevant.

Computation of the relevant cost of picking up Mr. M

Particulars	Nature and computation	Amount (₹)
Daily travel cost	₹250 per day is incurred for the proprietor’s own regular work. It is a committed cost — it does not change whether or not Mr. M is picked up. <i>given ₹250 — sunk / committed</i>	Nil
Apportionment on distance	The tempting answer is $\text{₹}250 \times \frac{5}{25} = \text{₹}50$. This is wrong . Apportioning a committed cost does not make it relevant — the total cash paid is still ₹250 either way. <i>a decoy — do not use</i>	—
Additional distance travelled	Marathahalli is on the way , so no extra kilometre is covered and no extra fuel, toll or time is incurred.	Nil
Total relevant cost of the pick-up		NIL

WHY THIS WORKS

Relevant cost is about **the difference between two futures** — one where you pick him up and one where you do not. Those two futures cost exactly the same ₹250. The difference is zero, so the relevant cost is zero.

Now change one word and the answer changes completely. Suppose Marathahalli were *5 km off* the route instead of on it. Then the decision causes 10 extra kilometres a day (there and back), and the running cost of those 10 km *would* be relevant. This is the examiner’s favourite variation, so read the words “on the way” carefully every time.

FINAL ANSWER

Cost of picking up Mr. M	NIL
Why the ₹250 is excluded	Committed cost — unchanged by the decision
Why ₹50 is <i>not</i> the answer	Apportioning a sunk cost does not create a relevant cost

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

ABC Ltd. has been approached by a customer who would like a special job to be done for him, and who is willing to pay ₹22,000 for it. The job would require the following materials:

Material	Total units required	Units already in stock	Book value of units in stock	Realisable value	Replacement cost
A	1,000	0	—	—	₹6 p.u.
B	1,000	600	₹2 p.u.	₹2.50 p.u.	₹5 p.u.
C	1,000	700	₹3 p.u.	₹2.50 p.u.	₹4 p.u.
D	200	200	₹4 p.u.	₹6.00 p.u.	₹9 p.u.

Material B is used regularly by ABC Ltd. and if units of B are used for this job, they would need to be replaced to meet other production demand.

Materials C and D are in stock as the result of previous over-buying and they have a restricted use. No other use could be found for material C, but the units of material D could be used in another job as a substitute for 300 units of material E, which currently costs ₹5 per unit (of which the company has no units in stock at the moment).

Compute the relevant costs of materials for deciding whether or not to accept the offer.

READ THE QUESTION FIRST — WHAT TO SPOT

Five columns of figures are given and **only one of them is used for each material**. Book value is never used. Walk the material tree once per material.

- **Material A** — nothing in stock, so all 1,000 units must be bought. Box ①: replacement cost.
- **Material B** — the words “used regularly ... would need to be replaced” put this in box ②. Take replacement cost on the **full 1,000 units**, not on the 400 short. Using stock triggers a purchase to refill it, so the cash outflow is the same as buying all 1,000.
- **Material C** — “no other use could be found” means the 700 in stock are in box ③, and the only sacrifice is the scrap sale at ₹2.50. The 300 short are in box ① at replacement cost. **Two rows for one material.**
- **Material D** — box ③ again, but there are now **two** alternative uses: sell it, or substitute it for material E. Value both and take the **higher**.
- Note the quantities in the D substitution: 200 units of D replace **300** units of E. Use 300, not 200. The workbook changes the quantity deliberately.

Calculation of the relevant cost of materials for the special order

Particulars	Nature and computation	Amount (₹)
Material A	1,000 units to be purchased. Nothing is in stock, so the whole quantity is a fresh cash outflow — out-of-pocket cost at the replacement price. <i>1,000 units × ₹6 ← replacement cost column</i>	6,000
Material B	Used regularly, so the 600 units drawn from stock must be bought back. Replacement cost is relevant on the full 1,000 units . <i>1,000 units × ₹5 ← replacement cost column</i>	5,000
Material C — 700 in stock	In stock, not regularly used and no other use exists. It will not be replaced, so the only sacrifice is the scrap proceeds — opportunity cost at net realisable value. <i>700 units × ₹2.50 ← realisable value column</i>	1,750

Particulars	Nature and computation	Amount (₹)
Material C — 300 short	The balance of 300 units must be bought. Out-of-pocket cost at the replacement price. <i>300 units × ₹4 ← replacement cost column</i>	1,200
Material D	In stock, not regularly used, but it has two alternative uses. Sell at NRV = ₹1,200; or substitute for 300 units of E = ₹1,500. The higher opportunity cost is relevant. <i>W1 ← higher of ₹1,200 and ₹1,500</i>	1,500
Book-value of stock	₹2, ₹3 and ₹4 per unit are historical costs already paid. Sunk, and never relevant.	—
Total relevant cost of materials		15,450

WORKING NOTES

W1 Material D — valuing the two alternative uses

D is in stock, not regularly used, and has two possible fates if it is not used on this job. The relevant cost is whichever benefit is **larger**, because that is what is actually being sacrificed.

Alternative use of the 200 units of D	Computation	Benefit forgone
Sell it at its realisable value	200 units × ₹6.00	₹1,200
Use it on another job in place of material E, which would otherwise have to be bought	300 units of E × ₹5.00	₹1,500
Relevant cost = the HIGHER of the two	the larger benefit is the one truly given up	₹1,500

Why the higher figure? A rational company would have chosen the best available alternative. By taking D for this job you deprive it of ₹1,500, not ₹1,200. Taking the lower figure is a standard one-mark error.

Decision

Particulars	Amount ₹
Price the customer is willing to pay	22,000
Less: Total relevant cost of materials <i>from the statement above</i>	(15,450)
Net benefit from accepting the job	6,550

WHY THIS WORKS

Notice that the answer used the **replacement cost** column for A and B, the **realisable value** column for C, and **neither** column for D — and never once used book value. That is the whole discipline of relevant costing: the right column depends on what happens next, not on what the accountant recorded.

The logic behind each choice is always the same question: *what does the company actually lose by putting this material into this job?*

- For A and B it loses **cash**, because a purchase is triggered. Replacement cost.
- For C it loses a **scrap sale**, because nothing else will ever use it. Realisable value.
- For D it loses **the best of its alternatives**, which happens to be a saving rather than a sale. Higher of the two.

Recommendation: accept the order. It brings in ₹22,000 against a relevant material cost of ₹15,450, a surplus of ₹6,550 — *provided* labour and overheads are unaffected, which is the assumption the question invites by giving no labour data.

FINAL ANSWER

Material A	₹ 6,000
Material B	₹ 5,000
Material C (1,750 + 1,200)	₹ 2,950
Material D	₹ 1,500
Total relevant cost of materials	₹ 15,450
Decision	ACCEPT — surplus of ₹ 6,550

Q3 Relevant cost of a machine already owned

Machine · p.39-40

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

ABC Ltd. is tendering for a six-month contract which would require the use of a specialised machine. The machine was purchased 4 years ago for ₹90,000 and now has a net book value of ₹35,000. The company was about to sell the machine for ₹40,000, but if they used it on this contract, they can sell it after 6 months for ₹25,000. The variable cost of operating the machine for 6 months would be ₹60,000. Ignoring interest costs, identify the relevant cost of using the machine on the contract.

READ THE QUESTION FIRST — WHAT TO SPOT

Four amounts are given. **Two are decoys.** Sort them before writing anything.

Figure in the question	Relevant?	Reason
₹90,000 — purchased 4 years ago	NO	Sunk. Paid four years ago; nothing decided today can change it.
₹35,000 — net book value	NO	A book record of that sunk cost. Not cash, not a sacrifice.
₹60,000 — variable operating cost	YES	Fresh cash outflow, caused only by taking the contract.
₹40,000 now vs ₹25,000 later	YES	Using the machine delays the sale and destroys ₹15,000 of resale value. That fall is the opportunity cost.

- **The trap is to write ₹25,000 as the opportunity cost.** It is not. The company still gets that ₹25,000 — only six months later. What it **loses** is the difference, ₹40,000 — ₹25,000.
- “Ignoring interest costs” is the examiner telling you not to discount the delayed receipt. Without that instruction you would also have to charge for six months of lost interest.

Computation of the relevant cost of the machine

Particulars	Nature and computation	Amount (₹)
Variable operating cost	Cash spent only because the contract is taken. Out-of-pocket cost, therefore relevant. <i>given, for the 6-month contract</i>	60,000
Reduction in net realisable value	The machine could be sold today for ₹40,000. If it is used on the contract it can only be sold for ₹25,000 in six months' time. The ₹15,000 of resale value destroyed by using the machine is an opportunity cost and is relevant. <i>₹40,000 — ₹25,000</i>	15,000
Original cost of the machine	₹90,000 was paid four years ago. Sunk cost — irrelevant.	—
Net book value	₹35,000 is an accounting figure derived from that sunk cost. It is neither a cash flow nor a sacrifice — irrelevant.	—
Total relevant cost of using the machine		75,000

WHY THIS WORKS

Compare the two futures side by side and the ₹75,000 falls out on its own:

	Do NOT take the contract ₹	Take the contract ₹
Sell the machine	40,000 (today)	25,000 (in 6 months)

	Do NOT take the contract ₹	Take the contract ₹
Operating cost of the machine	—	(60,000)
Net cash from the machine	40,000	(35,000)
Difference — the relevant cost		75,000

The ₹90,000 and the ₹35,000 appear nowhere in that comparison, which is precisely why they are irrelevant. They are identical in both columns, so they cannot influence the choice.

How to use the answer: ₹75,000 is the floor for the tender. Any price above ₹75,000 (plus the contract's other relevant costs) leaves the company better off than selling the machine today and walking away.

FINAL ANSWER

Variable operating cost	₹ 60,000
Opportunity cost — fall in resale value	₹ 15,000
Total relevant cost of the machine	₹ 75,000
Excluded as sunk	₹ 90,000 original cost • ₹ 35,000 NBV

Q4 Discussing the relevance of each item of a special order

Special order · p.40

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Noida Camera Company has received a special order for photographic equipment it does not normally produce. The company has excess capacity, and the order could be manufactured without reducing production of the firm's regular products. Discuss the relevance of each of the following items in computing the cost of the special order:

(a) Equipment to be used in producing the order has a book value of ₹2,000. The equipment has no other use for Noida Camera Company. If the order is not accepted, the equipment will be sold for ₹1,500. If the equipment is used in producing the order, it can be sold in three months for ₹800.

(b) If the special order is accepted, the operation will require some of the storage space in the company's plant. If the space is used for this purpose, the company will rent storage space temporarily in a nearby warehouse at a cost of ₹18,000. The building depreciation allocated to the storage space to be used in producing the special order is ₹12,000.

(c) If the special order is accepted, it will require a sub-assembly. Noida Camera can purchase the sub-assembly for ₹24 per unit from outside, or make it for ₹30 per unit. The ₹30 per unit was determined as follows:

Particulars	₹
Direct material	10.00
Direct labour	6.00
Variable overheads	6.00
Apportioned fixed overhead	8.00
Total unit cost of sub-assembly	30.00

READ THE QUESTION FIRST — WHAT TO SPOT

The word “**discuss**” tells you the marks are in the reasoning, not the arithmetic. Answer each part with a classification and a figure.

- “**The company has excess capacity**” — the standard signal that no existing contribution is sacrificed. Without it, part (c) would need an opportunity cost too.
- **Part (a) is Q3 again in miniature.** Book value ₹2,000 is sunk. The relevant amount is the **fall** in resale value, ₹1,500 – ₹800.
- **Part (b) pairs a real cost with a fake one on purpose.** The ₹18,000 warehouse rent is new cash caused by the order. The ₹12,000 depreciation is allocated, non-cash, and incurred whether or not the order is taken.
- **Part (c) is a make-or-buy.** Strip the apportioned fixed overhead of ₹8 out of the ₹30 before comparing with the ₹24 buying price. That single step reverses the decision, which is exactly why the question is set.

(a) Equipment

Particulars	Nature and computation	Amount (₹)
Equipment	Reduction in net realisable value is an opportunity cost and is relevant: the equipment could be sold for ₹1,500 today, but only for ₹800 after being used on the order. ₹1,500 – ₹800	700
Book value of equipment	₹2,000 is a sunk historical cost. Irrelevant.	—
Relevant cost of the equipment		700

(b) Storage space

Particulars	Nature and computation	Amount (₹)
Storage — warehouse rent	Additional rent that will be paid only if the order is accepted. An out-of-pocket cost , therefore relevant. given — new cash outflow	18,000
Building depreciation allocated	₹12,000 is a non-cash allocation of the cost of a building the company already owns. It will be charged whether or not the order is accepted. Irrelevant.	—
Relevant cost of storage space		18,000

(c) Sub-assembly — make or buy, per unit

Particulars	Nature	Make ₹
Direct material	Variable — relevant	10.00
Direct labour	Variable — relevant	6.00
Variable overheads	Variable — relevant	6.00
Apportioned fixed overhead	Apportioned share of a cost incurred anyway — irrelevant	—
Relevant cost of MAKING	10 + 6 + 6	22.00
Cost of BUYING	quoted by the outside supplier	24.00
Saving by making — so MAKE	₹24 – ₹22	2.00

WHY THIS WORKS

Each part turns on a different one of the three ideas in this module, which is why this problem is worth learning as a set:

- **(a) is opportunity cost.** No cash is spent on the equipment, yet ₹700 of value is destroyed. A cost need not be a payment.
- **(b) is the sunk-versus-incremental distinction.** Both figures relate to storage; one is new cash and one is book-keeping. Only the first can change.
- **(c) is the fixed-overhead trap.** At ₹30 making looks dearer than buying at ₹24, and a careless answer buys. Once the ₹8 apportioned overhead is removed — it is incurred either way — making costs ₹22 and is the cheaper option by ₹2 per unit.

Summary of the relevant cost of the special order: ₹700 for the equipment plus ₹18,000 for storage, and the sub-assembly should be **made** in-house at a relevant cost of ₹22 per unit. The order's price must cover ₹18,700 plus ₹22 for every unit of sub-assembly required.

FINAL ANSWER

(a) Equipment — fall in resale value	₹ 700 relevant
Book value ₹ 2,000	irrelevant — sunk
(b) Warehouse rent	₹ 18,000 relevant
Building depreciation ₹ 12,000	irrelevant — allocated, non-cash
(c) Relevant cost of making	₹ 22 per unit
Cost of buying	₹ 24 per unit
(c) Decision	MAKE — saves ₹ 2 per unit
Total identifiable relevant cost	₹ 18,700 + ₹ 22 per sub-assembly

Q5 Opportunity cost of three mutually exclusive products

Opportunity cost · p.40–41

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company can make any one of the 3 products X, Y or Z in a year. It can exercise its option only at the beginning of each year. Relevant information about the products for the next year is given below:

Particulars	X	Y	Z
Selling price	₹10 p.u.	₹12 p.u.	₹12 p.u.
Variable cost	₹6 p.u.	₹9 p.u.	₹7 p.u.
Market demand (units)	3,000	2,000	1,000
Production capacity (units)	2,000	3,000	900
Fixed cost	₹30,000 (common to all three)		

You are required to compute the opportunity costs for each of the products.

READ THE QUESTION FIRST — WHAT TO SPOT

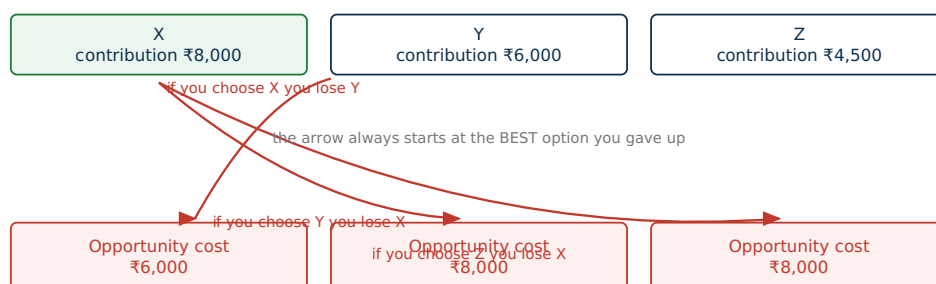
“Opportunity cost **for each** of the products” means three separate answers, not one. The opportunity cost of *choosing a product* is the contribution of the best alternative you had to give up.

- **“Can make any one of the 3”** — the options are mutually exclusive. Choose one and both others are forgone.

- **Effective production is the LOWER of market demand and production capacity.** There is no point making 3,000 units of X if only 2,000 can be produced, and no point having capacity for 3,000 units of Y if only 2,000 can be sold. This one line is where most marks are lost.
- The fixed cost of ₹30,000 is **common to all three** options. It cannot help you choose, so it plays no part in the opportunity cost. Do not deduct it product by product.
- The opportunity cost of X is the best of {Y, Z}; of Y the best of {X, Z}; of Z the best of {X, Y}. Because X is the overall best, it appears as the opportunity cost of both Y and Z.

Computation of opportunity cost

Particulars	X	Y	Z
Selling price per unit	10	12	12
Less: Variable cost per unit	(6)	(9)	(7)
Contribution per unit	4	3	5
Market demand (units)	3,000	2,000	1,000
Production capacity (units)	2,000	3,000	900
Effective production (units) <i>whichever is LOWER of demand and capacity</i>	2,000	2,000	900
Total contribution ₹ <i>effective units × contribution per unit</i>	8,000 <i>2,000 × 4</i>	6,000 <i>2,000 × 3</i>	4,500 <i>900 × 5</i>
Opportunity cost ₹ <i>best contribution GIVEN UP by choosing this product</i>	6,000 <i>← Y, the next best</i>	8,000 <i>← X, the best forgone</i>	8,000 <i>← X, the best forgone</i>



Reading the opportunity cost. Choose X and the best thing forgone is Y (₹6,000). Choose Y or Z and the best thing forgone is X (₹8,000) in both cases.

WHY THIS WORKS

Opportunity cost answers the question “what did this choice cost me in things I could have had instead?” It is not a cost the accountant will ever record, which is why it has to be computed deliberately.

The three answers together tell management exactly what it needs to know:

- **Make X.** Its opportunity cost of ₹6,000 is the *lowest* of the three, which is the same thing as saying its own contribution of ₹8,000 is the highest.
- The rule to remember: **the best option is always the one with the lowest opportunity cost.** You can use either test and you will get the same answer.
- Y and Z share an opportunity cost of ₹8,000 because both sacrifice the same thing — X. Identical opportunity costs do not make them equally good; Y still earns ₹6,000 and Z only ₹4,500.

WORTH ADDING FOR THE EXTRA MARK

Even the best option earns a contribution of only ₹8,000 against a fixed cost of ₹30,000 — a loss of ₹22,000 whichever product is chosen. Opportunity cost analysis **ranks** alternatives; it does not tell you the business is worth being in. Saying so shows the examiner you understand the limits of the technique.

FINAL ANSWER

Contribution per unit — X / Y / Z	₹ 4 / ₹ 3 / ₹ 5
Effective production — X / Y / Z	2,000 / 2,000 / 900 units
Total contribution — X / Y / Z	₹ 8,000 / ₹ 6,000 / ₹ 4,500
Opportunity cost of X	₹ 6,000
Opportunity cost of Y	₹ 8,000
Opportunity cost of Z	₹ 8,000
Decision	Make X — lowest opportunity cost

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A machine manufactures 10,000 units of a part at a total cost of ₹21 of which ₹18 is variable. This part is readily available in the market at ₹19 per unit. If the part is purchased from the market then the machine can either be utilised to manufacture a component of the same quantity contributing ₹2 per component, or it can be hired out at ₹21,000. Recommend which of the alternatives is profitable.

READ THE QUESTION FIRST — WHAT TO SPOT

There are **three** alternatives here, not two. Students routinely compare only “make” against “buy” and lose half the marks.

- Alternative I — Make the part.** Relevant cost is the variable cost only: ₹18 × 10,000. The ₹3 of fixed cost per unit (21 – 18) is incurred whether or not the part is made, so it is excluded.
 - Alternative II — Buy the part and use the freed machine to make the component.** Pay ₹19 × 10,000, then *credit* the contribution the component earns.
 - Alternative III — Buy the part and hire the machine out.** Pay ₹19 × 10,000, then credit the hire income of ₹21,000.
- The freed machine is the whole point.** Buying the part does not merely cost ₹1 more per unit — it releases capacity that can earn money. That benefit must be brought into the comparison, and it is what reverses the decision.
 - Present the benefit as a **deduction from cost** (“net relevant cost”) and then choose the **lowest** column. That is the format in your notes and it avoids sign errors.
 - Note the ₹2 is *per component* and the quantity is *the same* — so 10,000 × ₹2 = ₹20,000, which is **less** than the ₹21,000 hire charge. Read this carefully; it is the deciding comparison.

Statement of net relevant cost under the three alternatives

Particulars	Alternative I Make	Alternative II Buy & make the component	Alternative III Buy & hire out the machine
Relevant (variable) cost <i>fixed ₹3 p.u. excluded</i>	1,80,000 <i>10,000 × ₹18</i>	1,90,000 <i>10,000 × ₹19</i>	1,90,000 <i>10,000 × ₹19</i>
Less: Relevant benefit <i>what the freed machine earns</i>	—	(20,000) <i>10,000 × ₹2</i>	(21,000) <i>hire charge, given</i>
Net relevant cost ₹	1,80,000	1,70,000	1,69,000
Ranking	3rd	2nd	1st — best

WHY THIS WORKS

On the face of it making is obviously right: ₹18 to make against ₹19 to buy. Making *is* cheaper — by ₹10,000 in total. But that comparison quietly assumes the machine is worth nothing when idle, and the question tells you it is worth ₹21,000.

Extra cost of buying instead of making = $10,000 \times (\text{₹}19 - \text{₹}18) = \text{₹}10,000$

Best earning of the machine once freed = higher of ₹20,000 and ₹21,000 = ₹21,000

Net gain from buying and hiring out = ₹21,000 – ₹10,000 = ₹11,000

That ₹11,000 is exactly the gap between ₹1,80,000 and ₹1,69,000 in the statement, which is your arithmetic check.

Recommendation: buy the part from the market at ₹19 and hire the machine out for ₹21,000. This is the cheapest of the three at a net relevant cost of ₹1,69,000 — ₹11,000 better than making the part in-house and ₹1,000 better than using the machine for the component.

A caution worth one mark: the decision depends on the hire income being dependable. If the hirer might not be found, or if buying the part surrenders control over quality and delivery, the ₹1,000 margin over Alternative II is too thin to rely on.

FINAL ANSWER

Alternative I — Make	₹ 1,80,000
Alternative II — Buy & make the component	₹ 1,70,000
Alternative III — Buy & hire out	₹ 1,69,000
Decision	Alternative III — lowest net relevant cost
Saving over making in-house	₹ 11,000
Why the ₹ 3 fixed cost is excluded	Incurred whether or not the part is made

Q7 Minimum price using relevant cost

Minimum price · p.42

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

S Ltd. is engaged in manufacturing activities. It has received a request from one of its important customers to supply a product which will require conversion of Material M, which is a non-moving item. The following details are available:

Particulars	₹
Book value of Material M	60
Realisable value of Material M	80
Replacement cost of Material M	100

It is estimated that conversion of one unit of M into one unit of the finished product will require 1 labour hour. At present, labour is paid at the rate of ₹20 per hour. Other costs are as follows:

Particulars	₹
Out-of-pocket expenses	30 per unit
Allocated overheads	10 per unit

The labour will be re-deployed from other activities. It is estimated that the temporary re-deployment will not result in loss of contribution. The employees to be re-deployed are permanent employees of the company.

Estimate the minimum price to be charged from the customer so that the company is not worse off by executing the order.

READ THE QUESTION FIRST — WHAT TO SPOT

“**Minimum price so that the company is not worse off**” is the examiner’s way of saying *total relevant cost*. Compute the relevant cost and that total **is** the answer — no mark-up, no profit margin.

Five figures are given and **three of them are decoys**. Each phrase in the question exists to send one figure to the bin:

The phrase in the question	What it tells you
“Material M ... is a non-moving item”	M will not be replaced. So the replacement cost of ₹100 is irrelevant, and the sacrifice is the sale forgone — realisable value ₹80. This is box ③ of the material tree.
“ Book value of Material M ₹60”	Historical cost, already paid. Sunk — irrelevant.

The phrase in the question	What it tells you
" permanent employees ... re-deployed ... will not result in loss of contribution "	The wages are paid anyway and nothing is given up elsewhere. Committed cost — relevant cost is NIL . This is box ③ of the labour tree, and the ₹20 per hour is a decoy.
" Out-of-pocket expenses ₹30"	The words say it: fresh cash caused by the order. Relevant .
" Allocated overheads ₹10"	The word "allocated" is the giveaway — an apportioned share of a cost incurred anyway. Irrelevant.

Computation of the minimum price to be charged (per unit)

Particulars	Nature and computation	Amount (₹)
Material M	Material M is a slow-moving / non-moving item, therefore it will not be replaced. The net realisable value forgone is the opportunity cost and is relevant. <i>realisable value ₹80 — NOT replacement cost ₹100</i>	80
Labour	Permanent employees, and the re-deployment does not result in loss of contribution. Hence the wages are a committed cost and are irrelevant. <i>1 hour × ₹20 = ₹20 — excluded</i>	Nil
Out-of-pocket expenses	Fresh cash outflow incurred only because the order is executed. Relevant. <i>given, ₹30 per unit</i>	30
Book-value of Material M	₹60 was paid in the past. Sunk cost — irrelevant.	—
Replacement cost of M	₹100 would matter only if M were regularly used and had to be bought again. It is non-moving, so it will not be replaced — irrelevant.	—
Allocated overheads	₹10 per unit is an apportioned share of overheads the company incurs anyway. Irrelevant.	—
Minimum price to be charged to the customer		110

WHY THIS WORKS

"Not worse off" has a precise meaning: the company must end up with at least as much cash as if it had refused the order. Refusing the order means selling M for ₹80 and paying the workers their wages regardless. Accepting means giving up the ₹80 sale and spending ₹30. So ₹110 is the exact break-even price.

$$\text{Minimum price} = \text{Total relevant cost} = ₹80 + \text{Nil} + ₹30 = ₹110$$

At exactly ₹110 the company is indifferent. Above ₹110 it gains. Below ₹110 it would do better to sell the material as scrap and turn the order down.

The labour line is the one to understand properly, because it feels wrong to write "Nil" against a cost that is obviously being incurred. The wages of ₹20 *are* paid — but they are paid whether the order is taken or not, and the question expressly says nothing is given up by moving the workers. Two futures, same wage bill, no difference, no relevant cost.

Change either condition and the answer changes: if the workers had to be hired specially, ₹20 becomes an out-of-pocket cost and the price rises to ₹130; if re-deployment sacrificed contribution elsewhere, that contribution would be added as an opportunity cost.

FINAL ANSWER

Material M — NRV as opportunity cost	₹ 80
Labour — committed cost	Nil
Out-of-pocket expenses	₹ 30
Minimum price per unit	₹ 110
Excluded as irrelevant	Book value ₹ 60 · Replacement ₹ 100 · Allocated OH ₹ 10

Machine replacement — relevant cost, benefit hours and break-even hours

Use these steps for Q8 and Q12

Q8 and Q12 are the **same problem with different numbers**. They have three parts, and part (iii) deliberately contradicts parts (i) and (ii) — understanding why is the whole lesson.

1. **(i) Identify the relevant cost.** List what changes if the new machine is bought: its purchase price, the disposal proceeds of the old machine, and the two operating costs per hour. The price already paid for the old machine is **sunk** and is excluded. Then compute the **incremental investment**:

$$\text{Incremental investment} = \text{Cost of the new machine} - \text{Disposal value of the old machine}$$

2. **(ii) Hours the new machine must run to benefit the company.** Divide that incremental investment by the contribution the new machine earns per hour:

$$\text{Hours required} = \frac{\text{Incremental investment}}{\text{Revenue per hour} - \text{Operating cost per hour of the NEW machine}}$$

3. **(iii) Break-even hours of each machine.** Now switch to the **full purchase cost** of each machine — including the sunk cost of the old one — and divide by that machine's own contribution per hour:

$$\text{Break-even hours} = \frac{\text{FULL cost of that machine}}{\text{Revenue per hour} - \text{that machine's operating cost per hour}}$$

WHY (II) AND (III) USE DIFFERENT NUMERATORS — READ THIS TWICE

This looks like an inconsistency and students often “correct” it into a wrong answer. It is not an inconsistency, because the two parts ask **different questions**:

	Part (ii)	Part (iii)
The question being asked	Should we switch from the old machine to the new one?	How long must a machine run to recover its own cost ?
Type of question	A relevant-cost decision — comparing two futures	A cost-recovery or break-even calculation for one asset
Numerator	Incremental investment — sunk cost excluded	Full purchase cost — sunk cost included

	Part (ii)	Part (iii)
Denominator	Contribution per hour of the NEW machine	Contribution per hour of the machine being examined

Break-even is a question about whether an investment *as a whole* pays for itself, and for that you must count every rupee it cost. Relevant costing is a question about *changing course from here*, and for that money already gone is beside the point.

THE PATTERN TO MEMORISE

- Contribution per hour = **revenue per hour – operating cost per hour**. Compute it for both machines first; every later step uses it.
- The new machine has the **lower** operating cost, so the **higher** contribution per hour. That is the entire case for buying it.
- The new machine costs more but breaks even **sooner** in Q8, and **later** in Q12. Do not assume the pattern — compute it, because the two problems genuinely differ here.

Q8 Machine replacement — relevant cost, benefit hours, break-even hours

Machine · p.41

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A machinery was purchased for ₹20,00,000; the operating cost is ₹100 per hour. When the machine was about to be installed it was found out that another machine which is more efficient was available in the market for ₹35,00,000. The operating cost of that machine is ₹70 per hour.

If this new machine is purchased for ₹35,00,000 the old machine could be disposed of for ₹12,00,000. Consider the revenue per hour from both the machines as ₹120.

(i) Identify the relevant cost. **(ii)** How many hours does the new machine have to run for the benefit of the company? **(iii)** What is the break-even hour for both these machines?

READ THE QUESTION FIRST — WHAT TO SPOT

- **“When the machine was about to be installed”** — the ₹20,00,000 has already been paid. It is **sunk** for parts (i) and (ii). This is uncomfortable because it is the second-largest number on the page.
- **The old machine can be sold for ₹12,00,000.** That inflow reduces the true cost of switching, so the incremental investment is ₹35,00,000 – ₹12,00,000, not ₹35,00,000.
- **Revenue is ₹120 per hour from BOTH machines.** So the machines differ only in operating cost — ₹100 against ₹70 — and therefore in contribution per hour: ₹20 against ₹50.
- **Part (iii) switches to full cost.** Read the playbook above before attempting it.

WORKING NOTES

W1 Contribution per hour of each machine

Particulars	Old machine ₹	New machine ₹
Revenue per hour	120	120
Less: Operating cost per hour	(100)	(70)
Contribution per hour	20	50

W2 Incremental investment in switching

Cost of the new machine = ₹35,00,000
Less: Disposal value of the old machine = (₹12,00,000)
Incremental investment = ₹23,00,000

(i) Identification of the relevant cost

Particulars	Nature and computation	Amount (₹)
Operating cost — old machine	₹100 per hour. Differs between the alternatives, so relevant.	₹100 / hr
Cost of the new machine	A fresh cash outflow caused solely by the decision to switch. Relevant. given	35,00,000
Operating cost — new machine	₹70 per hour. Differs between the alternatives, so relevant.	₹70 / hr
Disposal value of the old machine	A cash inflow that arises only if the new machine is bought. Relevant — deduct it from the investment. given, NRV of the old machine	12,00,000
Revenue per hour	₹120 from either machine. Needed to compute contribution per hour.	₹120 / hr
Purchase price of the old machine	₹20,00,000 has already been committed and cannot be recovered by any decision taken now. Sunk — irrelevant for parts (i) and (ii).	—
Incremental investment (35,00,000 – 12,00,000)		23,00,000

(ii) Hours the new machine must run for the benefit of the company

$$\text{Hours required} = \frac{\text{Incremental investment}}{\text{Contribution per hour of the new machine}} = \frac{₹23,00,000}{₹50} = \mathbf{46,000 \text{ hours}}$$

Beyond 46,000 hours of running, the extra ₹23,00,000 laid out on the switch has been fully recovered out of the new machine's contribution, and every hour after that is pure gain.

(iii) Break-even hours of each machine

Here the **full** cost of each machine is recovered out of its own contribution per hour — see the method page for why the sunk cost comes back in.

Machine	Computation	Break-even hours
Machine 1 — old	$\frac{₹20,00,000}{₹120 - ₹100} = \frac{₹20,00,000}{₹20}$	1,00,000 hours
Machine 2 — new	$\frac{₹35,00,000}{₹120 - ₹70} = \frac{₹35,00,000}{₹50}$	70,000 hours

WHY THIS WORKS

The new machine is better on every count, which is what makes this problem a clean illustration:

- It earns ₹50 an hour instead of ₹20 — two and a half times as much contribution, purely because it burns ₹30 less an hour in operating cost.
- It recovers its **full** cost in 70,000 hours against the old machine's 1,00,000 hours, even though it cost 75% more to buy.
- The switch itself pays for itself in 46,000 hours — sooner still, because the ₹12,00,000 from selling the old machine funds part of the purchase.

Recommendation: buy the new machine, provided it will be run for more than 46,000 hours over its life. The ₹20,00,000 already spent on the old machine is painful but irrelevant — refusing to switch in order to “get

the value out of" the old machine would simply add a second mistake to the first. That instinct has a name in cost accounting: **throwing good money after bad**.

FINAL ANSWER

Contribution per hour — old / new	₹ 20 / ₹ 50
(i) Incremental investment (relevant)	₹ 23,00,000
Excluded as sunk	₹ 20,00,000 cost of the old machine
(ii) Hours for the benefit of the company	46,000 hours
(iii) Break-even hours — old machine	1,00,000 hours
(iii) Break-even hours — new machine	70,000 hours
Decision	Buy the new machine if it will run > 46,000 hours

Q12 Machine replacement — the same method, different numbers

Machine · p.42

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A machinery was purchased for ₹26,00,000. The operating cost is ₹130 per hour. When the machine was about to be installed it was found out that another machine which is more efficient was available in the market for ₹45,50,000. The operating cost of that machine is ₹91 per hour.

If this new machine is purchased for ₹45,50,000, the old machine could be disposed of for ₹15,60,000. Consider revenue per hour from the machine as ₹195.

(i) Identify the relevant cost. **(ii)** How many hours does the new machine have to run for the benefit of the company? **(iii)** Calculate the break-even hours of both the machines.

READ THE QUESTION FIRST — WHAT TO SPOT

Structurally this is Q8 with every figure multiplied by 1.3. Use exactly the same four steps — that is the point of the method page, and it is why the two problems are set together.

- Contribution per hour: old = ₹195 – ₹130 = ₹65; new = ₹195 – ₹91 = ₹104.
- Incremental investment = ₹45,50,000 – ₹15,60,000 = ₹29,90,000.
- The ₹26,00,000 already paid for the old machine is **sunk** for parts (i) and (ii), and comes back in only for part (iii).
- **Watch the ending of this one.** In Q8 the new machine broke even sooner than the old; here it breaks even **later** (43,750 hours against 40,000). Compute, do not assume.

WORKING NOTES

W1 Contribution per hour of each machine

Particulars	Old machine ₹	New machine ₹
Revenue per hour	195	195
Less: Operating cost per hour	(130)	(91)
Contribution per hour	65	104

W2 Incremental investment in switching

Cost of the new machine = ₹45,50,000

Less: Disposal value of the old machine = (₹15,60,000)

Incremental investment = ₹29,90,000

(i) Identification of the relevant cost

Particulars	Nature and computation	Amount (₹)
Operating cost — old machine	₹130 per hour. Differs between the alternatives — relevant.	₹130 / hr
Cost of the new machine	Fresh cash outflow caused only by the decision to switch. Relevant. given	45,50,000
Operating cost — new machine	₹91 per hour. Differs between the alternatives — relevant.	₹91 / hr
NRV of the old machine	Cash inflow available only if the switch is made. Relevant — deduct from the investment. given, disposal value	15,60,000
Revenue per hour	₹195 from either machine, needed for contribution per hour.	₹195 / hr
Purchase price of the old machine	₹26,00,000 is already committed and unrecoverable by any present decision. Sunk — irrelevant for parts (i) and (ii).	—
Incremental investment (45,50,000 – 15,60,000)		29,90,000

(ii) Hours the new machine must run for the benefit of the company

$$\text{Hours required} = \frac{\text{Incremental investment}}{\text{Contribution per hour of the new machine}} = \frac{₹29,90,000}{₹104} = \mathbf{28,750 \text{ hours}}$$

Run the new machine for more than 28,750 hours and the ₹29,90,000 net outlay on switching has been recovered out of its superior contribution.

(iii) Break-even hours of each machine

Full purchase cost of each machine, recovered out of its own contribution per hour.

Machine	Computation	Break-even hours
Machine 1 — old	$\frac{₹26,00,000}{₹195 - ₹130} = \frac{₹26,00,000}{₹65}$	40,000 hours
Machine 2 — new	$\frac{₹45,50,000}{₹195 - ₹91} = \frac{₹45,50,000}{₹104}$	43,750 hours

WHY THIS WORKS

Compare the two answers in part (iii) with Q8 and something interesting appears. Here the new machine takes **longer** to break even — 43,750 hours against the old machine's 40,000 — yet buying it is still the right decision. Those two facts sit together uncomfortably, and reconciling them is the real content of this problem:

- **Break-even hours answer a question about one machine in isolation:** “how long before this asset has paid for itself?” The new machine costs 75% more, so on that test it looks slower.
- **The switching decision is about the margin:** the extra ₹29,90,000 buys an extra ₹39 of contribution every hour (104 – 65), and recovers itself in 28,750 hours — well before either machine's own break-even point.

- **And the ₹26,00,000 is gone regardless.** Keeping the old machine does not refund it. The only live choice is which machine to run from here, and the new one earns ₹39 more an hour.

Recommendation: buy the new machine, provided the expected running time exceeds 28,750 hours. Report the break-even hours of 40,000 and 43,750 as answers to part (iii), but make clear in one sentence that they are a cost-recovery measure and **not** the basis of the replacement decision — that distinction is what part (iii) is really testing.

FINAL ANSWER

Contribution per hour — old / new	₹ 65 / ₹ 104
(i) Incremental investment (relevant)	₹ 29,90,000
Excluded as sunk	₹ 26,00,000 cost of the old machine
(ii) Hours for the benefit of the company	28,750 hours
(iii) Break-even hours — old machine	40,000 hours
(iii) Break-even hours — new machine	43,750 hours
Decision	Buy the new machine if it will run > 28,750 hours

Rectify or sell as rejects — further processing decisions

Use these steps for Q9 and Q11

Q9 and Q11 are the **same problem with different numbers**. Goods have already been made, they are defective, and there is a choice: dump them at a low price now, or spend more and sell them at a higher price. Both questions ask four sub-parts in the same order.

THE ONE THING THAT DECIDES THESE PROBLEMS

The cost of manufacturing the goods is **SUNK**. Ignore it entirely.

The shirts exist. Nothing you decide now can un-spend that money. It is identical under both options, so it cannot affect the choice — it only affects how large the loss is, not which option is better.

1. **(i) State the relevant costs.** Only the **rectification cost** per unit is relevant, because it is the only cash that depends on the decision. Say in words that the manufacturing cost is sunk — this is a marked point, not a throwaway line.
2. **(ii) Should the rectification be done?** Compare the two *increments*, per unit:

Incremental revenue = Price after rectification – Price as rejects
 Incremental cost = Rectification cost per unit
Rectify if incremental revenue > incremental cost

3. **(iii) Gain or loss if NOT rectified and (iv) if rectified.** Answer these on the **incremental** basis, exactly as your class notes do: the net gain per unit multiplied by the number of units. Then add the absolute figures as a cross-check, because they show *why* the recommendation still holds even though the company loses money either way.

TWO WAYS TO ANSWER (III) AND (IV) — GIVE BOTH

	Incremental view — the relevant one	Absolute view — the cross-check
What it measures	How much better or worse off you are by choosing one option over the other	The actual profit or loss reported, after charging the sunk manufacturing cost
Manufacturing cost	Excluded — sunk	Included — it did happen
What it tells management	Which option to pick. This is the answer the question wants.	That the order is a loss-maker either way — the choice is only about the size of the loss.

	Incremental view — the relevant one	Absolute view — the cross-check
Risk of using it alone	None for decision-making	Tempts you to reject both options because both show a loss, which would be worse still

Both views always agree on **the difference** between the two options — that is the arithmetic check to run before you write your conclusion.

THE TRAP / WHERE MARKS ARE LOST

- **Deducting the manufacturing cost from the rectification decision.** If you do, both options look catastrophic and students conclude “do not rectify”, which is wrong.
- **Comparing the new price against the manufacturing cost** instead of against the reject price. The alternative to rectifying is *selling as rejects*, not *not having made them*.
- **Forgetting that both prices are per shirt and must be multiplied by the quantity.** The per-unit answer earns part marks; the total earns the rest.

Q9 Rectify the defectives or sell them as rejects

Further processing · p.41–42

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

20,000 shirts have been manufactured for export at a cost of ₹150 per shirt. These shirts were rejected because of defects and can be sold for ₹120 per shirt in the local market. If the defects are rectified by spending ₹30 per shirt, the same can be sold for ₹160.

(i) What are the relevant costs? **(ii)** Should the rectification be done? **(iii)** What is the gain or loss if not rectified? **(iv)** What is the gain or loss if rectified?

READ THE QUESTION FIRST — WHAT TO SPOT

- **The ₹150 per shirt is already spent.** The shirts exist and are defective. That cost is **sunk** and takes no part in the decision.
- **The real choice is between two selling prices**, ₹120 now and ₹160 after spending ₹30. So compare ₹40 of extra revenue against ₹30 of extra cost.
- **Do not compare ₹160 with ₹150.** The alternative to rectifying is selling the shirts as rejects — not un-making them.
- Both options lose money against the ₹150 already spent. That is unavoidable now; the only question is which loses **less**.

(i) Identification of the relevant costs

Particulars	Nature and computation	Amount (₹)
Rectification cost	₹30 per shirt is spent only if rectification is undertaken. It is the one cash flow that changes with the decision — relevant . <i>20,000 shirts × ₹30</i>	6,00,000
Incremental revenue	Selling price rises from ₹120 to ₹160, a gain of ₹40 per shirt. A relevant benefit . <i>20,000 shirts × ₹40</i>	8,00,000
Cost of manufacture	₹150 per shirt, ₹30,00,000 in total, has already been incurred. The shirts are made. Sunk cost — irrelevant.	—
Net relevant benefit of rectifying		2,00,000

(ii) Should the rectification be done? — incremental statement

Particulars	Per shirt ₹	20,000 shirts ₹
Selling price after rectification	160	32,00,000
Less: Selling price as rejects <i>the alternative given up</i>	(120)	(24,00,000)

Particulars	Per shirt ₹	20,000 shirts ₹
Incremental revenue	40	8,00,000
Less: Rectification cost the only relevant cost	(30)	(6,00,000)
Net gain from rectifying	10	2,00,000
Decision	YES — rectify the shirts	

(iii) and (iv) Gain or loss under each option

On the incremental basis — the answer expected, and the one your class notes give:

	Per shirt ₹	Total ₹
(iii) If rectification is NOT done — the company gives up a net ₹10 a shirt, so relative to rectifying it is worse off by	(10)	(2,00,000)
(iv) If rectification IS done — a net gain of	10	2,00,000

Cross-check on the absolute basis — bringing the sunk ₹30,00,000 back in, purely to show the size of the loss. Note that the difference between the two columns is still ₹2,00,000, which proves the incremental answer above:

Particulars	Not rectified ₹	Rectified ₹
Sales revenue	24,00,000	32,00,000
Less: Cost of manufacture already incurred sunk — shown only for the check	(30,00,000)	(30,00,000)
Less: Rectification cost	—	(6,00,000)
Net loss	(6,00,000)	(4,00,000)
Rectifying reduces the loss by		2,00,000

WHY THIS WORKS

The whole problem reduces to one line: spending ₹30 brings in ₹40, so every shirt rectified is ₹10 better off. Multiply by 20,000 shirts and rectification is worth ₹2,00,000.

Why the sunk cost must be ignored, stated for the marks: the ₹30,00,000 spent on manufacture appears in *both* columns of the absolute statement above, unchanged. A figure that is identical under both options cannot possibly discriminate between them. Including it makes both options look like disasters and tempts you into the wrong conclusion — that neither should be pursued. But the shirts cannot be un-made, and refusing to rectify does not recover the money; it merely locks in the larger loss of ₹6,00,000 instead of the smaller ₹4,00,000.

Recommendation: rectify the defects. The loss on this consignment falls from ₹6,00,000 to ₹4,00,000 — an improvement of ₹2,00,000. The manufacturing cost is history; the only live question was whether ₹30 could buy more than ₹30 of extra revenue, and it buys ₹40.

FINAL ANSWER

(i) Relevant cost — rectification	₹ 30 per shirt = ₹ 6,00,000
Irrelevant — cost of manufacture	₹ 150 per shirt = ₹ 30,00,000 <i>sunk</i>
Incremental revenue per shirt	₹ 40
Net gain per shirt	₹ 10
(ii) Should rectification be done?	YES — gain of ₹ 2,00,000
(iii) If NOT rectified	Worse off by ₹ 2,00,000

(iv) If rectified**Gain of ₹ 2,00,000**

Absolute loss — not rectified / rectified

₹ 6,00,000 / ₹ 4,00,000**Q11 Rectify the defectives or sell them as rejects**

Further processing · p.42

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

10,000 shirts have been manufactured for export at a cost of ₹300 per shirt. These shirts were rejected because of defects and can be sold for ₹240 per shirt in the local market. If the defects are rectified by spending ₹60 per shirt, the same can be sold for ₹320.

(i) What are the relevant costs? **(ii)** Should the rectification be done? **(iii)** What is the gain or loss if not rectified? **(iv)** What is the gain or loss if rectified?

READ THE QUESTION FIRST — WHAT TO SPOT

- **The ₹300 per shirt is already spent.** The shirts exist and are defective. That cost is **sunk** and takes no part in the decision.
- **The real choice is between two selling prices**, ₹240 now and ₹320 after spending ₹60. So compare ₹80 of extra revenue against ₹60 of extra cost.
- **Do not compare ₹320 with ₹300.** The alternative to rectifying is selling the shirts as rejects — not un-making them.
- Both options lose money against the ₹300 already spent. That is unavoidable now; the only question is which loses **less**.

(i) Identification of the relevant costs

Particulars	Nature and computation	Amount (₹)
Rectification cost	₹60 per shirt is spent only if rectification is undertaken. It is the one cash flow that changes with the decision — relevant . <i>10,000 shirts × ₹60</i>	6,00,000
Incremental revenue	Selling price rises from ₹240 to ₹320, a gain of ₹80 per shirt. A relevant benefit . <i>10,000 shirts × ₹80</i>	8,00,000
Cost of manufacture	₹300 per shirt, ₹30,00,000 in total, has already been incurred. The shirts are made. Sunk cost — irrelevant.	—
Net relevant benefit of rectifying		2,00,000

(ii) Should the rectification be done? — incremental statement

Particulars	Per shirt ₹	10,000 shirts ₹
Selling price after rectification	320	32,00,000
Less: Selling price as rejects <i>the alternative given up</i>	(240)	(24,00,000)
Incremental revenue	80	8,00,000
Less: Rectification cost <i>the only relevant cost</i>	(60)	(6,00,000)
Net gain from rectifying	20	2,00,000
Decision	YES — rectify the shirts	

(iii) and (iv) Gain or loss under each option

On the incremental basis — the answer expected, and the one your class notes give:

	Per shirt ₹	Total ₹
(iii) If rectification is NOT done — the company gives up a net ₹20 a shirt, so relative to rectifying it is worse off by	(20)	(2,00,000)
(iv) If rectification IS done — a net gain of	20	2,00,000

Cross-check on the absolute basis — bringing the sunk ₹30,00,000 back in, purely to show the size of the loss. Note that the *difference* between the two columns is still ₹2,00,000, which proves the incremental answer above:

Particulars	Not rectified ₹	Rectified ₹
Sales revenue	24,00,000	32,00,000
Less: Cost of manufacture already incurred <i>sunk — shown only for the check</i>	(30,00,000)	(30,00,000)
Less: Rectification cost	—	(6,00,000)
Net loss	(6,00,000)	(4,00,000)
Rectifying reduces the loss by		2,00,000

WHY THIS WORKS

The whole problem reduces to one line: spending ₹60 brings in ₹80, so every shirt rectified is ₹20 better off. Multiply by 10,000 shirts and rectification is worth ₹2,00,000.

Why the sunk cost must be ignored, stated for the marks: the ₹30,00,000 spent on manufacture appears in *both* columns of the absolute statement above, unchanged. A figure that is identical under both options cannot possibly discriminate between them. Including it makes both options look like disasters and tempts you into the wrong conclusion — that neither should be pursued. But the shirts cannot be un-made, and refusing to rectify does not recover the money; it merely locks in the larger loss of ₹6,00,000 instead of the smaller ₹4,00,000.

Recommendation: rectify the defects. The loss on this consignment falls from ₹6,00,000 to ₹4,00,000 — an improvement of ₹2,00,000. The manufacturing cost is history; the only live question was whether ₹60 could buy more than ₹60 of extra revenue, and it buys ₹80.

FINAL ANSWER

(i) Relevant cost — rectification	₹ 60 per shirt = ₹ 6,00,000
Irrelevant — cost of manufacture	₹ 300 per shirt = ₹ 30,00,000 <i>sunk</i>
Incremental revenue per shirt	₹ 80
Net gain per shirt	₹ 20
(ii) Should rectification be done?	YES — gain of ₹ 2,00,000
(iii) If NOT rectified	Worse off by ₹ 2,00,000
(iv) If rectified	Gain of ₹ 2,00,000
Absolute loss — not rectified / rectified	₹ 6,00,000 / ₹ 4,00,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A printer has received a special order for the printing of a brochure. A brochure requires a special type of paper that is not regularly used. A limited quantity of paper (100 reams) is remaining from a previous job completed in the previous year. The cost of the paper last year was \$15 per ream.

The brochure requires 250 reams; the current market price is \$26 per ream. The resale value of the paper in inventory is \$10 per ream. Identify the relevant cost of the paper used in the printing of the brochure.

(a) What are the relevant costs in the problem given? **(b)** Calculate the total relevant cost.

READ THE QUESTION FIRST — WHAT TO SPOT

This is Q2's material C in miniature, and in dollars. Walk the material tree twice, because the 250 reams required come from two different places.

- **"not regularly used"** — the 100 reams in stock will **not** be replaced. Box ③: the sacrifice is the resale value of \$10 a ream.
- **"The cost of the paper last year was \$15"** — last year's money. Sunk, and the largest decoy in the question.
- **250 required – 100 in stock = 150 reams short.** These must be bought at today's \$26. Box ①: out-of-pocket cost.
- Do not price all 250 reams at \$26; and do not price the 100 in stock at \$15. Each of those errors is worth a mark.

Computation of the relevant cost of the paper

Particulars	Nature and computation	Amount (\$)
Paper in stock — 100 reams	A special paper that is not regularly used, so it will not be replaced. The only sacrifice in using it is the resale proceeds forgone — opportunity cost at net realisable value. $100 \text{ reams} \times \10	1,000
Paper to be purchased — 150 reams	250 reams are required and only 100 are in stock, so 150 reams must be bought at the current market price. Out-of-pocket cost. $(250 - 100) = 150 \text{ reams} \times \26	3,900
Historical cost of the paper in stock	\$15 per ream was paid last year. Sunk cost — irrelevant.	—
Total relevant cost of the paper		4,900

WHY THIS WORKS

Two rows, two different rules, and the reason is simply that the two lots of paper have different futures:

- The **100 reams already owned** cost nothing more to obtain. Using them destroys only the \$1,000 the printer could have raised by selling them. That is the opportunity cost.
- The **150 reams not owned** must be bought with real money at \$26. That is an out-of-pocket cost.

The \$15 historical cost belongs to neither calculation. It measures what the paper *cost*, and relevant costing is only ever interested in what a decision *changes*.

$$\text{Total relevant cost} = \$1,000 + \$3,900 = \$4,900$$

This is the floor for quoting the brochure job so far as paper is concerned. Any quotation must recover at least \$4,900 of paper cost, plus ink, labour and other relevant costs.

FINAL ANSWER

Opportunity cost — 100 reams at NRV	\$ 1,000
Out-of-pocket cost — 150 reams at market price	\$ 3,900
(b) Total relevant cost of the paper	\$ 4,900
Excluded as sunk	\$ 15 per ream historical cost

Q13 Make or buy — splitting a semi-variable overhead

Make or buy · p.42-43

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The cost of making component Q, which forms part of Product Y, is given below:

Particulars	Cost per unit in ₹
Raw materials	4
Direct labour	8
Production overheads (60% variable)	16
Total	28

Component Q could be bought from an outside supplier for ₹20 per unit. State whether the company should make or buy component Q.

READ THE QUESTION FIRST — WHAT TO SPOT

A three-line question with one instruction hidden in a bracket. Everything depends on “(60% variable)”.

- **Split the production overhead before doing anything else:** 60% of ₹16 = ₹9.60 is variable and relevant; the remaining 40% = ₹6.40 is fixed and continues whether or not the component is made in-house. Strike it out.
- **Never compare the ₹28 total with the ₹20 buying price.** That comparison says “buy, saving ₹8” and it is wrong, because ₹6.40 of that ₹28 will be incurred anyway.
- Raw materials and direct labour are variable in full, so both are relevant.
- Here, unusually for a textbook make-or-buy, the answer really is **buy** — but for a saving of ₹1.60, not ₹8.

Cost of making component Q — relevant cost only

Particulars	Nature	Amount ₹
Raw materials	Variable — relevant	4.00
Direct labour	Variable — relevant	8.00
Production overheads — variable portion 60% of ₹16	Variable — relevant	9.60
Production overheads — fixed portion 40% of ₹16 = ₹6.40	Fixed — incurred whether the component is made or bought. Irrelevant.	—
Relevant cost of MAKING component Q	4 + 8 + 9.60	21.60
Cost of BUYING component Q	quoted by the outside supplier	20.00
Saving per unit by buying	₹21.60 – ₹20.00	1.60

WHY THIS WORKS

The whole problem is the bracket “(60% variable)”, and it works in the opposite direction from the usual textbook case. Normally stripping out fixed overhead makes *making* look cheaper and reverses a naive “buy” decision. Here the fixed element is small enough that buying still wins — but the **size** of the advantage changes completely, and that is what is being marked.

	Naive comparison	Correct comparison
Cost of making	₹28.00 — the full total	₹21.60 — variable only

	Naive comparison	Correct comparison
Cost of buying	₹20.00	₹20.00
Apparent saving by buying	₹8.00	₹1.60
Decision	Buy	Buy
Is it safe?	Overstates the gain five-fold	A thin margin — report it as such

Recommendation: buy component Q from the outside supplier at ₹20, saving ₹1.60 per unit. But say plainly that the margin is narrow. Two things follow from that, and stating either earns credit:

- The ₹6.40 of fixed overhead does not disappear on buying — it must be reabsorbed by the remaining products, so the reported cost of Product Y will rise even though the company is better off in cash terms.
- If the released capacity can earn anything at all — even ₹2 per unit of contribution from another component — buying becomes clearly correct rather than marginally so. Conversely, a small rise in the supplier's price wipes the advantage out.

FINAL ANSWER

Relevant cost of making (4 + 8 + 9.60)	₹ 21.60 per unit
Cost of buying	₹ 20.00 per unit
Fixed overhead excluded	₹ 6.40 per unit (40% of ₹ 16)
Decision	BUY component Q
Saving	₹ 1.60 per unit

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The selling price of Product X is ₹22 and the production cost for one unit is:

Particulars	Cost per unit in ₹
Raw materials	8
Direct labour	4
Production overheads	8
Total	20

Variable production overhead, which is part of production overhead, is calculated at 50% of labour cost. There is a possibility of supplying a special order for 2,000 units of Product X at ₹16 each. If this order is accepted, the normal budgeted sales would not get affected and the company has the necessary capacity to produce the additional units. State whether the company should accept the special order.

READ THE QUESTION FIRST — WHAT TO SPOT

The special-order price of ₹16 is below the total cost of ₹20, which is precisely why the question is set. Accepting it is still correct, and the reason is one line of arithmetic buried in the words.

- **“Variable production overhead ... is calculated at 50% of labour cost”** — so variable overhead = $50\% \times ₹4 = ₹2$, and the fixed part of the production overhead is $₹8 - ₹2 = ₹6$. Note the percentage is applied to *labour cost*, not to the ₹8 overhead.
- **“the normal budgeted sales would not get affected”** — no existing contribution is sacrificed, so there is no opportunity cost.
- **“the company has the necessary capacity”** — spare capacity, so the fixed overhead of ₹6 per unit will be incurred anyway and does not increase. It is therefore irrelevant to this order.
- With those three phrases in hand, the relevant cost is $₹8 + ₹4 + ₹2 = ₹14$, comfortably below the ₹16 offered.

Computation of the relevant cost of the special order

Particulars	Nature	Per unit ₹
Raw materials	Variable — relevant	8
Direct labour	Variable — relevant	4
Production overheads — variable portion 50% of direct labour ₹4	Variable — relevant	2
Production overheads — fixed portion ₹8 - ₹2 = ₹6	Spare capacity exists, so this is incurred anyway. Irrelevant.	—
Total relevant cost per unit	8 + 4 + 2	14
Selling price per unit of the special order	given	16
Additional contribution per unit	₹16 - ₹14	2

Effect of accepting the order on total profit

Particulars	Amount ₹
Additional contribution per unit $16 - 14$	2
Number of units in the special order	2,000
Increase in total profit $₹2 \times 2,000 \text{ units}$	4,000

WHY THIS WORKS

At first sight the order is absurd: ₹16 for something that costs ₹20 to produce. The resolution is that ₹6 of that ₹20 is fixed overhead which the company pays whether or not it accepts the order. Against the costs that **actually change** — ₹14 of materials, labour and variable overhead — the ₹16 leaves ₹2 over.

$$\text{Additional profit} = (\text{₹}16 - \text{₹}14) \times 2,000 \text{ units} = \text{₹}4,000$$

Total profit rises by exactly this amount, because total fixed cost is unchanged. Nothing else in the accounts moves.

Recommendation: accept the special order. It generates an additional contribution of ₹4,000, and since fixed costs do not change, total profit rises by the same ₹4,000.

The two conditions that make this true — state them. They are given in the question precisely so that you will quote them back:

- **Spare capacity exists**, so no extra fixed cost is incurred and no existing production is displaced. Without spare capacity the contribution lost on displaced regular sales would be an opportunity cost, and at ₹22 – ₹14 = ₹8 of contribution per regular unit the order would be plainly unprofitable.
- **Normal budgeted sales are unaffected**, so there is no risk of the ₹16 price leaking into the regular market. If regular customers learned of it and demanded the same price, the ₹4,000 gain would be dwarfed by the loss on ordinary sales. Segregating the special order — typically an export or own-label market — is what makes it safe.

FINAL ANSWER

Relevant cost per unit (8 + 4 + 2)	₹ 14
Fixed overhead excluded	₹ 6 per unit
Special order price	₹ 16 per unit
Additional contribution per unit	₹ 2
Increase in total profit	₹ 4,000 (₹ 2 × 2,000 units)
Decision	ACCEPT the special order

Q15 The complete relevant-cost decision — every rule in one problem

Special order · p.43-44

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The production manager of your organisation has approached you for some costing advice on project X, a special order from overseas for ₹30,000. The costs associated with the project are as follows:

Particulars	Cost in ₹
Raw Material — A	4,000
Raw Material — B	8,000
Direct Labour	6,000
Supervision	2,000
Overheads	12,000
Total	32,000

You ascertain the following:

- **i)** Material A is in stock and the above was the cost. This material is of no use to the organisation other than the above project. If this material is not used, it could be realised at ₹1,750.
- **ii)** The organisation has no stock of material B. The required material for the project has to be ordered at the cost shown above.
- **iii)** Direct labour cost of ₹6,000 relates to workers that will be transferred to this project from another project. Extra labour will need to be recruited to the other project at a cost of ₹7,000.
- **iv)** Supervision costs have been charged to the project on the basis of 1/3 of labour cost and will be carried out by existing staff within their normal duties.
- **v)** Overheads have been charged to the project at the rate of 200% on direct labour.
- **vi)** A new machine has to be purchased specially for the project at a cost of ₹8,000 and then disposed of at ₹5,000 at the end of the project.

a) Find out the relevant costs for the special order, whether to accept or to reject the special order. **b)** Should the special order be accepted or rejected?

READ THE QUESTION FIRST — WHAT TO SPOT

This is the module's summary problem: **every rule appears exactly once**. The printed total of ₹32,000 exceeds the ₹30,000 offered, so a careless answer rejects the order. Work through notes (i) to (vi) in order — each one is a complete instruction about a single line.

Line	The note tells you	Rule applied
Material A	In stock, no other use, realisable at ₹1,750	Box ③ — opportunity cost at NRV, not the ₹4,000 book cost
Material B	No stock, must be ordered	Box ① — out-of-pocket cost, take the full ₹8,000
Direct labour	Workers transferred ; replacements cost ₹7,000	The ₹6,000 is committed; the ₹7,000 caused elsewhere is the real out-of-pocket cost
Supervision	Existing staff, within normal duties, charged as 1/3 of labour	Apportioned and committed — Nil
Overheads	Charged at 200% of direct labour	Absorption, not incremental cash — Nil

Line	The note tells you	Rule applied
New machine	Bought for ₹8,000, sold for ₹5,000 after	A specific fixed cost — relevant, net of residual value

The direct labour line is the hardest in the module and it is worth dwelling on. The ₹6,000 is not relevant, because those workers are already employed and paid. But the decision does cause a cash outflow — taking them off the other project forces the recruitment of replacements at ₹7,000. So the relevant figure is ₹7,000: *higher* than the amount printed, and located in a different project altogether. Relevant cost follows the consequences of a decision wherever they land, not the line in the cost sheet.

(a) Computation of the relevant cost of the special order

Particulars	Nature and computation	Amount (₹)
Raw Material A	In stock and of no use to the organisation on any other work. It will not be replaced, so the sacrifice is the realisable value forgone — opportunity cost . The ₹4,000 already paid is sunk. <i>realisable value ₹1,750, NOT the ₹4,000 cost</i>	1,750
Raw Material B	No stock is held, so the whole requirement must be ordered and paid for. Out-of-pocket cost — relevant. <i>given, ₹8,000</i>	8,000
Direct Labour	The ₹6,000 relates to workers already employed who are merely transferred — a committed cost , hence irrelevant. But the transfer forces the recruitment of extra labour for the other project at ₹7,000, and that is an out-of-pocket cost caused by accepting this order. <i>extra recruitment ₹7,000, not the ₹6,000 transferred</i>	7,000
Supervision	Charged at 1/3 of labour cost as a book apportionment, and will be carried out by existing staff within their normal duties. No additional cash — irrelevant . <i>₹2,000 excluded</i>	Nil
Overheads	Absorbed at 200% of direct labour. A recovery rate, not an incremental cost; total overhead expenditure does not change. Irrelevant . <i>₹12,000 excluded</i>	Nil
New machinery	Bought specially for this project, so it is a specific fixed cost and is relevant. Net it against the disposal proceeds at the end of the project. <i>₹8,000 – ₹5,000</i>	3,000
Total relevant cost of the special order		19,750

(b) Decision

Particulars	Amount ₹
Price offered for the special order <i>given, overseas order</i>	30,000
Less: Total relevant cost <i>from the statement above</i>	(19,750)
Net benefit from accepting the order	10,250
Decision	ACCEPT the special order

Why the printed cost sheet gives the wrong answer

Particulars	As printed ₹	Relevant ₹	Why it changed
Raw Material A	4,000	1,750	Sunk cost replaced by opportunity cost at NRV
Raw Material B	8,000	8,000	Genuine out-of-pocket cost — unchanged

Particulars	As printed ₹	Relevant ₹	Why it changed
Direct Labour	6,000	7,000	Committed cost replaced by the higher cost of replacement labour elsewhere
Supervision	2,000	—	Apportioned to the project; existing staff, no extra cash
Overheads	12,000	—	Absorbed at 200% of labour; total overhead does not change
New machinery	—	3,000	Omitted from the cost sheet altogether, yet caused entirely by this decision
Total	32,000	19,750	
Against the price of ₹30,000	REJECT	ACCEPT	the two approaches give opposite decisions

WHY THIS WORKS

This problem is the best argument in the whole syllabus for learning relevant costing. The cost sheet the production manager brought says the project costs ₹32,000 and earns ₹30,000 — a loss of ₹2,000, so refuse it. The relevant-cost statement says it costs ₹19,750 and earns ₹30,000 — a gain of ₹10,250, so take it. **The two answers are opposite, and only one of them is right.**

Three things went wrong in the printed sheet, and each is a lesson:

- 1. It charged costs that will not change.** Supervision and overheads together account for ₹14,000 of the ₹32,000, and not one rupee of it is caused by this project. Both are internal allocations designed for routine product costing, where spreading fixed costs is exactly right — and for one-off decisions is exactly wrong.
- 2. It used a historical cost where a sacrifice was needed.** Material A is valued at the ₹4,000 paid for it, when the only thing actually given up is the ₹1,750 it could be sold for. That overstated the cost by ₹2,250.
- 3. It missed a cost entirely.** The net ₹3,000 on the machine bought specially for the project appears nowhere in the cost sheet, although it is the most obviously decision-driven figure in the whole question. Absorption systems record what has been spent; they do not ask what a decision will cause.

Recommendation: accept the special order. It yields a net benefit of ₹10,250 in cash terms. Add the standard caution for full marks: the ₹30,000 price should not be allowed to set a precedent for other customers, since it recovers no share of general fixed overhead and would be unsustainable as a general price.

FINAL ANSWER

Raw Material A — NRV as opportunity cost	₹ 1,750
Raw Material B — out-of-pocket	₹ 8,000
Direct Labour — extra recruitment elsewhere	₹ 7,000
Supervision	Nil — apportioned, existing staff
Overheads	Nil — absorbed at 200% of labour
New machinery (8,000 – 5,000)	₹ 3,000
(a) Total relevant cost	₹ 19,750
Price offered	₹ 30,000
Net benefit	₹ 10,250
(b) Decision	ACCEPT the special order

Budgetary Control

15 problems · workbook pages 49–55

What Module 5 is really about

A budget is a plan expressed in numbers. Module 5 asks you to build those plans, and every problem is one of a small number of shapes. Once you can name the shape, the arithmetic is easy — and naming the shape is what this opening page is for.

THE THREE FORMULAS THAT CARRY THIS WHOLE MODULE

PRODUCTION BUDGET = Budgeted Sales + Desired Closing Stock – Opening Stock

Read it as: how much must I MAKE, given what I want to SELL and the stock I want to be left holding? Used in Q8, Q10, Q11, Q12 and Q15.

PURCHASE BUDGET = Materials Consumed + Closing Stock – Opening Stock

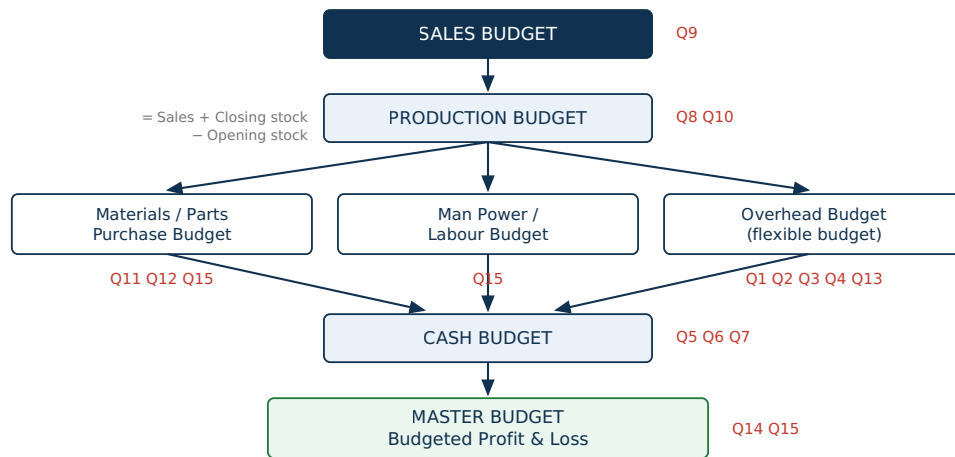
The same shape one level down — and “materials consumed” comes from the production budget, never from the sales budget. Used in Q11, Q12 and Q15.

CLOSING BALANCE = Opening Balance + Receipts – Payments

The cash budget. Its whole difficulty is deciding WHICH MONTH each receipt and payment falls in. Used in Q5, Q6 and Q7.

Notice that the first two are the same formula. Both say: *what I need to obtain equals what I will use up, plus what I want left over, less what I already have.* Learn it once and it covers finished goods, raw materials and parts alike.

How the budgets fit together



The budget hierarchy, and where each problem in this module sits. Everything begins with the sales budget, because sales is normally the limiting factor.

The five problem types in this module

Type	How to recognise it	What you produce	Problems
1. Flexible budget	"Prepare a budget at 60%, 70% and 90% capacity", or costs given at one level and wanted at another.	A column per activity level. Split every cost into fixed and variable first .	1, 2, 3, 4
2. Cost segregation (high-low)	Costs given at two volumes, and a third volume asked for.	Variable rate per unit from the two points, then the fixed element by back-substitution.	13
3. Cash budget	A table of monthly sales, purchases and wages plus credit terms.	Working notes for collections first, then a receipts-and-payments statement month by month.	5, 6, 7
4. Production & purchase budget	Opening and closing stock figures are given in units .	Sales + closing – opening, in quantity. Then the same again for materials.	8, 10, 11, 12
5. Sales & master budget	"Prepare a quantitative-cum-financial budget", or "prepare a master budget".	Quantity × price by product and zone; or a full budgeted profit statement.	9, 14, 15

The behaviour of costs — the idea behind every flexible budget

Type of cost	Total amount when output rises	Amount PER UNIT when output rises	Examples in this module
Fixed	Stays the same	Falls	Salaries, rent, depreciation, insurance, works manager's salary
Variable	Rises in proportion	Stays the same	Materials, direct labour, indirect materials, stores and spares
Semi-variable	Rises, but not in proportion	Falls, but not to a simple pattern	Repairs, maintenance, supervision, electricity, indirect labour

Write the two lines from your notes at the foot of every flexible budget — they are worth a mark and they prove you understand what you have just prepared:

Fixed cost **remains the same** in total; fixed cost **per unit changes**.

Variable cost **changes** in total; variable cost **per unit remains the same**.

THE TRAP / WHERE MARKS ARE LOST

- **Scaling a fixed cost.** The single commonest error in the module. If fixed overhead is ₹50,000 at 5,000 units it is still ₹50,000 at 7,000 units — only the per-unit figure moves, from ₹10 to ₹7.14.
- **Splitting a semi-variable cost by guesswork.** Use the percentages if the question gives them (Q3), the stated step-rules if it gives those (Q2), or the high-low method if it gives two volumes (Q13). Never invent a split.
- **Basing the purchase budget on SALES units.** Materials are consumed by **production**, not by sales. Always compute the production budget first, then feed it into the materials budget.
- **Putting a receipt in the wrong month.** In cash budgets, write out the collection working note in full before touching the budget itself. Every cash-budget problem in this module is decided by that working note.
- **Bringing depreciation into a cash budget.** It is not a cash flow. It belongs in a flexible budget and in the master budget, but never in a cash budget.

Q1 Flexible budget from a per-unit cost sheet

Flexible budget · p.49

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The expenses for the production of 5,000 units in a factory are given as follows:

Particulars	Per unit (₹)
Materials	50
Labour	20
Variable overheads	15
Fixed overheads (₹50,000)	10
Administrative expenses (5% variable)	10
Selling expenses (20% fixed)	6
Distribution expenses (10% fixed)	5
Total cost of sales per unit	116

You are required to prepare a budget for the production of 7,000 units.

READ THE QUESTION FIRST — WHAT TO SPOT

Everything is given **per unit at 5,000 units**, and three of the seven lines are semi-variable. Convert each line into total money at 5,000 units first — that is the only safe way to keep the fixed elements fixed.

- **The bracketed notes are instructions.** “(5% variable)” means 5% of that line is variable and **95% is fixed**. “(20% fixed)” means 20% fixed and **80% variable**. Read which way round each one is written — the workbook alternates deliberately.
- **Fixed overhead is given twice over** — as ₹10 per unit *and* as a total of ₹50,000. The total is the real figure; ₹10 is merely $50,000 \div 5,000$ and will change to ₹7.14 at 7,000 units.
- **Method: split, then scale.** For every line, work out the fixed rupees and the variable rupees per unit at 5,000 units. Carry the fixed rupees across unchanged; multiply the variable rate by 7,000.
- The total cost per unit will **fall** from ₹116 to ₹109.94, because the same fixed cost is being spread over 2,000 more units. That fall is the whole point of the exercise.

WORKING NOTES

W1 Splitting the three semi-variable lines at 5,000 units

Line	Total at 5,000 units	Given split	Fixed ₹	Variable ₹	Variable rate per unit
Administrative expenses	50,000 $5,000 \times ₹10$	5% variable, so 95% fixed	47,500	2,500	0.50
Selling expenses	30,000 $5,000 \times ₹6$	20% fixed, so 80% variable	6,000	24,000	4.80
Distribution expenses	25,000 $5,000 \times ₹5$	10% fixed, so 90% variable	2,500	22,500	4.50

Each variable rate is the variable rupees divided by 5,000 units — for example $₹24,000 \div 5,000 = ₹4.80$ for selling. That rate stays the same at 7,000 units; it is the fixed rupees beside it that stay the same in *total*.

Flexible Budget

Particulars	Output 5,000 units		Output 7,000 units	
	Per unit ₹	Total ₹	Per unit ₹	Total ₹
Materials	50.00	2,50,000	50.00	3,50,000
Labour	20.00	1,00,000	20.00	1,40,000
Variable overheads	15.00	75,000	15.00	1,05,000
Fixed overheads <i>total unchanged; per unit $50,000 \div 7,000 = 7.14$</i>	10.00	50,000	7.14	50,000
<i>Administrative expenses</i>				
Variable (5%)	0.50	2,500	0.50	3,500
Fixed (95%)	9.50	47,500	6.79	47,500
<i>Selling expenses</i>				
Fixed (20%)	1.20	6,000	0.86	6,000
Variable (80%)	4.80	24,000	4.80	33,600
<i>Distribution expenses</i>				
Fixed (10%)	0.50	2,500	0.36	2,500
Variable (90%)	4.50	22,500	4.50	31,500
TOTAL COST OF SALES	116.00	5,80,000	109.94	7,69,600

Every figure in bold is one that **changed its per-unit amount** while its total stayed the same. Those are exactly the fixed elements.

WHY THIS WORKS

Look down the two “per unit” columns and the logic of a flexible budget is visible at a glance. Materials stayed at ₹50, labour at ₹20, variable overhead at ₹15 — and every fixed line fell. Nothing became cheaper to buy; the fixed cost is simply spread thinner.

Total fixed cost = $50,000 + 47,500 + 6,000 + 2,500 = ₹1,06,000$ at either level

Total variable cost per unit = $50 + 20 + 15 + 0.50 + 4.80 + 4.50 = ₹94.80$ at either level

Check at 7,000 units = $(7,000 \times 94.80) + 1,06,000 = 6,63,600 + 1,06,000 = ₹7,69,600$ ✓

That two-line check — total variable rate plus total fixed — will verify any flexible budget in seconds, and it is the fastest way to catch a scaled fixed cost.

On the rounding: ₹7,69,600 ÷ 7,000 = ₹109.94 exactly. Some solutions round the column to ₹110. Show 109.94, because the total of ₹7,69,600 is what the marks are on and 109.94 × 7,000 reproduces it.

FINAL ANSWER

Total fixed cost (unchanged at both levels)	₹ 1,06,000
Total variable cost per unit	₹ 94.80
Total cost at 5,000 units	₹ 5,80,000 (₹ 116.00 p.u.)
Total cost at 7,000 units	₹ 7,69,600 (₹ 109.94 p.u.)

Q2 Flexible budget across four capacity levels with step-rule semi-variables

Flexible budget · p.49

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information at 50% capacity is given. Prepare a flexible budget and forecast the profit or loss at 60%, 70% and 90% capacity.

Particulars	Expenses at 50% capacity (₹)
Fixed expenses	
Salaries	50,000
Rent and taxes	40,000
Depreciation	60,000
Administrative expenses	70,000
Variable expenses	
Materials	2,00,000
Labour	2,50,000
Others	40,000
Semi-variable expenses	
Repairs	1,00,000
Indirect labour	1,50,000
Others	90,000

It is estimated that fixed expenses will remain constant at all capacities. Semi-variable expenses will not change between 45% and 61% capacity, will rise by 10% between 61% and 75% capacity, and a further increase of 5% when capacity crosses 75%.

Estimated sales at various levels of capacity are: 60% — ₹11,00,000; 70% — ₹13,00,000; 90% — ₹15,00,000.

READ THE QUESTION FIRST — WHAT TO SPOT

The question has already classified every cost for you. The only thinking needed is on the semi-variable step rules, and they are stated in words that must be read very precisely.

Capacity asked for	Which band it falls in	Cumulative uplift on the 50% figures
60%	Inside “will not change between 45% and 61%”	no change — use the 50% amounts as they stand
70%	Inside “rise by 10% between 61% and 75%”	× 1.10
90%	Above 75%, so the 10% rise and the further 5%	× 1.10 × 1.05 = × 1.155

- **The 90% uplift compounds.** “A further increase of 5%” means 5% on top of the 10% already applied, not 15% on the original. So the multiplier is $1.10 \times 1.05 = 1.155$, giving ₹1,15,500 for repairs rather than ₹1,15,000.
- **Semi-variable does not mean it needs splitting here.** The question replaces the usual fixed/variable split with a step rule, so apply the step rule and nothing else.
- **Variable expenses scale on capacity, not on the step rule.** Divide the 50% figure by 50 to get the cost per 1% of capacity, then multiply by 60, 70 and 90.
- Sales are given only at the three higher levels, so the profit line begins at 60%. Leave the 50% profit cell blank rather than inventing a sales figure.

WORKING NOTES

W1 Variable expenses — cost per 1% of capacity

Variable expense	At 50% capacity ₹	Per 1% of capacity ₹	At 60%	At 70%	At 90%
Materials	2,00,000	4,000 $2,00,000 \div 50$	2,40,000	2,80,000	3,60,000
Labour	2,50,000	5,000 $2,50,000 \div 50$	3,00,000	3,50,000	4,50,000
Others	40,000	800 $40,000 \div 50$	48,000	56,000	72,000

W2 Semi-variable expenses — applying the step rules

Semi-variable expense	At 50% ₹	At 60% no change	At 70% × 1.10	At 90% × 1.155
Repairs	1,00,000	1,00,000	1,10,000	1,15,500
Indirect labour	1,50,000	1,50,000	1,65,000	1,73,250
Others	90,000	90,000	99,000	1,03,950
Total semi-variable	3,40,000	3,40,000	3,74,000	3,92,700

Flexible Budget and forecast of profit / loss

Particulars	50% ₹	60% ₹	70% ₹	90% ₹
Fixed expenses				
Salaries	50,000	50,000	50,000	50,000
Rent and taxes	40,000	40,000	40,000	40,000
Depreciation	60,000	60,000	60,000	60,000
Administrative expenses	70,000	70,000	70,000	70,000

Particulars	50% ₹	60% ₹	70% ₹	90% ₹
Total fixed <i>constant at all capacities</i>	2,20,000	2,20,000	2,20,000	2,20,000
Variable expenses <i>W1</i>				
Materials	2,00,000	2,40,000	2,80,000	3,60,000
Labour	2,50,000	3,00,000	3,50,000	4,50,000
Others	40,000	48,000	56,000	72,000
Total variable	4,90,000	5,88,000	6,86,000	8,82,000
Semi-variable expenses <i>W2</i>				
Repairs	1,00,000	1,00,000	1,10,000	1,15,500
Indirect labour	1,50,000	1,50,000	1,65,000	1,73,250
Others	90,000	90,000	99,000	1,03,950
Total semi-variable	3,40,000	3,40,000	3,74,000	3,92,700
TOTAL COST	10,50,000	11,48,000	12,80,000	14,94,700
SALES <i>given</i>	—	11,00,000	13,00,000	15,00,000
PROFIT / (LOSS)	—	(48,000)	20,000	5,300

WHY THIS WORKS

The profit line is the interesting part of this answer, because it does **not** rise steadily with capacity:

Capacity	Sales ₹	Total cost ₹	Profit / (Loss) ₹	What happened
60%	11,00,000	11,48,000	(48,000)	Fixed and semi-variable costs are not yet covered
70%	13,00,000	12,80,000	20,000	Sales rose ₹2,00,000 while cost rose only ₹1,32,000 — the firm moves into profit
90%	15,00,000	14,94,700	5,300	Sales rose only ₹2,00,000 but cost rose ₹2,14,700 — profit falls

Profit peaks somewhere around 70% and then declines. That is the finding management needs, and stating it earns the last mark. Two things cause it:

- **The semi-variable step at 75%.** Crossing three-quarters capacity adds a further 5% to repairs, indirect labour and other semi-variable costs — ₹18,700 of extra cost that arrives all at once, whether or not the extra output is worth having.
- **Sales are not keeping pace.** From 70% to 90% is twenty points of extra capacity for only ₹2,00,000 more revenue, whereas the previous ten points brought in the same ₹2,00,000. The selling price is evidently being cut to move the extra volume.

Recommendation: operate at about 70% capacity. Pushing to 90% adds a fifth again to output and yet *reduces* profit by ₹14,700. Nothing in the cost structure rewards the extra volume.

FINAL ANSWER

Total fixed cost — all levels **₹ 2,20,000**
Total cost — 50% / 60% **₹ 10,50,000 / ₹ 11,48,000**

Total cost — 70% / 90%	₹ 12,80,000 / ₹ 14,94,700
Loss at 60% capacity	(₹ 48,000)
Profit at 70% capacity	₹ 20,000
Profit at 90% capacity	₹ 5,300
Best level of operation	70% — profit falls beyond it

Q3 Flexible budget of overheads and the overhead recovery rate

Flexible budget · p.50

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information relates to a flexible budget at 60% capacity. Find out the overhead costs at 50% and 70% capacity and also determine the overhead rates:

Particulars	Expenses at 60% capacity (₹)
Variable overheads	
Indirect labour	10,500
Indirect materials	8,400
Semi-variable overheads	
Repairs and maintenance (70% fixed, 30% variable)	7,000
Electricity (50% fixed, 50% variable)	25,200
Fixed overheads	
Office expenses including salaries	70,000
Insurance	4,000
Depreciation	20,000
Estimated direct labour hours	1,20,000

READ THE QUESTION FIRST — WHAT TO SPOT

Two things make this problem different from Q2. The base level is **60%**, not 50%, so every scaling is a fraction of sixty. And the semi-variable costs come with **explicit percentages**, so they must be split into a fixed half and a variable half before scaling.

- **Scale on $\frac{\text{required \%}}{60}$, not on $\frac{\text{required \%}}{50}$.** To go from 60% to 50% multiply by $\frac{50}{60}$; to reach 70% multiply by $\frac{70}{60}$. Getting this base wrong corrupts every figure in the answer.
- **Split the semi-variables at the 60% level, then scale only the variable half.** Repairs: 70% of ₹7,000 = ₹4,900 fixed and ₹2,100 variable. Electricity: 50/50, so ₹12,600 each way.
- **The 1,20,000 direct labour hours is not an expense.** It sits in the middle of the cost list, which is a deliberate trap. It is the **base for the overhead rate**, and it scales with capacity: 1,00,000 hours at 50% and 1,40,000 at 70%.
- **“Determine the overhead rates”** means the recovery rate per direct labour hour — total overhead divided by labour hours, at each of the three levels.

WORKING NOTES

W1 Splitting the semi-variable overheads at 60% capacity

Semi-variable overhead	At 60% ₹	Split given	Fixed portion ₹	Variable portion ₹
Repairs and maintenance	7,000	70% fixed / 30% variable	4,900 70% of 7,000	2,100 30% of 7,000
Electricity	25,200	50% fixed / 50% variable	12,600 50% of 25,200	12,600 50% of 25,200

Only the **variable portion** is then scaled. For repairs at 50% capacity: ₹4,900 fixed + $(\text{₹}2,100 \times \frac{50}{60}) = \text{₹}4,900 +$

₹1,750 = **₹6,650**. At 70%: ₹4,900 + $(\text{₹}2,100 \times \frac{70}{60}) = \text{₹}4,900 + \text{₹}2,450 = \text{₹}7,350$.

W2 Direct labour hours at each capacity

At 60% capacity = 1,20,000 hours (given)

At 50% capacity = $1,20,000 \times \frac{50}{60} = \text{1,00,000 hours}$

At 70% capacity = $1,20,000 \times \frac{70}{60} = \text{1,40,000 hours}$

Flexible Budget of overheads and overhead recovery rates

Particulars	50% ₹	60% ₹	70% ₹
Variable overheads			
Indirect labour 10,500 × 50/60 and 70/60	8,750	10,500	12,250
Indirect materials 8,400 × 50/60 and 70/60	7,000	8,400	9,800
Semi-variable overheads W1 — fixed half held, variable half scaled			
Repairs and maintenance	6,650	7,000	7,350
Electricity	23,100	25,200	27,300
Fixed overheads			
Office expenses including salaries	70,000	70,000	70,000
Insurance	4,000	4,000	4,000
Depreciation	20,000	20,000	20,000
TOTAL OVERHEAD COST	1,39,500	1,45,100	1,50,700
Estimated direct labour hours W2	1,00,000	1,20,000	1,40,000
OVERHEAD RECOVERY RATE PER HOUR total overhead ÷ labour hours	1.395	1.209	1.076

WHY THIS WORKS

The three recovery rates are the real answer to this question, and they fall as capacity rises — ₹1.395, ₹1.209, ₹1.076 per hour. That is the flexible-budget idea expressed as a rate rather than as a total.

Total overhead rises from ₹1,39,500 to ₹1,50,700 — an increase of only **8.0%**

Direct labour hours rise from 1,00,000 to 1,40,000 — an increase of **40%**
 So overhead *per hour* must fall, from ₹1.395 to ₹1.076 — a drop of **22.9%**

Why this matters in practice. If the company sets its absorption rate using the 50% figure of ₹1.395 and then actually operates at 70%, it will over-absorb overhead and overstate its product costs by nearly 30%. Quotations based on that rate would be too high, and the company would lose tenders it could profitably have won. Choosing the right activity level for the absorption rate is one of the practical reasons flexible budgets exist — worth a sentence in the answer.

FINAL ANSWER

Total overhead — 50% capacity	₹ 1,39,500 (rate ₹ 1.395 / hour)
Total overhead — 60% capacity	₹ 1,45,100 (rate ₹ 1.209 / hour)
Total overhead — 70% capacity	₹ 1,50,700 (rate ₹ 1.076 / hour)
Direct labour hours — 50% / 60% / 70%	1,00,000 / 1,20,000 / 1,40,000

Q4 Flexible budget at three activity levels, in total and per unit

Flexible budget · p.50

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

With the following data for a 60% activity, prepare a budget for production at 80% and 100% capacity:

Particulars	Amount
Production at 60% activity	600 units
Materials	₹100 per unit
Labour	₹40 per unit
Direct expenses	₹10 per unit
Factory overheads	₹40,000 (40% fixed)
Administrative expenses	₹30,000 (60% fixed)

READ THE QUESTION FIRST — WHAT TO SPOT

Start by converting capacity into **units**, because every per-unit figure has to be multiplied by a quantity.

60% activity = 600 units, so 1% = 10 units
 80% activity = **800 units** 100% activity = **1,000 units**

- **Materials, labour and direct expenses are given per unit**, so they are variable in full. Multiply straight through: ₹100, ₹40 and ₹10 stay constant per unit at every level.
- **Factory overhead: 40% fixed** → ₹16,000 fixed and ₹24,000 variable at 600 units, which is ₹40 per unit variable.
- **Administrative expenses: 60% fixed** → ₹18,000 fixed and ₹12,000 variable at 600 units, which is ₹20 per unit variable.
- Present both a **total** and a **per unit** column at each level, as the notes do. The per-unit column is where the falling fixed cost becomes visible, and it is what makes the answer a flexible budget rather than three separate budgets.

WORKING NOTES

W1 Splitting the two overhead lines at 600 units

Overhead	Total at 600 units ₹	Split given	Fixed ₹	Variable ₹	Variable per unit ₹
Factory overheads	40,000	40% fixed, 60% variable	16,000 40% of 40,000	24,000 60% of 40,000	40.00 24,000 ÷ 600
Administrative expenses	30,000	60% fixed, 40% variable	18,000 60% of 30,000	12,000 40% of 30,000	20.00 12,000 ÷ 600

Flexible Budget

Particulars	60% — 600 units		80% — 800 units		100% — 1,000 units	
	Per unit ₹	Total ₹	Per unit ₹	Total ₹	Per unit ₹	Total ₹
Materials	100.00	60,000	100.00	80,000	100.00	1,00,000
Labour	40.00	24,000	40.00	32,000	40.00	40,000
Direct expenses	10.00	6,000	10.00	8,000	10.00	10,000
<i>Factory overheads</i> W1						
Fixed (40%)	26.67	16,000	20.00	16,000	16.00	16,000
Variable (60%)	40.00	24,000	40.00	32,000	40.00	40,000
<i>Administrative expenses</i> W1						
Fixed (60%)	30.00	18,000	22.50	18,000	18.00	18,000
Variable (40%)	20.00	12,000	20.00	16,000	20.00	20,000
TOTAL COST	266.67	1,60,000	252.50	2,02,000	244.00	2,44,000

The four bold per-unit figures are the fixed elements. Their totals never move; their per-unit amounts fall as output rises.

WHY THIS WORKS

Verify the answer with the two-line check, which works on any flexible budget:

Total variable cost per unit = 100 + 40 + 10 + 40 + 20 = **₹210**

Total fixed cost = 16,000 + 18,000 = **₹34,000**

Check at 800 units = (800 × 210) + 34,000 = 1,68,000 + 34,000 = **₹2,02,000 ✓**

Check at 1,000 units = (1,000 × 210) + 34,000 = 2,10,000 + 34,000 = **₹2,44,000 ✓**

The falling unit cost is the finding to report: ₹266.67 at 600 units, ₹252.50 at 800, ₹244.00 at 1,000.

Cost per unit falls by ₹22.67 — a little over 8% — purely from spreading ₹34,000 of fixed cost across more units. No supplier gave a discount and no worker became faster.

This is also the arithmetic behind every “economies of scale” argument, and it is why a firm quoting a price must always be asked *at what volume?* A quotation built on the ₹266.67 figure and delivered at 1,000 units carries ₹22,670 of hidden margin.

FINAL ANSWER

Units at 60% / 80% / 100% **600 / 800 / 1,000 units**

Total variable cost per unit **₹ 210**

Total fixed cost (all levels) **₹ 34,000**

Total cost at 60% — 600 units ₹ 1,60,000 (₹ 266.67 p.u.)

Total cost at 80% — 800 units ₹ 2,02,000 (₹ 252.50 p.u.)

Total cost at 100% — 1,000 units ₹ 2,44,000 (₹ 244.00 p.u.)

Q13 High-low method — segregating semi-variable overheads

Cost segregation · p.53

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The budget manager of Cosmetics Ltd. is preparing a budget for the accounting year starting from 1st July. As part of the budget operations, some items of factory overhead costs have been estimated by him under specified conditions of volume as follows:

Volume of production (units)	1,20,000	1,50,000
Indirect materials	2,64,000	3,30,000
Indirect labour	1,50,000	1,87,500
Maintenance	84,000	1,02,000
Supervision	1,98,000	2,34,000
Engineering service	94,000	94,000

Calculate the cost of factory overhead items given above at 1,40,000 units of production.

READ THE QUESTION FIRST — WHAT TO SPOT

Costs at **two** volumes and a third volume asked for — that combination always means the **high-low method**. But before reaching for it, classify each line, because three of the five need no high-low work at all.

1. **Compute the cost per unit at each of the two volumes.** This one step classifies every line for you, and it is the step students skip.
2. **If cost per unit is the SAME at both volumes → purely VARIABLE.** Just multiply by 1,40,000.
3. **If the TOTAL is the same at both volumes → purely FIXED.** Carry it across unchanged.
4. **If cost per unit FALLS but the total rises → SEMI-VARIABLE.** Now use high-low:

$$\text{Variable rate per unit} = \frac{\text{Change in total cost}}{\text{Change in volume}}$$

$$\text{Fixed element} = \text{Total cost at either volume} - (\text{Variable rate} \times \text{that volume})$$

- The change in volume is always 1,50,000 – 1,20,000 = **30,000 units** in this problem, so every variable rate is the cost difference divided by 30,000.
- **Check the fixed element from both volumes.** If it does not come out the same from the high point and the low point, the arithmetic is wrong. That is a free self-check.

W1 Classifying the five overhead lines from the cost per unit

Overhead item	Total at 1,20,000 ₹	Cost p.u. ₹	Total at 1,50,000 ₹	Cost p.u. ₹	Classification
Indirect materials	2,64,000	2.20	3,30,000	2.20	VARIABLE — rate identical
Indirect labour	1,50,000	1.25	1,87,500	1.25	VARIABLE — rate identical
Maintenance	84,000	0.70	1,02,000	0.68	SEMI-VARIABLE — rate falls

Overhead item	Total at 1,20,000 ₹	Cost p.u. ₹	Total at 1,50,000 ₹	Cost p.u. ₹	Classification
Supervision	1,98,000	1.65	2,34,000	1.56	SEMI-VARIABLE — rate falls
Engineering service	94,000	0.783	94,000	0.627	FIXED — total identical

WORKING NOTES

W2 High-low segregation of the two semi-variable items

Maintenance

$$\text{Variable rate} = \frac{₹1,02,000 - ₹84,000}{1,50,000 - 1,20,000} = \frac{₹18,000}{30,000} = \text{₹0.60 per unit}$$

$$\text{Fixed element} = 84,000 - (0.60 \times 1,20,000) = 84,000 - 72,000 = \text{₹12,000}$$

$$\text{Check from the high point} = 1,02,000 - (0.60 \times 1,50,000) = 1,02,000 - 90,000 = ₹12,000 \checkmark$$

Supervision

$$\text{Variable rate} = \frac{₹2,34,000 - ₹1,98,000}{1,50,000 - 1,20,000} = \frac{₹36,000}{30,000} = \text{₹1.20 per unit}$$

$$\text{Fixed element} = 1,98,000 - (1.20 \times 1,20,000) = 1,98,000 - 1,44,000 = \text{₹54,000}$$

$$\text{Check from the high point} = 2,34,000 - (1.20 \times 1,50,000) = 2,34,000 - 1,80,000 = ₹54,000 \checkmark$$

Factory Overhead Budget at 1,40,000 units of production

Particulars	Basis of computation	Amount ₹
Indirect materials	Variable at ₹2.20 per unit → 1,40,000 × 2.20	3,08,000
Indirect labour	Variable at ₹1.25 per unit → 1,40,000 × 1.25	1,75,000
Maintenance	semi-variable — W2	
Fixed element	constant at every volume	12,000
Variable element	₹0.60 × 1,40,000	84,000
Supervision	semi-variable — W2	
Fixed element	constant at every volume	54,000
Variable element	₹1.20 × 1,40,000	1,68,000
Engineering service	Wholly fixed — unchanged at any volume	94,000
TOTAL FACTORY OVERHEAD at 1,40,000 units		8,95,000

WHY THIS WORKS

The high-low method rests on one observation: between two volumes, the **only** thing that can make the total change is the variable element, because the fixed element by definition did not move. So the whole of the cost increase must be variable, and dividing it by the volume increase gives the rate per unit. Everything left over is fixed.

$$\text{The change in cost is entirely variable} \rightarrow \frac{\text{Change in cost}}{\text{Change in volume}} = \text{variable rate per unit}$$

Then substitute that rate back at either volume; whatever the variable element does not explain must be the fixed element. Doing it at BOTH volumes is a free check on your arithmetic.

Cross-check the whole answer by rebuilding it as fixed plus variable:

Total variable rate = $2.20 + 1.25 + 0.60 + 1.20 = \text{₹}5.25 \text{ per unit}$

Total fixed cost = $12,000 + 54,000 + 94,000 = \text{₹}1,60,000$

At 1,40,000 units = $(1,40,000 \times 5.25) + 1,60,000 = 7,35,000 + 1,60,000 = \text{₹}8,95,000 \checkmark$

And at 1,20,000 units = $(1,20,000 \times 5.25) + 1,60,000 = \text{₹}7,90,000$, which is exactly the sum of the first column of the question $\checkmark$

A limitation worth stating. High-low uses only two observations and assumes the cost behaves in a straight line between them. It says nothing about what happens outside the 1,20,000–1,50,000 range, and 1,40,000 units falls safely inside it — which is why the answer can be relied on here. Extrapolating the same equation to, say, 2,50,000 units would not be safe, because a step in supervision cost would almost certainly occur first.

FINAL ANSWER

Indirect materials — variable	₹ 2.20 p.u. → ₹ 3,08,000
Indirect labour — variable	₹ 1.25 p.u. → ₹ 1,75,000
Maintenance — semi-variable	Fixed ₹ 12,000 + ₹ 0.60 p.u. → ₹ 96,000
Supervision — semi-variable	Fixed ₹ 54,000 + ₹ 1.20 p.u. → ₹ 2,22,000
Engineering service — fixed	₹ 94,000
Total factory overhead at 1,40,000 units	₹ 8,95,000

Cash budgets — get the timing right and the rest is addition

Use these steps for Q5, Q6 and Q7

A cash budget is not a profit statement. It records **only money actually moving**, and it records it in the month it moves. Nothing else about these problems is difficult; everything depends on getting the timing right.

THE RULE THAT MAKES CASH BUDGETS EASY

Never write a figure straight from the question into the cash budget.

Build a WORKING NOTE for every item that has a credit period, get the month right there, and only then transfer the monthly totals into the budget. Every cash-budget mark in this module is won or lost in the working notes.

- Working Note 1 — collections from customers.** Draw a grid with the sales months down the side and the budget months across the top. Take each month's sales, split it by the credit terms, and write each piece under the month it will be **received**. Then total each column.
- Working Note 2 — any other item with a lag**, typically wages when part is "paid in arrears". Same grid, same idea.
- Payments to suppliers.** A credit period of one month means April pays for March's purchases; two months means April pays for February's. Just shift the row.
- Now build the budget** in this fixed order — and keep it in this order every time, because the examiner is looking for it:

Opening balance

Add: Receipts — cash sales, collections from debtors, other income → **Total (A)**

Less: Payments — suppliers, wages, expenses, tax, capital items → **Total (B)**

Closing balance = (A) – (B), which becomes next month's opening balance

- Carry the closing balance forward — including a negative one.** If April closes at (₹47,000), May opens at (₹47,000). This is where Q6 catches people out.

WHAT GOES IN, AND WHAT NEVER DOES

Item	In a cash budget?	Note
Cash sales	YES	In the month of sale, less any cash discount allowed
Collections from debtors	YES	In the month received — use the working note
Payments to creditors	YES	In the month paid, per the credit period

Item	In a cash budget?	Note
Wages and expenses	YES	Watch for “paid a month in arrears” and for part-payment splits
Income tax, advance tax	YES	In the single month it falls due
Rent paid quarterly in advance	YES	The whole quarter in the due month — not one month’s worth
Income from investments	YES	Only in the months it is actually received
Depreciation	NEVER	Not a cash flow. The commonest error.
Provisions, write-offs, bad debt provisions	NEVER	Book entries only
Credit sales not yet collected	NO	It is a debtor, not cash, until the month it is received
Opening / closing stock	NO	Stock movements are not cash; the purchase that created the stock is

THE TRAP / WHERE MARKS ARE LOST

- **Forgetting the cash discount.** If 20% of sales are collected immediately “discount allowed 2%”, the cash received is $20\% \times 98\%$, not 20%. Q7 turns on this.
- **Treating a quarterly advance payment as monthly.** Rent of ₹1,000 a month payable quarterly in advance in April means **₹3,000 in April** and nothing in May or June.
- **Losing the sign on a negative closing balance.** Show it in brackets and carry it forward as a negative opening balance.
- **Not answering the actual question.** Q6 asks for the *extent of the bank facilities required*. The answer is the largest negative closing balance — say so explicitly, in words, at the end.

Q5 Cash budget with four different credit periods

Cash budget · p.50

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company is expecting to have ₹32,000 cash in hand on 1.4.2008 and it requests you to prepare a cash budget for the three months, April to June 2008. The following information is supplied to you:

Month	Sales ₹	Purchases ₹	Wages ₹	Expenses ₹
February	70,000	44,000	6,000	5,000
March	80,000	56,000	9,000	6,000
April	96,000	60,000	9,000	7,000
May	1,00,000	68,000	11,000	9,000
June	1,20,000	62,000	14,000	9,000

Other information:

- **a)** Period of credit allowed by suppliers is two months.
- **b)** 25% of sales are for cash and the period of credit allowed to customers for credit sales is one month.
- **c)** Delay in payment of wages and expenses one month.
- **d)** Income tax ₹28,000 is to be paid in June 2008.

READ THE QUESTION FIRST — WHAT TO SPOT

Four credit terms, and each one shifts a different row by a different number of months. Tabulate them before starting — this little table is the solution.

Item	Term given	So April's figure comes from ...
Cash sales (25%)	received immediately	April's own sales × 25%
Credit sales (75%)	one month's credit	March's sales × 75%
Purchases	two months' credit	February's purchases
Wages	one month's delay	March's wages
Expenses	one month's delay	March's expenses
Income tax	due in June	June only — nothing in April or May

- **Sales split into two rows, not one.** 25% arrives in the month of sale and 75% one month later, so April receives money from two different months' sales. Keep them on separate lines in the budget — it is clearer and it earns the marks.
- **Two months' credit on purchases is why February data is given.** If a month appears in the question but seems unused, you have missed a lag.
- February and March sales are needed only for their credit portion; February purchases only for the April payment. Nothing else from those two months is used.

WORKING NOTES

W1 Collection from debtors — 75% of sales, one month later

Month of sale	Sales ₹	Credit portion 75% ₹	Received in	April ₹	May ₹	June ₹
February	70,000	52,500	March	—	—	—
March	80,000	60,000	April	60,000	—	—
April	96,000	72,000	May	—	72,000	—
May	1,00,000	75,000	June	—	—	75,000
June	1,20,000	90,000	July	—	—	—
Collected from debtors				60,000	72,000	75,000

February's ₹52,500 is collected in March, which is before the budget period, and June's ₹90,000 in July, which is after it. Both are shown so that the grid is complete and the examiner can see the logic.

W2 Cash sales — 25% of the same month's sales

April = 25% of ₹96,000 = **₹24,000**
 May = 25% of ₹1,00,000 = **₹25,000**
 June = 25% of ₹1,20,000 = **₹30,000**

Cash Budget for the three months from April to June 2008

Particulars	April ₹	May ₹	June ₹
Opening cash balance 32,000 given; thereafter the previous closing balance	32,000	57,000	82,000
<i>Receipts</i>			

Particulars	April ₹	May ₹	June ₹
Cash sales (25%) W2 — 25% of the same month's sales	24,000	25,000	30,000
Collection from debtors (75%) W1 — last month's sales	60,000	72,000	75,000
TOTAL (A)	1,16,000	1,54,000	1,87,000
Payments			
Payment to suppliers 2 months' credit → Feb, Mar, Apr purchases	(44,000)	(56,000)	(60,000)
Wages 1 month's delay → Mar, Apr, May wages	(9,000)	(9,000)	(11,000)
Expenses 1 month's delay → Mar, Apr, May expenses	(6,000)	(7,000)	(9,000)
Income tax payable in June only	—	—	(28,000)
TOTAL (B)	(59,000)	(72,000)	(1,08,000)
CLOSING BALANCE (A – B)	57,000	82,000	79,000

WHY THIS WORKS

The company is comfortably liquid throughout, closing at ₹57,000, ₹82,000 and ₹79,000. No overdraft is needed.

The June dip is worth commenting on. Cash grew for two months and then fell by ₹3,000 in June — and it fell in the month with the *highest* sales of the whole period, ₹1,20,000. That apparent contradiction is the most useful thing a cash budget shows:

- June collected only 25% of its own record sales, because 75% will not arrive until July. Sales growth **consumes** cash before it produces any.
- The ₹28,000 income tax fell entirely in June. A single lumpy payment can reverse a healthy trend, and knowing the month in advance is exactly why the budget is prepared.

Recommendation: no borrowing is required for April-June. But the pattern shows that each further rise in sales will absorb cash for a month before releasing it, so if the growth in sales continues beyond June the position should be re-tested for July onwards, when the ₹90,000 owing from June is collected but a still larger sales figure may again outrun it.

FINAL ANSWER

Opening balance, 1 April	₹ 32,000
Total receipts — April / May / June	₹ 84,000 / ₹ 97,000 / ₹ 1,05,000
Total payments — April / May / June	₹ 59,000 / ₹ 72,000 / ₹ 1,08,000
Closing balance — April	₹ 57,000
Closing balance — May	₹ 82,000
Closing balance — June	₹ 79,000
Overdraft required	None — positive throughout

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

XY Co. wishes to arrange overdraft facilities with its bankers during the period April to June of a particular year, when it will be manufacturing mostly for stock. Prepare a cash budget for the above period from the following data, indicating the extent of the bank facilities the company will require at the end of each month:

Month	Sales ₹	Purchases ₹	Wages ₹
February	1,80,000	1,24,800	12,000
March	1,92,000	1,44,000	14,000
April	1,08,000	2,43,000	11,000
May	1,74,000	2,46,000	10,000
June	1,26,000	2,68,000	15,000

- **b)** 50% of the credit sales are realised in the month following the sales and the remaining sales in the second month following.
- **c)** Creditors are paid in the following month of purchase.
- **d)** Cash at bank on 1st April ₹25,000.

READ THE QUESTION FIRST — WHAT TO SPOT

Read the story the numbers tell before you compute anything. “**Manufacturing mostly for stock**” means purchases are rising steeply while sales fall — purchases go from ₹1,44,000 to ₹2,68,000 as sales drop from ₹1,92,000 to ₹1,26,000. A cash crisis is being built deliberately, and the question is how large it becomes.

- **There are no cash sales in this problem.** Every rupee of sales is on credit, collected 50% in the next month and 50% the month after. So each budget month collects from **two** earlier months.
- **Wages are paid in the same month** — no lag is mentioned, so do not invent one.
- **Creditors paid in the following month**, so April pays March’s purchases of ₹1,44,000, not April’s ₹2,43,000.
- **The closing balance will go negative and must be carried forward as a negative.** May opens with April’s ₹56,000 but June opens with May’s **minus** ₹47,000. This is the step that decides the problem.
- **Answer the question asked.** It wants the extent of the bank facility required at the end of each month — so state the overdraft figures in words at the end, not just the closing balances.

WORKING NOTES

W1 Collection from debtors — 50% in the next month, 50% in the second month

Month of sale	Sales ₹	50% received in the next month	50% received two months later	April ₹	May ₹	June ₹
February	1,80,000	March — 90,000	April — 90,000	90,000	—	—
March	1,92,000	April — 96,000	May — 96,000	96,000	96,000	—
April	1,08,000	May — 54,000	June — 54,000	—	54,000	54,000
May	1,74,000	June — 87,000	July — 87,000	—	—	87,000
June	1,26,000	July — 63,000	August — 63,000	—	—	—

Month of sale	Sales ₹	50% received in the next month	50% received two months later	April ₹	May ₹	June ₹
Total collected from debtors				1,86,000	1,50,000	1,41,000

Read a column downwards to see the answer: April collects ₹90,000 from February and ₹96,000 from March. June collects ₹54,000 from April and ₹87,000 from May.

Cash Budget for April to June

Particulars	April ₹	May ₹	June ₹
Opening balance <i>25,000 given; then the previous closing balance, negative included</i>	25,000	56,000	(47,000)
<i>Receipts</i>			
Collection from debtors <i>W1 — 50% + 50% from the two prior months</i>	1,86,000	1,50,000	1,41,000
TOTAL (A)	2,11,000	2,06,000	94,000
<i>Payments</i>			
Payment to suppliers <i>following month → Mar, Apr, May purchases</i>	(1,44,000)	(2,43,000)	(2,46,000)
Wages paid <i>same month, no lag given</i>	(11,000)	(10,000)	(15,000)
TOTAL (B)	(1,55,000)	(2,53,000)	(2,61,000)
CLOSING BALANCE (A – B)	56,000	(47,000)	(1,67,000)

A negative closing balance is an **overdraft**. Note carefully that May's closing balance of (₹47,000) becomes June's **opening** balance, which is why June's total receipts of only ₹94,000 are less than the ₹1,41,000 collected.

Extent of the bank facilities required — the answer to the question asked

End of month	Closing balance ₹	Overdraft required ₹	Position
April	56,000	Nil	Cash surplus — no facility needed
May	(47,000)	47,000	Overdraft required
June	(1,67,000)	1,67,000	Maximum facility needed

WHY THIS WORKS

The company must arrange an overdraft facility of at least ₹1,67,000, and it should be in place before the end of May. That is the answer the question asks for, and it should be stated in exactly those words.

Why the position deteriorates so fast. Follow the two forces working against each other:

	March	April	May	June
Purchases — cash goes out one month later	1,44,000	2,43,000	2,46,000	2,68,000
Sales — cash comes in over the next two months	1,92,000	1,08,000	1,74,000	1,26,000
Purchases in excess of sales	(48,000)	1,35,000	72,000	1,42,000

The company is buying far more than it is selling — exactly what “manufacturing mostly for stock” means. Cash is being converted into inventory. Worse, the timing compounds it: each month’s purchases are paid for in **one** month, while each month’s sales are collected over **two**. Money leaves twice as fast as it returns.

What to advise management. The overdraft is the treatment, not the cure. If the stock being built is genuinely needed for a seasonal peak, ₹1,67,000 of borrowing is a reasonable cost of doing business and should be negotiated early — a facility arranged in advance is always cheaper than one requested in crisis. If it is not, the buying programme should be slowed, or the collection period shortened by offering a cash discount, which is the remedy Q7 illustrates.

FINAL ANSWER

Opening balance, 1 April	₹ 25,000
Collections — April / May / June	₹ 1,86,000 / ₹ 1,50,000 / ₹ 1,41,000
Closing balance — April	₹ 56,000 surplus
Closing balance — May	(₹ 47,000) overdrawn
Closing balance — June	(₹ 1,67,000) overdrawn
Bank facility required	Nil in April, ₹ 47,000 in May, ₹ 1,67,000 in June
Maximum facility to arrange	₹ 1,67,000

Q7 Cash budget with cash discount, split wages and quarterly items

Cash budget · p.51-52

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following budget data, forecast the cash position at the end of April, May and June 2016:

Month	Sales ₹	Purchases ₹	Wages ₹	Misc. expenses ₹
February	1,20,000	84,000	10,000	7,000
March	1,30,000	1,00,000	12,000	8,000
April	80,000	1,04,000	8,000	6,000
May	1,16,000	1,06,000	10,000	12,000
June	88,000	80,000	8,000	6,000

Additional information:

- **Sales:** 20% realised in the month of sales, discount allowed 2%. Balance realised equally in two subsequent months.
- **Purchases:** paid in the month following the month of supply.
- **Wages:** 25% paid in arrears in the following month.
- **Miscellaneous expenses:** paid a month in arrears.
- **Rent:** ₹1,000 per month, paid quarterly in advance, due in April.
- **Income tax:** first instalment of advance tax ₹25,000 due on or before 15th June.
- **Income from investments:** ₹5,000 received quarterly in April, July, etc.
- **Cash in hand:** ₹5,000 on 1st April 2016.

READ THE QUESTION FIRST — WHAT TO SPOT

This is the hardest cash budget of the three because **seven** different timing rules apply at once, and two of them are easy to get subtly wrong.

Item	The rule	The subtlety
Cash sales	20% in the month of sale, less 2% discount	Cash received is $20\% \times 98\% = \mathbf{19.6\%}$ of sales. Deducting the discount is the point of the sub-question.
Credit sales	80% split equally over the two following months	"Equally" means 40% each , not 80% in one month.
Purchases	paid the following month	April pays March's ₹1,00,000
Wages	25% in arrears → 75% in the same month	Two wage lines are needed in the budget: 75% of this month plus 25% of last month.
Misc. expenses	a month in arrears	April pays March's ₹8,000
Rent	quarterly in advance, due April	₹3,000 in April for the whole quarter, and nothing in May or June.
Advance tax	due by 15 June	June only
Investment income	quarterly, in April	₹5,000 in April only

WORKING NOTES

W1 Collection from debtors — 80% split 40% and 40%

Month of sale	Sales ₹	Credit 80% ₹	40% in the next month	40% two months later	April ₹	May ₹	June ₹
February	1,20,000	96,000	March — 48,000	April — 48,000	48,000	—	—
March	1,30,000	1,04,000	April — 52,000	May — 52,000	52,000	52,000	—
April	80,000	64,000	May — 32,000	June — 32,000	—	32,000	32,000
May	1,16,000	92,800	June — 46,400	July — 46,400	—	—	46,400
June	88,000	70,400	July — 35,200	August — 35,200	—	—	—
Collected from debtors					1,00,000	84,000	78,400

W2 Cash sales — 20% of sales less 2% cash discount

Month	Sales ₹	20% ₹	Less 2% discount ₹	Cash received ₹
April	80,000	16,000	(320)	15,680
May	1,16,000	23,200	(464)	22,736
June	88,000	17,600	(352)	17,248

Effectively 19.6% of sales is received in cash. The 2% discount is the price of getting the money early — it never appears as a payment, it simply reduces the receipt.

W3 Wages — 75% in the month, 25% in the following month

Month	Wages ₹	75% paid in the month ₹	25% paid the next month ₹	Paid in
February	10,000	7,500	2,500	March

Month	Wages ₹	75% paid in the month ₹	25% paid the next month ₹	Paid in
March	12,000	9,000	3,000	April
April	8,000	6,000	2,000	April / May
May	10,000	7,500	2,500	May / June
June	8,000	6,000	2,000	June / July
Total wages paid	April 9,000	May 9,500	June 8,500	

April pays ₹6,000 (75% of April) + ₹3,000 (25% of March) = ₹9,000. May pays ₹7,500 + ₹2,000 = ₹9,500. June pays ₹6,000 + ₹2,500 = ₹8,500.

Cash Budget for April to June 2016

Particulars	April ₹	May ₹	June ₹
Opening balance <i>5,000 given; then previous closing, negatives carried forward</i>	5,000	5,680	(7,084)
<i>Receipts</i>			
Cash sales, net of 2% discount <i>W2</i>	15,680	22,736	17,248
Collection from debtors <i>W1 — 40% + 40%</i>	1,00,000	84,000	78,400
Income from investments <i>quarterly — April only</i>	5,000	—	—
TOTAL (A)	1,25,680	1,12,416	88,564
<i>Payments</i>			
Payment to creditors <i>following month → Mar, Apr, May purchases</i>	(1,00,000)	(1,04,000)	(1,06,000)
Miscellaneous expenses <i>one month in arrears → Mar, Apr, May</i>	(8,000)	(6,000)	(12,000)
Wages — 75% of the current month <i>W3</i>	(6,000)	(7,500)	(6,000)
Wages — 25% outstanding from last month <i>W3</i>	(3,000)	(2,000)	(2,500)
Rent <i>quarterly in advance — ₹1,000 × 3 in April</i>	(3,000)	—	—
Advance income tax <i>first instalment due by 15 June</i>	—	—	(25,000)
TOTAL (B)	(1,20,000)	(1,19,500)	(1,51,500)
CLOSING BALANCE (A – B)	5,680	(7,084)	(62,936)

WHY THIS WORKS

The forecast cash position is **₹5,680 at the end of April, an overdraft of ₹7,084 at the end of May, and an overdraft of ₹62,936 at the end of June.**

June is the month that breaks the position, and three things arrive together:

- **The ₹25,000 advance tax**, a single unavoidable payment that alone exceeds a third of the month's receipts.
- **April's poor sales of ₹80,000 finally reach the cash book.** Because 40% of each month's sales arrives two months later, the weak April is still depressing June collections — only ₹32,000 of the ₹78,400 comes from April.
- **May's purchases of ₹1,06,000 are paid in June** against June sales of only ₹88,000. The company pays for a strong month out of the receipts of a weak one.

Recommendation: arrange a facility of at least ₹65,000 before the end of May. Three things would ease the position, and any of them is worth mentioning:

1. Negotiate two months' credit from suppliers instead of one. That single change would defer ₹1,06,000 out of June and turn the closing balance positive on its own.
2. Extend the 2% cash discount to a larger share of sales. It costs 2% but converts a 40%/40% two-month wait into immediate cash, and the ₹1,136 of discount given in June is trivial against a ₹62,936 shortfall.
3. Ask whether the advance tax can be met from a short-term facility rather than working capital, since it is a known, dated, one-off payment — precisely the kind of item a cash budget exists to give warning of.

FINAL ANSWER

Opening balance, 1 April	₹ 5,000
Cash sales net of discount — Apr / May / Jun	₹ 15,680 / ₹ 22,736 / ₹ 17,248
Collections from debtors — Apr / May / Jun	₹ 1,00,000 / ₹ 84,000 / ₹ 78,400
Total payments — Apr / May / Jun	₹ 1,20,000 / ₹ 1,19,500 / ₹ 1,51,500
Closing balance — April	₹ 5,680
Closing balance — May	(₹ 7,084) overdrawn
Closing balance — June	(₹ 62,936) overdrawn
Facility to arrange	At least ₹ 65,000 by end-May

Production and purchase budgets — one formula, applied twice

Use these steps for Q8, Q10, Q11, Q12 and Q15

Four problems in this module (Q8, Q10, Q11, Q12) and two parts of Q15 use the **same single formula**, applied first to finished goods and then to materials. Learn it once.

ONE FORMULA, USED TWICE

What I must OBTAIN = What I will USE UP + What I want LEFT OVER – What I ALREADY HAVE

Applied to ...	What I will use up	The formula becomes
Finished goods → the PRODUCTION budget	Budgeted sales in units	Production = Sales + Closing stock of finished goods – Opening stock
Raw materials / parts → the PURCHASE budget	Materials consumed , which comes from the production budget	Purchases = Consumption + Closing stock of materials – Opening stock

The order is compulsory. You cannot compute the material budget until the production budget is finished, because consumption depends on units *produced*, never on units *sold*.

- 1. Production budget, in units.** Sales + closing stock – opening stock, one column per product (or one column per month, if the question is month-wise).
- 2. Determine the closing stock properly.** The question always tells you how, and it is never the same twice: a flat figure (Q10, Q11, Q12), **50% of next month's sales** (Q8), or **90% of opening stock** (Q15). Read it carefully; this is the marked step.
- 3. Consumption of each material.** Units produced × the quantity of that material per unit, summed across all products.
- 4. Purchase budget, in units.** Consumption + closing stock of the material – opening stock. If the question also gives **materials on order**, extend it:

Quantity to be ORDERED = Consumption + Closing stock + Closing orders outstanding – Opening stock – Opening orders outstanding

Stock in hand is material you HAVE; material on order is material you have already ARRANGED to have. Both reduce what you must order afresh. Only Q11 needs this version.

- 5. Value it, if asked.** Multiply the purchase quantity by the price per unit to get the purchase budget in value.

THE TRAP / WHERE MARKS ARE LOST

- **Adding the opening stock and subtracting the closing.** It is the other way round. Sense-check it: if you want to *end* with more stock than you started with, you must produce **more** than you sell.
- **Using sales units to compute material consumption.** Always production units.
- **In a month-wise budget, forgetting that this month's closing stock is next month's opening stock.** They must chain, and if they do not, the columns will not add up.
- **In Q8, needing October's sales to finish September.** The closing stock for September is 50% of *October's* sales. The question gives seven months of data for a six-month budget precisely for this reason — if a figure looks surplus, you have missed a step.

Q8 Month-wise production budget and production cost budget

Production budget · p.51-52

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company is drawing its production plan for the year 2002-03 in respect of two of its products 'Gamma' and 'Delta'. The company's policy is not to carry any closing WIP at the end of any month. However, its policy is to hold a closing stock of finished goods at 50% of the anticipated quantity of sales of the succeeding month. For the year 2002-03 the company's budgeted production is 20,000 units of 'Gamma' and 25,000 of 'Delta'. The following is the estimated cost data:

Particulars	Gamma	Delta
Direct material per unit	₹50	₹80
Direct labour per unit	₹20	₹30
Other manufacturing expenses apportionable to each type of product based on production	₹2,00,000	₹3,75,000

The estimated units to be sold in the first 7 months of the year 2002-03 are as under:

	April	May	June	July	Aug	Sept	Oct
Gamma	900	1,100	1,400	1,800	2,200	2,200	1,800
Delta	2,900	2,900	2,500	2,100	1,700	1,700	1,900

(a) Prepare a production budget showing month-wise number of units to be manufactured. **(b)** Present a summarised production cost budget for the half-year ending 30-09-2002.

READ THE QUESTION FIRST — WHAT TO SPOT

Two subtleties, and both are signposted by figures that look surplus.

- **Seven months of sales are given for a six-month budget.** October is needed because September's closing stock is 50% of **October's** sales. Whenever a question gives one extra period, that is why.
- **The opening stock of April is 50% of April's own sales** — because March's closing stock was set at 50% of the succeeding month's (that is, April's) sales. For Gamma that is $50\% \times 900 = 450$ units. Students often leave April's opening stock blank.
- **Each month's closing stock is the next month's opening stock.** Write the closing row first, then copy it down one column to form the opening row. That chaining is a free self-check.
- **The 20,000 and 25,000 annual budgeted production figures are used only for part (b),** to convert the lump-sum manufacturing expenses into a rate per unit:

$$\begin{aligned} \text{Gamma} &= \frac{\text{₹}2,00,000}{20,000 \text{ units}} = \text{₹}10 \text{ per unit} \\ \text{Delta} &= \frac{\text{₹}3,75,000}{25,000 \text{ units}} = \text{₹}15 \text{ per unit} \end{aligned}$$

- “No closing WIP” simply removes a complication — every unit started is finished, so units produced equals units completed.

(a) Production Budget — GAMMA, in units

Particulars	April	May	June	July	Aug	Sept
Budgeted sales given	900	1,100	1,400	1,800	2,200	2,200
Add: Desired closing stock 50% of NEXT month's sales	550	700	900	1,100	1,100	900
Total requirement	1,450	1,800	2,300	2,900	3,300	3,100
Less: Opening stock = previous month's closing stock	(450)	(550)	(700)	(900)	(1,100)	(1,100)
BUDGETED PRODUCTION	1,000	1,250	1,600	2,000	2,200	2,000

Closing stock working: April $50\% \times 1,100$ (May) = 550 · May $50\% \times 1,400$ = 700 · June $50\% \times 1,800$ = 900 · July $50\% \times 2,200$ = 1,100 · Aug $50\% \times 2,200$ = 1,100 · **Sept $50\% \times 1,800$ (October) = 900.** Total production for the half-year = **10,050 units.**

(a) Production Budget — DELTA, in units

Particulars	April	May	June	July	Aug	Sept
Budgeted sales given	2,900	2,900	2,500	2,100	1,700	1,700
Add: Desired closing stock 50% of NEXT month's sales	1,450	1,250	1,050	850	850	950
Total requirement	4,350	4,150	3,550	2,950	2,550	2,650
Less: Opening stock = previous month's closing stock	(1,450)	(1,450)	(1,250)	(1,050)	(850)	(850)
BUDGETED PRODUCTION	2,900	2,700	2,300	1,900	1,700	1,800

Closing stock working: April $50\% \times 2,900$ = 1,450 · May $50\% \times 2,500$ = 1,250 · June $50\% \times 2,100$ = 1,050 · July $50\% \times 1,700$ = 850 · Aug $50\% \times 1,700$ = 850 · **Sept $50\% \times 1,900$ (October) = 950.** Total production for the half-year = **13,300 units.**

(b) Summarised Production Cost Budget for the half-year ending 30-09-2002

Particulars	Gamma — 10,050 units		Delta — 13,300 units	
	Cost p.u. ₹	Total ₹	Cost p.u. ₹	Total ₹
Direct material	50	5,02,500 $10,050 \times 50$	80	10,64,000 $13,300 \times 80$
Direct labour	20	2,01,000 $10,050 \times 20$	30	3,99,000 $13,300 \times 30$
Other manufacturing expenses	10 $2,00,000 \div 20,000$	1,00,500 $10,050 \times 10$	15 $3,75,000 \div 25,000$	1,99,500 $13,300 \times 15$
TOTAL PRODUCTION COST	80	8,04,000	125	16,62,500
GRAND TOTAL — both products				₹24,66,500

WHY THIS WORKS

Compare the two products and the production budget shows something the sales budget alone cannot:

	Gamma	Delta
Sales trend over the half-year	900 → 2,200 rising	2,900 → 1,700 falling
Total sales for the half-year	9,600 units	13,700 units
Total production for the half-year	10,050 units	13,300 units
Production compared with sales	450 units MORE	400 units LESS

Gamma must be over-produced and Delta under-produced, and the reason is the stock policy working in opposite directions. Because closing stock is tied to *next* month's sales, a product with rising sales has to keep building stock, while a product with falling sales can be allowed to run its stock down.

Gamma = sales 9,600 + closing stock 900 – opening stock 450 = **10,050 units** ✓

Delta = sales 13,700 + closing stock 950 – opening stock 1,450 = **13,300 units** ✓

That two-line check is worth doing every time: the six monthly columns must reconcile to the same answer as the half-year taken as a whole. If they do not, a closing stock has been mis-chained.

A practical observation. Total production of 23,350 units is fairly steady month to month (Gamma rising, Delta falling), which is exactly what a factory wants — level loading of labour and machinery. The stock policy, whatever its cost in working capital, is smoothing production. Noting that shows you understand what a production budget is *for*.

FINAL ANSWER

(a) Gamma production — Apr to Sept **1,000 / 1,250 / 1,600 / 2,000 / 2,200 / 2,000**

Total Gamma **10,050 units**

(a) Delta production — Apr to Sept **2,900 / 2,700 / 2,300 / 1,900 / 1,700 / 1,800**

Total Delta **13,300 units**

Other manufacturing expenses per unit **Gamma ₹ 10 · Delta ₹ 15**

(b) Production cost — Gamma **₹ 8,04,000 (₹ 80 p.u.)**

(b) Production cost — Delta **₹ 16,62,500 (₹ 125 p.u.)**

(b) Grand total **₹ 24,66,500**

Q10 Production budget for three products

Production budget · p.52-53

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following particulars, prepare a Production Budget of a company for the year ended June 30, 2010.

Product	Sales in units (as per sales budget)	Estimated stock 1 July 2009	Estimated stock 30 June 2010
A	1,50,000	14,000	15,000
B	1,00,000	5,000	14,500
C	70,000	8,000	8,000

READ THE QUESTION FIRST — WHAT TO SPOT

The most direct application of the formula in the whole module. One line of arithmetic per product — but be sure which stock figure is which.

- **1 July 2009 is the OPENING stock** (the year begins) and **30 June 2010 is the CLOSING stock** (the year ends). Sort the dates before you sort the numbers.
- **Add the closing, subtract the opening.** Product B must build stock from 5,000 to 14,500, so it has to produce 9,500 units *more* than it sells.
- **Product C is the check on your understanding:** opening and closing stock are both 8,000, so they cancel and production equals sales exactly. If you get anything other than 70,000 for C, the signs are the wrong way round.

Production Budget for the year ended 30 June 2010 (in units)

Particulars	Product A	Product B	Product C
Budgeted sales <i>as per the sales budget</i>	1,50,000	1,00,000	70,000
Add: Desired closing stock <i>estimated stock at 30 June 2010</i>	15,000	14,500	8,000
Total requirement	1,65,000	1,14,500	78,000
Less: Opening stock <i>estimated stock at 1 July 2009</i>	(14,000)	(5,000)	(8,000)
BUDGETED PRODUCTION	1,51,000	1,09,500	70,000

WHY THIS WORKS

The three products illustrate the three possible relationships between production and sales, which is almost certainly why the question was set this way:

Product	Stock movement	Production vs sales	Why
A	14,000 → 15,000, a rise of 1,000	1,000 units more	A small stock build must be produced in addition to sales
B	5,000 → 14,500, a rise of 9,500	9,500 units more	A large stock build — production must exceed sales by nearly 10%
C	8,000 → 8,000, no change	exactly equal	With no change in stock, everything produced is sold

$$\text{Production} - \text{Sales} = \text{Closing stock} - \text{Opening stock}$$

This is just the main formula rearranged, and it is the fastest way to check any production budget: the amount by which production differs from sales must equal the change in stock, nothing else.

Worth adding for the extra mark: Product B's stock is being taken from 5,000 units to 14,500 — nearly tripled. That is a substantial commitment of working capital and storage space, and a budget manager would ask why. If it is preparation for an expected rise in demand it is sound; if it is simply the result of over-production it should be challenged. A production budget is a plan to be interrogated, not an arithmetic exercise.

FINAL ANSWER

Product A ($1,50,000 + 15,000 - 14,000$) **1,51,000 units**

Product B ($1,00,000 + 14,500 - 5,000$) **1,09,500 units**

Product C ($70,000 + 8,000 - 8,000$) **70,000 units**

Total budgeted production **3,30,500 units**

Q11 Material procurement budget, allowing for materials on order

Purchase budget · p.53

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Draw a material procurement budget (quantitative) from the following information:

Estimated sales of a product are 40,000 units. Each unit of the product requires 3 units of material A and 4 units of material B.

Estimated opening balances at the commencement of the next year

Particulars	Units
Finished products	5,000
Material A	12,000
Material B	20,000
<i>Materials on order</i>	
Material A	7,000
Material B	11,000

The desirable closing balances at the end of the next year

Particulars	Units
Finished products	7,000
Material A	15,000
Material B	25,000
<i>Materials on order</i>	
Material A	8,000
Material B	10,000

READ THE QUESTION FIRST — WHAT TO SPOT

The word “**procurement**” is the key. This is not asking how much material to *receive*; it is asking how much to **order**. That is why the question gives “materials on order” as well as materials in hand, and it is the only problem in the module that does.

- **Three steps, in this order.** Production budget in units of finished product → material consumption → procurement budget. You cannot skip to the third.
- **Materials on order are material you have already arranged to have.** The 7,000 units of A on order at the start will arrive during the year without any fresh order being placed, so they reduce what must be ordered — exactly like opening stock. And the 8,000 you wish to have on order at the year end must be ordered during the year, so they increase it — exactly like closing stock.
- **Hence four adjustments, not two:**

$$\begin{aligned} \text{Quantity to be PROCURED} &= \text{Consumption} + \text{Closing stock in hand} + \text{Closing quantity on order} \\ &\quad - \text{Opening stock in hand} - \text{Opening quantity on order} \end{aligned}$$

- **Note that material B’s orders on hand FALL**, from 11,000 to 10,000. So B’s procurement is reduced by 1,000, while A’s is increased by 1,000. The question changes the direction deliberately to test whether you have understood the sign rather than memorised a layout.

WORKING NOTES

W1 Budgeted production of the finished product

$$\begin{aligned} \text{Budgeted production} &= \text{Budgeted sales} + \text{Closing stock} - \text{Opening stock} \\ &= 40,000 + 7,000 - 5,000 = \mathbf{42,000 \text{ units}} \end{aligned}$$

W2 Materials consumed by that production

$$\begin{aligned} \text{Material A} &= 42,000 \text{ units} \times 3 \text{ units per unit} = \mathbf{1,26,000 \text{ units}} \\ \text{Material B} &= 42,000 \text{ units} \times 4 \text{ units per unit} = \mathbf{1,68,000 \text{ units}} \end{aligned}$$

Note the multiplication is by **42,000 units produced**, not by the 40,000 units sold. Materials are consumed in the factory, not in the sales office.

Material Procurement Budget (quantitative), in units

Particulars	Material A	Material B
Materials consumed <i>W2 — 42,000 × 3 and 42,000 × 4</i>	1,26,000	1,68,000
Add: Closing stock of material in hand	15,000	25,000
Add: Closing quantity on order <i>must be ordered during the year</i>	8,000	10,000
Total to be arranged	1,49,000	2,03,000
Less: Opening stock of material in hand	(12,000)	(20,000)
Less: Opening quantity on order <i>already ordered, will arrive anyway</i>	(7,000)	(11,000)
MATERIALS TO BE PROCURED	1,30,000	1,72,000

WHY THIS WORKS

The distinction this problem is really testing is between material **received** during the year and material **ordered** during the year. They are different numbers, and only one of them is a procurement budget:

	Material A	Material B	Comment
Consumption + closing stock – opening stock	1,29,000	1,73,000	the quantity that must be received during the year
Add: closing orders outstanding	8,000	10,000	ordered this year but arriving next year
Less: opening orders outstanding	(7,000)	(11,000)	ordered last year but arriving this year
Quantity to be ORDERED – the procurement budget	1,30,000	1,72,000	what the purchase department must actually do

Notice that the two adjustments push in opposite directions for the two materials. A ends the year with *more* on order than it began with, so 1,000 extra units must be ordered. B ends with *fewer*, so 1,000 fewer need be ordered. Both answers differ from the “received” figure, and both differ in a different direction.

Why a firm holds orders outstanding at all. Material on order is a buffer against lead time. If material A takes six weeks to arrive, the company must always have roughly six weeks’ consumption in the pipeline, or the factory stops. The procurement budget therefore has to plan the pipeline as well as the shelf — which is precisely what the four adjustments above do.

FINAL ANSWER

Budgeted production $(40,000 + 7,000 - 5,000)$ **42,000 units**

Material A consumed $(42,000 \times 3)$ **1,26,000 units**

Material B consumed $(42,000 \times 4)$ **1,68,000 units**

Material A to be procured **1,30,000 units**

Material B to be procured **1,72,000 units**

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The Sales Director of a manufacturing company reports that next year he expects to sell 50,000 units of a particular product. The Production Manager consults the store-keeper and casts his figures as follows.

Two kinds of raw materials A and B are required for manufacturing the product. Each unit of the product requires 2 units of A and 3 units of B. The estimated opening balances at the commencement of the next year are:

Particulars	Units
Finished products	10,000
Material A	12,000
Material B	15,000

The desirable closing balances at the end of the next year are:

Particulars	Units
Finished products	14,000
Material A	13,000
Material B	16,000

Draw up a quantitative chart showing the materials purchase budget for the next year.

READ THE QUESTION FIRST — WHAT TO SPOT

Q11 without the “materials on order” complication — so the formula has two adjustments, not four. Same three steps in the same order.

- **Production first.** $50,000 + 14,000 - 10,000 = 54,000$ units. The stock of finished goods is being increased by 4,000 units, so production must exceed sales by 4,000.
- **Then consumption**, at 2 units of A and 3 units of B for every unit produced — 54,000, not 50,000.
- **Then purchases**, adjusting only for the opening and closing stocks of the materials themselves.
- Both materials need a small stock increase of 1,000 units, so both purchase figures are 1,000 above consumption. That symmetry makes the answer easy to check.

WORKING NOTES

W1 Budgeted production

$$\begin{aligned} \text{Budgeted production} &= \text{Budgeted sales} + \text{Closing stock} - \text{Opening stock} \\ &= 50,000 + 14,000 - 10,000 = \mathbf{54,000 \text{ units}} \end{aligned}$$

W2 Materials consumed

$$\begin{aligned} \text{Material A} &= 54,000 \times 2 = \mathbf{1,08,000 \text{ units}} \\ \text{Material B} &= 54,000 \times 3 = \mathbf{1,62,000 \text{ units}} \end{aligned}$$

Materials Purchase Budget (quantitative), in units

Particulars	Material A	Material B
Materials consumed <i>W2 — 54,000 × 2 and 54,000 × 3</i>	1,08,000	1,62,000
Add: Desired closing stock of material	13,000	16,000
Total requirement	1,21,000	1,78,000
Less: Opening stock of material	(12,000)	(15,000)
MATERIALS TO BE PURCHASED	1,09,000	1,63,000

WHY THIS WORKS

Trace the chain from the Sales Director's single figure to the purchase order, because that chain is budgetary control:

Sales Director says → **50,000 units** will be sold
 Stock policy adds 4,000 → **54,000 units** must be produced
 Bill of materials multiplies by 2 and 3 → **1,08,000 A** and **1,62,000 B** will be consumed
 Material stock policy adds 1,000 each → **1,09,000 A** and **1,63,000 B** must be bought

Notice how the error would propagate. If you had used the 50,000 sales figure instead of the 54,000 production figure, material A would come out at 1,01,000 instead of 1,09,000 — a shortfall of 8,000 units, enough to stop production of 4,000 finished units. This is the single most common error in the topic, and the reason the two figures are never allowed to be confused.

The link worth stating in the answer: every figure above traces back to one sales forecast. That is the strength of budgetary control — one plan, consistently followed through every department — and also its weakness, because an error in the sales forecast is faithfully reproduced in the purchase order. This is why sales budgets are prepared with the most care and revised most often.

FINAL ANSWER

Budgeted production (50,000 + 14,000 – 10,000)	54,000 units
Material A consumed (54,000 × 2)	1,08,000 units
Material B consumed (54,000 × 3)	1,62,000 units
Material A to be purchased	1,09,000 units
Material B to be purchased	1,63,000 units

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company produces and sells three items: (a) Snow Cream, (b) Talcum Powder and (c) Cold Cream. The company has divided its market into two zones: Zone A and Zone B. The actual figures for the previous year's sales were as under:

Product	Zone A — Units	Zone A — Unit price ₹	Zone B — Units	Zone B — Unit price ₹
Snow Cream	4,00,000	12.00	2,50,000	12.00
Talcum Powder	2,50,000	15.00	3,50,000	15.00
Cold Cream	3,00,000	16.00	3,00,000	16.00

For the current year, it is estimated that sales of Snow Cream will go up by 10% in Zone B and of Cold Cream by 25,000 units in Zone A. The company plans to introduce a publicity film for Talcum Powder on the T.V. network. The budgeted figures for Talcum Powder are to be increased by 20% in both zones.

The prices of the two creams are to be maintained but for Talcum Powder a bonus cut of ₹1 will be announced.

You are required to prepare a quantitative-cum-financial budget for sales in the current year.

READ THE QUESTION FIRST — WHAT TO SPOT

“Quantitative-cum-financial” means show **both** the units and the value — two columns for every product. The difficulty is that each of the three products changes in a *different* way, and one of the changes applies to only one zone.

Product	Quantity change	Which zone	Price change
Snow Cream	up 10%	Zone B only — Zone A unchanged	none — “prices of the two creams are to be maintained”
Talcum Powder	up 20%	both zones	down ₹1 — the bonus cut, so ₹15 becomes ₹14
Cold Cream	up 25,000 units (not a percentage)	Zone A only — Zone B unchanged	none — a cream, so price maintained

- “**The two creams**” means Snow Cream and Cold Cream. Only Talcum Powder gets the price cut — and the phrase is the examiner's way of telling you so without saying it directly.
- **Cold Cream rises by an absolute 25,000 units, not by 25%.** Read the units of measurement; the workbook mixes percentages and absolutes on purpose.
- A “**bonus cut**” is a **price reduction**. ₹15 – ₹1 = ₹14, applied in **both** zones.
- Set out the answer zone by zone, with a total row. The zone totals are what a sales manager is actually held responsible for.

Quantitative-cum-Financial Budget for Sales — current year

Zone	Snow Cream at ₹12		Talcum Powder at ₹14		Cold Cream at ₹16	
	Units	Value ₹	Units	Value ₹	Units	Value ₹
Zone A	4,00,000 no change	48,00,000	3,00,000 2,50,000 × 1.20	42,00,000	3,25,000 3,00,000 + 25,000	52,00,000

Zone	Snow Cream at ₹12		Talcum Powder at ₹14		Cold Cream at ₹16	
Zone B	2,75,000 2,50,000 × 1.10	33,00,000	4,20,000 3,50,000 × 1.20	58,80,000	3,00,000 no change	48,00,000
TOTAL	6,75,000	81,00,000	7,20,000	1,00,80,000	6,25,000	1,00,00,000

Summary of the sales budget

Particulars	Units	Value ₹
Snow Cream	6,75,000	81,00,000
Talcum Powder	7,20,000	1,00,80,000
Cold Cream	6,25,000	1,00,00,000
TOTAL BUDGETED SALES	20,20,000	2,81,80,000
Zone A total	10,25,000	1,42,00,000
Zone B total	9,95,000	1,39,80,000

WHY THIS WORKS

The Talcum Powder decision deserves comment, because that is where the management question lies. The company is spending money on a television campaign *and* cutting the price by ₹1. Was it worth it?

Talcum Powder	Previous year	Current year budget	Change
Units — both zones	6,00,000	7,20,000	+1,20,000 (+20%)
Price per unit	15.00	14.00	−1.00 (−6.7%)
Sales value	90,00,000	1,00,80,000	+10,80,000 (+12%)

Sales value rises by ₹10,80,000, so on the face of it the plan works. But two cautions belong in the answer:

- **The revenue gain is not the profit gain.** Against the ₹10,80,000 the company must set the cost of the television film and the variable cost of making 1,20,000 extra units. Only if contribution exceeds the publicity cost is the plan sound — and this budget does not tell us, because a sales budget stops at revenue.
- **The price cut applies to ALL units, not only the extra ones.** The 6,00,000 units that would have sold anyway now yield ₹1 less each — ₹6,00,000 of revenue given away to win ₹16,80,000 of new business. That is the real trade the ₹10,80,000 conceals.

Also worth observing: the two zones are now almost exactly balanced — ₹1,42,00,000 against ₹1,39,80,000 — whereas the products are not. Cold Cream sells 3,25,000 units in Zone A but only 3,00,000 in Zone B, while Talcum Powder sells 4,20,000 in Zone B against 3,00,000 in Zone A. A zone-wise budget of this kind exists precisely to expose such differences, so that selling effort can be directed where a product is weak.

FINAL ANSWER

Snow Cream — Zone A / Zone B	4,00,000 / 2,75,000 units
Talcum Powder — Zone A / Zone B	3,00,000 / 4,20,000 units at ₹ 14
Cold Cream — Zone A / Zone B	3,25,000 / 3,00,000 units
Snow Cream total	6,75,000 units = ₹ 81,00,000
Talcum Powder total	7,20,000 units = ₹ 1,00,80,000
Cold Cream total	6,25,000 units = ₹ 1,00,00,000
Total budgeted sales	20,20,000 units = ₹ 2,81,80,000

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company requires you to prepare a Master Budget from the following:

Sales per annum: Product A — ₹40,00,000; Product B — ₹60,00,000.

Other information:

- Direct material cost = 40% of sales
- Direct wages: 15 workers @ ₹12,000 per month per worker
- Factory overheads: works manager's salary at ₹20,000 per month, foreman's salary at ₹15,000 per month
- Stores and spares: 2½% on sales; depreciation on machinery ₹1,26,000 per annum
- Light and power: ₹50,000 p.a.; Repairs and maintenance: ₹80,000 per annum; Miscellaneous expenses: 10% of direct wages
- Administration, selling and distribution expenses: ₹1,40,000 p.a.

READ THE QUESTION FIRST — WHAT TO SPOT

A master budget is simply a **budgeted profit and loss account**. Every item is given; the work is in converting each one to an annual figure on the right base.

Item as given	Base	Annual computation
Sales	given annually	₹40,00,000 + ₹60,00,000 = ₹1,00,00,000
Direct material	% of sales	40% × 1,00,00,000 = ₹40,00,000
Direct wages	per worker per MONTH	15 × 12,000 × 12 = ₹21,60,000
Works manager's salary	per month	20,000 × 12 = ₹2,40,000
Foreman's salary	per month	15,000 × 12 = ₹1,80,000
Stores and spares	% of sales	2.5% × 1,00,00,000 = ₹2,50,000
Miscellaneous expenses	% of direct WAGES	10% × 21,60,000 = ₹2,16,000
Depreciation, light and power, repairs, admin	already annual	take as given — 1,26,000 + 50,000 + 80,000 + 1,40,000

- **The two ×12 conversions are where the marks are lost.** Both salaries and the wages are quoted per month; everything else is already annual. Mixing them understates cost by lakhs.
- **Miscellaneous expenses are 10% of direct WAGES, not of sales.** Read the base attached to every percentage — this question uses three different bases (sales, sales, wages) in three consecutive lines.
- **Depreciation belongs in a master budget**, unlike in a cash budget. A master budget measures profit, and depreciation is a charge against profit.
- Product A and B are given only as sales values, with no separate cost data, so the master budget is prepared for the company as a whole. Show the two products in the sales line and then work with the total.

MASTER BUDGET — Budgeted Profit and Loss Account for the year

Particulars	Basis	₹	₹
Sales			
Product A	given	40,00,000	

Particulars	Basis	₹	₹
Product B	given	60,00,000	
Total Sales			1,00,00,000
Less: Cost of Goods Sold			
Direct materials	40% of sales	40,00,000	
Direct wages	15 workers × 12,000 × 12 months	21,60,000	
<i>Factory overheads</i>			
Works manager's salary	20,000 × 12	2,40,000	
Foreman's salary	15,000 × 12	1,80,000	
Stores and spares	2½% of sales	2,50,000	
Depreciation on machinery	given, per annum	1,26,000	
Light and power	given, per annum	50,000	
Repairs and maintenance	given, per annum	80,000	
Miscellaneous expenses	10% of direct wages	2,16,000	
Administration, selling and distribution	given, per annum	1,40,000	
Total Cost			(74,42,000)
BUDGETED PROFIT			25,58,000

WHY THIS WORKS

The master budget is the point at which every other budget in the module comes together. It answers the question the board actually asks: **if all these plans are carried out, what profit will we make?** Here the answer is ₹25,58,000, a margin of 25.58% on sales.

Analysis of the budgeted cost	₹	% of sales
Direct materials	40,00,000	40.00%
Direct wages	21,60,000	21.60%
Prime cost	61,60,000	61.60%
Factory overheads	11,42,000	11.42%
Administration, selling and distribution	1,40,000	1.40%
Total cost	74,42,000	74.42%
Budgeted profit	25,58,000	25.58%

What the percentage column is for. Once the budget is expressed as a percentage of sales it becomes a control document rather than a forecast. If the actual results come in with materials at 43% of sales instead of 40%, that is a ₹3,00,000 problem which can be investigated in the month it arises — and Module 6 is entirely about measuring such differences.

One structural observation worth a mark. Only two of the eleven cost lines vary with sales — direct materials and stores and spares, together 42.5% of sales. Everything else, including all the wages, is fixed for the year. So a 10% shortfall in sales would cut revenue by ₹10,00,000 but costs by only ₹4,25,000, taking profit down by ₹5,75,000 — more than a fifth. A high proportion of fixed cost makes budgeted profit very sensitive to the sales forecast, which is exactly the risk the master budget exists to make visible.

FINAL ANSWER

Total sales	₹ 1,00,00,000
Direct materials (40% of sales)	₹ 40,00,000
Direct wages (15 × 12,000 × 12)	₹ 21,60,000
Factory overheads	₹ 11,42,000
Administration, selling and distribution	₹ 1,40,000
Total budgeted cost	₹ 74,42,000
BUDGETED PROFIT	₹ 25,58,000
Profit as a percentage of sales	25.58%

Q15 The complete set of functional budgets, worked back from a profit target

Master budget · p.55

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

A company manufactures three products A, B and C and sells at ₹450, ₹550 and ₹650 per unit respectively. The ratio of sales in quantity of A, B and C is **1 : 2 : 4**. Other information for the three products:

Product	D	E	F	G	Skilled hours	Unskilled hours	Variable overheads p.u. ₹
A	1	10	2	8	6	8	9
B	1	2	14	10	4	6	11
C	1	6	10	2	3	6	7

The present purchase price per part is ₹45, ₹15, ₹15 and ₹5 for D, E, F and G respectively. The wage rate per hour is skilled worker ₹6 and unskilled ₹5.

The opening stock is 500, 1,000, 3,000, 1,500, 1,000, 20,000 and 10,000 for A, B, C, D, E, F and G. The closing stock of the product and the parts is **90% of the opening stock**.

The workers work for 8 hours a day for 25 days in a month. The share of fixed overhead per month for A, B and C comes to ₹15,75,000, ₹5,80,000 and ₹8,45,000 respectively. The yearly profit projected is ₹1,20,00,000.

Prepare the following budgets for the next fiscal month: **(a)** Sales budget in quantity and value; **(b)** Production budget; **(c)** Parts usage budget; **(d)** Purchase budget in quantity and value; **(e)** Man power budget showing labour hours and wages payable; **(f)** Master budget.

READ THE QUESTION FIRST — WHAT TO SPOT

The hardest problem in the module, but only because of its length. The trick is that **you are not told the sales quantity** — you must work it out backwards from the profit target. Do that first and the remaining five budgets fall out in order.

- 1. Build the cost per unit of each product** from the parts and the labour hours. Three small tables: raw material cost, labour cost, then total variable cost.
 - 2. Find the sales quantity from the profit target.** Let $A = x$, so $B = 2x$ and $C = 4x$. Compute total contribution in terms of x , set it equal to monthly fixed overhead plus monthly profit, and solve.
 - 3. Then the five budgets in strict order:** sales → production → parts usage → purchases → man power → master budget. Each one feeds the next.
- **“The yearly profit projected” but “budgets for the next fiscal MONTH”.** Divide ₹1,20,00,000 by 12 to get a monthly profit of ₹10,00,000. The fixed overhead figures are already monthly. Mixing the two periods is the single biggest trap in the question.
 - **Closing stock is 90% of OPENING stock** — so stock is being run down by 10%, and production will be slightly *less* than sales for every product.

- **8 hours × 25 days = 200 hours per worker per month.** Needed only for part (e), to convert hours into a number of workers.
- **Every product uses exactly 1 unit of part D,** which is a useful check: total D consumption must equal total units produced.

WORKING NOTES

W1 Raw material (parts) cost per unit of product

Part	Price ₹	A — nos.	A — ₹	B — nos.	B — ₹	C — nos.	C — ₹
D	45	1	45	1	45	1	45
E	15	10	150	2	30	6	90
F	15	2	30	14	210	10	150
G	5	8	40	10	50	2	10
Raw material cost per unit			265		335		295

W2 Labour cost per unit of product

Labour	Rate per hour ₹	A — hrs	A — ₹	B — hrs	B — ₹	C — hrs	C — ₹
Skilled	6	6	36	4	24	3	18
Unskilled	5	8	40	6	30	6	30
Labour cost per unit		14	76	10	54	9	48

W3 Total variable cost and contribution per unit

Particulars	A ₹	B ₹	C ₹
Raw material cost W1	265	335	295
Labour cost W2	76	54	48
Variable overheads given	9	11	7
Total variable cost per unit	350	400	350
Selling price per unit given	450	550	650
CONTRIBUTION PER UNIT	100	150	300

WORKING NOTES

W4 Finding the sales quantity from the profit target

Let the sales quantity of A be x . The ratio 1 : 2 : 4 then gives $B = 2x$ and $C = 4x$.

Product	Quantity	Contribution per unit ₹	Total contribution ₹
A	x	100	$100x$

Product	Quantity	Contribution per unit ₹	Total contribution ₹
B	2x	150	300x
C	4x	300	1,200x
Total	7x		1,600x

Total fixed overhead per month = 15,75,000 + 5,80,000 + 8,45,000 = **₹30,00,000**

Profit required per month = $\frac{₹1,20,00,000}{12 \text{ months}}$ = **₹10,00,000**

Contribution needed = Fixed overhead + Profit = 30,00,000 + 10,00,000 = **₹40,00,000**

$$1,600x = ₹40,00,000 \rightarrow x = \frac{40,00,000}{1,600} = \mathbf{2,500 \text{ units}}$$

So A = 2,500 units, B = 5,000 units and C = 10,000 units for the month.

(a) SALES BUDGET in quantity and value

Particulars	A	B	C	Total
Sales ratio given, 1 : 2 : 4	x	2x	4x	7x
Sales quantity (units) W4 — x = 2,500	2,500	5,000	10,000	17,500
Selling price per unit ₹	450	550	650	—
TOTAL BUDGETED SALES ₹	11,25,000	27,50,000	65,00,000	1,03,75,000

(b) PRODUCTION BUDGET, in units

Particulars	A	B	C
Budgeted sales from the sales budget	2,500	5,000	10,000
Add: Closing stock 90% of opening stock	450	900	2,700
Total requirement	2,950	5,900	12,700
Less: Opening stock given	(500)	(1,000)	(3,000)
BUDGETED PRODUCTION	2,450	4,900	9,700

Closing stock = 90% of opening: A = 90% × 500 = 450 · B = 90% × 1,000 = 900 · C = 90% × 3,000 = 2,700. Because stock is being reduced, production is **below** sales for every product.

(c) PARTS USAGE BUDGET, in numbers of parts

Product	Units produced	D — nos.	D — total	E — nos.	E — total	F — nos.	F — total	G — nos.	G — total
A	2,450	1	2,450	10	24,500	2	4,900	8	19,600
B	4,900	1	4,900	2	9,800	14	68,600	10	49,000
C	9,700	1	9,700	6	58,200	10	97,000	2	19,400
TOTAL PARTS REQUIRED	17,050		17,050		92,500		1,70,500		88,000

(d) PURCHASE BUDGET in quantity and value

Particulars	Part D	Part E	Part F	Part G
Parts consumed from the parts usage budget	17,050	92,500	1,70,500	88,000
Add: Closing stock 90% of opening stock	1,350	900	18,000	9,000
Total requirement	18,400	93,400	1,88,500	97,000
Less: Opening stock given: 1,500 / 1,000 / 20,000 / 10,000	(1,500)	(1,000)	(20,000)	(10,000)
PARTS TO BE PURCHASED (nos.)	16,900	92,400	1,68,500	87,000
Purchase price per part ₹	45	15	15	5
PURCHASE VALUE ₹	7,60,500	13,86,000	25,27,500	4,35,000
Total purchase value — all four parts				₹51,09,000

(e) MAN POWER BUDGET — labour hours and wages payable

Product	Units produced	Skilled — hrs p.u.	Skilled — total hrs	Unskilled — hrs p.u.	Unskilled — total hrs
A	2,450	6	14,700	8	19,600
B	4,900	4	19,600	6	29,400
C	9,700	3	29,100	6	58,200
TOTAL LABOUR HOURS			63,400		1,07,200
Wage rate per hour ₹			6		5
WAGES PAYABLE ₹			3,80,400		5,36,000
Hours available per worker per month 8 hours × 25 days			200		200
Number of workers required			317		536

Total wages payable = ₹3,80,400 + ₹5,36,000 = **₹9,16,400**. Total workers required = 317 skilled + 536 unskilled = **853**.
 Skilled workers: 63,400 ÷ 200 = 317; unskilled: 1,07,200 ÷ 200 = 536.

(f) MASTER BUDGET for the month

Particulars	A ₹	B ₹	C ₹	Total ₹
Sales (a) — quantity × selling price	11,25,000	27,50,000	65,00,000	1,03,75,000
Less: Variable cost of sales sales quantity × W3 variable cost	(8,75,000)	(20,00,000)	(35,00,000)	(63,75,000)
CONTRIBUTION 2,500×100, 5,000×150, 10,000×300	2,50,000	7,50,000	30,00,000	40,00,000
Less: Fixed overheads given, per month	(15,75,000)	(5,80,000)	(8,45,000)	(30,00,000)
BUDGETED PROFIT FOR THE MONTH	(13,25,000)	1,70,000	21,55,000	10,00,000

Particulars	A ₹	B ₹	C ₹	Total ₹
Budgeted profit for the year 10,00,000 × 12				1,20,00,000

The annual profit of ₹1,20,00,000 is reproduced exactly, which confirms that the sales quantity of 2,500 : 5,000 : 10,000 derived in W4 is correct. **That is the check to state.**

WHY THIS WORKS

Everything in this problem hangs on W4, and it is worth seeing why the question had to be built that way. A budget normally starts with a sales forecast; here the company has instead fixed a **profit target** and asked what sales are needed to reach it. So the usual chain is run in reverse for one step, and then forwards for five.

$$\text{Required contribution} = \text{Fixed cost} + \text{Target profit}$$

This is the Module 2 formula reappearing in a Module 5 setting. Contribution must cover the fixed cost before it can produce any profit, so the contribution needed is simply the two added together — here ₹30,00,000 + ₹10,00,000 = ₹40,00,000.

The master budget by product is the finding management most needs to see, and it is uncomfortable:

	A	B	C
Contribution ₹	2,50,000	7,50,000	30,00,000
Fixed overhead charged ₹	(15,75,000)	(5,80,000)	(8,45,000)
Profit / (Loss) ₹	(13,25,000)	1,70,000	21,55,000
Contribution per unit ₹	100	150	300
Share of total contribution	6.25%	18.75%	75.00%

Product A shows a loss of ₹13,25,000, and it is carrying more than half of the total fixed overhead while generating only 6.25% of the contribution. Two things must be said about this, and saying both is what separates a full-mark answer:

- 1. Do not conclude that A should be dropped.** A still earns a positive contribution of ₹2,50,000 a month. The ₹15,75,000 of fixed overhead is an *apportionment*, and most of it would simply be re-charged to B and C if A were discontinued — wiping out B's profit entirely. This is the Module 4 lesson applied to a Module 5 statement.
- 2. But do ask why A absorbs so much fixed overhead** for so little return. Either the apportionment basis is wrong, or A genuinely consumes a great deal of capacity — it uses 14 labour hours per unit against C's 9, and ₹265 of parts to earn ₹100 of contribution. If the ratio 1 : 2 : 4 could be shifted further towards C, profit would rise sharply without any change in cost structure.

Finally, note the manpower figure. 853 workers are needed for a single month — 317 skilled and 536 unskilled. A budget of this kind is what allows a personnel department to recruit and train in time, and it is the reason the man power budget is prepared in *hours* first and converted to *people* second: hours come from production, but people come from hours divided by the 200 hours each worker can offer.

FINAL ANSWER

Contribution per unit — A / B / C	₹ 100 / ₹ 150 / ₹ 300
Sales quantity (W4) — A / B / C	2,500 / 5,000 / 10,000 units
(a) Sales budget	17,500 units = ₹ 1,03,75,000
(b) Production budget — A / B / C	2,450 / 4,900 / 9,700 units
(c) Parts usage — D / E / F / G	17,050 / 92,500 / 1,70,500 / 88,000
(d) Purchases (nos.) — D / E / F / G	16,900 / 92,400 / 1,68,500 / 87,000
(d) Purchase value	₹ 51,09,000
(e) Labour hours — skilled / unskilled	63,400 / 1,07,200 hours

(e) Wages payable	₹ 9,16,400 (317 + 536 = 853 workers)
(f) Contribution / Fixed / Profit	₹ 40,00,000 – ₹ 30,00,000 = ₹ 10,00,000
(f) Profit for the year	₹ 1,20,00,000 ✓

Standard Costing and Variance Analysis

17 problems · workbook pages 79–84

What Module 6 is really about

A standard cost is what a thing *ought* to cost. Actual cost is what it *did* cost. The difference is a **variance**, and this module is entirely about splitting that one difference into its causes, so that each cause can be put in front of the manager responsible for it.

THE TWO CONVENTIONS THAT MUST BE RIGHT BEFORE ANYTHING ELSE

For COSTS: $\text{Variance} = \text{STANDARD} - \text{ACTUAL}$

A positive answer means actual cost was LOWER than standard, which is good — mark it **(F) favourable**. A negative answer means actual cost was higher — mark it **(A) adverse**.

For SALES and PROFIT: $\text{Variance} = \text{ACTUAL} - \text{BUDGET}$

The subtraction reverses, because more revenue is good whereas more cost is not. Q16 and Q17 use this convention; Q1 to Q15 use the one above.

Every single figure you write must carry an (F) or an (A). An untagged variance is meaningless — and in an examination it earns no marks, because the sign is the answer.

The seven symbols — learn these and the formulas write themselves

Symbol	Stands for	How to compute it
SQ / SH	Standard Quantity / Hours for the actual output	Standard per unit × actual units produced. Never the standard for the standard output — this is the single most common error in the module.
SP / SR	Standard Price / Rate	Given
AQ / AH	Actual Quantity / Hours	Given. For labour, hours paid .
AP / AR	Actual Price / Rate	Given, or actual cost ÷ actual quantity
RSQ / RSH	Revised Standard Quantity / Hours	Total actual input re-split in the standard proportion . It answers: “if we had used this much input but in the right mix, how much of each would it be?”

Symbol	Stands for	How to compute it
BH	Budgeted Hours	Only for fixed overhead — the hours in the original budget
RBQ	Revised Budgeted Quantity	Sales version of RSQ — total actual quantity sold, re-split in the budgeted mix

RSQ is the one that has to be understood rather than memorised, because it is what makes mix and yield separable:

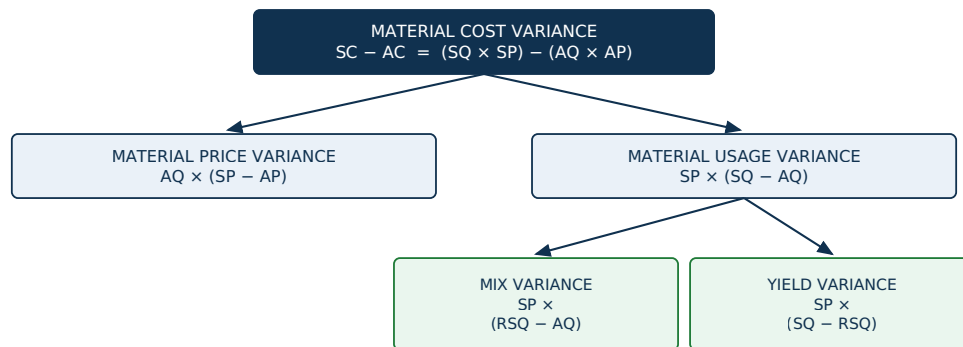
$$\text{RSQ for a material} = \frac{\text{Total actual quantity of ALL materials} \times \text{Standard quantity of THIS material}}{\text{Total standard quantity of ALL materials}}$$

The total of the RSQ column always equals the total of the AQ column. If it does not, you have made an arithmetic error - and that is a free self-check on every mix problem.

The variance trees

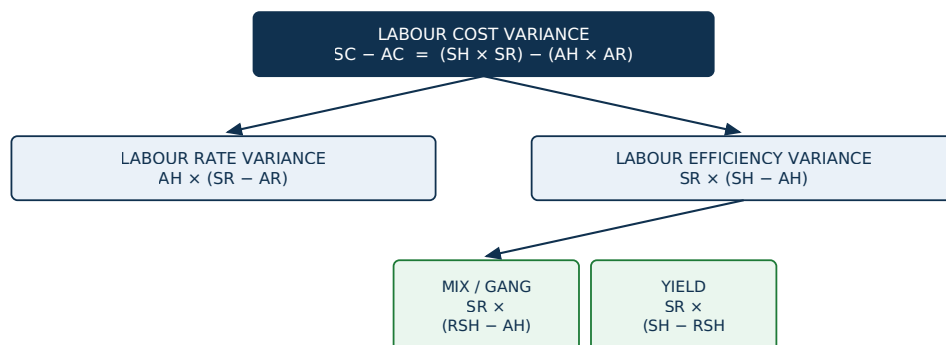
Each tree is read the same way: the box at the top is the total, and it always equals the sum of the boxes below it. Those identities are the **reconciliation** that every question in this module asks for.

Material variances — Q1 to Q7



Material variances. Price is a purchasing responsibility; usage is a production responsibility. Usage then splits into mix and yield.

Labour variances — Q8 to Q10

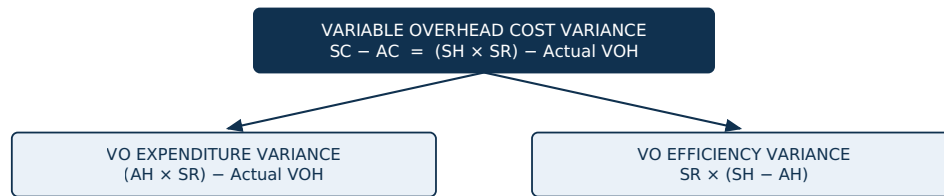


hours × SR(always adverse)

worked)IDLE TIMEIdle

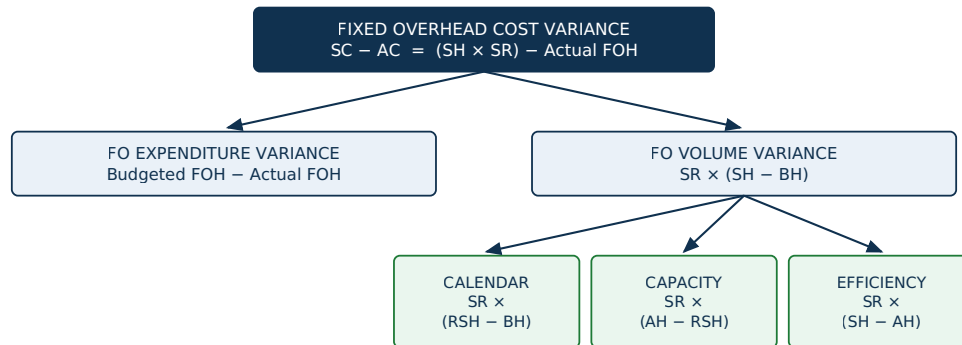
Labour variances. Note that labour has one extra branch that material does not have — idle time, because labour is paid for hours it does not work.

Variable overhead variances – Q11 to Q13



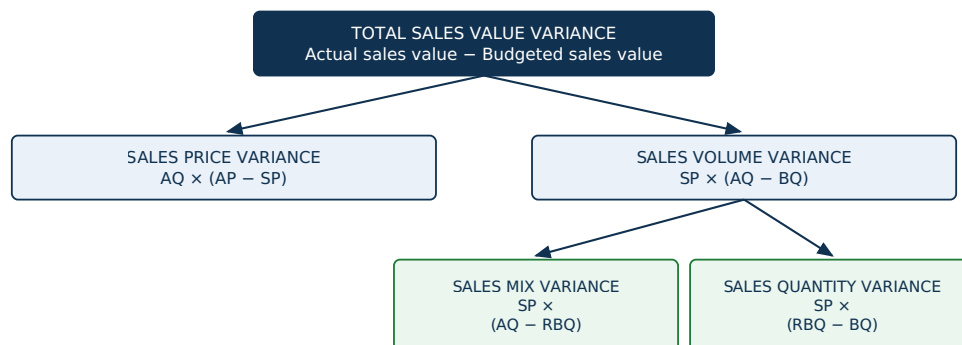
Variable overhead has only two branches, because variable overhead is assumed to move with hours.

Fixed overhead variances – Q14 and Q15



Fixed overhead variances. The volume variance exists only because fixed overhead is absorbed per hour although the total does not change with hours.

Sales variances – Q16 and Q17



Sales variances. Note the subtraction is reversed — ACTUAL minus BUDGET — because more revenue is favourable, whereas more cost is not.

The reconciliations you must be able to write from memory

Family	Reconciliation	And then
Material	MCV = MPV + MUV	MUV = MMV + MYV
Labour	LCV = LRV + LEV	LEV = LMV + LYV + LITV
Variable overhead	VOCV = VOEXPV + VOEFFV	—
Fixed overhead	FOCV = FOEXPV + FOVV	FOVV = Calendar + Capacity + Efficiency
Sales value	Total = Price + Volume	Volume = Mix + Quantity
Sales margin	Total = Margin price + Margin volume	Margin volume = Margin mix + Margin quantity

Use the reconciliation as your checking tool, not as an afterthought. Compute the total variance first and the branches second; if the branches do not add to the total you have made an error, and you will know before the examiner does.

The five problem types

Type	How to recognise it	Problems
Material variances	Materials in a mix, with a standard loss or a standard yield.	1-7
Labour variances	A “gang” of skilled / semi-skilled / unskilled workers, usually with idle time.	8-10
Variable overhead variances	Budgeted and actual variable overhead, with a standard time per unit.	11-13
Fixed overhead variances	Budgeted and actual fixed overhead; working days or man-hours per day.	14, 15
Sales variances	Budgeted and actual quantities and selling prices by product.	16, 17

THE TRAP / WHERE MARKS ARE LOST

- **Using the standard quantity for the STANDARD output instead of for the ACTUAL output.** If the standard is 1,100 kg of input for 1,000 kg of output and actual output was 20,000 kg, then SQ is 22,000 kg — twenty times the standard. Everything downstream is wrong if this is wrong.
- **Forgetting the (F) / (A) tag, or getting it backwards.** Remember: for costs, a positive answer is favourable because standard exceeded actual.
- **Rounding RSQ too early.** RSQ is often a recurring decimal. Carry at least two decimals, or the mix and yield variances will not add back to the usage variance. This is the reason some published solutions to Q1 show ₹26,365 instead of the correct ₹26,363.64.
- **Confusing hours PAID with hours WORKED in labour problems.** AH is hours **paid**. Hours worked = hours paid – idle hours, and it is used only for the yield variance and to show that the idle-time variance closes the reconciliation.
- **Treating an idle-time variance as favourable.** It is **always adverse** — the company paid for hours in which nothing was produced. Even when the arithmetic yields a positive number, the situation is adverse; say so.
- **Keeping the cost convention for sales variances.** Q16 and Q17 reverse it to actual minus budget. A favourable sales price variance means you charged MORE than standard.

Q1 All five material variances, with a standard loss

Material · p.79

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The standard material inputs required for 1,000 kg of a finished product are given below:

Material	Quantity in kg	Standard rate per kg ₹
P	450	20
Q	400	40
R	250	60
Total	1,100	
Standard loss	100	
Standard output	1,000	

Actual production in a period was 20,000 kg of the finished product, for which the actual quantities of material used and the prices paid thereof are as under:

Material	Quantity in kg	Purchase price per kg ₹
P	10,000	19
Q	8,500	42
R	4,500	65
Total	23,000	

Calculate the: **(a)** Material Cost Variance **(b)** Material Price Variance **(c)** Material Usage Variance **(d)** Material Mix Variance **(e)** Material Yield Variance.

READ THE QUESTION FIRST — WHAT TO SPOT

Before any formula, build the **Basic Calculation** table. Every one of the five variances is then read straight out of it.

- **The standard is for 1,000 kg of output but actual output was 20,000 kg**, so every standard quantity must be multiplied by 20. SQ is 9,000, 8,000 and 5,000 — **not** 450, 400 and 250. Getting this wrong invalidates the whole answer, and it is the reason the column is headed “SQ for actual output”.
- **The standard loss of 100 kg is already built into the numbers.** 1,100 kg of input gives 1,000 kg of output, so the standard is 1.1 kg of input per kg of output. Do not adjust for the loss a second time.
- **Total AQ is 23,000 kg against total SQ of 22,000 kg.** A thousand kilos more input was used than the standard allows, so the usage variance will be adverse before you compute anything.
- **RSQ splits the 23,000 kg actually used in the standard ratio 450 : 400 : 250.** It answers “if we had used 23,000 kg but in the correct proportions, how much of each?” The answers are recurring decimals — keep two places.

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
P	450	9,000	20	10,000	19	9,409.09
Q	400	8,000	40	8,500	42	8,363.64

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
R	250	5,000	60	4,500	65	5,227.27
Total	1,100	22,000		23,000		23,000

Two checks before going on. The RSQ column totals 23,000, which equals the AQ total of 23,000 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 20, the ratio of actual output 20,000 to standard output 1,000.

1. MATERIAL COST VARIANCE = SC – AC = (SQ × SP) – (AQ × AP)

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
P	9,000	20	1,80,000	10,000	19	1,90,000
Q	8,000	40	3,20,000	8,500	42	3,57,000
R	5,000	60	3,00,000	4,500	65	2,92,500
Total			8,00,000			8,39,500

$$\text{MCV} = ₹8,00,000 - ₹8,39,500 = \mathbf{39,500 \text{ (A)}}$$

MATERIAL PRICE VARIANCE = AQ × (SP – AP)

Material	SP	AP	SP – AP	AQ	Amount ₹
P	20	19	1	10,000	10,000 (F)
Q	40	42	(2)	8,500	17,000 (A)
R	60	65	(5)	4,500	22,500 (A)
MATERIAL PRICE VARIANCE					29,500 (A)

MATERIAL USAGE VARIANCE = SP × (SQ – AQ)

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
P	9,000	10,000	(1,000)	20	20,000 (A)
Q	8,000	8,500	(500)	40	20,000 (A)
R	5,000	4,500	500	60	30,000 (F)
MATERIAL USAGE VARIANCE					10,000 (A)

MATERIAL MIX VARIANCE = SP × (RSQ – AQ)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
P	9,409.09	10,000	(590.91)	20	11,818.18 (A)
Q	8,363.64	8,500	(136.36)	40	5,454.55 (A)
R	5,227.27	4,500	727.27	60	43,636.36 (F)
MATERIAL MIX VARIANCE					26,363.64 (F)

MATERIAL YIELD VARIANCE = SP × (SQ – RSQ)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
P	9,000	9,409.09	(409.09)	20	8,181.82 (A)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
Q	8,000	8,363.64	(363.64)	40	14,545.45 (A)
R	5,000	5,227.27	(227.27)	60	13,636.36 (A)
MATERIAL YIELD VARIANCE					36,363.64 (A)

WORKING NOTES

Independent check on the yield variance

$$\text{Standard cost per unit of output} = \frac{\text{standard cost of one standard batch}}{\text{standard output of that batch}} = ₹40$$

$$\text{Output the actual input SHOULD have yielded} = 23,000 \times \frac{1,000}{1,100} = 20,909.09 \text{ units}$$

$$\text{Actual output} = 20,000 \text{ units}$$

$$\text{MYV} = ₹40 \times (20,000 - 20,909.09) = 36,363.64 \text{ (A)}$$

This route never touches RSQ, so it is free of rounding error and is the best way to **check** the yield variance computed material by material above. The two agree exactly.

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	29,500 (A) + 10,000 (A)	39,500 (A)
and MCV computed directly was		39,500 (A) ✓
MUV = MMV + MYV	26,363.64 (F) + 36,363.64 (A)	10,000 (A)
and MUV computed directly was		10,000 (A) ✓

WHY THIS WORKS

Read the five answers together and they tell a single connected story about the period.

Variance	Result	What it says
Price	29,500 (A)	Materials cost more than standard overall. P was bought ₹1 cheaper, but Q was ₹2 dearer and R was ₹5 dearer, and those two swamped the saving on P.
Mix	26,363.64 (F)	Favourable. The mix was shifted towards the cheap material — more P at ₹20 and less R at ₹60 than the standard proportions require. Substituting cheap material for expensive saves money.
Yield	36,363.64 (A)	Adverse, and larger than the mix gain. 23,000 kg of input produced only 20,000 kg of output, whereas the standard yield would be 20,909.09 kg.
Usage	10,000 (A)	The net of the two above — the mix saving did not cover the yield loss.
Cost	39,500 (A)	Price plus usage. The overall verdict on materials for the period.

The mix and yield variances point at the same decision. Someone economised by using more of the cheap material P and less of the expensive material R, saving ₹26,363.64. But the resulting mixture yielded badly, and the shortfall in output cost ₹36,363.64 — nearly ₹10,000 more than the saving. **That is the finding to report:** the cheaper mix was a false economy, and the standard proportions should be restored.

A note on the arithmetic. Some published solutions to this problem give the mix variance as ₹26,365.60 (F) and the yield variance as ₹36,365.60 (A), because they round RSQ for Q to 8,363.69 instead of 8,363.64. The correct figures are **26,363.64 (F)** and **36,363.64 (A)**, and you can prove it: they must differ by exactly the usage variance of ₹10,000, and only the unrounded pair does.

FINAL ANSWER

SQ for actual output — P / Q / R	9,000 / 8,000 / 5,000 kg
RSQ for actual input — P / Q / R	9,409.09 / 8,363.64 / 5,227.27 kg
Standard cost of actual output	₹ 8,00,000
Actual cost	₹ 8,39,500
Material Cost Variance	39,500 (A)
Material Price Variance	29,500 (A)
Material Usage Variance	10,000 (A)
Material Mix Variance	26,363.64 (F)
Material Yield Variance	36,363.64 (A)
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

Q2 Material variances when the mix is given in percentages

Material · p.79

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

XYZ Company manufactures a product ABC by mixing three raw materials. For every 100 kg of ABC, 125 kg of materials are used. In April 2017 there was an output of 5,600 kg of ABC. The standard and actual particulars of April 2017 are as under:

Raw material	Standard — mix %	Standard — price per kg ₹	Actual — mix %	Actual — price per kg ₹
X	50	40	60	42
Y	30	20	20	16
Z	20	10	20	12

Calculate all the material variances.

READ THE QUESTION FIRST — WHAT TO SPOT

The quantities are given as **percentages**, not kilograms. Convert them before anything else, and the conversion needs one number the question hides in a sentence.

- Find the total actual input.** “For every 100 kg of ABC, 125 kg of materials are used”, and output was 5,600 kg. So input = $5,600 \times \frac{125}{100} = 7,000 \text{ kg}$. This single figure unlocks the problem.
 - Apply the actual mix percentages to 7,000 kg** to get AQ: X = 60% = 4,200 kg, Y = 20% = 1,400 kg, Z = 20% = 1,400 kg.
 - Apply the standard mix percentages to the standard 125 kg batch** to get the standard mix: X = 62.5, Y = 37.5, Z = 25 kg per 100 kg of output.
 - Scale that standard up to the actual output of 5,600 kg** (a factor of 56) to get SQ: X = 3,500, Y = 2,100, Z = 1,400 kg — a total of 7,000 kg.
- Note what has just happened:** total SQ = 7,000 kg and total AQ = 7,000 kg. The yield was exactly standard, so the **yield variance will be nil** and the whole of the usage variance is mix. That is a strong check on your arithmetic before you compute anything.
 - The standard loss is 25 kg per 125 kg of input**, i.e. 20% of input. It is already inside the 125 : 100 relationship — do not deduct it again.

WORKING NOTES

W1 Converting the percentages into kilograms

$$\text{Total actual input} = 5,600 \text{ kg output} \times \frac{125}{100} = \mathbf{7,000 \text{ kg}}$$

Scale factor from the standard 100 kg batch to the actual 5,600 kg = **56 times**

Material	Standard mix % of 125 kg	Standard kg per 100 kg output	SQ for 5,600 kg (× 56)	Actual mix % of 7,000 kg	AQ in kg
X	50%	62.50	3,500	60%	4,200
Y	30%	37.50	2,100	20%	1,400
Z	20%	25.00	1,400	20%	1,400
Total	100%	125.00	7,000	100%	7,000

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
X	62.50	3,500	40	4,200	42	3,500
Y	37.50	2,100	20	1,400	16	2,100
Z	25	1,400	10	1,400	12	1,400
Total	125	7,000		7,000		7,000

Two checks before going on. The RSQ column totals 7,000, which equals the AQ total of 7,000 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 56, the ratio of actual output 5,600 to standard output 100.

$$\mathbf{1. \text{ MATERIAL COST VARIANCE} = \text{SC} - \text{AC} = (\text{SQ} \times \text{SP}) - (\text{AQ} \times \text{AP})}$$

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
X	3,500	40	1,40,000	4,200	42	1,76,400
Y	2,100	20	42,000	1,400	16	22,400
Z	1,400	10	14,000	1,400	12	16,800
Total			1,96,000			2,15,600

$$\mathbf{MCV = ₹1,96,000 - ₹2,15,600 = 19,600 (A)}$$

$$\mathbf{\text{MATERIAL PRICE VARIANCE} = \text{AQ} \times (\text{SP} - \text{AP})}$$

Material	SP	AP	SP - AP	AQ	Amount ₹
X	40	42	(2)	4,200	8,400 (A)
Y	20	16	4	1,400	5,600 (F)
Z	10	12	(2)	1,400	2,800 (A)
MATERIAL PRICE VARIANCE					5,600 (A)

$$\mathbf{\text{MATERIAL USAGE VARIANCE} = \text{SP} \times (\text{SQ} - \text{AQ})}$$

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
X	3,500	4,200	(700)	40	28,000 (A)
Y	2,100	1,400	700	20	14,000 (F)
Z	1,400	1,400	—	10	Nil
MATERIAL USAGE VARIANCE					14,000 (A)

MATERIAL MIX VARIANCE = SP × (RSQ – AQ)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
X	3,500	4,200	(700.00)	40	28,000 (A)
Y	2,100	1,400	700.00	20	14,000 (F)
Z	1,400	1,400	—	10	Nil
MATERIAL MIX VARIANCE					14,000 (A)

MATERIAL YIELD VARIANCE = SP × (SQ – RSQ)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
X	3,500	3,500	—	40	Nil
Y	2,100	2,100	—	20	Nil
Z	1,400	1,400	—	10	Nil
MATERIAL YIELD VARIANCE					Nil

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	5,600 (A) + 14,000 (A)	19,600 (A)
and MCV computed directly was		19,600 (A) ✓
MUV = MMV + MYV	14,000 (A) + Nil	14,000 (A)
and MUV computed directly was		14,000 (A) ✓

WHY THIS WORKS

This problem is built to isolate the **mix** variance, and the way it does so is worth understanding because it makes the concept unmistakable.

Total SQ for actual output = **7,000 kg**

Total AQ actually used = **7,000 kg**

So the total input was exactly right → **Yield variance = Nil**

Therefore the whole usage variance is mix → MUV = MMV = **14,000 (A)**

The company used exactly the right *amount* of material and got exactly the standard output from it. What it got wrong was the **proportions**: 60% X instead of 50%, and only 20% Y instead of 30%. Since X costs ₹40 a kg and Y only ₹20, substituting the dearer material for the cheaper one cost money even though not one extra kilogram was consumed.

Material	Standard mix	Actual mix	Price ₹	Effect
X	50%	60%	40	10% more of the dearest material — adverse
Y	30%	20%	20	10% less of a mid-priced material — favourable

Material	Standard mix	Actual mix	Price ₹	Effect
Z	20%	20%	10	unchanged — no mix effect

The price variance is also adverse, but much smaller, at **5,600 (A)** against **14,000 (A)** on the mix. X and Z were both bought above standard price, but Y was bought ₹4 below it and 1,400 kg of Y was used, so the saving on Y recovered most of the loss:

$$X = 4,200 \text{ kg} \times (\text{₹}40 - \text{₹}42) = \textbf{8,400 (A)}$$

$$Y = 1,400 \text{ kg} \times (\text{₹}20 - \text{₹}16) = \textbf{5,600 (F)}$$

$$Z = 1,400 \text{ kg} \times (\text{₹}10 - \text{₹}12) = \textbf{2,800 (A)}$$

$$\textbf{Net} = \textbf{5,600 (A)}$$

So the purchasing department did comparatively well — only ₹5,600 adverse on ₹2,15,600 of spend — while the mixing operation lost more than twice as much. That is exactly the separation of responsibility that variance analysis exists to produce.

Overall verdict: a cost variance of **19,600 (A)**. Report the mix variance of **14,000 (A)** to the production manager and ask why the recipe was altered; report the price variance of **5,600 (A)** to the purchasing manager as a point in their favour.

FINAL ANSWER

Total actual input (5,600 × 125/100)	7,000 kg
SQ — X / Y / Z	3,500 / 2,100 / 1,400 kg
AQ — X / Y / Z	4,200 / 1,400 / 1,400 kg
Standard cost of actual output	₹ 1,96,000
Actual cost	₹ 2,15,600
Material Cost Variance	19,600 (A)
Material Price Variance	5,600 (A)
Material Usage Variance	14,000 (A)
Material Mix Variance	14,000 (A)
Material Yield Variance	Nil
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The standard cost of a certain chemical mixture is as under:

- 40% of Material A at ₹20 per tonne
- 60% of Material B at ₹30 per tonne
- A standard loss of 10% is expected in production

The following actual cost data is given for the period:

Material	Quantity in tonnes	Cost per tonne ₹
A	180	18
B	220	34
Total input	400	

The weight produced is 364 tonnes. Calculate the: **(a)** Material Cost Variance **(b)** Material Price Variance **(c)** Material Usage Variance **(d)** Material Mix Variance **(e)** Material Yield Variance.

READ THE QUESTION FIRST — WHAT TO SPOT

The standard is given only as percentages and a loss rate, so you must **invent a convenient standard batch** to work with. Take 100 tonnes of input — it makes every percentage a whole number.

Standard batch of input = **100 tonnes** → A = 40 tonnes, B = 60 tonnes

Standard loss = 10% of input = 10 tonnes

Standard output = 100 – 10 = 90 tonnes

Scale factor to the actual output of 364 tonnes = $\frac{364}{90} = \mathbf{4.0444 \text{ times}}$

- **The standard loss is 10% of INPUT, not of output.** So 100 tonnes in gives 90 tonnes out, and the standard is $\frac{100}{90} = 1.1111$ tonnes of input per tonne of output. Reading it as 10% of output would give 90.9 tonnes and every answer would be wrong.
- **SQ therefore comes out in decimals** — A = $40 \times 4.0444 = 161.78$ tonnes and B = $60 \times 4.0444 = 242.67$ tonnes, a total of 404.44 tonnes. Keep two decimals throughout.
- **Actual input was 400 tonnes for 364 tonnes of output**, a loss of 36 tonnes or 9% of input — *better* than the 10% standard. So expect a **favourable yield variance**.
- **But the mix was wrong:** 180 : 220 is 45% : 55% against a standard 40% : 60%. More of the cheap A and less of the dear B, so expect a **favourable mix variance** too.

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
A	40	161.78	20	180	18	160
B	60	242.67	30	220	34	240
Total	100	404.44		400		400

Two checks before going on. The RSQ column totals 400, which equals the AQ total of 400 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 4.04, the ratio of actual output 364 to standard output 90.

1. MATERIAL COST VARIANCE = SC – AC = (SQ × SP) – (AQ × AP)

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
A	161.78	20	3,235.56	180	18	3,240
B	242.67	30	7,280	220	34	7,480
Total			10,515.56			10,720

$$\text{MCV} = ₹10,515.56 - ₹10,720 = \text{204.44 (A)}$$

MATERIAL PRICE VARIANCE = AQ × (SP – AP)

Material	SP	AP	SP – AP	AQ	Amount ₹
A	20	18	2	180	360 (F)
B	30	34	(4)	220	880 (A)
MATERIAL PRICE VARIANCE					520 (A)

MATERIAL USAGE VARIANCE = SP × (SQ – AQ)

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
A	161.78	180	(18.22)	20	364.44 (A)
B	242.67	220	22.67	30	680 (F)
MATERIAL USAGE VARIANCE					315.56 (F)

MATERIAL MIX VARIANCE = SP × (RSQ – AQ)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
A	160	180	(20.00)	20	400 (A)
B	240	220	20.00	30	600 (F)
MATERIAL MIX VARIANCE					200 (F)

MATERIAL YIELD VARIANCE = SP × (SQ – RSQ)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
A	161.78	160	1.78	20	35.56 (F)
B	242.67	240	2.67	30	80 (F)
MATERIAL YIELD VARIANCE					115.56 (F)

WORKING NOTES

Independent check on the yield variance

$$\text{Standard cost per unit of output} = \frac{\text{standard cost of one standard batch}}{\text{standard output of that batch}} = ₹28.89$$

$$\text{Output the actual input SHOULD have yielded} = 400 \times \frac{90}{100} = \text{360 units}$$

$$\text{Actual output} = \text{364 units}$$

$$\text{MYV} = ₹28.89 \times (364 - 360) = 115.56 \text{ (F)}$$

This route never touches RSQ, so it is free of rounding error and is the best way to **check** the yield variance computed material by material above. The two agree exactly.

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	520 (A) + 315.56 (F)	204.44 (A)
and MCV computed directly was		204.44 (A) ✓
MUV = MMV + MYV	200 (F) + 115.56 (F)	315.56 (F)
and MUV computed directly was		315.56 (F) ✓

WHY THIS WORKS

This is the one problem in the set where **both** parts of the usage variance are favourable, and it is instructive to see why.

	Standard	Actual	Verdict
Proportion of A — the cheap material at ₹20	40%	45%	more of the cheap one
Proportion of B — the dear material at ₹30	60%	55%	less of the dear one
Loss as a percentage of input	10%	9%	better yield than standard
Input consumed for 364 tonnes of output	404.44 t	400.00 t	4.44 tonnes less input than allowed

So the works manager did well twice over: a cheaper recipe *and* less waste. Together the mix and yield variances are worth **315.56 (F)**.

The cost variance is nevertheless 204.44 (A), because the price variance of **520 (A)** more than cancels the usage gain:

$$\text{Material A} = 180 \text{ t} \times (\text{₹}20 - \text{₹}18) = 360 \text{ (F)}$$

$$\text{Material B} = 220 \text{ t} \times (\text{₹}30 - \text{₹}34) = 880 \text{ (A)}$$

$$\text{Net price variance} = 520 \text{ (A)}$$

B was bought ₹4 a tonne above standard — a 13% overrun on the material that makes up more than half the mixture. That single fact turns a well-run production month into an adverse result overall.

The management point worth making. There may be a connection between the two. If B became scarce and its price rose ₹4, the works manager may have substituted A deliberately — and if so, the favourable mix variance is a *response* to the adverse price variance, not an independent achievement. Variances are computed separately but they are not always caused separately, and saying so is the mark of a real answer rather than an arithmetic one.

FINAL ANSWER

Standard batch assumed	100 tonnes input → 90 tonnes output
SQ for 364 tonnes — A / B	161.78 / 242.67 tonnes
Actual loss vs standard loss	9% of input vs 10% — better
Standard cost of actual output	₹ 10,515.56
Actual cost	₹ 10,720
Material Cost Variance	204.44 (A)
Material Price Variance	520 (A)
Material Usage Variance	315.56 (F)

Material Mix Variance	200 (F)
Material Yield Variance	115.56 (F)
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

Q4 Material variances when only the total cost is given

Material · p.80

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information has been extracted from the records of a chemical company:

- **Standard price:** Raw Material A = ₹2 per kg; Raw Material B = ₹10 per kg
- **Standard mix:** A = 75% and B = 25% (by weight)
- **Standard yield:** 90%

In a period of actual costs, usages and output were as follows:

Used	Quantity	Cost ₹
Material A	2,200 kg	4,650
Material B	800 kg	7,850
Total	3,000 kg	12,500

Output: 2,850 kg of product. Calculate the material cost variances. Also reconcile the material cost variances.

READ THE QUESTION FIRST — WHAT TO SPOT

Two conversions are needed before the standard tables can be built, and each is a common stumbling point.

1. **The actual PRICE is not given — only the total cost.** Divide:

$$A = \frac{₹4,650}{2,200 \text{ kg}} = \text{₹2.1136 per kg} \text{ (against a standard of ₹2)}$$

$$B = \frac{₹7,850}{800 \text{ kg}} = \text{₹9.8125 per kg} \text{ (against a standard of ₹10)}$$

2. **“Standard yield 90%” means 90% of INPUT.** Take a standard batch of 100 kg of input: A = 75 kg, B = 25 kg, and output = 90 kg. So the scale factor to the actual output of 2,850 kg is $\frac{2,850}{90} = \mathbf{31.6667}$.

- **SQ therefore comes out in decimals** — A = 2,375 kg and B = 791.67 kg, totalling 3,166.67 kg. Note that this is **more** than the 3,000 kg actually used, so the overall usage variance will be favourable.
- **But look at the mix.** Actual was 2,200 : 800, which is 73.33% : 26.67% against a standard of 75% : 25%. More of the **expensive** B at ₹10 and less of the cheap A at ₹2 — so expect an **adverse mix variance**.
- **And the yield.** 3,000 kg of input should have yielded 2,700 kg; it yielded 2,850 kg, which is 95% against a standard 90%. A clearly **favourable yield variance**.
- So this problem has a favourable yield fighting an adverse mix — the exact opposite of Q1.

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
A	75	2,375	2	2,200	2.11	2,250

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
B	25	791.67	10	800	9.81	750
Total	100	3,166.67		3,000		3,000

Two checks before going on. The RSQ column totals 3,000, which equals the AQ total of 3,000 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 31.67, the ratio of actual output 2,850 to standard output 90.

1. MATERIAL COST VARIANCE = SC – AC = (SQ × SP) – (AQ × AP)

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
A	2,375	2	4,750	2,200	2.11	4,650
B	791.67	10	7,916.67	800	9.81	7,850
Total			12,666.67			12,500

$$\text{MCV} = ₹12,666.67 - ₹12,500 = 166.67 \text{ (F)}$$

MATERIAL PRICE VARIANCE = AQ × (SP – AP)

Material	SP	AP	SP – AP	AQ	Amount ₹
A	2	2.11	(0.11)	2,200	250 (A)
B	10	9.81	0.19	800	150 (F)
MATERIAL PRICE VARIANCE					100 (A)

MATERIAL USAGE VARIANCE = SP × (SQ – AQ)

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
A	2,375	2,200	175	2	350 (F)
B	791.67	800	(8.33)	10	83.33 (A)
MATERIAL USAGE VARIANCE					266.67 (F)

MATERIAL MIX VARIANCE = SP × (RSQ – AQ)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
A	2,250	2,200	50.00	2	100 (F)
B	750	800	(50.00)	10	500 (A)
MATERIAL MIX VARIANCE					400 (A)

MATERIAL YIELD VARIANCE = SP × (SQ – RSQ)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
A	2,375	2,250	125.00	2	250 (F)
B	791.67	750	41.67	10	416.67 (F)
MATERIAL YIELD VARIANCE					666.67 (F)

WORKING NOTES

Independent check on the yield variance

$$\text{Standard cost per unit of output} = \frac{\text{standard cost of one standard batch}}{\text{standard output of that batch}} = \text{₹4.44}$$

$$\text{Output the actual input SHOULD have yielded} = 3,000 \times \frac{90}{100} = \text{2,700 units}$$

$$\text{Actual output} = \text{2,850 units}$$

$$\text{MYV} = \text{₹4.44} \times (2,850 - 2,700) = \text{666.67 (F)}$$

This route never touches RSQ, so it is free of rounding error and is the best way to **check** the yield variance computed material by material above. The two agree exactly.

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	100 (A) + 266.67 (F)	166.67 (F)
<i>and MCV computed directly was</i>		166.67 (F) ✓
MUV = MMV + MYV	400 (A) + 666.67 (F)	266.67 (F)
<i>and MUV computed directly was</i>		266.67 (F) ✓

WHY THIS WORKS

The overall result is **166.67 (F)** — a rare favourable cost variance, and it is worth tracing where it comes from because three of the five variances are adverse.

Variance	Result	Cause
Price	100 (A)	A cost ₹0.1136 a kg above standard (250 (A)), but B was bought ₹0.1875 below standard (150 (F)). The net is small.
Mix	400 (A)	Too much of the expensive B. Only 73.33% A instead of 75%, and B is five times the price of A.
Yield	666.67 (F)	The whole story. 3,000 kg yielded 2,850 kg — a 95% yield against the 90% standard, so 150 kg of extra output came free.
Usage	266.67 (F)	Yield gain net of the mix loss
Cost	166.67 (F)	Usage gain net of the small price loss

$$\text{Standard yield from 3,000 kg of input} = 3,000 \times 90\% = \text{2,700 kg}$$

$$\text{Actual output} = \text{2,850 kg} \rightarrow \text{an actual yield of 95\%}$$

$$\text{Extra output obtained free} = \text{150 kg, worth } 150 \times \text{₹4.4444} = \text{666.67 (F)}$$

The lesson of this problem is that the yield variance dominates. In a chemical process the proportion of raw material converted into finished product matters far more than the price paid for it or the exact recipe used. A five-point improvement in yield was worth ₹666.67, whereas the entire adverse effect of the mix was only ₹400 and of the price only ₹100.

What to recommend. Investigate *why* the yield improved to 95% and whether it can be held — if the improvement is genuine and repeatable, the **standard itself should be revised** from 90% to something nearer 95%. A standard that is consistently beaten has stopped being a control and becomes an excuse. This is the essential maintenance work of any standard costing system.

FINAL ANSWER

Actual price — A / B

₹ 2.1136 / ₹ 9.8125 per kg

SQ for 2,850 kg — A / B

2,375.00 / 791.67 kg

Actual yield vs standard	95% vs 90% — better by 150 kg
Standard cost of actual output	₹ 12,666.67
Actual cost	₹ 12,500
Material Cost Variance	166.67 (F)
Material Price Variance	100 (A)
Material Usage Variance	266.67 (F)
Material Mix Variance	400 (A)
Material Yield Variance	666.67 (F)
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

Q5 Material variances with no loss — the mechanics in their simplest form Material · p.80

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following data compute the material cost variances and reconcile the material variances.

Materials	Standard — units	Standard — price per unit ₹	Actual — units	Actual — price per unit ₹
Zee	3,500	10	3,700	12
Wee	1,500	21	1,650	20
Tee	1,000	33	1,250	36
Total	6,000		6,600	

READ THE QUESTION FIRST — WHAT TO SPOT

The simplest material problem in the set, because **no loss and no separate output figure are given**. The standard quantities are already the standard for the actual output, so there is no scaling to do — SQ is simply 3,500, 1,500 and 1,000.

- **Total AQ of 6,600 units against total SQ of 6,000.** Ten per cent more material was consumed than the standard allows, so the usage variance is going to be substantially adverse.
- **RSQ splits the 6,600 units actually used in the standard ratio 3,500 : 1,500 : 1,000**, which simplifies to 7 : 3 : 2. So RSQ = 3,850, 1,650 and 1,100 — all whole numbers, which makes this a good problem on which to learn the mechanics.
- **Notice Wee.** Actual 1,650 units is exactly its RSQ of 1,650, so Wee contributes **nothing** to the mix variance. Its entire usage variance is yield. That is a neat illustration of what the split actually means.
- **Two prices rose and one fell.** Zee up ₹2, Tee up ₹3, Wee down ₹1.

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
Zee	3,500	3,500	10	3,700	12	3,850
Wee	1,500	1,500	21	1,650	20	1,650
Tee	1,000	1,000	33	1,250	36	1,100
Total	6,000	6,000		6,600		6,600

Two checks before going on. The RSQ column totals 6,600, which equals the AQ total of 6,600 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 1, the ratio of actual output 6,000 to standard output 6,000.

1. MATERIAL COST VARIANCE = SC – AC = (SQ × SP) – (AQ × AP)

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
Zee	3,500	10	35,000	3,700	12	44,400
Wee	1,500	21	31,500	1,650	20	33,000
Tee	1,000	33	33,000	1,250	36	45,000
Total			99,500			1,22,400

$$\text{MCV} = ₹99,500 - ₹1,22,400 = \text{22,900 (A)}$$

MATERIAL PRICE VARIANCE = AQ × (SP – AP)

Material	SP	AP	SP – AP	AQ	Amount ₹
Zee	10	12	(2)	3,700	7,400 (A)
Wee	21	20	1	1,650	1,650 (F)
Tee	33	36	(3)	1,250	3,750 (A)
MATERIAL PRICE VARIANCE					9,500 (A)

MATERIAL USAGE VARIANCE = SP × (SQ – AQ)

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
Zee	3,500	3,700	(200)	10	2,000 (A)
Wee	1,500	1,650	(150)	21	3,150 (A)
Tee	1,000	1,250	(250)	33	8,250 (A)
MATERIAL USAGE VARIANCE					13,400 (A)

MATERIAL MIX VARIANCE = SP × (RSQ – AQ)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
Zee	3,850	3,700	150.00	10	1,500 (F)
Wee	1,650	1,650	—	21	Nil
Tee	1,100	1,250	(150.00)	33	4,950 (A)
MATERIAL MIX VARIANCE					3,450 (A)

MATERIAL YIELD VARIANCE = SP × (SQ – RSQ)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
Zee	3,500	3,850	(350.00)	10	3,500 (A)
Wee	1,500	1,650	(150.00)	21	3,150 (A)
Tee	1,000	1,100	(100.00)	33	3,300 (A)
MATERIAL YIELD VARIANCE					9,950 (A)

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	9,500 (A) + 13,400 (A)	22,900 (A)
and MCV computed directly was		22,900 (A) ✓
MUV = MMV + MYV	3,450 (A) + 9,950 (A)	13,400 (A)
and MUV computed directly was		13,400 (A) ✓

WHY THIS WORKS

Every variance here is adverse except the price variance on Wee. This is a straightforwardly bad month, and the value of the analysis is that it shows **where** the damage was done rather than merely that it happened.

	Amount	Share of the total	Who is answerable
Price variance	9,500 (A)	41%	Purchasing. Zee and Tee were both bought above standard price.
Mix variance	3,450 (A)	15%	Production. Too much Tee, the dearest material at ₹33.
Yield variance	9,950 (A)	43%	Production. 6,600 units of input produced only what 6,000 should have.
Total cost variance	22,900 (A)	100%	

Look at Wee to understand what mix and yield really separate. Wee's usage variance is **3,150 (A)** in total, and it divides as **Nil** mix and **3,150 (A)** yield — the mix part is exactly nil.

Wee: RSQ = 1,650 units and AQ = 1,650 units → the proportion used was **exactly standard**, so no mix effect
Wee: SQ = 1,500 units but RSQ = 1,650 units → 150 extra units were needed because the batch as a whole over-consumed — that is **pure yield**

So mix asks “were the proportions right?” and yield asks “was the total right?”. Wee's proportion was perfect; it simply got dragged along by a batch that used 10% too much material overall.

Recommendation: the yield variance of **9,950 (A)** is the largest single item and points to a process problem — wastage, spoilage, or a specification that is no longer achievable. Investigate that first. The mix variance of **3,450 (A)** is a secondary matter of using too much Tee at ₹33 when Zee at ₹10 was available.

FINAL ANSWER

SQ — Zee / Wee / Tee	3,500 / 1,500 / 1,000 units
RSQ — Zee / Wee / Tee	3,850 / 1,650 / 1,100 units
Total input — standard vs actual	6,000 vs 6,600 units
Standard cost of actual output	₹ 99,500
Actual cost	₹ 1,22,400
Material Cost Variance	22,900 (A)
Material Price Variance	9,500 (A)
Material Usage Variance	13,400 (A)
Material Mix Variance	3,450 (A)
Material Yield Variance	9,950 (A)
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The standard mix of product A2 is as follows:

Kgs	Materials	Price per kg ₹
45	X	6.00
25	Y	4.50
30	Z	9.50
100	Total input	

The standard loss in production is 10% of input. There is no scrap value. Actual production for a month was 7,425 kg of A2 from 80 mixes. Actual consumption and purchases of material during the month were:

Kgs	Materials	Price per kg ₹
4,200	X	6.50
1,700	Y	4.25
2,600	Z	9.75
8,500	Total input	

You are required to calculate the material variances and reconcile the same.

READ THE QUESTION FIRST — WHAT TO SPOT

The standard batch is conveniently 100 kg, so the kilogram figures *are* the percentages. One phrase in the question is a distraction and one is essential.

- **Essential: “standard loss 10% of input”.** 100 kg in gives **90 kg out**. The scale factor to the actual 7,425 kg of output is $\frac{7,425}{90} = 82.5$.
- **Essential: SQ = 45 × 82.5, 25 × 82.5, 30 × 82.5 = 3,712.50, 2,062.50 and 2,475 kg — a total of 8,250 kg**, against 8,500 kg actually used.
- **A distraction: “from 80 mixes”.** Eighty standard mixes would be 8,000 kg of input and 7,200 kg of output. Neither figure is the one to use — actual input was 8,500 kg and actual output 7,425 kg. But the phrase is a useful reality check: **8,500 kg was drawn for what was planned as 80 mixes of 100 kg**, so 500 kg of extra material went in.
- **“No scrap value”** means the loss cannot be credited back. It removes a complication rather than adding one.
- Actual mix is 4,200 : 1,700 : 2,600 on 8,500 kg = 49.4% : 20.0% : 30.6% against a standard of 45% : 25% : 30%. So **more X and much less Y**.

WORKING NOTES

W1 The three quantities, and the 80-mixes check

Standard output per 100 kg batch = 100 – 10% = **90 kg**

Scale factor = $\frac{\text{actual output } 7,425}{\text{standard output } 90} = 82.5$

Standard input allowed for 7,425 kg of output = 100 × 82.5 = **8,250 kg**

Actual input used = **8,500 kg** → **250 kg more than allowed**

The 80-mixes cross-check. 80 mixes should have consumed 8,000 kg and produced 7,200 kg. Actual output of 7,425 kg is 103% of that, but actual input of 8,500 kg is 106% — so input rose faster than output, which is another way of seeing that the yield was poor.

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
X	45	3,712.50	6	4,200	6.50	3,825
Y	25	2,062.50	4.50	1,700	4.25	2,125
Z	30	2,475	9.50	2,600	9.75	2,550
Total	100	8,250		8,500		8,500

Two checks before going on. The RSQ column totals 8,500, which equals the AQ total of 8,500 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 82.50, the ratio of actual output 7,425 to standard output 90.

1. MATERIAL COST VARIANCE = SC – AC = (SQ × SP) – (AQ × AP)

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
X	3,712.50	6	22,275	4,200	6.50	27,300
Y	2,062.50	4.50	9,281.25	1,700	4.25	7,225
Z	2,475	9.50	23,512.50	2,600	9.75	25,350
Total			55,068.75			59,875

$$\text{MCV} = ₹55,068.75 - ₹59,875 = \mathbf{4,806.25 (A)}$$

MATERIAL PRICE VARIANCE = AQ × (SP – AP)

Material	SP	AP	SP – AP	AQ	Amount ₹
X	6	6.50	(0.50)	4,200	2,100 (A)
Y	4.50	4.25	0.25	1,700	425 (F)
Z	9.50	9.75	(0.25)	2,600	650 (A)
MATERIAL PRICE VARIANCE					2,325 (A)

MATERIAL USAGE VARIANCE = SP × (SQ – AQ)

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
X	3,712.50	4,200	(487.50)	6	2,925 (A)
Y	2,062.50	1,700	362.50	4.50	1,631.25 (F)
Z	2,475	2,600	(125)	9.50	1,187.50 (A)
MATERIAL USAGE VARIANCE					2,481.25 (A)

MATERIAL MIX VARIANCE = SP × (RSQ – AQ)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
X	3,825	4,200	(375.00)	6	2,250 (A)

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
Y	2,125	1,700	425.00	4.50	1,912.50 (F)
Z	2,550	2,600	(50.00)	9.50	475 (A)
MATERIAL MIX VARIANCE					812.50 (A)

MATERIAL YIELD VARIANCE = SP × (SQ – RSQ)

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
X	3,712.50	3,825	(112.50)	6	675 (A)
Y	2,062.50	2,125	(62.50)	4.50	281.25 (A)
Z	2,475	2,550	(75.00)	9.50	712.50 (A)
MATERIAL YIELD VARIANCE					1,668.75 (A)

WORKING NOTES

Independent check on the yield variance

Standard cost per unit of output = $\frac{\text{standard cost of one standard batch}}{\text{standard output of that batch}}$ = ₹7.42

Output the actual input SHOULD have yielded = $8,500 \times \frac{90}{100}$ = 7,650 units

Actual output = 7,425 units

MYV = ₹7.42 × (7,425 – 7,650) = 1,668.75 (A)

This route never touches RSQ, so it is free of rounding error and is the best way to **check** the yield variance computed material by material above. The two agree exactly.

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	2,325 (A) + 2,481.25 (A)	4,806.25 (A)
and MCV computed directly was		4,806.25 (A) ✓
MUV = MMV + MYV	812.50 (A) + 1,668.75 (A)	2,481.25 (A)
and MUV computed directly was		2,481.25 (A) ✓

WHY THIS WORKS

Every variance in this problem is adverse except the price of Y and the mix contribution of Y. The overall result is **4,806.25 (A)**.

Material	Standard mix	Actual mix	Price ₹	Mix effect	Comment
X	45.0%	49.4%	6.00	2,250 (A)	4.4 points too much of a mid-priced material
Y	25.0%	20.0%	4.50	1,912.50 (F)	5 points too little of the CHEAPEST material — the only favourable line
Z	30.0%	30.6%	9.50	475 (A)	slightly too much of the dearest material

Material	Standard mix	Actual mix	Price ₹	Mix effect	Comment
Total	100%	100%		812.50 (A)	

The mix variance is favourable only because Y was under-used, and that is the interesting subtlety. Y is the cheapest material at ₹4.50, so using less of it should be *adverse* for the mix — yet the line shows **1,912.50 (F)**. The reason is that the mix variance measures the deviation of each material from **its own** revised standard: RSQ for Y is 2,125 kg but only 1,700 kg was used, so 425 kg of the cheapest material was *saved*, and the formula credits that saving. What the mix variance cannot see is that the 425 kg of Y had to be replaced by something dearer — and that shows up as the adverse lines on X and Z.

The yield variance of **1,668.75 (A)** is the real problem. 8,500 kg of input should have yielded 7,650 kg of A2 but produced only 7,425 kg — a shortfall of 225 kg. The actual loss was 12.6% of input against a standard 10%.

Recommendation: report the price variance of **2,325 (A)** to purchasing — X and Z were both bought above standard, and X at 4,200 kg is the largest single item of spend. Report the yield shortfall to production, and note that reducing Y from 25% to 20% of the mixture may itself be the cause of the poor yield. As in Q1, an apparent economy in the recipe has probably cost more than it saved.

FINAL ANSWER

Scale factor (7,425 ÷ 90)	82.5
SQ — X / Y / Z	3,712.50 / 2,062.50 / 2,475.00 kg
Standard input allowed vs actual	8,250 vs 8,500 kg
Actual loss vs standard loss	12.6% of input vs 10%
Standard cost of actual output	₹ 55,068.75
Actual cost	₹ 59,875
Material Cost Variance	4,806.25 (A)
Material Price Variance	2,325 (A)
Material Usage Variance	2,481.25 (A)
Material Mix Variance	812.50 (A)
Material Yield Variance	1,668.75 (A)
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

Q7 Material variances when the actual quantity must be deduced

Material · p.81

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The standard cost for producing 180 kg of a product whose raw material inputs are A and B is given below:

Material	Standard	₹
A	60 kg at ₹10 per kg	600
B	140 kg at ₹2 per kg	280
Total cost	200 kg of input → 180 kg of output	880

The actual prices of A and B were ₹12 and ₹8 per kg respectively. Consumption of B was 108 kg. The actual output at 80% yield was 144 kg. Calculate all the material variances and reconcile them.

READ THE QUESTION FIRST — WHAT TO SPOT

This question deliberately withholds the actual quantity of A. You must deduce it, and the deduction is the whole difficulty.

1. **Find the actual input from the yield.** “Actual output at 80% yield was 144 kg” means 144 kg is 80% of the input:

$$\text{Actual input} = \frac{144 \text{ kg}}{80\%} = \mathbf{180 \text{ kg}}$$

2. **Then find A by subtraction.** Total input 180 kg less B’s 108 kg gives **A = 72 kg**.
3. **Now the standard.** 200 kg of input gives 180 kg of output, so the standard yield is 90%. Actual output was 144 kg, so the scale factor is $\frac{144}{180} = \mathbf{0.8}$ and SQ = 48 kg of A and 112 kg of B.
 - **Standard yield 90%, actual yield 80%.** A ten-point fall — expect a firmly adverse yield variance.
 - **The mix is badly wrong.** Standard is 60 : 140, i.e. 30% A and 70% B. Actual is 72 : 108, i.e. **40% A and 60% B** — ten points more of the material that costs five times as much.
 - **Both prices rose sharply.** A from ₹10 to ₹12 (+20%) and B from ₹2 to ₹8 — a **fourfold** increase. B’s price alone will dominate the price variance.
 - Every single variance in this problem will be adverse. It is the worst month in the set, and a good one on which to practise writing a critical commentary.

WORKING NOTES

W1 Deducing the actual quantities

Actual output = 144 kg, at an actual yield of 80% of input

$$\text{Actual input} = \frac{144}{0.80} = \mathbf{180 \text{ kg}}$$

Consumption of B = 108 kg (given)

Therefore consumption of A = 180 – 108 = 72 kg

W2 The standard, scaled to the actual output

Standard: 200 kg of input → 180 kg of output, a standard yield of **90%**

$$\text{Scale factor} = \frac{\text{actual output } 144}{\text{standard output } 180} = \mathbf{0.8}$$

SQ = A: 60 × 0.8 = **48 kg** B: 140 × 0.8 = **112 kg** total **160 kg**

Actual input **180 kg** against 160 kg allowed → **20 kg too much**

Basic Calculation

Material	SQ standard mix	SQ for actual output	SP	AQ	AP	RSQ for actual input
A	60	48	10	72	12	54
B	140	112	2	108	8	126
Total	200	160		180		180

Two checks before going on. The RSQ column totals 180, which equals the AQ total of 180 — it must, because RSQ is the same total input in a different mix. And SQ for actual output = standard quantity × 0.80, the ratio of actual output 144 to standard output 180.

$$\mathbf{1. \text{ MATERIAL COST VARIANCE } = \text{SC} - \text{AC} = (\text{SQ} \times \text{SP}) - (\text{AQ} \times \text{AP})}$$

Material	SQ for actual output	SP	SC = SQ × SP	AQ	AP	AC = AQ × AP
A	48	10	480	72	12	864
B	112	2	224	108	8	864
Total			704			1,728

$$\text{MCV} = ₹704 - ₹1,728 = \text{1,024 (A)}$$

$$\text{MATERIAL PRICE VARIANCE} = \text{AQ} \times (\text{SP} - \text{AP})$$

Material	SP	AP	SP – AP	AQ	Amount ₹
A	10	12	(2)	72	144 (A)
B	2	8	(6)	108	648 (A)
MATERIAL PRICE VARIANCE					792 (A)

$$\text{MATERIAL USAGE VARIANCE} = \text{SP} \times (\text{SQ} - \text{AQ})$$

Material	SQ	AQ	SQ – AQ	SP	Amount ₹
A	48	72	(24)	10	240 (A)
B	112	108	4	2	8 (F)
MATERIAL USAGE VARIANCE					232 (A)

$$\text{MATERIAL MIX VARIANCE} = \text{SP} \times (\text{RSQ} - \text{AQ})$$

Material	RSQ	AQ	RSQ – AQ	SP	Amount ₹
A	54	72	(18.00)	10	180 (A)
B	126	108	18.00	2	36 (F)
MATERIAL MIX VARIANCE					144 (A)

$$\text{MATERIAL YIELD VARIANCE} = \text{SP} \times (\text{SQ} - \text{RSQ})$$

Material	SQ	RSQ	SQ – RSQ	SP	Amount ₹
A	48	54	(6.00)	10	60 (A)
B	112	126	(14.00)	2	28 (A)
MATERIAL YIELD VARIANCE					88 (A)

WORKING NOTES

Independent check on the yield variance

$$\text{Standard cost per unit of output} = \frac{\text{standard cost of one standard batch}}{\text{standard output of that batch}} = ₹4.89$$

$$\text{Output the actual input SHOULD have yielded} = 180 \times \frac{180}{200} = 162 \text{ units}$$

$$\text{Actual output} = 144 \text{ units}$$

$$\text{MYV} = ₹4.89 \times (144 - 162) = 88 \text{ (A)}$$

This route never touches RSQ, so it is free of rounding error and is the best way to **check** the yield variance computed material by material above. The two agree exactly.

Reconciliation of the material variances

Identity	Working	Result
MCV = MPV + MUV	792 (A) + 232 (A)	1,024 (A)
and MCV computed directly was		1,024 (A) ✓
MUV = MMV + MYV	144 (A) + 88 (A)	232 (A)
and MUV computed directly was		232 (A) ✓

WHY THIS WORKS

Every variance is adverse, and the total cost variance of **1,024 (A)** is enormous relative to a standard cost of only ₹704 — an overrun of 145%. Three separate things went wrong.

What went wrong	Evidence	Cost
1. Prices rose steeply	A up 20% from ₹10 to ₹12; B up 300% from ₹2 to ₹8	792 (A)
2. The mix moved to the dear material	40% A instead of 30% — and A costs five times B at standard prices	144 (A)
3. The yield collapsed	80% against a standard 90%; 180 kg of input yielded 144 kg where 162 kg was expected	88 (A)
Total material cost variance		1,024 (A)

The price variance is by far the largest, and it is almost entirely B.

Material A = 72 kg × (₹10 – ₹12) = **144 (A)**

Material B = 108 kg × (₹2 – ₹8) = **648 (A)** ← 82% of the whole price variance

Total price variance = 792 (A)

A material quadrupling in price is not a purchasing failure; it is a change in market conditions. **The correct response is to revise the standard**, not to reprimand the buyer. A standard price of ₹2 for a material now costing ₹8 makes every subsequent variance meaningless, because the whole of the reported adverse variance is simply the obsolescence of the standard.

There may also be a connection between the three failures, and pointing it out is what a full answer does. If B quadrupled in price, the works manager had an incentive to substitute A — which is exactly what the mix variance shows. And if the substitution was made hurriedly, using 40% A in a process designed for 30%, the collapse in yield from 90% to 80% may be its direct consequence. On that reading there is **one** root cause, the price of B, and the mix and yield variances are its downstream effects.

Recommendation: revise the standard price of B immediately; investigate whether the altered mix caused the yield loss; and if B is permanently dearer, re-examine whether the 30 : 70 recipe is still the economically correct one rather than merely the standard one.

FINAL ANSWER

Actual input deduced (144 ÷ 80%)	180 kg
Actual quantity of A (180 – 108)	72 kg
SQ — A / B	48 / 112 kg
Yield — standard vs actual	90% vs 80%
Standard cost of actual output	₹ 704
Actual cost	₹ 1,728
Material Cost Variance	1,024 (A)
Material Price Variance	792 (A)

Material Usage Variance	232 (A)
Material Mix Variance	144 (A)
Material Yield Variance	88 (A)
Reconciliation	MCV = MPV + MUV · MUV = MMV + MYV ✓

Q8 All six labour variances, with abnormal idle time

Labour · p.81

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Following information is given regarding standard composition and standard rates of a gang of workers:

Standard composition	Standard hourly rate ₹
10 Men	0.625
5 Women	0.400
5 Boys	0.350

According to given specifications, a week consists of 40 hours and standard output for a week is 1,000 units. In a particular week, the gang consisted of 13 men, 4 women and 3 boys and actual wages were paid as follows: Men at ₹0.600 per hour, Women at ₹0.425 per hour, Boys at ₹0.325 per hour.

Two hours were lost in the week due to abnormal idle time. Actual production was 960 units in the week.

Find out: **(a)** Labour Rate Variance **(b)** Labour Mix Variance **(c)** Labour Idle Time Variance **(d)** Labour Yield Variance **(e)** Labour Efficiency Variance **(f)** Labour Cost Variance. You are also required to reconcile the above variances.

READ THE QUESTION FIRST — WHAT TO SPOT

Labour problems have one complication that material problems do not: **hours paid are not the same as hours worked**. Set out four different hour figures before starting.

Figure	Meaning	Computation	Total
SH for standard output	Hours the standard gang works in a week	(10 + 5 + 5) workers × 40 hours	800 hrs
SH for actual output	Hours the standard gang <i>should</i> take for 960 units	$800 \times \frac{960}{1,000}$	768 hrs
AH — hours PAID	Hours the actual gang was paid for	(13 + 4 + 3) workers × 40 hours	800 hrs
Hours WORKED	Hours in which production actually happened	800 paid – 40 idle	760 hrs

- **“Two hours were lost” means two hours PER WORKER.** With 20 workers that is **40 idle hours** in total. Reading it as two hours altogether is a standard error and it makes the reconciliation fail.
- **By coincidence AH = 800 hours and SH for standard output = 800 hours.** The gang is the same size — 20 workers — but composed differently: 13 men instead of 10, and fewer women and boys. So there is **no capacity difference, only a composition difference**, which is exactly what the mix variance measures.
- **The gang produced 960 units in 800 paid hours** where the standard is 1,000 units in 800 hours. So efficiency is down — and part of that is the idle time, which is why the efficiency variance splits three ways.

- **Men were paid BELOW standard and women and boys ABOVE.** ₹0.600 against ₹0.625 for men; ₹0.425 against ₹0.400 for women; ₹0.325 against ₹0.350 for boys. Mixed signs, so compute each line separately.

THE METHOD FOR THIS TYPE — ALWAYS THESE STEPS

1. **Compute the four hour figures** in the table above: SH for actual output, AH paid, hours worked, and total idle hours.
2. **Compute RSH twice** — once on hours **paid** (for the mix variance) and once on hours **worked** (for the yield variance), both in the standard composition 10 : 5 : 5.
3. **Rate variance** = $AH \times (SR - AR)$, grade by grade.
4. **Efficiency variance** = $SR \times (SH - AH)$, using hours **paid**.
5. **Split the efficiency variance three ways:** mix on hours paid, yield on hours worked, and idle time on the standard composition.
6. **Reconcile.** $LCV = LRV + LEV$, and $LEV = LMV + LYV + LITV$. If the second does not close, the idle hours have been apportioned on the wrong composition.

WORKING NOTES

W1 Idle hours — two hours per worker

Men = 13 workers $\times$ 2 hours = 26 idle hours

Women = 4 workers $\times$ 2 hours = 8 idle hours

Boys = 3 workers $\times$ 2 hours = 6 idle hours

Total idle hours = 40 $\rightarrow$ hours worked = 800 – 40 = **760**

For the **idle time variance** these 40 hours are re-apportioned on the *standard* composition 10 : 5 : 5, giving 20 : 10 : 10 hours. That is the apportionment which makes the reconciliation close, and the reason is algebraic: the idle-time variance must equal $SR \times (RSH \text{ on hours worked} - RSH \text{ on hours paid})$, and both RSH figures are built on the standard proportions.

Basic Calculation — hours

Grade	Standard workers	SH std output	SH for actual output	SR	Actual workers	AH paid	AR	RSH on hours paid	RSH on hours worked
Men	10	400	384	0.625	13	520	0.600	400	380
Women	5	200	192	0.400	4	160	0.425	200	190
Boys	5	200	192	0.350	3	120	0.325	200	190
Total	20	800	768		20	800		800	760

Checks. RSH on hours paid totals 800 = total AH of 800. RSH on hours worked totals 760 = hours paid 800 less idle hours 40. Both must hold.

1. LABOUR COST VARIANCE = $SC - AC = (SH \times SR) - (AH \times AR)$

Grade	SH for actual output	SR	SC	AH	AR	AC
Men	384	0.625	240.00	520	0.600	312.00
Women	192	0.400	76.80	160	0.425	68.00
Boys	192	0.350	67.20	120	0.325	39.00
Total			384.00			419.00

$$LCV = ₹384.00 - ₹419.00 = \mathbf{35 (A)}$$

$$\text{LABOUR RATE VARIANCE} = \text{AH} \times (\text{SR} - \text{AR})$$

Grade	SR	AR	SR – AR	AH	Amount ₹
Men	0.625	0.600	0.025	520	13 (F)
Women	0.400	0.425	(0.025)	160	4 (A)
Boys	0.350	0.325	0.025	120	3 (F)
LABOUR RATE VARIANCE					12 (F)

$$\text{LABOUR EFFICIENCY VARIANCE} = \text{SR} \times (\text{SH} - \text{AH})$$

Grade	SH	AH	SH – AH	SR	Amount ₹
Men	384	520	(136)	0.625	85 (A)
Women	192	160	32	0.400	12.80 (F)
Boys	192	120	72	0.350	25.20 (F)
LABOUR EFFICIENCY VARIANCE					47 (A)

$$\text{LABOUR MIX (GANG COMPOSITION) VARIANCE} = \text{SR} \times (\text{RSH} - \text{AH})$$

Grade	RSH on hours paid	AH	RSH – AH	SR	Amount ₹
Men	400	520	(120)	0.625	75 (A)
Women	200	160	40	0.400	16 (F)
Boys	200	120	80	0.350	28 (F)
LABOUR MIX (GANG COMPOSITION) VARIANCE					31 (A)

$$\text{LABOUR YIELD VARIANCE} = \text{SR} \times (\text{SH} - \text{RSH on hours worked})$$

Grade	SH	RSH on hours worked	SH – RSH	SR	Amount ₹
Men	384	380	4	0.625	2.50 (F)
Women	192	190	2	0.400	0.80 (F)
Boys	192	190	2	0.350	0.70 (F)
LABOUR YIELD VARIANCE					4 (F)

$$\text{LABOUR IDLE TIME VARIANCE} = \text{Idle hours} \times \text{SR} - \text{always ADVERSE}$$

Grade	Idle hours on standard composition	SR	Amount ₹
Men	20	0.625	12.50 (A)
Women	10	0.400	4 (A)
Boys	10	0.350	3.50 (A)
LABOUR IDLE TIME VARIANCE			20 (A)

Total idle hours of 40 apportioned on the **standard** gang composition. Apportioning on the standard rather than the actual composition is what makes the reconciliation $\text{LEV} = \text{LMV} + \text{LYV} + \text{LITV}$ close exactly.

Reconciliation of the labour variances

Identity	Working	Result
LCV = LRV + LEV	12 (F) + 47 (A)	35 (A)

Identity	Working	Result
and LCV computed directly was		35 (A) ✓
LEV = LMV + LYV + LITV	31 (A) + 4 (F) + 20 (A)	47 (A)
and LEV computed directly was		47 (A) ✓

WHY THIS WORKS

The labour cost variance is **35 (A)**, and it divides into a small favourable rate effect and a larger adverse efficiency effect.

Variance	Result	What it says
Rate	12 (F)	Favourable overall. The saving on 520 hours of men at ₹0.025 below standard (13 (F)) outweighed the small overpayments to women and boys.
Mix / gang	31 (A)	Adverse. 13 men instead of 10 — 120 more man-hours at the highest rate of ₹0.625, only partly offset by fewer women and boys.
Idle time	20 (A)	Adverse, and unavoidably so. 40 hours were paid for in which nothing was produced.
Yield	4 (F)	Favourable. In the 760 hours actually worked the gang did slightly better than the standard rate of production.
Efficiency	47 (A)	The three above, netted.
Cost	35 (A)	Rate plus efficiency.

The idle time variance is the single largest adverse item, and it is the useful one. Note carefully that it is described as *abnormal* idle time — a machine breakdown, a power failure, a material shortage. It is not the workers' fault, and it is not the gang leader's fault. Isolating it in its own variance is precisely so that the production supervisor is not held answerable for it.

$$\text{Idle time as a proportion of hours paid} = \frac{40 \text{ hours}}{800 \text{ hours}} = \mathbf{5\% \text{ of the payroll}}$$

Cost of that lost time = **20 (A)**

Had there been no idle time, LEV would have been **27 (A)** instead of **47 (A)**

The mix variance carries a real management message. Substituting men for women and boys costs money at standard rates, because men are the dearest grade. Unless the extra men were genuinely needed, the gang should be restored to its 10 : 5 : 5 composition — that alone is worth ₹31.00 a week.

And notice what the favourable yield variance tells us: in the hours they actually worked, the gang was *more* productive than standard. The problem in this week was not the workers' effort. It was 40 hours of stoppage and an over-expensive gang composition — both management matters.

FINAL ANSWER

Hours — SH for actual output	768 hours
Hours — AH paid	800 hours
Hours — idle (2 per worker × 20)	40 hours
Hours — actually worked	760 hours
Standard cost of actual output	₹ 384.00
Actual cost	₹ 419.00
(a) Labour Rate Variance	12 (F)
(b) Labour Mix / Gang Variance	31 (A)

(c) Labour Idle Time Variance	20 (A)
(d) Labour Yield Variance	4 (F)
(e) Labour Efficiency Variance	47 (A)
(f) Labour Cost Variance	35 (A)
Reconciliation	LCV = LRV + LEV · LEV = LMV + LYV + LITV ✓

Q9 Labour variances with machine breakdown idle time

Labour · p.81-82

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following was the composition of a gang of workers in a factory during a particular month, in one of the production departments. The standard composition of workers and wage rate per hour were as follows:

Grade	Standard composition	Standard rate per hour ₹
Skilled	Two workers	20
Semi-skilled	Four workers	12
Unskilled	Four workers	8

The standard output of the gang was **four units per hour** of the product. During the month in question, however, the actual composition of the gang and hourly rates paid were as under:

Nature of worker	No. of workers	Wage rate paid per worker per hour ₹
Skilled	2	20
Semi-skilled	3	14
Unskilled	5	10

The gang was engaged for **200 hours** during the month, which included **12 hours when no production was possible due to machine break-down**. **810 units** of the product were recorded as output of the gang during the month.

You are required to compute the Labour Rate, Mix, Idle Time, Yield, Efficiency and Cost Variances, and to reconcile them.

READ THE QUESTION FIRST — WHAT TO SPOT

Same structure as Q8, but the output standard is expressed differently and that is where the work lies.

1. **Convert “four units per hour of the gang” into a standard output.** The gang was engaged for 200 hours, so the standard output for that engagement is $200 \times 4 = 800$ units. Actual output was 810 units — slightly **above** standard.
 2. **Standard hours of the gang** = 10 workers $\times$ 200 hours = **2,000 hours** for those 800 units.
 3. **SH for the actual output of 810 units** = $2,000 \times \frac{810}{800} = 2,025$ hours.
 4. **AH paid** = 10 workers $\times$ 200 hours = **2,000 hours**. Idle hours = 10 workers $\times$ 12 = **120 hours**, so hours worked = **1,880**.
- **The gang is again the same size — 10 workers — but differently composed:** 2 : 3 : 5 against a standard 2 : 4 : 4. One semi-skilled worker has been replaced by an unskilled one.
 - **Replacing semi-skilled (₹12) with unskilled (₹8) is FAVOURABLE for the mix**, because the cheaper grade was substituted. Expect a favourable mix variance — the opposite of Q8.

- **But look at the rates paid.** Semi-skilled were paid ₹14 against a standard ₹12, and unskilled ₹10 against ₹8 — both 2 rupees over. Only the skilled grade was paid at standard. Expect a firmly **adverse rate variance**.
- **810 units in 200 hours is 4.05 units an hour** against a standard 4.00 — and that was achieved despite losing 12 hours to breakdown. In the 188 productive hours the rate was 4.31 units an hour. So the gang worked well; expect a favourable yield variance.

WORKING NOTES

W1 Converting the output standard into hours

Standard output of the gang = 4 units per hour × 200 hours engaged = **800 units**

Standard gang hours for those 800 units = 10 workers × 200 hours = **2,000 hours**

SH for the actual 810 units = $2,000 \times \frac{810}{800}$ = **2,025 hours**

AH paid = 10 × 200 = **2,000 hours**

Idle hours = 10 workers × 12 hours = **120 hours** → hours worked = **1,880**

Note the direction of travel. The standard allows 2,025 hours for 810 units but only 2,000 were paid for — so on hours alone the gang was **efficient**. The adverse pressure in this problem comes from the rates paid, not from the hours taken.

Basic Calculation — hours

Grade	Standard workers	SH std output	SH for actual output	SR	Actual workers	AH paid	AR	RSH on hours paid	RSH on hours worked
Skilled	2	400	405	20.000	2	400	20.000	400	376
Semi-skilled	4	800	810	12.000	3	600	14.000	800	752
Unskilled	4	800	810	8.000	5	1,000	10.000	800	752
Total	10	2,000	2,025		10	2,000		2,000	1,880

Checks. RSH on hours paid totals 2,000 = total AH of 2,000. RSH on hours worked totals 1,880 = hours paid 2,000 less idle hours 120. Both must hold.

1. LABOUR COST VARIANCE = SC – AC = (SH × SR) – (AH × AR)

Grade	SH for actual output	SR	SC	AH	AR	AC
Skilled	405	20.000	8,100.00	400	20.000	8,000.00
Semi-skilled	810	12.000	9,720.00	600	14.000	8,400.00
Unskilled	810	8.000	6,480.00	1,000	10.000	10,000.00
Total			24,300.00			26,400.00

$$\text{LCV} = ₹24,300.00 - ₹26,400.00 = \mathbf{2,100 (A)}$$

LABOUR RATE VARIANCE = AH × (SR – AR)

Grade	SR	AR	SR – AR	AH	Amount ₹
Skilled	20.000	20.000	—	400	Nil
Semi-skilled	12.000	14.000	(2.000)	600	1,200 (A)

Grade	SR	AR	SR – AR	AH	Amount ₹
Unskilled	8.000	10.000	(2.000)	1,000	2,000 (A)
LABOUR RATE VARIANCE					3,200 (A)

LABOUR EFFICIENCY VARIANCE = SR × (SH – AH)

Grade	SH	AH	SH – AH	SR	Amount ₹
Skilled	405	400	5	20.000	100 (F)
Semi-skilled	810	600	210	12.000	2,520 (F)
Unskilled	810	1,000	(190)	8.000	1,520 (A)
LABOUR EFFICIENCY VARIANCE					1,100 (F)

LABOUR MIX (GANG COMPOSITION) VARIANCE = SR × (RSH – AH)

Grade	RSH on hours paid	AH	RSH – AH	SR	Amount ₹
Skilled	400	400	—	20.000	Nil
Semi-skilled	800	600	200	12.000	2,400 (F)
Unskilled	800	1,000	(200)	8.000	1,600 (A)
LABOUR MIX (GANG COMPOSITION) VARIANCE					800 (F)

LABOUR YIELD VARIANCE = SR × (SH – RSH on hours worked)

Grade	SH	RSH on hours worked	SH – RSH	SR	Amount ₹
Skilled	405	376	29	20.000	580 (F)
Semi-skilled	810	752	58	12.000	696 (F)
Unskilled	810	752	58	8.000	464 (F)
LABOUR YIELD VARIANCE					1,740 (F)

LABOUR IDLE TIME VARIANCE = Idle hours × SR — always ADVERSE

Grade	Idle hours on standard composition	SR	Amount ₹
Skilled	24	20.000	480 (A)
Semi-skilled	48	12.000	576 (A)
Unskilled	48	8.000	384 (A)
LABOUR IDLE TIME VARIANCE			1,440 (A)

Total idle hours of 120 apportioned on the **standard** gang composition. Apportioning on the standard rather than the actual composition is what makes the reconciliation $LEV = LMV + LYV + LITV$ close exactly.

Reconciliation of the labour variances

Identity	Working	Result
LCV = LRV + LEV	3,200 (A) + 1,100 (F)	2,100 (A)
and LCV computed directly was		2,100 (A) ✓
LEV = LMV + LYV + LITV	800 (F) + 1,740 (F) + 1,440 (A)	1,100 (F)

Identity	Working	Result
and LEV computed directly was		1,100 (F) ✓

WHY THIS WORKS

The cost variance is **2,100 (A)**, and this problem is the mirror image of Q8: there the rate variance was favourable and the mix adverse; here the rate variance is heavily adverse and the mix favourable.

Variance	Result	Cause
Rate	3,200 (A)	The dominant item. Semi-skilled paid ₹14 not ₹12 (1,200 (A)) and unskilled ₹10 not ₹8 (2,000 (A)). Skilled were at standard, so contributed nothing.
Mix / gang	800 (F)	Favourable. One semi-skilled worker at ₹12 replaced by an unskilled one at ₹8 — 200 hours moved to a cheaper grade.
Idle time	1,440 (A)	Adverse. 120 hours paid for during the machine breakdown.
Yield	1,740 (F)	Favourable. 810 units were produced in 1,880 working hours where the standard rate would have given fewer.
Efficiency	1,100 (F)	Mix + yield + idle time
Cost	2,100 (A)	Rate + efficiency

Two management conclusions, and they point in opposite directions.

1. **The rate variance of 3,200 (A) needs explaining.** Paying semi-skilled and unskilled workers ₹2 an hour above standard is a 17% and 25% overrun respectively. Either the standard rates are out of date — in which case they must be revised — or overtime or a higher grade of worker was used on lower-grade work. Both are worth investigating, and the answer determines whether anyone is at fault.
2. **The idle time of 1,440 (A) is a maintenance failure, not a labour failure.** 120 hours — 6% of the payroll — were paid for during a machine breakdown. That is a cost of poor plant maintenance and it should be reported to whoever is responsible for the machine, not to the gang leader.
3. **The gang itself performed well.** Mix and yield together are **2,540 (F)** favourable: a cheaper composition *and* above-standard productivity in the hours available. Had the breakdown not occurred, the efficiency variance would have been **2,540 (F)** rather than **1,100 (F)**.

The point of the whole exercise is visible here. A single cost variance of **2,100 (A)** would tell management only that labour cost more than planned. The decomposition tells them that the workforce did well, the maintenance department did badly, and the wage rates in the standard are probably obsolete — three different actions for three different people.

FINAL ANSWER

Standard output (4 units/hr × 200 hrs)	800 units
Actual output	810 units
SH for actual output	2,025 hours
AH paid / idle / worked	2,000 / 120 / 1,880 hours
Standard cost of actual output	₹ 24,300.00
Actual cost	₹ 26,400.00
(a) Labour Rate Variance	3,200 (A)
(b) Labour Mix / Gang Variance	800 (F)
(c) Labour Idle Time Variance	1,440 (A)
(d) Labour Yield Variance	1,740 (F)
(e) Labour Efficiency Variance	1,100 (F)
(f) Labour Cost Variance	2,100 (A)

Q10 Labour variances for a fifty-worker gang

Labour · p.82

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The standard labour complement and the actual labour complement engaged during the month are given below:

Particulars	Skilled	Semi-skilled	Unskilled
Standard number of workers in a group	30	10	10
Standard wage rate per hour ₹	5	3	2
Actual number of workers employed during the month in the group	24	15	12
Actual wage rate per hour ₹	6	2.50	2

During the month of **200 working hours** (which included **15 hours when no production was possible due to machine break-down**), the group produced **9,600 units**. Standard output of the gang was **50 units per hour**.

Calculate all the labour variances and reconcile them.

READ THE QUESTION FIRST — WHAT TO SPOT

The largest gang in the module — 50 standard workers against 51 actual — so the arithmetic is bigger but the method is identical.

- Standard output** = 50 units per hour × 200 hours = **10,000 units**. Actual output was 9,600 units, so output was **4% short of standard**.
- Standard gang hours** = (30 + 10 + 10) = 50 workers × 200 hours = **10,000 hours** for those 10,000 units — conveniently one hour per unit.
- SH for the actual 9,600 units** = $10,000 \times \frac{9,600}{10,000} = \mathbf{9,600 \text{ hours}}$.
- AH paid** = (24 + 15 + 12) = 51 workers × 200 = **10,200 hours**. Idle hours = 51 × 15 = **765 hours**, so hours worked = **9,435**.
 - This gang is BIGGER than standard** — 51 workers instead of 50, and 10,200 hours paid against 9,600 allowed. So unlike Q8 and Q9 there is a genuine capacity difference as well as a composition difference.
 - The composition shifted away from skilled labour:** 24 skilled instead of 30, but 15 semi-skilled instead of 10 and 12 unskilled instead of 10. Fewer of the dearest grade at ₹5 and more of the cheaper grades — expect a **favourable mix variance**.
 - Skilled workers were paid ₹6 against a standard ₹5**, a 20% overrun on the dearest grade. Semi-skilled were paid ₹0.50 *below* standard and unskilled exactly at standard.
 - Output fell 4% short while hours paid ran 6% over.** That combination guarantees a substantial adverse efficiency variance.

WORKING NOTES

W1 The four hour figures

Standard output = 50 units per hour × 200 hours = **10,000 units**

Standard gang hours = 50 workers × 200 hours = **10,000 hours**

$$\text{SH for the actual 9,600 units} = 10,000 \times \frac{9,600}{10,000} = \mathbf{9,600 \text{ hours}}$$

$$\text{AH paid} = 51 \text{ workers} \times 200 \text{ hours} = \mathbf{10,200 \text{ hours}}$$

$$\text{Idle hours} = 51 \text{ workers} \times 15 \text{ hours} = \mathbf{765 \text{ hours}} \rightarrow \text{hours worked} = \mathbf{9,435}$$

Basic Calculation — hours

Grade	Standard workers	SH std output	SH for actual output	SR	Actual workers	AH paid	AR	RSH on hours paid	RSH on hours worked
Skilled	30	6,000	5,760	5.000	24	4,800	6.000	6,120	5,661
Semi-skilled	10	2,000	1,920	3.000	15	3,000	2.500	2,040	1,887
Unskilled	10	2,000	1,920	2.000	12	2,400	2.000	2,040	1,887
Total	50	10,000	9,600		51	10,200		10,200	9,435

Checks. RSH on hours paid totals 10,200 = total AH of 10,200. RSH on hours worked totals 9,435 = hours paid 10,200 less idle hours 765. Both must hold.

$$\mathbf{1. \text{ LABOUR COST VARIANCE} = \text{SC} - \text{AC} = (\text{SH} \times \text{SR}) - (\text{AH} \times \text{AR})}$$

Grade	SH for actual output	SR	SC	AH	AR	AC
Skilled	5,760	5.000	28,800.00	4,800	6.000	28,800.00
Semi-skilled	1,920	3.000	5,760.00	3,000	2.500	7,500.00
Unskilled	1,920	2.000	3,840.00	2,400	2.000	4,800.00
Total			38,400.00			41,100.00

$$\mathbf{LCV = ₹38,400.00 - ₹41,100.00 = 2,700 (A)}$$

$$\mathbf{\text{LABOUR RATE VARIANCE} = \text{AH} \times (\text{SR} - \text{AR})}$$

Grade	SR	AR	SR - AR	AH	Amount ₹
Skilled	5.000	6.000	(1.000)	4,800	4,800 (A)
Semi-skilled	3.000	2.500	0.500	3,000	1,500 (F)
Unskilled	2.000	2.000	—	2,400	Nil
LABOUR RATE VARIANCE					3,300 (A)

$$\mathbf{\text{LABOUR EFFICIENCY VARIANCE} = \text{SR} \times (\text{SH} - \text{AH})}$$

Grade	SH	AH	SH - AH	SR	Amount ₹
Skilled	5,760	4,800	960	5.000	4,800 (F)
Semi-skilled	1,920	3,000	(1,080)	3.000	3,240 (A)
Unskilled	1,920	2,400	(480)	2.000	960 (A)
LABOUR EFFICIENCY VARIANCE					600 (F)

$$\mathbf{\text{LABOUR MIX (GANG COMPOSITION) VARIANCE} = \text{SR} \times (\text{RSH} - \text{AH})}$$

Grade	RSH on hours paid	AH	RSH – AH	SR	Amount ₹
Skilled	6,120	4,800	1,320	5.000	6,600 (F)
Semi-skilled	2,040	3,000	(960)	3.000	2,880 (A)
Unskilled	2,040	2,400	(360)	2.000	720 (A)
LABOUR MIX (GANG COMPOSITION) VARIANCE					3,000 (F)

LABOUR YIELD VARIANCE = SR × (SH – RSH on hours worked)

Grade	SH	RSH on hours worked	SH – RSH	SR	Amount ₹
Skilled	5,760	5,661	99	5.000	495 (F)
Semi-skilled	1,920	1,887	33	3.000	99 (F)
Unskilled	1,920	1,887	33	2.000	66 (F)
LABOUR YIELD VARIANCE					660 (F)

LABOUR IDLE TIME VARIANCE = Idle hours × SR — always ADVERSE

Grade	Idle hours on standard composition	SR	Amount ₹
Skilled	459	5.000	2,295 (A)
Semi-skilled	153	3.000	459 (A)
Unskilled	153	2.000	306 (A)
LABOUR IDLE TIME VARIANCE			3,060 (A)

Total idle hours of 765 apportioned on the **standard** gang composition. Apportioning on the standard rather than the actual composition is what makes the reconciliation $LEV = LMV + LYV + LITV$ close exactly.

Reconciliation of the labour variances

Identity	Working	Result
LCV = LRV + LEV	3,300 (A) + 600 (F)	2,700 (A)
and LCV computed directly was		2,700 (A) ✓
LEV = LMV + LYV + LITV	3,000 (F) + 660 (F) + 3,060 (A)	600 (F)
and LEV computed directly was		600 (F) ✓

WHY THIS WORKS

The cost variance is **2,700 (A)**. This is the most instructive of the three labour problems because the mix variance and the yield variance point in **opposite** directions, and the reason is a real operational trade-off.

Variance	Result	What happened
Rate	3,300 (A)	Skilled paid ₹6 not ₹5 on 4,800 hours (4,800 (A)), partly recovered by paying semi-skilled ₹0.50 below standard (1,500 (F)).
Mix / gang	3,000 (F)	Favourable. Six skilled workers at ₹5 replaced by five semi-skilled at ₹3 and two unskilled at ₹2 — a cheaper gang.
Idle time	3,060 (A)	Adverse. 765 hours paid for during the breakdown — 7.5% of the payroll.

Variance	Result	What happened
Yield	660 (F)	Favourable. In the 9,435 hours actually worked the gang produced 9,600 units — slightly better than the standard rate of production.
Efficiency	600 (F)	Mix + yield + idle time
Cost	2,700 (A)	Rate + efficiency

The de-skilling worked. Management took six skilled workers out and put in five semi-skilled and two unskilled. Mix and yield together come to **3,660 (F)** favourable — the gang was made cheaper *and* it held its productivity.

Favourable mix variance from the cheaper gang = **3,000 (F)**

Favourable yield variance from productivity in the hours worked = **660 (F)**

Net gain from the de-skilling decision = 3,660 (F)

So why is the cost variance 2,700 (A)? Because of two things that have nothing to do with the gang's composition or its effort:

1. **765 hours of machine breakdown, costing 3,060 (A).** That is equivalent to nearly four workers being paid to do nothing for the whole month. It is a **maintenance** failure and must be reported as such, not charged against the gang.
2. **Skilled workers paid ₹6 an hour against a standard ₹5, costing 4,800 (A).** A 20% overrun on every one of 4,800 skilled hours, partly recovered by paying semi-skilled ₹0.50 below standard (**1,500 (F)**), giving a net rate variance of **3,300 (A)**.

Those two items together are **6,360 (A)** adverse, against **3,660 (F)** favourable from the gang itself. **That is the finding to report: the workforce performed well and was let down by plant breakdown and by wage rates that no longer match the standard.**

A SUBTLETY WORTH THE EXTRA MARK

The gang used **more hours in total** than the standard allows — 10,200 paid against 9,600 allowed for the output achieved, 600 hours over — and yet the efficiency variance is **600 (F) favourable**. That is not a contradiction. The hours *saved* were in the expensive skilled grade and the hours *overspent* were in the cheap grades:

Grade	SH allowed	AH used	Difference	SR ₹	Money effect
Skilled	5,760	4,800	960	5.00	4,800 (F)
Semi-skilled	1,920	3,000	(1,080)	3.00	3,240 (A)
Unskilled	1,920	2,400	(480)	2.00	960 (A)
Total	9,600	10,200	(600)		600 (F)

960 skilled hours were saved at ₹5 each, which is worth more than the 1,560 cheap hours added at ₹3 and ₹2. **Counting hours and counting money give opposite answers, and money is the one that matters.** This is the best illustration in the module of why variances are expressed in rupees rather than in hours.

FINAL ANSWER

Standard output (50 units/hr × 200 hrs)	10,000 units
Actual output	9,600 units
SH for actual output	9,600 hours
AH paid / idle / worked	10,200 / 765 / 9,435 hours
Workers — standard vs actual	50 vs 51

Standard cost of actual output	₹ 38,400.00
Actual cost	₹ 41,100.00
(a) Labour Rate Variance	3,300 (A)
(b) Labour Mix / Gang Variance	3,000 (F)
(c) Labour Idle Time Variance	3,060 (A)
(d) Labour Yield Variance	660 (F)
(e) Labour Efficiency Variance	600 (F)
(f) Labour Cost Variance	2,700 (A)
Reconciliation	LCV = LRV + LEV · LEV = LMV + LYV + LITV ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Following information is obtained from M/s Will and Urvish Co. Ltd.

Particulars	Amount
Budgeted production for the period	600 units
Budgeted variable overhead	₹15,600
Standard time for one unit	20 hours
Actual production for the period	500 units
Actual variable overhead	₹14,000
Actual hours worked	9,000

Calculate: (a) Variable overhead expenditure variance (b) Variable overhead efficiency variance (c) Variable overhead variance. You are also required to reconcile the same.

READ THE QUESTION FIRST — WHAT TO SPOT

Variable overhead is the easiest of the four cost families, because it has only **two** branches. But the standard rate must be derived first, and it must be derived **per hour**, not per unit.

1. **Budgeted hours** = 600 units × 20 hours = **12,000 hours**.

2. **Standard rate** = $\frac{₹15,600}{12,000 \text{ hours}}$ = **₹1.30 per hour**. This is the figure every later step uses.

3. **Standard hours for the ACTUAL output** = 500 units × 20 hours = **10,000 hours**. Note this is not the 12,000 budgeted hours — output fell short, so fewer hours are allowed.

4. **Then the two variances**, and the reconciliation.

- **Actual hours of 9,000 against 10,000 allowed for the output achieved.** The company produced 500 units in 9,000 hours where the standard permits 10,000 — so it was **efficient**, and the efficiency variance will be favourable.
- **But actual overhead of ₹14,000 against ₹11,700 that 9,000 hours should have cost.** Spending per hour was far above standard, so the expenditure variance will be adverse and larger.
- **No idle time is mentioned**, so there are only two branches. Q12 adds the third.

Basic Calculation

Particulars	Computation	Amount
Budgeted hours (BH)	600 units × 20 hours	12,000 hours
Standard variable overhead rate (SR)	$\frac{\text{budgeted VOH ₹15,600}}{\text{budgeted hours 12,000}}$	₹1.30 per hour
Standard hours for actual output (SH)	500 units × 20 hours	10,000 hours
Actual hours (AH)	given	9,000 hours
Standard variable overhead for actual output (SC)	10,000 hours × ₹1.30	₹13,000
Actual variable overhead	given	₹14,000

The variable overhead variances

Variance	Working	Result
1. Variable Overhead Cost Variance <i>SC – Actual VOH</i>	₹13,000 – ₹14,000	1,000 (A)
2. VO Expenditure (Spending) Variance <i>(AH × SR) – Actual VOH</i>	(9,000 × 1.30) – 14,000 = 11,700 – 14,000	2,300 (A)
3. VO Efficiency Variance <i>SR × (SH – AH)</i>	1.30 × (10,000 – 9,000)	1,300 (F)

Reconciliation

Identity	Working	Result
VOCV = Expenditure + Efficiency	2,300 (A) + 1,300 (F)	1,000 (A)
<i>and VOCV computed directly was</i>		1,000 (A) ✓

WHY THIS WORKS

The two variances point in opposite directions, and that is the whole information content of the answer.

	Result	Meaning
Efficiency	1,300 (F)	Good. 500 units were made in 9,000 hours where the standard allows 10,000 — the factory saved 1,000 hours, worth ₹1.30 each.
Expenditure	2,300 (A)	Bad, and worse. Those 9,000 hours should have carried ₹11,700 of variable overhead but actually carried ₹14,000 - ₹1.56 an hour instead of ₹1.30.
Cost variance	1,000 (A)	The overspend outweighed the time saved.

$$\begin{aligned}\text{Actual variable overhead per hour} &= \frac{\text{₹14,000}}{9,000 \text{ hours}} = \text{₹1.5556} \\ \text{Standard variable overhead per hour} &= \text{₹1.30} \\ \text{Overspend per hour} &= \text{₹0.2556} \rightarrow \text{₹0.2556} \times 9,000 \text{ hours} = \text{2,300 (A)}\end{aligned}$$

What to report to management. The production department worked efficiently — it used 1,000 fewer hours than the standard allows — and should be told so. The problem is entirely on the spending side: variable overhead cost ₹0.26 an hour more than it should have. Since variable overhead means power, consumables, indirect materials and the like, the investigation should look at prices and at consumption per hour, **not** at the workforce.

A subtlety worth noticing. There is a possible connection: working faster often means running machines harder, which consumes more power and more consumables per hour. If so, part of the **2,300 (A)** is the price of the **1,300 (F)** — and the net **1,000 (A)** says the trade was not worth making. Whether that is the explanation can only be settled by investigation, but raising the question is what a real answer does.

FINAL ANSWER

Budgeted hours (600 × 20)	12,000 hours
Standard rate (15,600 ÷ 12,000)	₹ 1.30 per hour
Standard hours for actual output (500 × 20)	10,000 hours
Standard VOH for actual output	₹ 13,000
Actual VOH	₹ 14,000
(a) VO Expenditure Variance	2,300 (A)
(b) VO Efficiency Variance	1,300 (F)

(c) VO Cost Variance**1,000 (A)**

Reconciliation

Expenditure + Efficiency = Cost ✓**Q12 Variable overhead variances with abnormal idle time**

Variable overhead · p.83

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following information pertaining to January, calculate the overhead variances:

Particulars	Production in units	Variable overheads ₹	Hours worked
Budgeted	300	7,800	see Note 1
Actual	250	7,000	4,500 see Note 2

- **Note 1:** Standard time for 1 unit is 20 hours.
- **Note 2:** This includes **300 hours of abnormal idle time**.

READ THE QUESTION FIRST — WHAT TO SPOT

Q11 with one addition: **abnormal idle time**. That turns the two-branch analysis into three branches, and the way the three fit together needs care.

1. **Budgeted hours** = 300 units × 20 hours = **6,000 hours**; standard rate = $\frac{₹7,800}{6,000} = ₹1.30 \text{ per hour}$.
2. **Standard hours for actual output** = 250 units × 20 = **5,000 hours**.
3. **Separate the actual hours**. 4,500 hours were paid for, of which 300 were abnormal idle time, so **4,200 hours were productive**.
4. **Use the two figures in different places** — and this is the point of the problem:
 - **Expenditure variance uses the TOTAL 4,500 hours**, because overhead was incurred on all of them.
 - **Efficiency variance uses the PRODUCTIVE 4,200 hours**, because efficiency means output per hour actually worked.
 - **Idle time variance bridges the gap** — 300 hours × ₹1.30.
- **If you use the same hours figure in both places the reconciliation still closes**, but the idle-time variance then has nowhere to go and the analysis loses its whole point. Use 4,500 for expenditure and 4,200 for efficiency.
- **250 units in 4,200 productive hours** is 16.8 hours a unit against a standard 20 — so the workforce was efficient once it had work to do.

Basic Calculation

Particulars	Computation	Amount
Budgeted hours (BH)	300 units × 20 hours	6,000 hours
Standard variable overhead rate (SR)	$\frac{\text{budgeted VOH ₹7,800}}{\text{budgeted hours 6,000}}$	₹1.30 per hour
Standard hours for actual output (SH)	250 units × 20 hours	5,000 hours
Actual hours (AH)	given	4,500 hours
<i>of which</i> abnormal idle time	given	300 hours
Productive hours worked	4,500 – 300	4,200 hours

Particulars	Computation	Amount
Standard variable overhead for actual output (SC)	5,000 hours × ₹1.30	₹6,500
Actual variable overhead	given	₹7,000

The variable overhead variances

Variance	Working	Result
1. Variable Overhead Cost Variance <i>SC – Actual VOH</i>	₹6,500 – ₹7,000	500 (A)
2. VO Expenditure (Spending) Variance <i>(AH × SR) – Actual VOH</i>	(4,500 × 1.30) – 7,000 = 5,850 – 7,000	1,150 (A)
3. VO Idle Time Variance <i>Idle hours × SR — always adverse</i>	300 hours × 1.30	390 (A)
4. VO Efficiency Variance <i>SR × (SH – productive hours)</i>	1.30 × (5,000 – 4,200)	1,040 (F)

Reconciliation

Identity	Working	Result
VOCV = Expenditure + Idle time + Efficiency	1,150 (A) + 390 (A) + 1,040 (F)	500 (A)
<i>and VOCV computed directly was</i>		500 (A) ✓

WHY THIS WORKS

The total variance of **500 (A)** looks trivial. The decomposition shows that it is the net of three substantial and quite different effects.

Variance	Result	Whose responsibility	Action
Expenditure	1,150 (A)	The department controlling overhead spending	4,500 hours should have carried ₹5,850 but carried ₹7,000 - investigate prices and consumption
Idle time	390 (A)	Maintenance / production planning	300 hours - 6.7% of the total - were paid for with nothing produced
Efficiency	1,040 (F)	The workforce - credit due	250 units in 4,200 hours = 16.8 hours a unit against a standard 20
Cost variance	500 (A)		the three above, netted

This is the clearest illustration in the module of why variances are decomposed at all. A manager told only that variable overhead was ₹500 over budget would reasonably conclude that nothing much had happened. In fact:

Overhead was overspent by **1,150 (A)** — 16% of the actual overhead incurred
A further **390 (A)** was wasted on 300 idle hours
And the workforce **earned back 1,040 (F)** by working faster than standard
Net: 500 (A) — which conceals all three

Recommendation: the **1,150 (A)** expenditure variance is the priority — actual variable overhead ran at ₹1.5556 an hour against a standard ₹1.30, an overrun of 20%. Then address the 300 hours of abnormal idle

time. And tell the production team that their efficiency saved ₹1,040, because a variance report that only ever reports bad news stops being read.

FINAL ANSWER

Budgeted hours (300×20)	6,000 hours
Standard rate ($7,800 \div 6,000$)	₹ 1.30 per hour
Standard hours for actual output (250×20)	5,000 hours
Actual hours — total / idle / productive	4,500 / 300 / 4,200 hours
Standard VOH for actual output	₹ 6,500
VO Expenditure Variance	1,150 (A)
VO Idle Time Variance	390 (A)
VO Efficiency Variance	1,040 (F)
VO Cost Variance	500 (A)
Reconciliation	Expenditure + Idle time + Efficiency = Cost ✓

Q13 Variable overhead variances when the standard time must be derived

Variable overhead · p.83

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following data is given. Calculate variable overhead variances.

Particulars	Production in units	Man hours to produce the output	Variable overheads ₹
Budgeted	400	8,000	10,000
Actual	360	7,000	9,150

READ THE QUESTION FIRST — WHAT TO SPOT

Here the standard time per unit is **not stated** — it must be derived from the budget, and that is the only real step.

1. **Standard time per unit** = $\frac{8,000 \text{ budgeted hours}}{400 \text{ budgeted units}} = \mathbf{20 \text{ hours per unit.}}$
2. **Standard rate per hour** = $\frac{₹10,000}{8,000 \text{ hours}} = \mathbf{₹1.25 \text{ per hour.}}$
3. **Standard hours for the actual output** = 360 units × 20 hours = **7,200 hours.**
4. **Then the two variances**, on 7,000 actual hours.
 - **Do not use the ₹25 per unit figure** (10,000 ÷ 400) as a rate. Variable overhead variances are computed **per hour**, because that is the basis on which variable overhead is assumed to vary. Per-unit rates give the right cost variance but the wrong split.
 - **7,000 actual hours against 7,200 allowed** for the output achieved — so 200 hours were saved and the efficiency variance is favourable.
 - **₹9,150 of actual overhead against ₹8,750 that 7,000 hours should have cost** — so the expenditure variance is adverse and slightly larger.
 - No idle time is given, so two branches again.

Basic Calculation

Particulars	Computation	Amount
Budgeted hours (BH)	400 units × 20 hours	8,000 hours
Standard variable overhead rate (SR)	$\frac{\text{budgeted VOH ₹10,000}}{\text{budgeted hours 8,000}}$	₹1.25 per hour
Standard hours for actual output (SH)	360 units × 20 hours	7,200 hours
Actual hours (AH)	given	7,000 hours
Standard variable overhead for actual output (SC)	7,200 hours × ₹1.25	₹9,000
Actual variable overhead	given	₹9,150

The variable overhead variances

Variance	Working	Result
1. Variable Overhead Cost Variance SC – Actual VOH	₹9,000 – ₹9,150	150 (A)

Variance	Working	Result
2. VO Expenditure (Spending) Variance <i>(AH × SR) – Actual VOH</i>	$(7,000 \times 1.25) - 9,150 = 8,750 - 9,150$	400 (A)
3. VO Efficiency Variance <i>SR × (SH – AH)</i>	$1.25 \times (7,200 - 7,000)$	250 (F)

Reconciliation

Identity	Working	Result
VOCV = Expenditure + Efficiency	400 (A) + 250 (F)	150 (A)
<i>and VOCV computed directly was</i>		150 (A) ✓

WHY THIS WORKS

The pattern is the same as Q11 and Q13 is the smallest of the three in magnitude: a cost variance of only **150 (A)** on ₹9,150 of actual overhead, which is a 1.6% overrun.

Efficiency = $₹1.25 \times (7,200 - 7,000) = 250 (F)$ — 200 hours saved
Expenditure = $(7,000 \times ₹1.25) - ₹9,150 = ₹8,750 - ₹9,150 = 400 (A)$
Net = 150 (A)

Two observations for the marks.

- **Actual overhead per hour was ₹1.3071 against a standard ₹1.25** — an overrun of 4.6%. Small in percentage terms, which is why the expenditure variance is modest.
- **The output shortfall does NOT of itself create an adverse variance.** Production was 360 units against 400 budgeted — a 10% shortfall — yet the variances are almost nil. That is because variable overhead is *flexed*: the standard allowed is 7,200 hours for 360 units, not the 8,000 budgeted for 400. This is the flexible budget of Module 5 doing its work inside a standard costing system, and saying so links the two modules.

Recommendation: variable overhead was essentially under control. Note the favourable efficiency variance of **250 (F)** and the small adverse expenditure variance of **400 (A)**; neither warrants investigation on its own, but if the expenditure variance recurs month after month the standard rate of ₹1.25 an hour should be reviewed.

FINAL ANSWER

Standard time per unit $(8,000 \div 400)$	20 hours
Standard rate $(10,000 \div 8,000)$	₹ 1.25 per hour
Standard hours for actual output (360×20)	7,200 hours
Standard VOH for actual output	₹ 9,000
Actual VOH	₹ 9,150
VO Expenditure Variance	400 (A)
VO Efficiency Variance	250 (F)
VO Cost Variance	150 (A)
Reconciliation	Expenditure + Efficiency = Cost ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following cost data, calculate the fixed overhead variances and reconcile the same.

Particulars	Budgeted	Actual
No. of working days	20	22
Man-hours per day	8,000	8,400
Output per man-hour in units	1.0	0.9
Overhead cost ₹	1,60,000	1,68,000

READ THE QUESTION FIRST — WHAT TO SPOT

Nothing in this question is given directly — every figure the formulas need must be built from the four rows. Build all of them first, in a table, before touching a formula.

Figure needed	Budgeted	Actual	How
Total hours	1,60,000	1,84,800	working days × man-hours per day
Total output in units	1,60,000	1,66,320	total hours × output per man-hour
Overhead cost ₹	1,60,000	1,68,000	given
Overhead rate per hour ₹	1.00	0.9091	overhead cost ÷ total hours
SH for actual output	—	1,66,320	1,66,320 units × 1 hour per unit
RSH — revised standard hours	—	1,76,000	22 actual days × 8,000 budgeted hours per day

- **Output per man-hour is 1.0 budgeted and 0.9 actual.** So budgeted output equals budgeted hours (1,60,000 of each, which is a coincidence that makes the sums easy) but actual output of 1,66,320 is only 90% of the 1,84,800 hours worked.
- **The standard rate is ₹1.00 per hour** — and because output per hour is 1.0, it is also ₹1.00 per unit. Convenient, but do not confuse the two: the formulas work in **hours**.
- **RSH uses ACTUAL days but BUDGETED hours per day.** $22 \times 8,000 = 1,76,000$. This is the figure that isolates the calendar variance — it asks “how many hours would we have expected simply because the month was longer?”
- **Three quite different reasons the volume changed**, and RSH is what separates them: the factory worked **more days** (calendar), it worked **more hours per day** (capacity), and it produced **less per hour** (efficiency).

WORKING NOTES

W1 Building every figure the formulas need

Budgeted hours = 20 days × 8,000 = **1,60,000 hours**

Actual hours = 22 days × 8,400 = **1,84,800 hours**

Budgeted output = 1,60,000 × 1.0 = **1,60,000 units**

Actual output = 1,84,800 × 0.9 = **1,66,320 units**

Standard rate = $\frac{₹1,60,000}{1,60,000 \text{ hours}}$ = **₹1.00 per hour**

SH for actual output = 1,66,320 units × 1 hour = **1,66,320 hours**

$$\text{RSH} = 22 \text{ actual days} \times 8,000 \text{ budgeted hours} = \mathbf{1,76,000 \text{ hours}}$$

Basic Calculation

Particulars	Computation	Amount
Budgeted hours (BH)	given / derived	1,60,000 hours
Standard fixed overhead rate (SR)	$\frac{\text{budgeted FOH ₹1,60,000}}{\text{budgeted hours 1,60,000}}$	₹1.00 per hour
Standard hours per unit	$\frac{1,60,000 \text{ hours}}{1,60,000 \text{ units}}$	1 hours
Standard hours for actual output (SH)	1,66,320 units × 1 hours	1,66,320 hours
Actual hours (AH)	given	1,84,800 hours
Revised standard hours (RSH)	$1,60,000 \times \frac{22 \text{ actual days}}{20 \text{ budgeted days}}$	1,76,000 hours
Standard fixed overhead for actual output (SC)	1,66,320 hours × ₹1.00	₹1,66,320
Actual fixed overhead	given	₹1,68,000

The fixed overhead variances

Variance	Working	Result
1. Fixed Overhead Cost Variance <i>SC – Actual FOH</i>	₹1,66,320 – ₹1,68,000	1,680 (A)
2. FO Expenditure Variance <i>Budgeted FOH – Actual FOH</i>	₹1,60,000 – ₹1,68,000	8,000 (A)
3. FO Volume Variance <i>SR × (SH – BH)</i>	1.00 × (1,66,320 – 1,60,000)	6,320 (F)

Analysis of the volume variance

Variance	Working	Result
3a. Calendar Variance <i>SR × (RSH – BH)</i>	1.00 × (1,76,000 – 1,60,000)	16,000 (F)
3b. Capacity Variance <i>SR × (AH – RSH)</i>	1.00 × (1,84,800 – 1,76,000)	8,800 (F)
3c. Efficiency Variance <i>SR × (SH – AH)</i>	1.00 × (1,66,320 – 1,84,800)	18,480 (A)

Reconciliation

Identity	Working	Result
FOCV = Expenditure + Volume	8,000 (A) + 6,320 (F)	1,680 (A)
<i>and FOCV computed directly was</i>		1,680 (A) ✓
Volume = Calendar + Capacity + Efficiency	16,000 (F) + 8,800 (F) + 18,480 (A)	6,320 (F)
<i>and the Volume Variance computed directly was</i>		6,320 (F) ✓

WHY THIS WORKS

The cost variance is only **1,680 (A)**, which on ₹1,68,000 of overhead looks like a quiet month. The decomposition shows it was anything but.

Variance	Result	What actually happened
Expenditure	8,000 (A)	The company spent ₹1,68,000 where it budgeted ₹1,60,000 — a 5% overspend on a cost that is supposed to be fixed . That is the item to investigate.
Calendar	16,000 (F)	The month had 22 working days instead of 20. Two extra days at 8,000 budgeted hours absorbed ₹16,000 more overhead. Nothing to do with performance — it is the calendar.
Capacity	8,800 (F)	The factory ran 8,400 hours a day instead of 8,000. Working the plant harder absorbed a further ₹8,800.
Efficiency	18,480 (A)	The bad news. Output fell to 0.9 units per hour from 1.0. Those 1,84,800 hours should have produced 1,84,800 units; they produced 1,66,320.
Volume	6,320 (F)	The three above, netted
Cost	1,680 (A)	Expenditure plus volume

The story is that a large efficiency failure was almost entirely hidden by working longer.

Output per hour fell from 1.0 to 0.9 — a **10% loss of productivity**

That cost **18,480 (A)** in unabsorbed fixed overhead

It was masked by working 2 extra days (**16,000 (F)**) and 400 extra hours a day (**8,800 (F)**)

Net volume variance: 6,320 (F) — apparently favourable

Had the factory worked its budgeted 20 days of 8,000 hours at the actual 0.9 productivity, output would have been only 1,44,000 units and the volume variance would have been a heavily adverse ₹16,000. The favourable **6,320 (F)** is bought with overtime and extra working days, not earned by good performance.

Recommendation: report the efficiency variance of **18,480 (A)** as the principal finding. It is by far the largest number in the analysis and it is entirely a performance matter. Report the expenditure overrun of **8,000 (A)** separately, since a fixed cost exceeding its budget by 5% needs explanation. Do **not** present the volume variance of **6,320 (F)** as an achievement.

A note on the arithmetic. Some solutions to this problem compute the actual overhead as 1,84,800 hours × ₹0.91 = ₹1,68,168 and report the cost variance as ₹1,848 (A). That is a rounding artefact: the actual overhead is **given** as ₹1,68,000, and ₹1,68,000 ÷ 1,84,800 is ₹0.9091, not ₹0.91. The correct cost variance is **1,680 (A)**, and it is provable — it must equal the expenditure variance of **8,000 (A)** plus the volume variance of **6,320 (F)**, and only ₹1,680 does.

FINAL ANSWER

Budgeted / actual hours	1,60,000 / 1,84,800 hours
Budgeted / actual output	1,60,000 / 1,66,320 units
Standard rate	₹ 1.00 per hour
SH for actual output	1,66,320 hours
RSH (22 days × 8,000)	1,76,000 hours
FO Expenditure Variance	8,000 (A)
FO Calendar Variance	16,000 (F)
FO Capacity Variance	8,800 (F)
FO Efficiency Variance	18,480 (A)
FO Volume Variance	6,320 (F)
FO Cost Variance	1,680 (A)

Q15 Fixed overhead variances without a calendar variance

Fixed overhead · p.83

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

Cost data given for Nishu & Co. for July 1993 is as follows:

Particulars	Budgeted	Actual
Fixed overheads	₹10,000	₹10,200
Units of production	5,000	5,200
Standard time for one unit	4 hours	—
Actual hours worked	—	20,100 hours

Find out the fixed overhead variances and reconcile them.

READ THE QUESTION FIRST — WHAT TO SPOT

Simpler than Q14 in one important respect: **no working-days data is given**, so there is **no calendar variance**. The volume variance splits two ways instead of three.

- Budgeted hours** = 5,000 units × 4 hours = **20,000 hours**.
- Standard rate** = $\frac{₹10,000}{20,000 \text{ hours}}$ = **₹0.50 per hour**.
- SH for actual output** = 5,200 units × 4 hours = **20,800 hours**.
- Capacity variance uses BH directly**, because there is no RSH: $SR \times (AH - BH)$.
 - **This is a genuinely good month** and the only favourable cost variance among the four overhead problems. Output beat budget by 200 units (4%) using only 100 extra hours (0.5%).
 - **20,800 standard hours allowed against 20,100 actually worked** — 700 hours better than standard, so a firmly **favourable efficiency variance**.
 - **Fixed overhead was ₹200 over budget**, so the expenditure variance is adverse — but only by 2%.
 - **Do not invent a calendar variance**. Without working-days data it cannot be computed, and saying so explicitly is worth a mark: the capacity variance then absorbs the whole difference between actual and budgeted hours.

Basic Calculation

Particulars	Computation	Amount
Budgeted hours (BH)	given / derived	20,000 hours
Standard fixed overhead rate (SR)	$\frac{\text{budgeted FOH ₹10,000}}{\text{budgeted hours 20,000}}$	₹0.50 per hour
Standard hours per unit	$\frac{20,000 \text{ hours}}{5,000 \text{ units}}$	4 hours
Standard hours for actual output (SH)	5,200 units × 4 hours	20,800 hours
Actual hours (AH)	given	20,100 hours

Particulars	Computation	Amount
Standard fixed overhead for actual output (SC)	20,800 hours × ₹0.50	₹10,400
Actual fixed overhead	given	₹10,200

The fixed overhead variances

Variance	Working	Result
1. Fixed Overhead Cost Variance <i>SC – Actual FOH</i>	₹10,400 – ₹10,200	200 (F)
2. FO Expenditure Variance <i>Budgeted FOH – Actual FOH</i>	₹10,000 – ₹10,200	200 (A)
3. FO Volume Variance <i>SR × (SH – BH)</i>	0.50 × (20,800 – 20,000)	400 (F)

Analysis of the volume variance

Variance	Working	Result
3a. Capacity Variance <i>SR × (AH – BH)</i>	0.50 × (20,100 – 20,000)	50 (F)
3b. Efficiency Variance <i>SR × (SH – AH)</i>	0.50 × (20,800 – 20,100)	350 (F)

Reconciliation

Identity	Working	Result
FOCV = Expenditure + Volume	200 (A) + 400 (F)	200 (F)
<i>and FOCV computed directly was</i>		200 (F) ✓
Volume = Capacity + Efficiency	50 (F) + 350 (F)	400 (F)
<i>and the Volume Variance computed directly was</i>		400 (F) ✓

WHY THIS WORKS

A favourable cost variance of **200 (F)** — the only one in the overhead problems — and the decomposition shows it was properly earned.

Variance	Result	Reason
Expenditure	200 (A)	Fixed overhead of ₹10,200 against a budget of ₹10,000 — a 2% overspend, small but real.
Capacity	50 (F)	20,100 hours were worked against 20,000 budgeted. 100 extra hours absorbed ₹50 more overhead.
Efficiency	350 (F)	The main favourable item. 5,200 units were produced in 20,100 hours where the standard allows 20,800 — 700 hours better than standard.
Volume	400 (F)	Capacity plus efficiency — ₹400 of fixed overhead over-absorbed
Cost	200 (F)	Volume gain less the small expenditure overrun

Contrast this with Q14 and the difference is instructive.

	Q14	Q15
Standard hours allowed for the actual output	1,66,320	20,800

	Q14	Q15
Actual hours worked	1,84,800	20,100
Hours saved / (wasted)	(18,480)	700
Efficiency variance	18,480 (A)	350 (F)
Volume variance	6,320 (F)	400 (F)
Why the volume variance was favourable	extra days and hours, despite bad efficiency	genuine efficiency

In Q14 the favourable volume variance came from working longer while producing less per hour. Here it comes from producing **more per hour**. The two look similar in the summary line and are completely different in substance — which is the whole argument for splitting the volume variance rather than reporting it as one figure.

Recommendation: report the efficiency variance of **350 (F)** as a genuine achievement. Investigate the **200 (A)** expenditure variance only if it recurs — a 2% overrun on a fixed cost is within normal tolerance. And state clearly that **no calendar variance can be computed**, because the number of working days is not given; the whole **50 (F)** difference between actual and budgeted hours is therefore reported as capacity.

FINAL ANSWER

Budgeted hours (5,000 × 4)	20,000 hours
Standard rate (10,000 ÷ 20,000)	₹ 0.50 per hour
SH for actual output (5,200 × 4)	20,800 hours
Actual hours	20,100 hours
Standard FOH for actual output	₹ 10,400
FO Expenditure Variance	200 (A)
FO Capacity Variance	50 (F)
FO Efficiency Variance	350 (F)
FO Volume Variance	400 (F)
FO Cost Variance	200 (F)
Calendar variance	Not applicable — no working-days data

Q16 All five sales value variances

Sales · p.84

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

From the following data, calculate: **(a)** Sales Value Variance **(b)** Sales Price Variance **(c)** Sales Mix Variance **(d)** Sales Quantity Variance **(e)** Sales Value Volume Variance.

Product	Budget — Qty (kg)	Budget — Std. sales price ₹ per kg	Budget — Total ₹	Actual — Qty (kg)	Actual — sales price ₹ per kg	Actual — Total ₹
X	500	5.00	2,500	500	5.00	2,500
Y	400	6.00	2,400	600	6.25	3,750
Z	300	7.00	2,100	400	6.75	2,700
Total	1,200		7,000	1,500		8,950

READ THE QUESTION FIRST — WHAT TO SPOT

Stop and change the convention. Every problem so far has used STANDARD minus ACTUAL. Sales variances reverse it.

$$\text{Sales variance} = \text{ACTUAL} - \text{BUDGET}$$

More revenue is favourable, so a positive answer is (F). If you keep the cost convention every sign in this answer will be wrong.

- **RBQ — revised budgeted quantity — is the sales twin of RSQ.** Take the total actual quantity of 1,500 kg and re-split it in the **budgeted** mix of 500 : 400 : 300, which is 5 : 4 : 3. So RBQ = 625, 500 and 375 kg, totalling 1,500 kg.
- **Product X is the control case.** Same quantity, same price — so X has no price variance and no volume variance. But it **does** have a mix variance and a quantity variance, and they cancel. That is worth understanding: X's share of the total *fell* even though its own sales did not change, because Y and Z grew.
- **Y sold above standard price and Z below.** Y at ₹6.25 against ₹6.00, Z at ₹6.75 against ₹7.00.
- **Total quantity rose 25%** from 1,200 to 1,500 kg, so the quantity variance will be large and favourable.

Basic Calculation

Product	BQ budgeted qty	SP std price	Budgeted value	AQ actual qty	AP actual price	Actual value	RBQ actual total in budgeted mix
X	500	5.00	2,500	500	5.00	2,500	625
Y	400	6.00	2,400	600	6.25	3,750	500
Z	300	7.00	2,100	400	6.75	2,700	375
Total	1,200		7,000	1,500		8,950	1,500

RBQ working: the budgeted mix is 500 : 400 : 300 = 5 : 4 : 3 of 12 parts. $X = 1,500 \times \frac{5}{12} = 625$ · $Y = 1,500 \times \frac{4}{12} = 500$ · $Z = 1,500 \times \frac{3}{12} = 375$. Total 1,500 kg = total AQ

(a) SALES VALUE VARIANCE = Actual sales value – Budgeted sales value

Product	Budgeted value ₹	Actual value ₹	Variance ₹
X	2,500	2,500	Nil
Y	2,400	3,750	1,350 (F)
Z	2,100	2,700	600 (F)
Sales Value Variance	7,000	8,950	1,950 (F)

(b) SALES PRICE VARIANCE = AQ × (AP – SP)

Product	AP	SP	AP – SP	AQ	Amount ₹
X	5.00	5.00	—	500	Nil
Y	6.25	6.00	0.25	600	150 (F)
Z	6.75	7.00	(0.25)	400	100 (A)
(b) SALES PRICE VARIANCE					50 (F)

(c) SALES MIX VARIANCE = SP × (AQ – RBQ)

Product	AQ	RBQ	AQ – RBQ	SP	Amount ₹
X	500	625	(125)	5.00	625 (A)
Y	600	500	100	6.00	600 (F)
Z	400	375	25	7.00	175 (F)
(c) SALES MIX VARIANCE					150 (F)

(d) SALES QUANTITY VARIANCE = SP × (RBQ – BQ)

Product	RBQ	BQ	RBQ – BQ	SP	Amount ₹
X	625	500	125	5.00	625 (F)
Y	500	400	100	6.00	600 (F)
Z	375	300	75	7.00	525 (F)
(d) SALES QUANTITY VARIANCE					1,750 (F)

(e) SALES VALUE VOLUME VARIANCE = SP × (AQ – BQ)

Product	AQ	BQ	AQ – BQ	SP	Amount ₹
X	500	500	—	5.00	Nil
Y	600	400	200	6.00	1,200 (F)
Z	400	300	100	7.00	700 (F)
(e) SALES VALUE VOLUME VARIANCE					1,900 (F)

Reconciliation of the sales variances

Identity	Working	Result
Sales Value Variance = Price + Volume	50 (F) + 1,900 (F)	1,950 (F)
and Sales Value Variance computed directly was		1,950 (F) ✓
Volume = Mix + Quantity	150 (F) + 1,750 (F)	1,900 (F)
and Volume computed directly was		1,900 (F) ✓

WHY THIS WORKS

Sales beat budget by **1,950 (F)** — a 27.9% increase on a budget of ₹7,000 — and the analysis shows that almost all of it came from selling **more**, not from charging more.

Variance	Result	Share	What it says
Price	50 (F)	3%	Y sold ₹0.25 above standard (150 (F)) but Z sold ₹0.25 below (100 (A)). Almost a wash.
Mix	150 (F)	8%	The mix moved towards Z at ₹7 and away from X at ₹5 — selling a higher proportion of the dearer products.
Quantity	1,750 (F)	90%	The whole story. Total volume rose from 1,200 to 1,500 kg — a 25% increase.
Volume	1,900 (F)	97%	Mix plus quantity
Sales value	1,950 (F)	100%	Price plus volume

Product X repays close attention, because it is the one product whose own figures did not change at all — 500 kg at ₹5 both budgeted and actual — and yet it produces two non-zero variances.

X mix variance = ₹5 × (500 – 625) = **625 (A)**

X quantity variance = ₹5 × (625 – 500) = **625 (F)**

X volume variance = Nil — they cancel exactly

The reason is that mix and quantity measure two different things and X moved on one but not the other. Total sales grew 25%, so X *should* have grown to 625 kg to keep its share; it stayed at 500, so it lost mix. But the quantity variance credits X with the share of the overall 25% growth that a product of its size would have contributed. The two effects are equal and opposite because X's own sales were flat. **Only the total volume variance of Nil is real for X**; the split is an accounting attribution.

What to report. A favourable sales value variance of **1,950 (F)** driven by a 25% volume increase is good news, and the quantity variance of **1,750 (F)** is the item to highlight. Two cautions belong with it:

- **Sales value variances say nothing about profit.** Selling 300 extra kilograms also cost money to produce. The company may have sold more and earned less — which is exactly why Q17 uses the **margin** basis instead.
- **Z's price was cut by ₹0.25** and its volume rose by a third. That may be cause and effect, and if so the **100 (A)** price variance is the price paid for the **700 (F)** volume gain — a good trade on these numbers, but one worth making consciously rather than discovering afterwards.

FINAL ANSWER

RBQ — X / Y / Z	625 / 500 / 375 kg
Budgeted / actual sales value	₹ 7,000 / ₹ 8,950
(a) Sales Value Variance	1,950 (F)
(b) Sales Price Variance	50 (F)
(c) Sales Mix Variance	150 (F)
(d) Sales Quantity Variance	1,750 (F)
(e) Sales Value Volume Variance	1,900 (F)
Reconciliation	Value = Price + Volume · Volume = Mix + Quantity ✓

THE QUESTION, AS PRINTED IN YOUR WORKBOOK

The following information of Lucky & Co. is given for December 2007.

Budgeted sales

Product	Units	Selling price ₹	Standard cost ₹
A	500	2.00	1.75
B	700	1.50	1.30

Actual sales

Product	Units	Selling price ₹
A	560	1.95
B	710	1.40

Find out: **(a)** Sales Margin Price Variance **(b)** Sales Margin Mix Variance **(c)** Sales Margin Quantity Variance **(d)** Total Sales Margin Variance **(e)** Sales Margin Volume Variance.

READ THE QUESTION FIRST — WHAT TO SPOT

The word **margin** changes everything. Q16 measured revenue; this question measures **profit**. Every formula is the same shape but the multiplier is the margin per unit, not the selling price.

$$\text{Margin per unit} = \text{Selling price} - \text{STANDARD cost}$$

The standard cost is used in BOTH the budgeted and the actual margin. That is deliberate: it keeps cost variances out of the sales analysis, so a sales manager is judged only on prices and volumes and not on the factory's efficiency.

Product	Selling price ₹	Standard cost ₹	Margin per unit ₹
A — standard margin	2.00	(1.75)	0.25
A — actual margin	1.95	(1.75)	0.20
B — standard margin	1.50	(1.30)	0.20
B — actual margin	1.40	(1.30)	0.10

- **Both products were sold below their standard price**, so both actual margins are lower than standard. A fell from ₹0.25 to ₹0.20; B fell from ₹0.20 to ₹0.10 — **B lost half its margin**.
- **Note how heavily a small price cut hits the margin**. B's price fell 6.7% (from ₹1.50 to ₹1.40) but its margin fell **50%**. That is the whole reason for analysing on a margin basis: a 6.7% discount does not cost 6.7% of profit.
- **RBQ = total actual quantity in the budgeted mix**. Total actual is 1,270 units and the budgeted mix is 500 : 700, so $\text{RBQ} = 1,270 \times \frac{5}{12} = 529.17$ and $1,270 \times \frac{7}{12} = 740.83$. Keep two decimals.
- **Volumes rose only slightly** — A up 60 units, B up 10 — so the volume variances will be small and the price effect will dominate.

Basic Calculation — on the MARGIN basis

Product	BQ	Standard margin ₹	Budgeted margin ₹	AQ	Actual margin ₹	Actual margin ₹	RBQ
A	500	0.25	125.00	560	0.20	112.00	529.17
B	700	0.20	140.00	710	0.10	71.00	740.83
Total	1,200		265.00	1,270		183.00	1,270

RBQ working: budgeted mix 500 : 700 = 5 : 7 of 12 parts. A = $1,270 \times \frac{5}{12} = 529.17$ · B = $1,270 \times \frac{7}{12} = 740.83$. Total 1,270 units = total AQ **Budgeted total margin ₹265.00** against an **actual total margin of ₹183.00**.

(a) SALES MARGIN PRICE VARIANCE = AQ × (Actual margin – Standard margin)

Product	Actual margin	Std margin	Difference	AQ	Amount ₹
A	0.20	0.25	(0.05)	560	28 (A)
B	0.10	0.20	(0.10)	710	71 (A)
(a) SALES MARGIN PRICE VARIANCE					99 (A)

(b) SALES MARGIN MIX VARIANCE = Standard margin × (AQ – RBQ)

Product	AQ	RBQ	AQ – RBQ	Std margin	Amount ₹
A	560	529.17	30.83	0.25	7.71 (F)
B	710	740.83	(30.83)	0.20	6.17 (A)
(b) SALES MARGIN MIX VARIANCE					1.54 (F)

(c) SALES MARGIN QUANTITY VARIANCE = Standard margin × (RBQ – BQ)

Product	RBQ	BQ	RBQ – BQ	Std margin	Amount ₹
A	529.17	500	29.17	0.25	7.29 (F)
B	740.83	700	40.83	0.20	8.17 (F)
(c) SALES MARGIN QUANTITY VARIANCE					15.46 (F)

(d) TOTAL SALES MARGIN VARIANCE = Actual margin – Budgeted margin

Product	Budgeted margin ₹	Actual margin ₹	Variance ₹
A	125.00	112.00	13 (A)
B	140.00	71.00	69 (A)
Total Sales Margin Variance	265.00	183.00	82 (A)

(e) SALES MARGIN VOLUME VARIANCE = Standard margin × (AQ – BQ)

Product	AQ	BQ	AQ – BQ	Std margin	Amount ₹
A	560	500	60	0.25	15 (F)
B	710	700	10	0.20	2 (F)
(e) SALES MARGIN VOLUME VARIANCE					17 (F)

Reconciliation of the sales variances

Identity	Working	Result
Total Sales Margin Variance = Margin Price + Margin Volume	99 (A) + 17 (F)	82 (A)
<i>and Total Sales Margin Variance computed directly was</i>		82 (A) ✓
Margin Volume = Margin Mix + Margin Quantity	1.54 (F) + 15.46 (F)	17 (F)
<i>and Margin Volume computed directly was</i>		17 (F) ✓

WHY THIS WORKS

This is the most important problem in the module, because it contradicts Q16. Sales volume rose and yet profit fell.

	Budget	Actual	Change
Units sold	1,200	1,270	+70 units (+5.8%)
Sales value ₹	2,050.00	2,086.00	+36.00
Total margin (profit) ₹	265.00	183.00	82 (A)

The company sold 70 more units and took in ₹36.00 more revenue — and its profit fell by ₹82.00. On a sales *value* basis this would have looked like a good month. On a **margin** basis it was a bad one.

Variance	Result	What it says
Margin price	99 (A)	The whole problem. Both products were discounted. A lost ₹0.05 of margin on 560 units (28 (A)); B lost ₹0.10 on 710 units (71 (A)).
Margin mix	1.54 (F)	Marginally favourable — the mix moved slightly towards A, which carries the higher standard margin of ₹0.25 against B's ₹0.20.
Margin quantity	15.46 (F)	Favourable — 70 more units were sold in total, each carrying a standard margin.
Margin volume	17 (F)	Mix plus quantity — the gain from selling more
Total sales margin	82 (A)	The discounting cost nearly six times what the extra volume earned.

The arithmetic of the trade-off, and this is the answer to write out:

Gained by selling 70 more units = **17 (F)**

Lost by discounting both products = **99 (A)**

Net = 82 (A)

The discount cost **5.8 times** what the extra volume earned

Why the discount was so expensive. Because the margins are thin, a small cut in price is a large cut in profit. Product B is the clearest case:

B's price fell from ₹1.50 to ₹1.40 — a cut of **6.7%**

But B's margin fell from ₹0.20 to ₹0.10 — a cut of **50%**

To stand still on profit, B's volume would have had to **double**. It rose by **1.4%**.

Recommendation: withdraw the discounts. On these margins the price reductions cannot be recovered by any realistic increase in volume — B would need to sell twice as many units simply to break even on the ₹0.10 cut. If the discounts were a response to competition, the company should recognise that at a standard cost of

₹1.30 against a price of ₹1.40 it has almost no room to compete on price, and should look at cost reduction instead.

The examinable point in one sentence: sales *value* variances measure revenue and sales *margin* variances measure profit, and when prices are cut to win volume the two give **opposite** answers. That is why margin variances are the ones that matter for decision-making.

FINAL ANSWER

Standard margin — A / B	₹ 0.25 / ₹ 0.20 per unit
Actual margin — A / B	₹ 0.20 / ₹ 0.10 per unit
RBQ — A / B	529.17 / 740.83 units
Budgeted total margin	₹ 265.00
Actual total margin	₹ 183.00
(a) Sales Margin Price Variance	99 (A)
(b) Sales Margin Mix Variance	1.54 (F)
(c) Sales Margin Quantity Variance	15.46 (F)
(d) Total Sales Margin Variance	82 (A)
(e) Sales Margin Volume Variance	17 (F)
Reconciliation	Total = Price + Volume · Volume = Mix + Quantity ✓